Satisficing: Data Cleaning & Variables Generation | Revision

anonymized

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0.1 Get data

- load data

```
## Wave 1

# drop all-zero/all-missing columns (keeps only variables with at least one
# value > 0); applied to every imported .dta file

# --- import: wave 1 main file (not shown in the output) ---
```

```
w1p_full <- w1p_full |>  select(where(~any(.x > 0, na.rm = TRUE)))



# w1 with breakoff-codes
# --- import: wave 1 break-off file ---
```

```
w1p_bo <- w1p_bo |>
  select(where(~any(.x > 0, na.rm = TRUE)))


# sample design weights (bland, gkpol7, gebjahr, age)
# --- import: design weights ---
```

```
# --- import: wave 1 raw (unverified) file; reused below for the w1 durations ---
```

```
w1p_up <- w1p_up_raw |>
  select(c(lfd, psundvote, psunpartoth, pedu01, peduoth01))


## Wave 2

# --- import: wave 2 break-off file ---
```

```
w2p_bo <- w2p_bo |>
  select(where(~any(.x > 0, na.rm = TRUE)))

# w2 durations
# --- import: wave 2 durations file ---
```

```
w2_dur <- w2_dur |>
  select(where(~any(.x > 0, na.rm = TRUE)))


# raw data
# --- import: wave 2 raw (unverified) file ---
```

```
w2up <- w2up |>
  select(
    lfd,
    ppartytend, pothpartend,
    pdiscrim12, pdiscrimoth)



## Wave 3

# import once; kept raw so later joins can reuse it
# --- import: wave 3 main file ---
```

```
w3p_full <- w3p_full_raw |>
  filter(ZG %in% c(1,2)) |>
  select(where(~any(.x > 0, na.rm = TRUE))) |>
  select(-c(cid, syear))
#pid, cid, hid
#pgmode

# --- import: wave 3 durations file ---
```

```
w3_dur <- w3_dur |>
  filter(ZG %in% c(1,2)) |>
  select(where(~any(.x > 0, na.rm = TRUE)))


# raw data
# --- import: wave 3 raw (unverified) file ---
```

```
w3_up <- w3_up |>
  filter(ZG %in% c(1,2)) |> select(pid, pedu01.raw = pedu01, peduoth01.raw = peduoth01,
                                   poccstat.raw = poccstat, poccupation_isco08,
                                   plifevent14.raw = plifevent14,
                                   plifevoth14.raw = plifevoth14,
                                   pnetmodwill02.raw = pnetmodwill02,
                                   pmember11.raw = pmember11,
                                   pmemberoth.raw = pmemberoth,
                                   pdevice05.raw = pdevice05,
                                   pothdevice.raw = pothdevice) |>
  left_join(
    w3p_full_raw |>
      # free-text item: network module: other reason for unwillingness (rds)
```

```
      select(pid, pnetmodwill03), by = "pid")



mode_palette <- c("PAPI" = "#1C7984", "CAWI" = "#702963")

posn_d <- position_dodge(0.4)
```

1 Wave 1

2 Adjust Variable psunpartoth and peduoth

```
w1p_up |>
  mutate(
    nonsub = case_when(
      psundvote == 8 &
      ((psunpartoth %in% c("...", "....", "xxx", "XXX", "abc", "ABC", "K.a.", "k.A.")) |
        grepl("-1", psunpartoth) |
        grepl("-9", psunpartoth)
       ) ~ 1,
      TRUE ~ 0)

    ) |> frq(nonsub)
```

```
nonsub <numeric>
# total N=17031 valid N=17031 mean=0.00 sd=0.07

Value |     N | Raw % | Valid % | Cum. %
--------------------------------------
    0 | 16956 | 99.56 |   99.56 |  99.56
    1 |    75 |  0.44 |    0.44 | 100.00
 <NA> |     0 |  0.00 |    <NA> |   <NA>
```

```
# 75 missings, 99 missings incl. nonsubstantial
```

```
w1p_up <-
w1p_up |>
  mutate(
    psundvoteOthWhich = case_when(

      # only applies to those who answered "Other"
      psundvote != 8 ~ -2,

      # responded by naming a political party
      psundvote == 8 &
        # longest: Partei Mensch, Umwelt, Tierschutz
        (nchar(psunpartoth) >= 3 & nchar(psunpartoth) <= 35 &

        !(psunpartoth %in% c("...", "....", "xxx", "XXX", "abc", "ABC", "K.a.", "k.A.")) &
        !grepl("-1", psunpartoth) &
        !grepl("-9", psunpartoth) &
        !grepl("KEIN", psunpartoth) &
        !grepl("Kein", psunpartoth) &
        !grepl("Frage", psunpartoth) &
```

```
      !grepl("nicht", psunpartoth) &
      !grepl("NICHT", psunpartoth) &
      !grepl("UMGÜLTIG", psunpartoth) &
      !grepl("GEHEIM", psunpartoth)) ~ 1,


    psundvote == 8 & !(

      (nchar(psunpartoth) >= 3 & nchar(psunpartoth) <= 35) &

      !(psunpartoth %in% c("...", "....", "xxx", "XXX", "abc", "ABC", "K.a.", "k.A.")) &
      !grepl("-1", psunpartoth) &
      !grepl("-9", psunpartoth) &
      !grepl("KEIN", psunpartoth) &
      !grepl("Kein", psunpartoth) &
      !grepl("Frage", psunpartoth) &
      !grepl("nicht", psunpartoth) &
      !grepl("NICHT", psunpartoth) &
      !grepl("UMGÜLTIG", psunpartoth) &
      !grepl("GEHEIM", psunpartoth)) ~ -1),




  peduOthWhich = case_when(
    pedu01 != 6  ~ -2,

    pedu01 == 6 & (nchar(peduoth01) >= 3 & #nchar(peduoth01) <= 20 &

      !(peduoth01 %in% c("...", "....", "xxx", "XXX", "abc", "ABC", "K.a.", "k.A.")) &
      !grepl("-1", peduoth01) &
      !grepl("-9", peduoth01)) ~ 1,


    pedu01 == 6 & !(
      nchar(peduoth01) >= 3 & #nchar(peduoth01) <= 20 &

      !(peduoth01 %in% c("...", "....", "xxx", "XXX", "abc", "ABC", "K.a.", "k.A.")) &
      !grepl("-1", peduoth01) &
      !grepl("-9", peduoth01)) ~ -1))
```

3 Join w1

3.1 w1 Age & Age_cat & exclude Age Inconsistency and No Panelconsent

```
w1p_complete <-
  # join the vars from df weights_full...
  weights_full |> select(lfd, gebjahr.weights = gebjahr) |>
  # ...to w1p_full
  right_join(w1p_full, by = "lfd") |>
  # only take those vars and...
  select(lfd, pid, hid, gebjahr.weights) |>
  # join them to w1p_bo, from which
```

```
right_join(w1p_bo |> rename(peduoth01.bo = peduoth01,
                            psunpartoth.bo = psunpartoth,
                            pedu01.bo = pedu01,
                            pedu02.bo = pedu02,
                            psundvote.bo = psundvote),
           by = "lfd") |>
# ... you rename one variable
rename(pbirthyear_w1 = pbirthyear

       ) |>

left_join(w2p_bo |>
            select(lfd,
                   pbirthyear_w2 = pbirthyear,
                   intvollst_w2 = intvollst,
                   plifesat01_w2 = plifesat01), by = "lfd") |>
left_join(w3p_full |>
            select(pid,
                   pbirthyear_w3 = pbirthyear,
                   intvollst_w3 = intvollst), by = "pid") |>
# join pnoinc from raw data
left_join(w1p_up |>
            rename(peduoth01.up = peduoth01,
                   pedu01.up = pedu01,
                   psundvote.up = psundvote,
                   psunpartoth.up = psunpartoth), by = "lfd") |>

# use the raw (unverified) data for the free text, but take the break-off code from w1p_bo
mutate(

   peduOthWhich = case_when(pedu01.bo == -10 | peduoth01.bo == -10  ~-10,
                        TRUE ~ peduOthWhich),

   psundvoteOthWhich = case_when(psundvote.bo == -10| psunpartoth.bo == -10 ~-10,
                            TRUE  ~ psundvoteOthWhich),

   psundvoteOthWhich.text = case_when(psundvoteOthWhich == -10 ~ "-10",
                           TRUE ~ psunpartoth.up),

   peduOthWhich.text = case_when(peduOthWhich == -10 ~ "-10",
                           TRUE ~ peduoth01.up),


       # bring ppanelcons to its original pattern
       ppanelcons = case_when(
         intvollst == 3 ~ -10,
         TRUE ~ ppanelcons),


    # if the respondent broke off before the first item, count as non-participation
       intvollst_w2 = case_when(
         plifesat01_w2 == -10 ~ NA,
         TRUE ~ intvollst_w2)
   ) |>

rename(
```

```
    pedu01 = pedu01.bo,
    psundvote = psundvote.bo,
    pedu02 = pedu02.bo
  ) |>

  select(
    - (contains("edu") & (ends_with(".bo") | ends_with(".up"))),
    - (contains("undvote") & (ends_with(".bo") | ends_with(".up"))),
    - contains("psunpartoth"))

# N = 33 obs. is.na(pbirthyear_w1) => substituted N = 7 obs. with information from
 ↪  gebjahr.weights, pbirthyear_w2, pbirthyear_w3

w1p_complete <-
  w1p_complete |> mutate(

     PartWaves = case_when(

      ((intvollst_w2 %in% c(1, 2, 3)) & (intvollst_w3 %in% c(1, 2, 3))) ~ "3 waves",
      intvollst_w2 %in% c(1, 2, 3) | intvollst_w3 %in% c(1, 2, 3) ~ "2 waves",

      TRUE ~ "1 wave"),

    PartWaves = factor(PartWaves, levels = c("1 wave", "2 waves", "3 waves")),


    byear = case_when(
        pbirthyear_w1 > 0 ~ as.double(pbirthyear_w1),
        (is.na(pbirthyear_w1) | (pbirthyear_w1 < 0) & !is.na(gebjahr.weights)) ~
 ↪  as.double(gebjahr.weights),

        (is.na(pbirthyear_w1) | (pbirthyear_w1 < 0)) &
          is.na(gebjahr.weights) &
          pbirthyear_w2 > 0 ~ as.double(pbirthyear_w2),

        (is.na(pbirthyear_w1) | (pbirthyear_w1 < 0)) &
          is.na(gebjahr.weights) &
          (is.na(pbirthyear_w2) | (pbirthyear_w2 < 0)) &
          pbirthyear_w3 > 0 ~ as.double(pbirthyear_w3),
        TRUE ~ NA
    ),

    bmnth = case_when(
      is.na(byear)       ~ NA,
      pbirthmnth > 0     ~ pbirthmnth,
      TRUE               ~ 1),

    myob = my(paste(bmnth, byear, sep = "-")),

    ymd_interview = ymd(interviewdatum),

  Age = floor(decimal_date(ymd_interview) - decimal_date(myob))) |>

  filter(!is.na(pid)) #|>
  # this removes 2 duplicate entries (lfd %in% c(...))
# nrow() #=17027
```

```
w1p_adults <-
  w1p_complete |>
# exclude n = 4 obs. of Age < 18 (2x16, 2x17yrs)
    filter(Age >= 18 | is.na(Age))  # n = 17,023


w1p_adults <-
  w1p_adults |>
# find  observations whose pbirthyear statements differ between the waves
  #AgeConsistency: helper variable
  mutate(AgeConsistency = case_when(
     (as.double(pbirthyear_w1) > 0 & as.double(pbirthyear_w2) > 0 & as.double(pbirthyear_w1) !=
 ↪  as.double(pbirthyear_w2)) |
     (as.double(pbirthyear_w1) > 0 & as.double(pbirthyear_w3) > 0 & as.double(pbirthyear_w1) !=
 ↪  as.double(pbirthyear_w3)) |
     (as.double(pbirthyear_w2) > 0 & as.double(pbirthyear_w3) > 0 & as.double(pbirthyear_w2) !=
 ↪  as.double(pbirthyear_w3)) ~ "Inconsistent",
      TRUE ~ "Not inconsistent")) #|>   frq(AgeConsistency) N = 279 inconsistent: THESE ARE
       ↪  STILL INCLUDED


w1p <-
  w1p_adults |>
  mutate(
    Sample =
      case_when(
        PartWaves %in% "1 wave" |
        AgeConsistency == "Inconsistent" ~ "sensan",
       PartWaves %in% c("2 waves", "3 waves") & AgeConsistency == "Not inconsistent" ~ "Sample"

      )
  )

w1p |> frq(Sample)
```

```
Sample <character>
# total N=17023 valid N=17023 mean=1.45 sd=0.50

Value  |    N | Raw % | Valid % | Cum. %
---------------------------------------
Sample | 9447 | 55.50 |   55.50 |  55.50
sensan | 7576 | 44.50 |   44.50 | 100.00
<NA>   |    0 |  0.00 |    <NA> |   <NA>
```

```
quantile(w1p$Age, prob = c(0.25, 0.5, 0.75), na.rm = TRUE)
```

```
25% 50% 75%
 35  53  65
```

```
#35, 53, 65
```

CAVE: - The data frame `w1p |> filter(Sample == "sensan")` is used for the sensitivity analysis. It contains all individuals aged 18 and older who participated in at least the first wave; only 2 duplicate entries were excluded. - Subsequent computations use only `w1p |> filter(Sample %in% c("Sample"))`

```r
# quantiles do not differ between the samples:

w1p |>
     filter(Sample == "Sample") |>
     pull(Age) |>
     summary()
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
  18.00   35.00   53.00   50.62   65.00   97.00       1
```

```r
#   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
#  18.00   35.00   53.00   50.62   65.00   97.00       1
 w1p |>
   #  filter(Sample == "Sample") |>
     pull(Age) |>
     summary()
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
  18.00   35.00   53.00   51.22   65.00  101.00      26
```

```r
#   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
#  18.00   35.00   53.00   51.22   65.00  101.00      26
```

```r
# function to create quartile groups with range labels
create_quantile_with_range <- function(x, varname) {
  # Compute quartiles
  quantile_group <- ntile(x, 3)

  # find the minimum/maximum values per group
  ranges <- tapply(x, quantile_group, function(vals) {
    c(min(vals, na.rm = TRUE), max(vals, na.rm = TRUE))
  })

  # create labels
  labels <- sapply(1:3, function(i) {
    if(!is.null(ranges[[i]])) {
      sprintf("%s_T%d[%.1f-%.1f]", varname, i, ranges[[i]][1], ranges[[i]][2])
    } else {
      paste0(varname, "_T", i)
    }
  })

  # create the factor with labels
  factor(quantile_group, levels = 1:3, labels = labels)
}



# apply to Age in the sensitivity analysis sample
w1p <-
  w1p |>
  mutate(
    Age_cat_sensan = create_quantile_with_range(Age, "Age")
  )

# apply to Age in the analysis sample
w1p <-
```

```
  w1p |>
  filter(Sample == "Sample") |>
  mutate(
    Age_cat_Sample = create_quantile_with_range(Age, "Age")
  ) |>
  select(Age_cat_Sample, pid) |>
  right_join(w1p, by = "pid")



cat("\n=== SUMMARY TABLE FOR PANELISTS ===\n")
```

```
=== SUMMARY TABLE FOR PANELISTS ===
```

```
frq(w1p$Age_cat_sensan)
```

```
x <categorical>
# total N=17023 valid N=16997 mean=2.00 sd=0.82

Value              |    N | Raw % | Valid % | Cum. %
--------------------------------------------------
Age_T1[18.0-41.0]  | 5666 | 33.28 |   33.34 |  33.34
Age_T2[41.0-61.0]  | 5666 | 33.28 |   33.34 |  66.67
Age_T3[61.0-101.0] | 5665 | 33.28 |   33.33 | 100.00
<NA>               |   26 |  0.15 |    <NA> |   <NA>
```

```
frq(w1p$Age_cat_Sample)
```

```
x <categorical>
# total N=17023 valid N=9446 mean=2.00 sd=0.82

Value             |    N | Raw % | Valid % | Cum. %
-------------------------------------------------
Age_T1[18.0-40.0] | 3149 | 18.50 |   33.34 |  33.34
Age_T2[40.0-60.0] | 3149 | 18.50 |   33.34 |  66.67
Age_T3[60.0-97.0] | 3148 | 18.49 |   33.33 | 100.00
<NA>              | 7577 | 44.51 |    <NA> |   <NA>
```

```
w1p <-
  w1p |>
  mutate(
    Age_cat_Sample = case_when(
      # derived via:
      #read_dta(paste0(datapath,"w1_up.dta")) |> filter(lfd == <case id>) |>
       ↪  select(contains("geb")| contains("yr") | contains("year")| contains("birth") ) |>
       ↪  ai()
      # NOTE: re-insert the real case id instead of the anonymization
      # placeholder "<case id>" before re-running this branch
      lfd == "<case id>" & is.na(Age_cat_Sample) ~"Age_T2[40.0-60.0]",
      TRUE ~Age_cat_Sample
    ),


    # 26 missings on Age_cat_sensan carried over to Age_cat

    Age_cat_sensan =
```

```
          case_when(is.na(Age_cat_sensan) ~"Age_T2[41.0-61.0]",
                    TRUE ~Age_cat_sensan)
  )

w1p |>
  frq(Age_cat_Sample, Age_cat_sensan)
```

```
Age_cat_Sample <character>
# total N=17023 valid N=9446 mean=2.00 sd=0.82

Value             |    N | Raw % | Valid % | Cum. %
--------------------------------------------------
Age_T1[18.0-40.0] | 3149 | 18.50 |   33.34 |  33.34
Age_T2[40.0-60.0] | 3149 | 18.50 |   33.34 |  66.67
Age_T3[60.0-97.0] | 3148 | 18.49 |   33.33 | 100.00
<NA>              | 7577 | 44.51 |    <NA> |   <NA>

Age_cat_sensan <character>
# total N=17023 valid N=17023 mean=2.00 sd=0.82

Value              |    N | Raw % | Valid % | Cum. %
---------------------------------------------------
Age_T1[18.0-41.0]  | 5666 | 33.28 |   33.28 |  33.28
Age_T2[41.0-61.0]  | 5692 | 33.44 |   33.44 |  66.72
Age_T3[61.0-101.0] | 5665 | 33.28 |   33.28 | 100.00
<NA>               |    0 |  0.00 |    <NA> |   <NA>
```

3.2 w1 BreakOff

```
w1p <-
  w1p |> mutate(BreakOff = case_when(
    intvollst %in% c(2,3) ~ "Break-off",
    qmethod == 3 ~ "PAPI",              #PAPI
    (qmethod == 4 & intvollst == 1) ~ "No Break-off"
  ),
    BreakOff = factor(BreakOff, levels = c("No Break-off", "Break-off", "PAPI")))



w1p |> count(Sample, BreakOff)
```

```
# A tibble: 6 x 3
  Sample BreakOff         n
  <chr>  <fct>        <int>
1 Sample No Break-off  3656
2 Sample Break-off      121
3 Sample PAPI          5670
4 sensan No Break-off  2269
5 sensan Break-off      421
6 sensan PAPI          4886
```

```
w1p |>  frq(BreakOff)
```

```
BreakOff <categorical>
# total N=17023 valid N=17023 mean=2.27 sd=0.95
```

```
Value        |     N | Raw % | Valid % | Cum. %
-----------------------------------------------
No Break-off |  5925 | 34.81 |   34.81 |  34.81
Break-off    |   542 |  3.18 |    3.18 |  37.99
PAPI         | 10556 | 62.01 |   62.01 | 100.00
<NA>         |     0 |  0.00 |    <NA> |   <NA>
```

```
#121 cases in Sample, 542 altogether
```

3.3 w1 Gender, Education, Employment, ChildHH, PartWave

```
w1p <-
  w1p |>
  left_join(
    sufw1t1,
    by = c("lfd"="pid")) |>
  mutate(
  Education3 = case_when(
      pgisced11_scp %in% c(0, 1, 2, 3, 4) ~ "Non-tertiary (ISCED 0-4)",
      pgisced11_scp %in% c(5, 6, 7, 8) ~ "Tertiary (ISCED 5-8)",
      TRUE ~"missing"
  ),
   Education3 = factor(Education3, levels = c("Non-tertiary (ISCED 0-4)", "Tertiary (ISCED
    ↪  5-8)", "missing")),

  Education_i = case_when(
    !(Education3 %in% "missing") ~ Education3,


    Education3 %in% "missing" &
    # Priority 1: unambiguous cases (CASMIN)
    #[4] Post-secondary non-tertiary education
  pgcasmin_scp %in% c(4) ~ "Non-tertiary (ISCED 0-4)",

  Education3 %in% "missing" &
    #[5] Short-cycle tertiary education
   pgcasmin_scp %in% c(5) ~ "Tertiary (ISCED 5-8)",


  Education3 %in% "missing" &
      # Priority 2: unambiguous cases (pemplstat)
      pgemplst_scp == 10 ~ "Tertiary (ISCED 5-8)",
  Education3 %in% "missing" &
      pgemplst_scp == 5 ~ "Non-tertiary (ISCED 0-4)",

  Education3 %in% "missing" &
      # Priority 3: pgisco08_scp
      substr(pgisco08_scp, 1, 1) %in% c("1", "2") ~ "Tertiary (ISCED 5-8)",
    Education3 %in% "missing"&
      substr(pgisco08_scp, 1, 1) %in% c("3", "4", "5", "6", "7", "8", "9") ~ "Non-tertiary
       ↪  (ISCED 0-4)",

  Education3 %in% "missing" &

      # Priority 4: conservative assumptions (minijob, homemaker)
      pgemplst_scp %in% c(4, 12) ~ "Non-tertiary (ISCED 0-4)",
```

```
    TRUE ~ "missing"
  ),

Education_i = factor(Education_i, levels = c("Non-tertiary (ISCED 0-4)", "Tertiary (ISCED
 ↪  5-8)", "missing")),


  Education_method = case_when(
    !(Education3 %in% "missing") ~ "original",
    !(Education_i %in% "missing") ~ "imputed",
    TRUE ~ "missing"
  ),

# COLLAPSE EDUCATION CATEGORIES
Education =

  case_when(
    Education3 %in% c("Non-tertiary (ISCED 0-4)", "Tertiary (ISCED 5-8)") ~Education3,

    Education3 %in% "missing" &
    Education_i  %in%  c("Non-tertiary (ISCED 0-4)", "Tertiary (ISCED 5-8)") ~Education_i,

Education3 %in% "missing" & Education_i %in% "missing" ~ "missing"),

Education = factor(Education, levels = c("Non-tertiary (ISCED 0-4)",
                                         "Tertiary (ISCED 5-8)",
                                         "missing")),


Gender = case_when(

  pgender == 2 ~ "Female",

  # identify as "male" or "non-binary"
  pgender %in% c(1, 3) ~ "Not female",
  # n = 136 -10, n = 37 -1
  TRUE ~ "Not female"),

  Gender = factor(Gender, levels = c("Female", "Not female")),



  Employment = case_when(

    pempstat1 == 1 ~ "Full-time",
    pempstat1 %in% c(2:6, 11) ~ "Part-time or flexible",
    pempstat1 == 9 ~ "Retirement",
    pempstat1 %in% c(7, 8, 10, 12, 13) ~ "Not currently employed, training",
    TRUE ~ "missing"),
  Employment = factor(Employment, levels = c("Full-time", "Part-time or flexible", "Not
   ↪  currently employed, training", "Retirement", "missing")),

  ChildHH = case_when(
    hhhage02 == 0 | hhhage03 == 1 ~ "No child in hh",
    hhhage02 > 0 & hhhage03 != 1 ~ "Child in hh",
```

```
      TRUE ~ "missing"),
    ChildHH = factor(ChildHH, levels = c("No child in hh", "Child in hh", "missing")),
    #1408 NA

    AllWavesComplete = case_when(

      #n = 7368
      (intvollst == 1 & intvollst_w2 == 1 & intvollst_w3 ==1) ~ "All waves complete",
      TRUE ~ "Not complete"),

    AllWavesComplete = factor(AllWavesComplete, levels = c("Not complete", "All waves
     ↪  complete"))


  )



w1p |> filter(Sample == "Sample") |> select(Education3, Education_i, Education_method,
 ↪  Education,
            Gender, AllWavesComplete, Employment, ChildHH) |> summary()
```

```
                    Education3                     Education_i
 Non-tertiary (ISCED 0-4):5084   Non-tertiary (ISCED 0-4):5148
 Tertiary (ISCED 5-8)    :4071   Tertiary (ISCED 5-8)    :4103
 missing                 : 292   missing                 : 196


 Education_method                      Education           Gender
 Length:9447        Non-tertiary (ISCED 0-4):5148   Female    :4685
 Class :character   Tertiary (ISCED 5-8)    :4103   Not female:4762
 Mode  :character   missing                 : 196


           AllWavesComplete                              Employment
 Not complete      :2079    Full-time                         :3778
 All waves complete:7368    Part-time or flexible             :1683
                            Not currently employed, training:1314
                            Retirement                        :2541
                            missing                           : 131
           ChildHH
 No child in hh:7236
 Child in hh   :1787
 missing       : 424
```

3.4 w1 Migration

- self or parents' birthcountry is not Germany, or no German nationality

```
w1p <-
  w1p |>
  mutate(
    Mig = case_when(
      # origin is not Germany, parents not born in Germany, or no German nationality
```

```
      porigin == 2 | pmigback01 == 2| pnation01 == 0 ~ "Yes",
      porigin < 0 & pmigback01 < 0 & pnation01 > 0 ~ "Not applicable",
      TRUE ~ "No"
    ),
    Mig = factor(Mig, levels = c( "No", "Yes"))
  )
```

3.5 w1 Income

```
w1p <-
  w1p |>

# Equivalized income:
  # according to OECD weighting procedure (<14yrs => weight factor 0.3, >14yrs => weight factor
   ↪  0.5)

    rowwise() |>

    mutate(
      hhsize = ifelse(hhhsize < 0, NA, hhhsize),
      u14 = ifelse(hhhage02 < 0, NA, hhhage02),
      ue14 = hhsize - u14,

      equivSize = (1 + 0.5 * (ue14 - 1) + 0.3 * u14),
      EquivIncome_num = case_when(
        hhhincnet < 0 ~ NA,
        TRUE ~ (12 * (hhhincnet/equivSize)))) |>
     ungroup() |>
  mutate(
        EquivIncome_sensan = create_quantile_with_range(EquivIncome_num, "EquivIncome"),
        EquivIncome_sensan = case_when(
          is.na(EquivIncome_sensan) ~ "missing",
          EquivIncome_sensan == "EquivIncome_T1[0.0-20800.0]" ~ "EquivIncome_T1min",
          EquivIncome_sensan == "EquivIncome_T2[20800.0-31200.0]" ~ "EquivIncome_T2medium",
          EquivIncome_sensan == "EquivIncome_T3[31200.0-56800480.0]" ~ "EquivIncome_T3max"),
        EquivIncome_sensan = factor(EquivIncome_sensan, levels = c("EquivIncome_T2medium",
                                                   "EquivIncome_T1min",
                                                   "EquivIncome_T3max",
                                                   "missing"))
        )

w1p <-
  w1p |>
  filter(Sample == "Sample") |>
  mutate(
   EquivIncome_Sample = create_quantile_with_range(EquivIncome_num, "EquivIncome"),
        EquivIncome_Sample = case_when(
          is.na(EquivIncome_Sample) ~ "missing",
          EquivIncome_Sample == "EquivIncome_T1[0.0-21600.0]" ~ "EquivIncome_T1min",
          EquivIncome_Sample == "EquivIncome_T2[21600.0-32000.0]" ~ "EquivIncome_T2medium",
          EquivIncome_Sample == "EquivIncome_T3[32000.0-56800480.0]" ~ "EquivIncome_T3max"),
        EquivIncome_Sample = factor(EquivIncome_Sample, levels = c("EquivIncome_T2medium",
                                                   "EquivIncome_T1min",
                                                   "EquivIncome_T3max",
                                                   "missing"))) |>
```

```
  select(lfd, EquivIncome_Sample) |>
  right_join(w1p, by = "lfd")


w1p |> group_by(Sample) |> slice(1:5) |>  ungroup() |> select(lfd, Sample, contains("Income"))
```

```
# A tibble: 10 x 5
        lfd Sample EquivIncome_Sample   EquivIncome_num EquivIncome_sensan
      <dbl> <chr>  <fct>                          <dbl> <fct>
 1 10000301 Sample EquivIncome_T1min              20000  EquivIncome_T1min
 2 10000901 Sample EquivIncome_T1min              17077. EquivIncome_T1min
 3 10001601 Sample EquivIncome_T2medium           24680  EquivIncome_T2medium
 4 10002001 Sample missing                           NA  missing
 5 10002101 Sample EquivIncome_T3max              36000  EquivIncome_T3max
 6 10001001 sensan <NA>                           44000  EquivIncome_T3max
 7 10001201 sensan <NA>                              NA  missing
 8 10003201 sensan <NA>                           10560  EquivIncome_T1min
 9 10003701 sensan <NA>                           40000  EquivIncome_T3max
10 10004401 sensan <NA>                           12000  EquivIncome_T1min
```

```
w1p |> frq(EquivIncome_Sample, EquivIncome_sensan) # 1,366 and 3,765, respectively
```

```
EquivIncome_Sample <categorical>
# total N=17023 valid N=9447 mean=2.29 sd=1.03

Value                |    N | Raw % | Valid % | Cum. %
-----------------------------------------------------------
EquivIncome_T2medium | 2694 | 15.83 |   28.52 |  28.52
EquivIncome_T1min    | 2694 | 15.83 |   28.52 |  57.03
EquivIncome_T3max    | 2693 | 15.82 |   28.51 |  85.54
missing              | 1366 |  8.02 |   14.46 | 100.00
<NA>                 | 7576 | 44.50 |    <NA> |   <NA>

EquivIncome_sensan <categorical>
# total N=17023 valid N=17023 mean=2.44 sd=1.10

Value                |    N | Raw % | Valid % | Cum. %
-----------------------------------------------------------
EquivIncome_T2medium | 4419 | 25.96 |   25.96 |  25.96
EquivIncome_T1min    | 4420 | 25.96 |   25.96 |  51.92
EquivIncome_T3max    | 4419 | 25.96 |   25.96 |  77.88
missing              | 3765 | 22.12 |   22.12 | 100.00
<NA>                 |    0 |  0.00 |    <NA> |   <NA>
```

```
w1p |> filter(EquivIncome_num > 1 & EquivIncome_num <= 200000) |>
  ggplot( aes(EquivIncome_num,fill = Sample)) +
  geom_histogram(color="#e9ecef") + #"#69b3a2", color="#e9ecef") +
    ggtitle("Distribution of Equivalized Income") +
  facet_wrap(~Sample) +
  labs(caption = "N = 62 > 200 000 excluded from this plot for clarity ") +
    papaja::theme_apa()
```

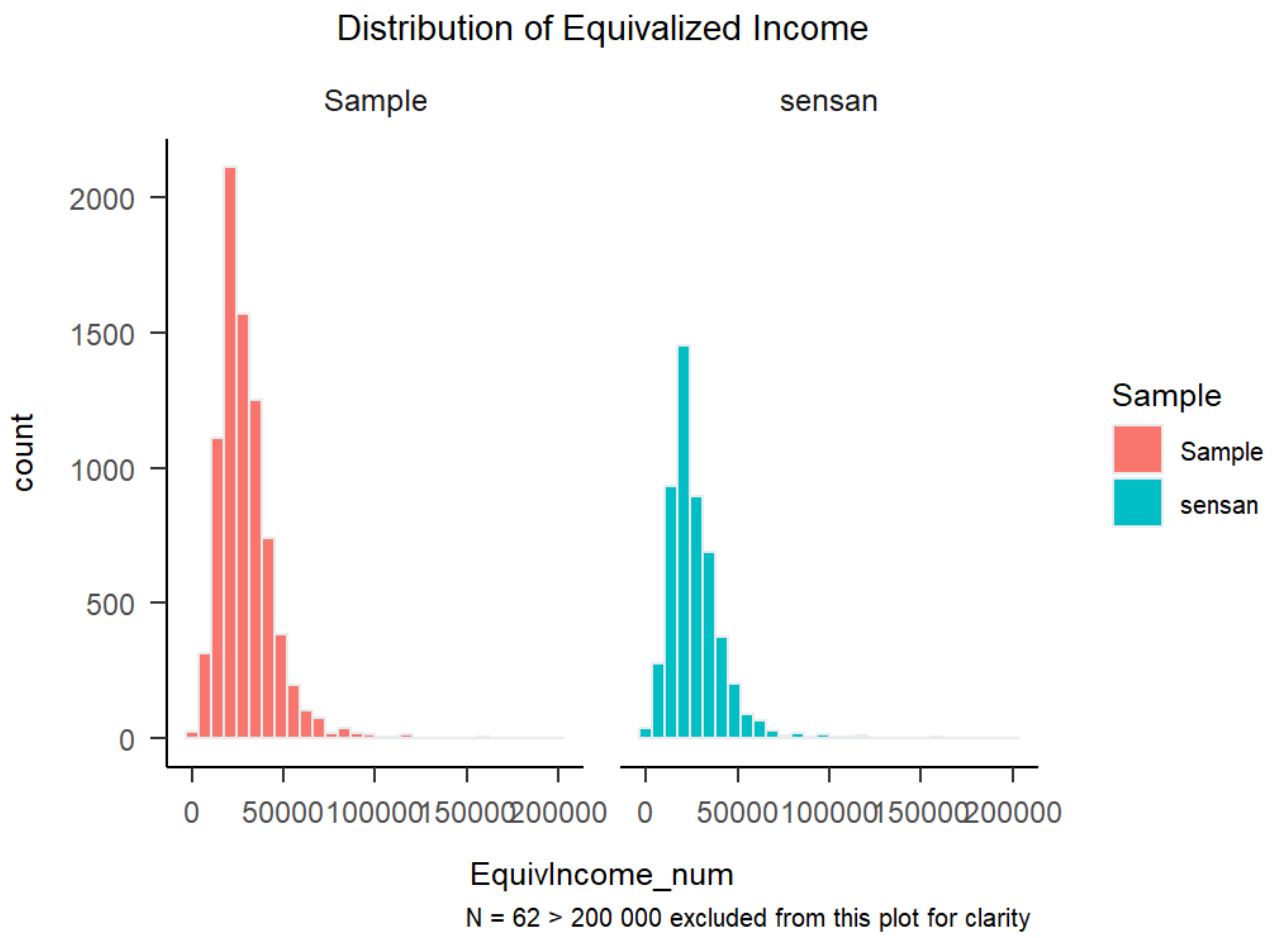


Figure 1

3.6 w1 Speeding

See Figure 2

```
# w1 durations


w1_dur <- w1_dur |>  select(where(~any(.x > 0, na.rm = TRUE)))
#lfd, hhlfd

# Visualization of choice of 10% border for speeders


quantiles_data <-
  w1p |> select(lfd, Sample, Age, Age_cat_sensan, BreakOff) |>
  inner_join(w1_dur, by = "lfd") |>

  mutate(



    # not include into calculation of quantile borders: N = 490 Break-off cases
  interviewdauer = ifelse(BreakOff == "Break-off", NA, interviewdauer)) |>
   ↪  #filter(is.na(interviewdauer)) |> nrow()
```

```r
  group_by(Sample, Age_cat_sensan) |>


  summarise(
    `1` = quantile(interviewdauer, probs = 0.01, na.rm = TRUE),
    `3` = quantile(interviewdauer, probs = 0.03, na.rm = TRUE),
    `5` = quantile(interviewdauer, probs = 0.05, na.rm = TRUE),
    `10` = quantile(interviewdauer, probs = 0.1, na.rm = TRUE),
    `15` = quantile(interviewdauer, probs = 0.15, na.rm = TRUE),
    `20` = quantile(interviewdauer, probs = 0.2, na.rm = TRUE),
    `25` = quantile(interviewdauer, probs = 0.25, na.rm = TRUE),
    `33` = quantile(interviewdauer, probs = 0.33, na.rm = TRUE),
    `50` = quantile(interviewdauer, probs = 0.5, na.rm = TRUE),
    `66` = quantile(interviewdauer, probs = 0.66, na.rm = TRUE),
    `70` = quantile(interviewdauer, probs = 0.7, na.rm = TRUE),
    `75` = quantile(interviewdauer, probs = 0.75, na.rm = TRUE),
    `80` = quantile(interviewdauer, probs = 0.80, na.rm = TRUE),
    `85` = quantile(interviewdauer, probs = 0.85, na.rm = TRUE),
    `95` = quantile(interviewdauer, probs = 0.95, na.rm = TRUE)
  )

# reshape the data for plotting
plot_data <- quantiles_data |>
  gather(key = "Quantile", value = "Value", -c(Sample, Age_cat_sensan)) |>
  mutate(Quantile = as.numeric(Quantile))
```

```r
ggplot(plot_data,
       aes(y = Quantile,
           x = Value,
         # group = interaction(Sample, Age_cat),
          color = Age_cat_sensan)) +
  geom_line() +
  #geom_hline(yintercept = 10, linetype = "dashed") +
  scale_y_continuous(breaks = seq(0,100, by = 5)) +
  facet_wrap(~ Sample, ncol = 4) +
  labs(x = "Interview duration", y = "Quantile",
       title = "Quantiles of interview duration per age group") +
  papaja::theme_apa()
```

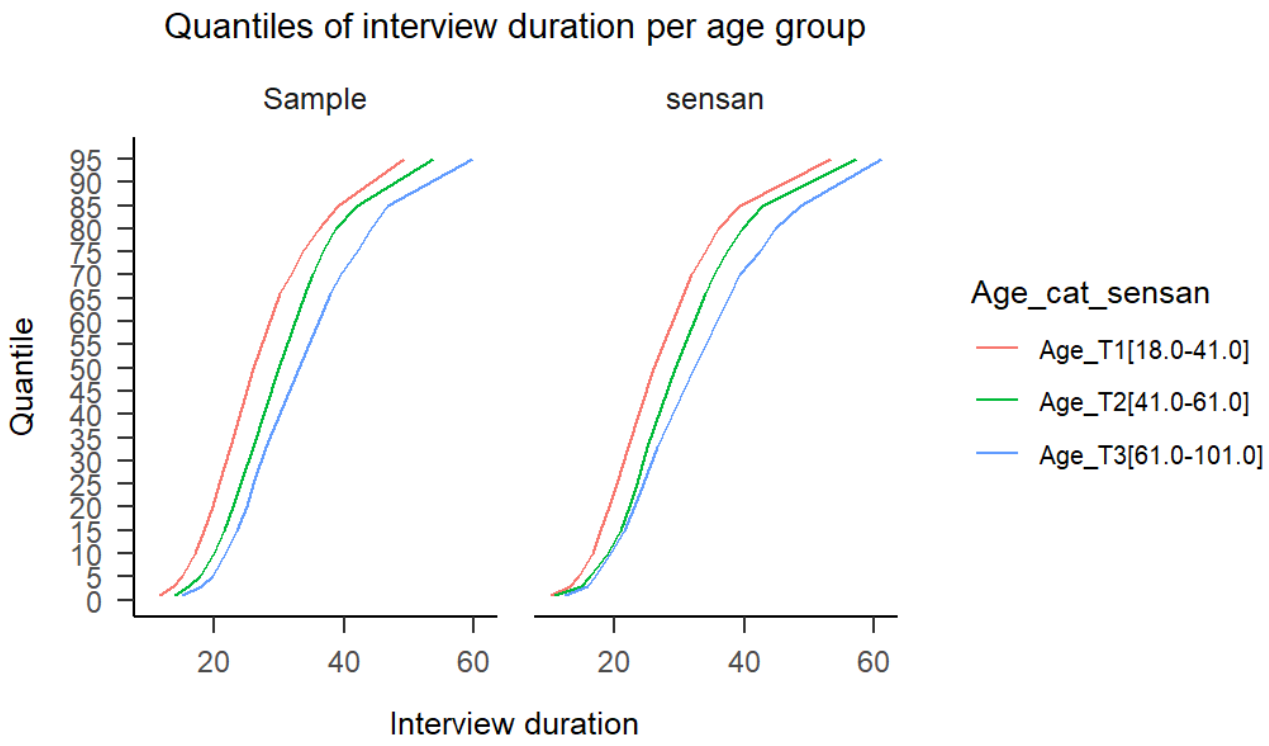


Figure 2: Speeding by Sample and Age Group

```
#|label: w1-build-speeding-variable
#|eval: true


w1_dur <-
  w1_dur |>
  left_join(w1p |> select(lfd, contains("Age_cat"), Sample)) |>
  # Step 1: set 0 to NA in all item columns
  mutate(across(ends_with("_dur"),
           ~ if_else(.x == 0, NA_real_, .x))
  )  |>


  group_by(Age_cat_sensan) |>
  # Step 2: compute medians (0/NA excluded)
  mutate(
    across(ends_with("_dur"),
           list(md = ~median(.x, na.rm = TRUE)),
           .names = "{.col}_sensan_md")
  ) |>

  # Step 3: speeding indicators (_s) with group-specific medians
  # speeded: item answered in less than 40% of the age-group median time
  mutate(
    across(
      .cols = ends_with("_dur"),
      .fns = ~ case_when(
        is.na(.x) ~ NA_real_,
        .x < 0.4 * get(paste0(cur_column(), "_sensan_md")) ~ 1,
```

```
        TRUE ~ 0
      ),
      .names = "{.col}_sensan_s"
    )
  ) |>
  ungroup() |>

  group_by(Age_cat_Sample) |>
  # Step 2: compute medians (0/NA excluded)
  mutate(
    across(ends_with("_dur"),
           list(md = ~median(.x, na.rm = TRUE)),
           .names = "{.col}_Sample_md")
  ) |>

  # Step 3: speeding indicators (_s) with group-specific medians
  mutate(
    across(
      .cols = ends_with("_dur"),
      .fns = ~ case_when(
        is.na(.x) ~ NA_real_,
        .x < 0.4 * get(paste0(cur_column(), "_Sample_md")) ~ 1,
        TRUE ~ 0
      ),
      .names = "{.col}_Sample_s"
    )
  ) |>
  ungroup() |>

  # Step 4: sum scores
  rowwise() |>
  mutate(
    Age_cat_sensan = if_else(
      is.na(Age_cat_sensan),
      factor("Age_T2[41.0-61.0]", levels = levels(Age_cat_sensan)),
      Age_cat_sensan),

    Age_cat_Sample = if_else(
      Sample == "Sample" & is.na(Age_cat_Sample),
      factor("Age_T2[40.0-60.0]", levels = levels(Age_cat_Sample)),
      Age_cat_Sample),

    # number of items whose duration is < 0.4 * the item's median duration across all
     ↪  respondents in the same age group (can be 0 if the respondent did not speed on any
     ↪  item)
    ItemsSpeeded_sensan = sum(c_across(ends_with("_sensan_s")), na.rm = TRUE),
    TotalItems_sensan = sum(!is.na(c_across(ends_with("_sensan_s")))),
    ProportionSpeeding_sensan = ItemsSpeeded_sensan / TotalItems_sensan)


w1_dur <-
  w1_dur |>
  filter(Sample == "Sample") |>

  mutate(
```

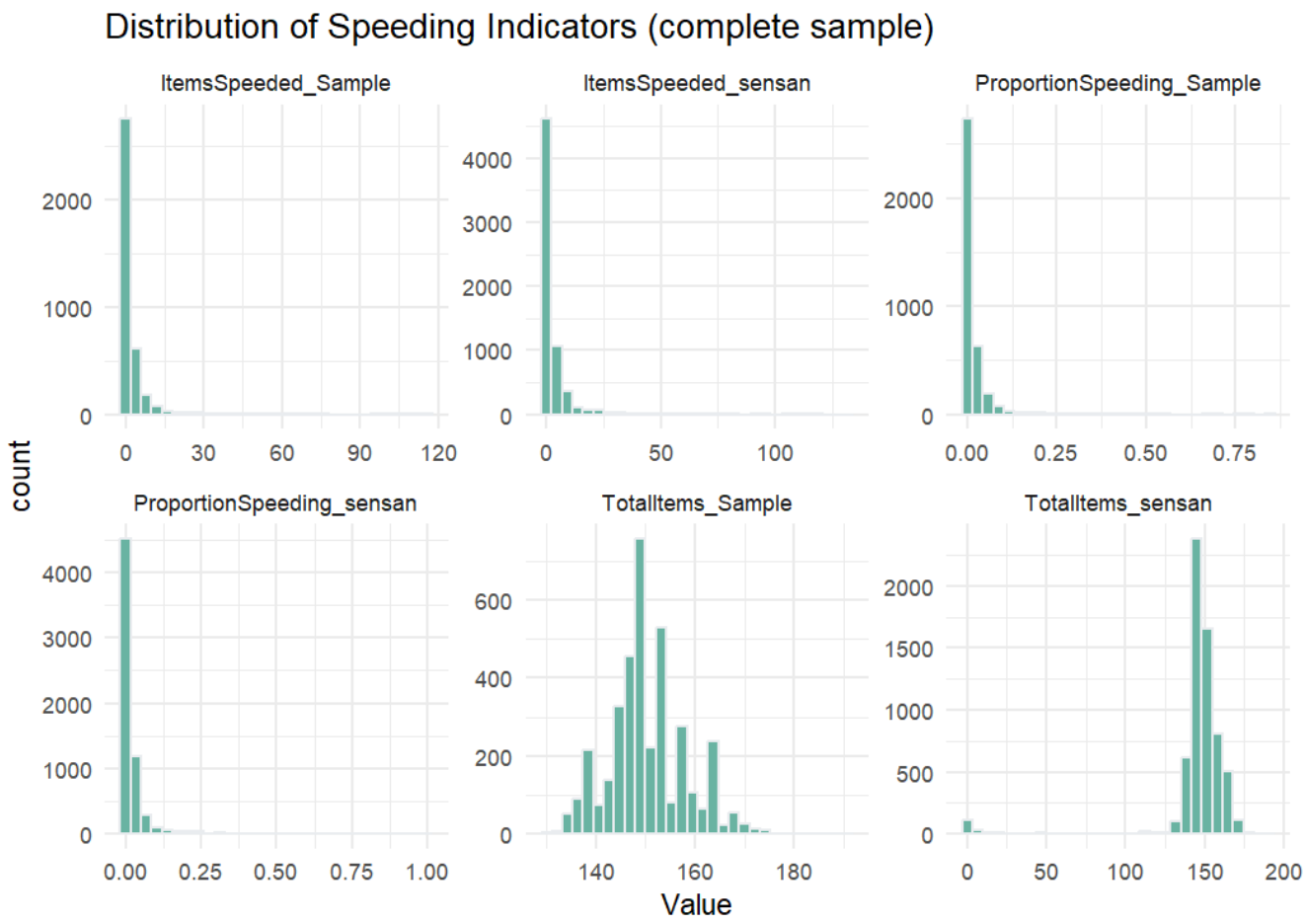
```
    ItemsSpeeded_Sample = sum(c_across(ends_with("_Sample_s")), na.rm = TRUE),
    TotalItems_Sample = sum(!is.na(c_across(ends_with("_Sample_s")))),
    ProportionSpeeding_Sample = ItemsSpeeded_Sample / TotalItems_Sample
  ) |>
  ungroup() |>
  select(lfd, ItemsSpeeded_Sample, TotalItems_Sample, ProportionSpeeding_Sample) |>
  right_join(w1_dur, by = "lfd")
#
# w1_dur |> pull(TotalItems_Sample) |> summary()
#    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
#   130.0   146.0   148.0   150.2   154.0   191.0    2692

w1_dur |> select(Sample, contains("pfrspcons07_du"), contains("Age_cat")) |> head(10) |>
 ↪  glimpse()
```

```
Rows: 10
Columns: 8
$ Sample                   <chr> "Sample", "Sample", "Sample", "Sample", "Sam~
$ pfrspcons07_dur          <dbl> 7.130, 5.954, 8.741, 3.345, 3.595, 4.748, 6.~
$ pfrspcons07_dur_sensan_md <dbl> 5.4240, 5.4240, 5.4240, 4.2370, 5.4240, 5.42~
$ pfrspcons07_dur_sensan_s  <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0
$ pfrspcons07_dur_Sample_md <dbl> 5.2705, 5.2705, 6.6780, 4.1475, 5.2705, 5.27~
$ pfrspcons07_dur_Sample_s  <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0
$ Age_cat_Sample           <chr> "Age_T2[40.0-60.0]", "Age_T2[40.0-60.0]", "A~
$ Age_cat_sensan           <chr> "Age_T2[41.0-61.0]", "Age_T2[41.0-61.0]", "A~
```

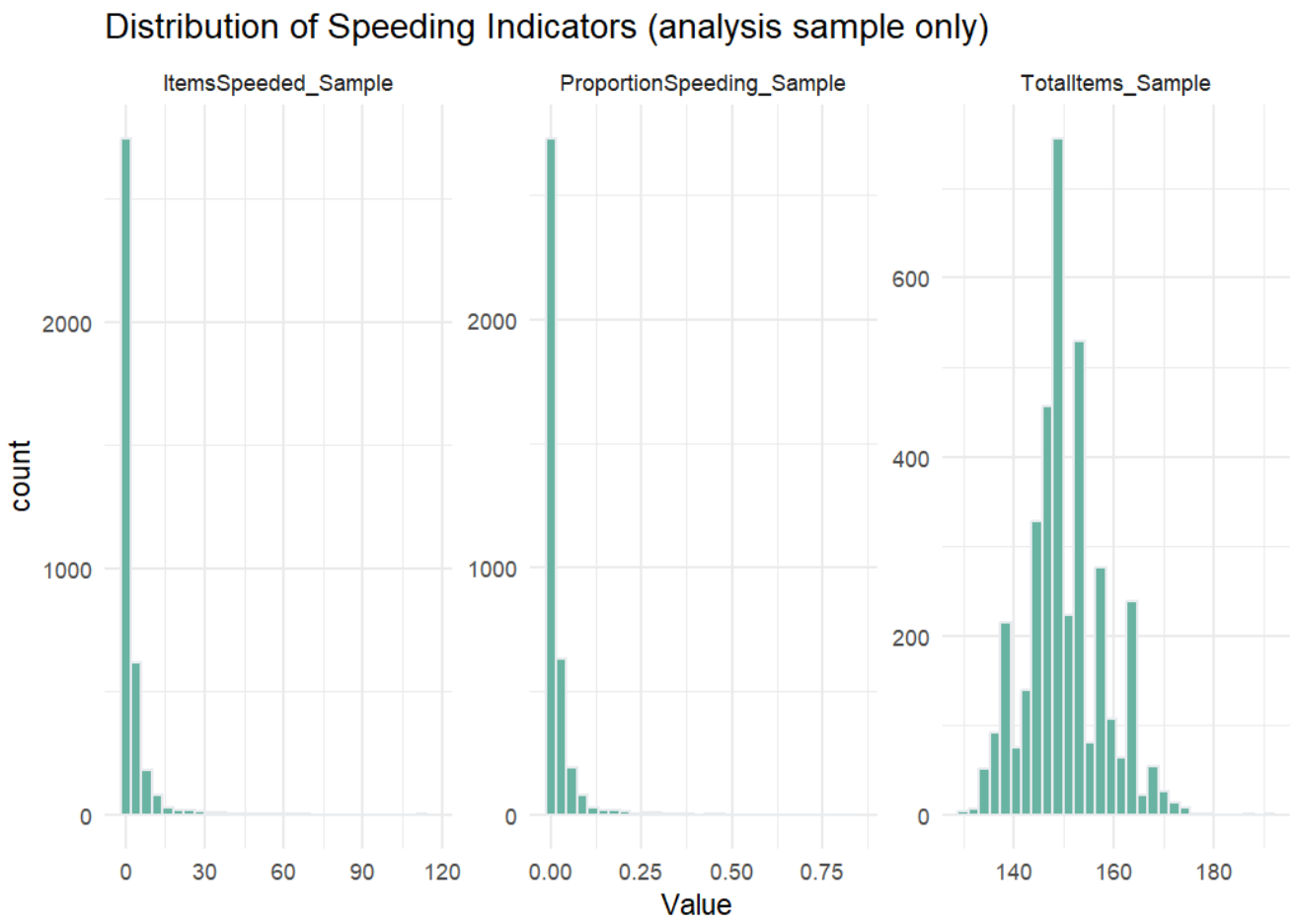
```
w1_dur |>
  select(ItemsSpeeded_sensan, TotalItems_sensan, ProportionSpeeding_sensan,
         ItemsSpeeded_Sample, TotalItems_Sample, ProportionSpeeding_Sample) |>
  pivot_longer(everything(), names_to = "Variable", values_to = "Value") |>
  ggplot(aes(x = Value)) +
      geom_histogram(fill="#69b3a2", color="#e9ecef") +

 # geom_density(fill = "steelblue", alpha = 0.5) +
  facet_wrap(~Variable, scales = "free") +
  theme_minimal() +
  labs(title = "Distribution of Speeding Indicators (complete sample)")
```




```
w1_dur |>
  filter(Sample == "Sample") |>
  select(ItemsSpeeded_Sample, TotalItems_Sample, ProportionSpeeding_Sample) |>
  pivot_longer(everything(), names_to = "Variable", values_to = "Value") |>
  ggplot(aes(x = Value)) +
      geom_histogram(fill="#69b3a2", color="#e9ecef") +

 # geom_density(fill = "steelblue", alpha = 0.5) +
  facet_wrap(~Variable, scales = "free") +
  theme_minimal() +
  labs(title = "Distribution of Speeding Indicators (analysis sample only)")
```




```
# Complete sample (sensitivity analysis only)
w1_dur |> filter(ProportionSpeeding_sensan>0.000) |> nrow()
```

```
[1] 3867
```

```
w1_dur |> filter(ProportionSpeeding_sensan >0 & ProportionSpeeding_sensan<0.014388) |>
 ↪  pull(ItemsSpeeded_sensan) |> summary()  #1-2 Items - marginal
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  1.000   1.000   1.000   1.325   2.000   2.000
```

```
# upper median range:
w1_dur |> filter(ProportionSpeeding_sensan>=0.014388) |> pull(ItemsSpeeded_sensan) |> summary()
 ↪  #1-132 Items
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   1.00    3.00    5.00   10.69   10.00  132.00
```

```
# Analysis sample
w1_dur |> filter(Sample =="Sample" & ProportionSpeeding_Sample>0.000) |> nrow()
```

```
[1] 2255
```

```
w1_dur |> filter(Sample == "Sample" & ProportionSpeeding_Sample >0 &
 ↪  ProportionSpeeding_Sample<0.013699) |> pull(ItemsSpeeded_Sample) |> summary()  #1-2 Items -
 ↪  marginal
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
```

```
  1.000   1.000   1.000   1.288   2.000   2.000
```

```
# upper median range
w1_dur |> filter(Sample == "Sample" & ProportionSpeeding_Sample>=0.013699) |>
 ↪  pull(ItemsSpeeded_Sample) |> summary()  #1-116 Items
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  2.000   3.000   5.000   9.424   9.000 116.000
```

```
#### TOP 10 MOST SPEEDED ITEMS

 # filter the _s columns
 s_sensan_cols <- w1_dur |>
     select(ends_with("_sensan_s")) |>
     names()

 # compute column sums (number of 1s, ignoring NAs)
 w1_dur |>
     summarise(across(all_of(s_sensan_cols), ~ sum(.x == 1, na.rm = TRUE))) |>
     pivot_longer(everything(), names_to = "variable", values_to = "sum_1er") |>
     arrange(desc(sum_1er)) |>
   head(10)
```

```
# A tibble: 10 x 2
   variable                              sum_1er
   <chr>                                   <int>
 1 ppanelcons_dur_sensan_s                   650
 2 pedu02_dur_sensan_s                       631
 3 pcohes03_dur_sensan_s                     468
 4 hhhincnet_dur_sensan_s                    467
 5 preligion_dur_sensan_s                    460
 6 pcohes02_dur_sensan_s                     430
 7 pgrpknow01_dur_sensan_s                   416
 8 plabincbrut_net_pnoinc_dur_sensan_s       375
 9 pladder01_dur_sensan_s                    371
10 pedu01_dur_sensan_s                       370
```

```
# variable  sum_1er
# ppanelcons_dur_sensan_s   648
# pedu02_dur_sensan_s   632
# pcohes03_dur_sensan_s 467
# hhhincnet_dur_sensan_s    466
# preligion_dur_sensan_s    460
# pcohes02_dur_sensan_s 430
# pgrpknow01_dur_sensan_s   416
# plabincbrut_net_pnoinc_dur_sensan_s   374
# pladder01_dur_sensan_s    371
# pedu01_dur_sensan_s   370

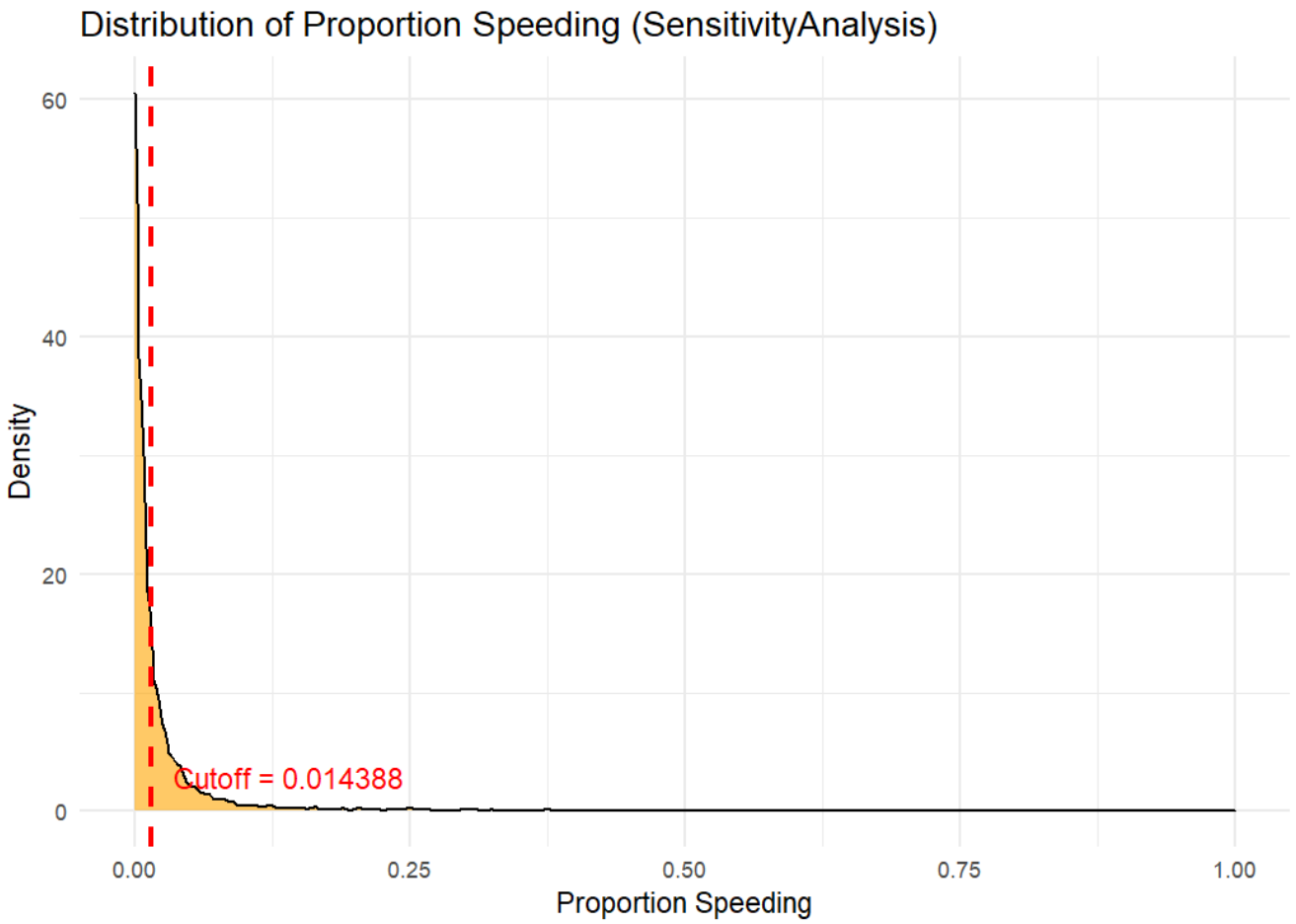
 # filter the _s columns
 s_Sample_cols <- w1_dur |>
     select(ends_with("_Sample_s")) |>
     names()

 # compute column sums (number of 1s, ignoring NAs)
 w1_dur |>
     summarise(across(all_of(s_Sample_cols), ~ sum(.x == 1, na.rm = TRUE))) |>
     pivot_longer(everything(), names_to = "variable", values_to = "sum_1er") |>
```

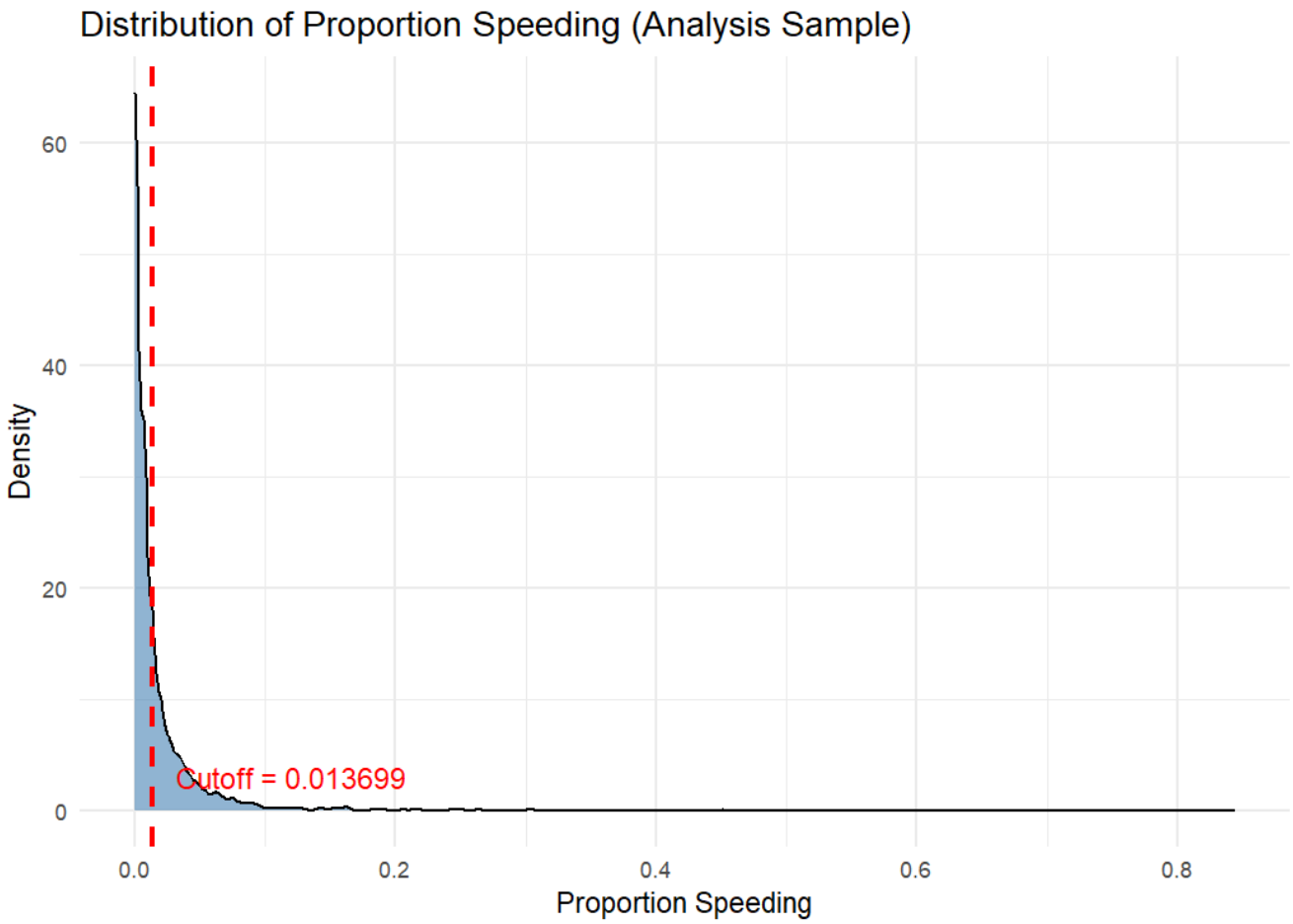
```
    arrange(desc(sum_1er)) |>
  head(10)
```

```
# A tibble: 10 x 2
   variable                              sum_1er
   <chr>                                   <int>
 1 ppanelcons_dur_Sample_s                   654
 2 pedu02_dur_Sample_s                       638
 3 pcohes03_dur_Sample_s                     469
 4 preligion_dur_Sample_s                    458
 5 hhhincnet_dur_Sample_s                    441
 6 pcohes02_dur_Sample_s                     433
 7 pgrpknow01_dur_Sample_s                   429
 8 plabincbrut_net_pnoinc_dur_Sample_s       380
 9 pladder01_dur_Sample_s                    372
10 pedu01_dur_Sample_s                       361
```

```
w1_dur |>
  #filter(Sample == "Sample") |>
  # with cutoff line
ggplot(aes(x = ProportionSpeeding_sensan)) +
  geom_density(alpha = 0.6, fill = "orange") +
  geom_vline(xintercept = 0.014388, linetype = "dashed", color = "red", linewidth = 1) +
  annotate("text", x = 0.014388, y = 0, label = "Cutoff = 0.014388",
           hjust = -0.1, vjust = -1, color = "red") +
  labs(title = "Distribution of Proportion Speeding (SensitivityAnalysis)",
       x = "Proportion Speeding",
       y = "Density") +
  theme_minimal()
```




```
w1_dur |>
  #filter(Sample == "Sample") |>
  # with cutoff line
ggplot(aes(x = ProportionSpeeding_Sample, fill = Sample)) +
  geom_density(alpha = 0.6, fill = "steelblue") +
  geom_vline(xintercept = 0.013699, linetype = "dashed", color = "red", linewidth = 1) +
  annotate("text", x = 0.013699, y = 0, label = "Cutoff = 0.013699",
           hjust = -0.1, vjust = -1, color = "red") +
  labs(title = "Distribution of Proportion Speeding (Analysis Sample)",
       x = "Proportion Speeding",
       y = "Density") +
  theme_minimal()
```




```
# consistent threshold: mean + 1 SD per group

thresholds_speeding_w1_sensan <-
   w1_dur |>
   group_by(Age_cat_sensan) |>
   summarise(
     thresh_Speed = mean(ProportionSpeeding_sensan, na.rm = TRUE) +
      ↪  sd(ProportionSpeeding_sensan, na.rm = TRUE),
     percentile_sp = mean(ProportionSpeeding_sensan <= thresh_Speed, na.rm = TRUE) * 100,
     prop_above_sp = mean(ProportionSpeeding_sensan > thresh_Speed, na.rm = TRUE) * 100) |>
      ↪  ungroup()

thresholds_speeding_w1_Sample <-
   w1_dur |>
   group_by(Age_cat_Sample) |>
   summarise(
     thresh_Speed = mean(ProportionSpeeding_Sample, na.rm = TRUE) +
      ↪  sd(ProportionSpeeding_Sample, na.rm = TRUE),
     percentile_sp = mean(ProportionSpeeding_Sample <= thresh_Speed, na.rm = TRUE) * 100,
     prop_above_sp = mean(ProportionSpeeding_Sample > thresh_Speed, na.rm = TRUE) * 100) |>
      ↪  ungroup()


w1_dur <-
  w1_dur |>
  mutate(
```

```
    Speeding_sensan = case_when(
      is.na(ProportionSpeeding_sensan) ~NA,
      Age_cat_sensan == "Age_T1[18.0-41.0]" & ProportionSpeeding_sensan >
↪   thresholds_speeding_w1_sensan |> filter(Age_cat_sensan == "Age_T1[18.0-41.0]") |>
↪   pull(thresh_Speed) ~ "Speeding",
      Age_cat_sensan == "Age_T2[41.0-61.0]" & ProportionSpeeding_sensan >
↪   thresholds_speeding_w1_sensan |> filter(Age_cat_sensan == "Age_T2[41.0-61.0]") |>
↪   pull(thresh_Speed) ~ "Speeding",
      Age_cat_sensan == "Age_T3[61.0-101.0]" & ProportionSpeeding_sensan >
↪   thresholds_speeding_w1_sensan |> filter(Age_cat_sensan == "Age_T3[61.0-101.0]") |>
↪   pull(thresh_Speed) ~ "Speeding",
      TRUE ~"No speeding"
    ),
    Speeding_sensan = factor(Speeding_sensan, levels = c("No speeding", "Speeding")),

    Speeding_Sample = case_when(
      is.na(ProportionSpeeding_Sample) ~NA,
      Age_cat_Sample == "Age_T1[18.0-40.0]" & ProportionSpeeding_Sample >
↪   thresholds_speeding_w1_Sample |> filter(Age_cat_Sample == "Age_T1[18.0-40.0]") |>
↪   pull(thresh_Speed) ~ "Speeding",
      Age_cat_Sample == "Age_T2[40.0-60.0]" & ProportionSpeeding_Sample >
↪   thresholds_speeding_w1_Sample |> filter(Age_cat_Sample == "Age_T2[40.0-60.0]") |>
↪   pull(thresh_Speed) ~ "Speeding",
      Age_cat_Sample == "Age_T3[60.0-97.0]" & ProportionSpeeding_Sample >
↪   thresholds_speeding_w1_Sample |> filter(Age_cat_Sample == "Age_T3[60.0-97.0]") |>
↪   pull(thresh_Speed) ~ "Speeding",
      TRUE ~"No speeding"
    ),
    Speeding_Sample = factor(Speeding_Sample, levels = c("No speeding", "Speeding"))
  )

w1_dur |> frq(Speeding_sensan, Speeding_Sample)
```

```
Speeding_sensan <categorical>
# total N=6469 valid N=6400 mean=1.05 sd=0.22

Value       |    N | Raw % | Valid % | Cum. %
--------------------------------------------
No speeding | 6059 | 93.66 |   94.67 |  94.67
Speeding    |  341 |  5.27 |    5.33 | 100.00
<NA>        |   69 |  1.07 |    <NA> |   <NA>

Speeding_Sample <categorical>
# total N=6469 valid N=3777 mean=1.05 sd=0.22

Value       |    N | Raw % | Valid % | Cum. %
--------------------------------------------
No speeding | 3586 | 55.43 |   94.94 |  94.94
Speeding    |  191 |  2.95 |    5.06 | 100.00
<NA>        | 2692 | 41.61 |    <NA> |   <NA>
```

```
# join to w1p
w1p <-
  w1p |>
  left_join(
    w1_dur |> select(lfd, ProportionSpeeding_sensan, ProportionSpeeding_Sample,
↪   Speeding_sensan, Speeding_Sample), by = "lfd"
```

```
  )
```

3.7 w1 FreeTextMis, Interviewmode, TargetGroup

```
FreeTextItems <-

  w1p |>

  select(
  pbirthmnth,
    pbirthyear_w1,
    pcohabgebyear,
    pothcount,
    pgermstay,
    peduOthWhich,
    poccupation_isco08,
    plabincbrut,
    plabincnet,
    psundvoteOthWhich,
    hhhsize,
    hhhage01,
    hhhage02,
  hhhincnet) |> colnames()


w1p <-
  w1p |>

  mutate(
    Interviewmode = case_when(
      qmethod == 3 ~ "PAPI",
      qmethod == 4 ~ "CAWI"
    ),
    Interviewmode = factor(Interviewmode, levels = c("PAPI", "CAWI")),


    TargetGroup = case_when(
      ZG == 1 ~ "Anchor",
      ZG == 2 ~ "HH member"
    ),
    TargetGroup = factor(TargetGroup, levels = c("Anchor", "HH member")),



    # apply standard missing codes to poccupation_isco08
    poccupation_isco08 = case_when(poccupation_isco08 == 0            ~ -2,
                                   poccupation_isco08 %in% c(8, 9)    ~ -1,
                                   TRUE                               ~ poccupation_isco08)) |>

  rowwise() |>
  mutate(

    freetextmis = sum(across(all_of(FreeTextItems)) %in% c(-1, -9, -4), na.rm = TRUE),

    N_allFreeTitems = sum(across(all_of(FreeTextItems), ~!. %in% c(-2, -10)), na.rm = TRUE),
```

```
    FrTMisRatio = freetextmis/N_allFreeTitems) |>

  ungroup()


# more groups of Ratio Free text items missings not useful
w1p |>
  reframe(Qu = quantile(FrTMisRatio, prob = c(0.33, 0.66), na.rm = TRUE))
```

```
# A tibble: 2 x 1
     Qu
  <dbl>
1     0
2     0
```

```
thresholds_FrTMis_w1_sensan <-
   w1p |>
   group_by(Interviewmode) |>
   summarise(
     thresh_FrTMis = mean(FrTMisRatio, na.rm = TRUE) + sd(FrTMisRatio, na.rm = TRUE),
     percentile_sp = mean(FrTMisRatio <= thresh_FrTMis, na.rm = TRUE) * 100,
     prop_above_sp = mean(FrTMisRatio > thresh_FrTMis, na.rm = TRUE) * 100) |> ungroup()

thresholds_FrTMis_w1_sensan
```

```
# A tibble: 2 x 4
  Interviewmode thresh_FrTMis percentile_sp prop_above_sp
  <fct>                 <dbl>         <dbl>         <dbl>
1 PAPI                  0.245          84.7          15.3
2 CAWI                  0.165          86.3          13.7
```

```
thresholds_FrTMis_w1_Sample <-
   w1p |>
   filter(Sample == "Sample") |>
   group_by(Interviewmode) |>
   summarise(
     thresh_FrTMis = mean(FrTMisRatio, na.rm = TRUE) + sd(FrTMisRatio, na.rm = TRUE),
     percentile_sp = mean(FrTMisRatio <= thresh_FrTMis, na.rm = TRUE) * 100,
     prop_above_sp = mean(FrTMisRatio > thresh_FrTMis, na.rm = TRUE) * 100) |> ungroup()

thresholds_FrTMis_w1_Sample
```

```
# A tibble: 2 x 4
  Interviewmode thresh_FrTMis percentile_sp prop_above_sp
  <fct>                 <dbl>         <dbl>         <dbl>
1 PAPI                  0.186          85.1          14.9
2 CAWI                  0.124          87.2          12.8
```

```
w1p <-
  w1p |>
  mutate(FrTMis_sensan =
           case_when(
             is.na(FrTMisRatio) ~ "Not applicable",
             Interviewmode == "PAPI" & FrTMisRatio > thresholds_FrTMis_w1_sensan |>
 ↪  filter(Interviewmode == "PAPI") |> pull(thresh_FrTMis) ~ "High",
             Interviewmode == "CAWI" & FrTMisRatio > thresholds_FrTMis_w1_sensan |>
 ↪  filter(Interviewmode == "CAWI") |> pull(thresh_FrTMis) ~ "High",
```

```
            TRUE ~ "Not high"),
        FrTMis_sensan = factor(FrTMis_sensan, levels = c("Not high", "High", "Not
         ↪  applicable")),

        FrTMis_Sample =
          case_when(

            Sample == "sensan" ~NA,
            is.na(FrTMisRatio) ~ "Not applicable",
            Interviewmode == "PAPI" & FrTMisRatio > thresholds_FrTMis_w1_Sample |>
 ↪  filter(Interviewmode == "PAPI") |> pull(thresh_FrTMis) ~ "High",
            Interviewmode == "CAWI" & FrTMisRatio > thresholds_FrTMis_w1_Sample |>
 ↪  filter(Interviewmode == "CAWI") |> pull(thresh_FrTMis) ~ "High",
            TRUE ~ "Not high"),
        FrTMis_Sample = factor(FrTMis_Sample, levels = c("Not high", "High", "Not
         ↪  applicable")))
```

```
plot_freetextmis_w1 <-
  w1p |> select(lfd, Interviewmode, FrTMisRatio) |>
  pivot_longer(cols = 3) |>

  ggplot(aes(Interviewmode, value, group = Interviewmode)) +


  geom_jitter(aes(Interviewmode, value, color = Interviewmode, group = Interviewmode),
              stat = "identity",
              position = "jitter",
              alpha = .3) +

  stat_summary(fun = mean,
               geom = "point",
               position = posn_d,
               size = 4,
               color = "black") +

  labs (y = "Ratio Free Text Item Missings",
        color = "Interviewmode", caption = "Free Text Missings by Interviewmode\n(12 items in
         ↪  w1)") +
  scale_color_manual(name = "Interviewmode", values = mode_palette) +
  #guides(color = Interviewmode) +
 # facet_wrap(~ name, ncol = 5) +

   papaja::theme_apa() +

   theme(axis.text.x = element_blank(), #element_text(size = 9, angle = 15),
        axis.ticks.x = element_blank(),
        axis.line.x = element_blank(),
        legend.position="bottom")

plot_freetextmis_w1

w1p |> group_by(Interviewmode) |>
  summarise(mean.FrTMisRatio = mean(FrTMisRatio, na.rm = TRUE),
            med.FrTMisRatio = median(FrTMisRatio, na.rm = TRUE),
            sd.FrTMisRatio = sd(FrTMisRatio, na.rm = TRUE))
```

```
# A tibble: 2 x 4
  Interviewmode mean.FrTMisRatio med.FrTMisRatio sd.FrTMisRatio
  <fct>                    <dbl>           <dbl>          <dbl>
1 PAPI                    0.0961               0          0.149
2 CAWI                    0.0512               0          0.114
```

```
w1p |> group_by(Sample, Interviewmode) |>
  summarise(mean.FrTMisRatio = mean(FrTMisRatio, na.rm = TRUE),
            med.FrTMisRatio = median(FrTMisRatio, na.rm = TRUE),
            sd.FrTMisRatio = sd(FrTMisRatio, na.rm = TRUE))
```

```
# A tibble: 4 x 5
# Groups:   Sample [2]
  Sample Interviewmode mean.FrTMisRatio med.FrTMisRatio sd.FrTMisRatio
  <chr>  <fct>                    <dbl>           <dbl>          <dbl>
1 Sample PAPI                    0.0671               0         0.119
2 Sample CAWI                    0.0348               0         0.0889
3 sensan PAPI                    0.130                0         0.171
4 sensan CAWI                    0.0757               0         0.140
```

Figure 3: Free text missings by Interviewmode (w1)




```
w1p |>
  group_by(Interviewmode) |>
   count(FrTMis_sensan, name="n") |>
     mutate(share = n/sum(n)) |>
```

```
     ggplot(aes(Interviewmode, share, fill=FrTMis_sensan)) +
     geom_col(position="stack")   +
labs(caption = "Free Text Missings by Sample (12 items over w1)",
       x = "",
     y = "Proportion") +

 papaja::theme_apa()  +
 theme(axis.text.x = element_text(angle = 30, vjust = 0.5),
       panel.grid.major.y = element_line(color = "grey", linewidth = 0.1))

w1p |>
  group_by(Interviewmode) |> filter(!is.na(FrTMis_Sample)) |>
   count(FrTMis_Sample, name="n") |>
     mutate(share = n/sum(n)) |>
     ggplot(aes(Interviewmode, share, fill=FrTMis_Sample)) +
     geom_col(position="stack")   +
labs(caption = "Free Text Missings by Sample (12 items over w1)",
       x = "",
     y = "Proportion") +

 papaja::theme_apa()  +
 theme(axis.text.x = element_text(angle = 30, vjust = 0.5),
       panel.grid.major.y = element_line(color = "grey", linewidth = 0.1))



w1p |>
  group_by(Sample, Interviewmode) |>
   count(FrTMisRatio, name="n") |>
     mutate(share = n/sum(n)) |>
     ggplot(aes(Interviewmode, share, fill=FrTMisRatio)) +
     geom_col(position="stack")   +
labs(caption = "Free Text Missings Ratio by Sample and Mode (12 items over w1)",
       x = "",
     y = "Proportion") +
  facet_wrap(~Sample) +
 papaja::theme_apa()  +
 theme(#axis.text.x = element_text(angle = 30, vjust = 0.5),
       panel.grid.major.y = element_line(color = "grey", linewidth = 0.1))
```

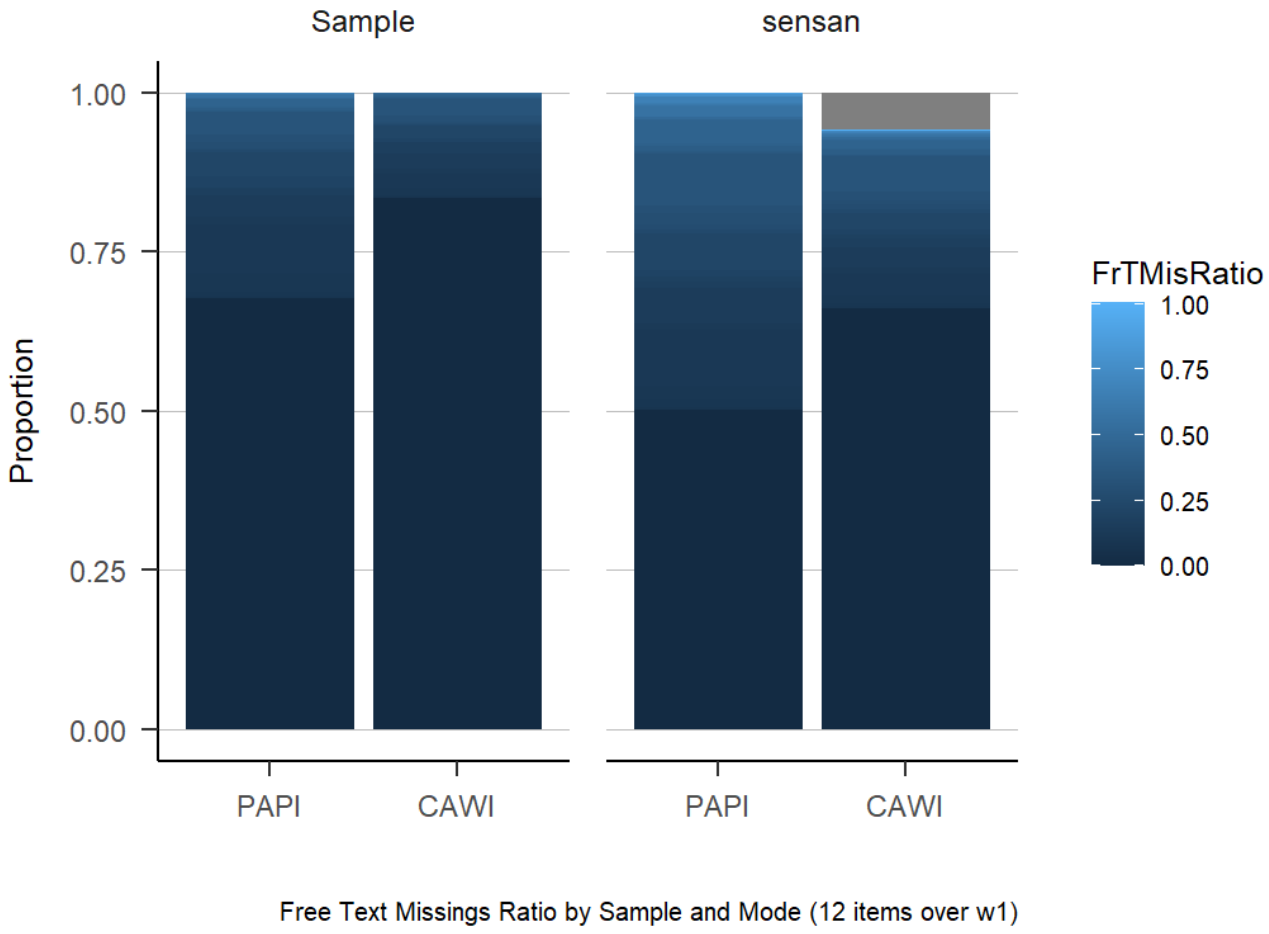
Figure 4: Free text missings by Interviewmode



Figure 5: Free text missings by Interviewmode



Figure 6: Free text missings by Interviewmode



3.8 w1 Item Nonresponse

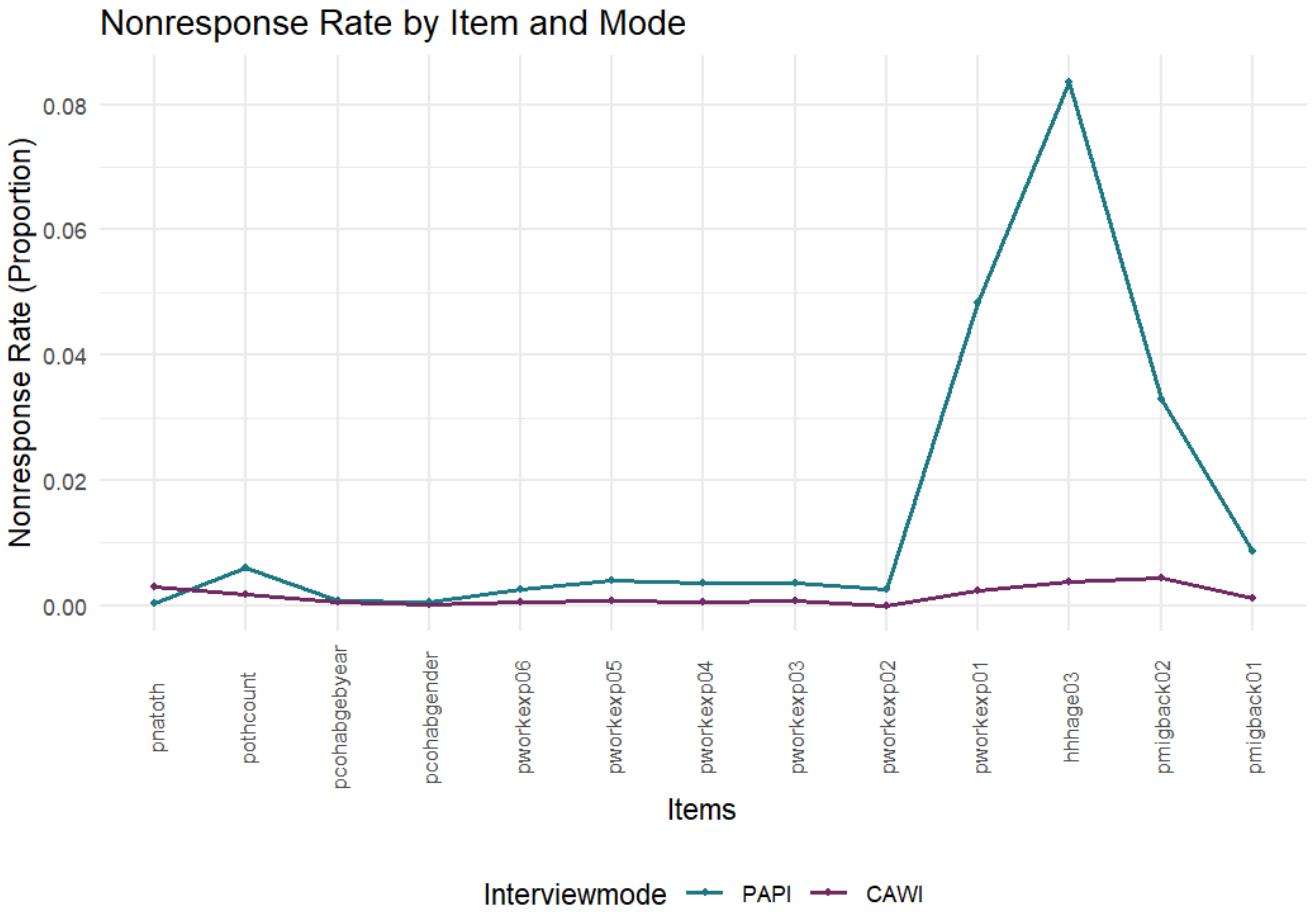
3.8.1 Problem Vars only

- so-called `problem vars` have been detected manually and have a significantly different appearance between modes
- to be sure not to measure mere mode effects, these items are tested for significance in their nonresponse rate between modes
  - if significant difference => item is removed from df

```
problem.vars <- c("pmigback01", "pmigback02",
          "hhhage03",
            "pworkexp01", "pworkexp02", "pworkexp03", "pworkexp04", "pworkexp05", "pworkexp06",
             ↪
            "pcohabgender", "pcohabgebyear",
            "pothcount", "pnatoth")
w1p |>
  select(lfd, Interviewmode,
         all_of(problem.vars)) |> #names()
 # select(all_of(mis_items_w1), Interviewmode) |>
  group_by(Interviewmode) |>
  summarise(across(pmigback01:pnatoth,
                   ~ sum(.x %in% c(-1, -9), na.rm = TRUE) / n(),
                   .names = "PropMiss_{.col}")) |>
```

```
  pivot_longer(-Interviewmode, names_to = c("dummy", "Variable"),
               names_pattern = "(PropMiss_)(.*)", values_to = "Prop") |>
  select(-dummy) |>
  mutate(Variable = factor(Variable, levels = rev(unique(Variable)))) |>

ggplot(aes(x = Variable, y = Prop, colour = Interviewmode, group = Interviewmode)) +
  geom_point(size = 1) +
  geom_line(linewidth = 0.8) +
  scale_colour_manual(values = mode_palette) +
  labs(
    x = "Items",
    y = "Nonresponse Rate (Proportion)",
    colour = "Interviewmode",
    title = "Nonresponse Rate by Item and Mode"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(size = 8, angle = 90, vjust = 0.5),
    legend.position = "bottom"
  )
```




```
w1p |>

    select(lfd, Interviewmode,
           all_of(problem.vars)) |>
```

```
  group_by(Interviewmode) |>
  summarise(across(pmigback01:pnatoth,
                   ~ sum(.x %in% c(-1, -9), na.rm = TRUE) / n(),
                   .names = "PropMiss_{.col}"))
```

```
# A tibble: 2 x 14
  Interviewmode PropMiss_pmigback01 PropMiss_pmigback02 PropMiss_hhhage03
  <fct>                       <dbl>               <dbl>             <dbl>
1 PAPI                      0.00872             0.0330            0.0836
2 CAWI                      0.00108             0.00433           0.00371
# i 10 more variables: PropMiss_pworkexp01 <dbl>, PropMiss_pworkexp02 <dbl>,
#   PropMiss_pworkexp03 <dbl>, PropMiss_pworkexp04 <dbl>,
#   PropMiss_pworkexp05 <dbl>, PropMiss_pworkexp06 <dbl>,
#   PropMiss_pcohabgender <dbl>, PropMiss_pcohabgebyear <dbl>,
#   PropMiss_pothcount <dbl>, PropMiss_pnatoth <dbl>
```

```
# Absolute counts + N per group

counts <- w1p |>
  select(lfd, Interviewmode, all_of(problem.vars)) |>
  group_by(Interviewmode) |>
  summarise(
    n = n(),
    across(all_of(problem.vars),
           ~ sum(.x %in% c(-1, -9), na.rm = TRUE),
           .names = "miss_{.col}")
  )

# Chi-square test per item
results <- map_dfr(problem.vars, function(var) {
  col <- paste0("miss_", var)

  miss  <- counts[[col]]        # c(n_miss_PAPI, n_miss_CAWI)
  n     <- counts$n             # c(n_PAPI, n_CAWI)
  notmiss <- n - miss

  mat <- matrix(c(miss, notmiss), nrow = 2,
                dimnames = list(c("PAPI", "CAWI"), c("Miss", "NotMiss")))

  # Fisher when expected cell counts < 5, otherwise chi-squared
  test <- if (any(chisq.test(mat)$expected < 5)) {
    fisher.test(mat)
  } else {
    chisq.test(mat, correct = FALSE)
  }

  tibble(
    Item       = var,
    p_value    = test$p.value,
    method     = test$method,
    prop_PAPI  = miss[1] / n[1],
    prop_CAWI  = miss[2] / n[2]
  )
}) |>
  mutate(
    p_adj      = p.adjust(p_value, method = "BH"),  # Benjamini-Hochberg
```

```
    significant = p_adj < 0.05
  ) |>
  arrange(p_adj)

print(results)
```

```
# A tibble: 13 x 7
   Item            p_value method      prop_PAPI prop_CAWI     p_adj significant
   <chr>             <dbl> <chr>           <dbl>     <dbl>     <dbl> <lgl>
 1 hhhage03      2.40e-112 Pearson's ~  0.0836    0.00371  3.12e-111 TRUE
 2 pworkexp01    1.10e- 63 Pearson's ~  0.0483    0.00232  7.15e- 63 TRUE
 3 pmigback02    5.56e- 35 Pearson's ~  0.0330    0.00433  2.41e- 34 TRUE
 4 pmigback01    2.06e- 10 Pearson's ~  0.00872   0.00108  6.69e- 10 TRUE
 5 pnatoth       1.03e-  5 Pearson's ~  0.000379  0.00294  2.67e-  5 TRUE
 6 pothcount     3.04e-  5 Pearson's ~  0.00606   0.00170  6.58e-  5 TRUE
 7 pworkexp02    4.69e-  5 Pearson's ~  0.00256   0        8.72e-  5 TRUE
 8 pworkexp05    1.09e-  4 Pearson's ~  0.00398   0.000773 1.78e-  4 TRUE
 9 pworkexp04    1.41e-  4 Pearson's ~  0.00360   0.000619 2.04e-  4 TRUE
10 pworkexp03    4.88e-  4 Pearson's ~  0.00351   0.000773 6.34e-  4 TRUE
11 pworkexp06    3.97e-  3 Pearson's ~  0.00256   0.000619 4.69e-  3 TRUE
12 pcohabgender  2.73e-  1 Fisher's E~  0.000663  0.000155 2.96e-  1 FALSE
13 pcohabgebyear 7.77e-  1 Fisher's E~  0.000853  0.000619 7.77e-  1 FALSE
```

```
# all significant except pcohabgender and pcohabgebyear


problem.vars <- c("pmigback01", "pmigback02",
          "hhhage03",
            "pworkexp01", "pworkexp02", "pworkexp03", "pworkexp04", "pworkexp05", "pworkexp06",
             ↪
           #"pcohabgender", "pcohabgebyear",
            "pothcount", "pnatoth"
            )
```

3.8.2 All other items

```
w1p |>

  select(lfd, Interviewmode,
         pcohes01:q46_check4,
       -(any_of(FreeTextItems))) |> #names()
 # select(all_of(mis_items_w1), Interviewmode) |>
  group_by(Interviewmode) |>
  summarise(across(pcohes01:q46_check4,
                   ~ sum(.x %in% c(-1, -9), na.rm = TRUE) / n(),
                   .names = "PropMiss_{.col}")) |>
  pivot_longer(-Interviewmode, names_to = c("dummy", "Variable"),
               names_pattern = "(PropMiss_)(.*)", values_to = "Prop") |>
  select(-dummy) |>
  mutate(Variable = factor(Variable, levels = rev(unique(Variable)))) |>

ggplot(aes(x = Variable, y = Prop, colour = Interviewmode, group = Interviewmode)) +
  geom_point(size = 1) +
  geom_line(linewidth = 0.8) +
  scale_colour_manual(values = mode_palette) +
  labs(
```

```
    x = "Items",
    y = "Nonresponse Rate (Proportion)",
    colour = "Interviewmode",
    title = "Nonresponse Rate by Item and Mode"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(size = 3, angle = 90, vjust = 0.5),
    legend.position = "bottom"
  )
```

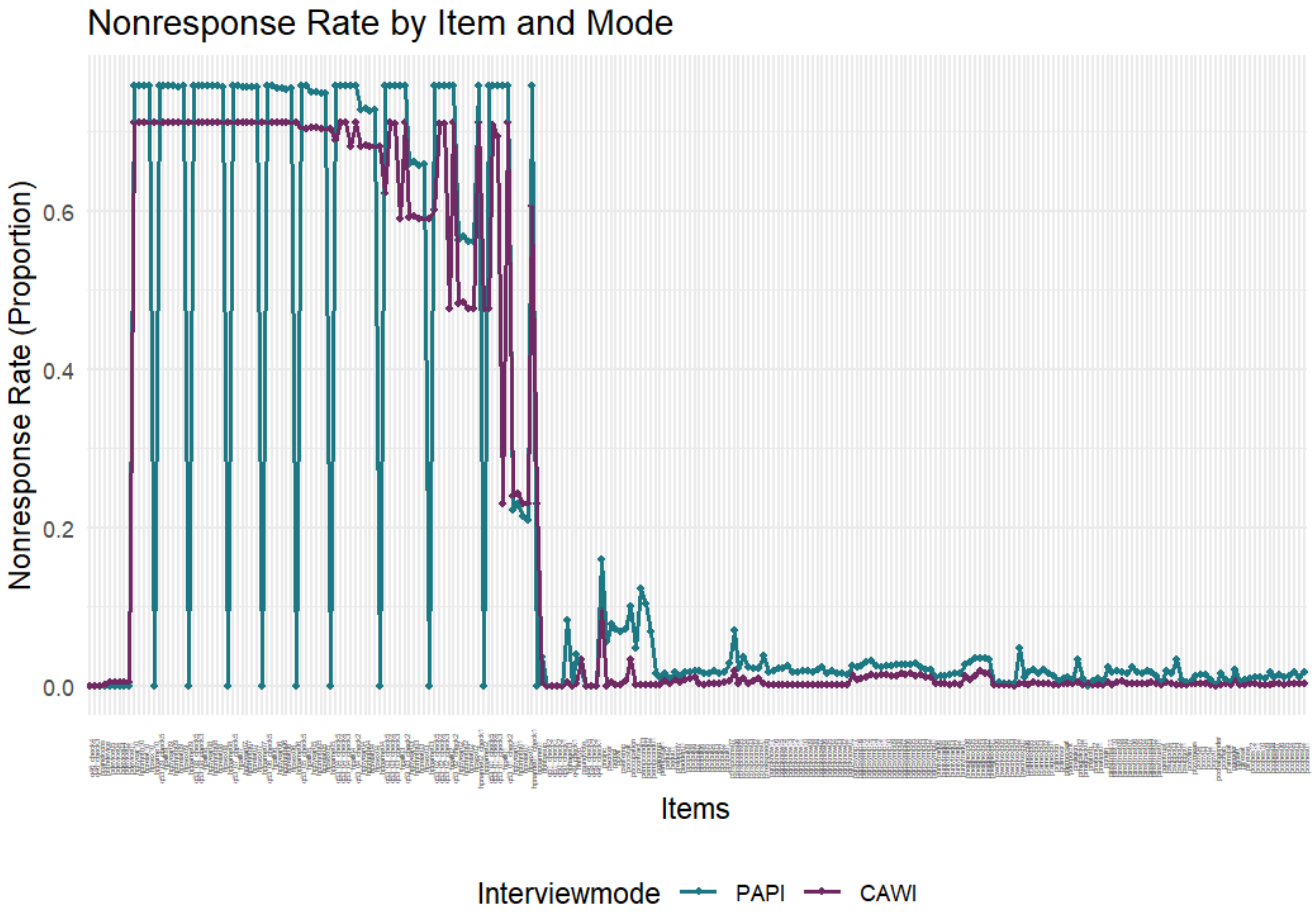


Figure 7: Nonresponse Items w1

```
mis_df_w1 <-
  w1p |>
  select(lfd, Interviewmode,
         pcohes01:q46_check4,
         -(any_of(FreeTextItems)),
         all_of(problem.vars))



# w1 items with high nonresponse rates
```

```
manymissings_CAWI <- mis_df_w1 |> filter(Interviewmode == "CAWI") |>
  summarise_all(~sum(.  %in% c(-1, -9))) |>
  pivot_longer(
    everything(), names_to = "Variable", values_to = "Numb.missings") |>
  rowwise() |>
  mutate(Prop.missings = Numb.missings/(mis_df_w1 |> filter(Interviewmode == "CAWI") |>
   ↪  nrow())) |>
  ungroup() |>
  arrange(desc(Prop.missings)) |>
  filter(Prop.missings > .15) |> pull(Variable) #|> paste(collapse = '", "')

manymissings_PAPI <- mis_df_w1 |> filter(Interviewmode == "PAPI") |>
  summarise_all(~sum(.  %in% c(-1, -9))) |>
  pivot_longer(
    everything(), names_to = "Variable", values_to = "Numb.missings") |>
  rowwise() |>
  mutate(Prop.missings = Numb.missings/(mis_df_w1 |> filter(Interviewmode == "PAPI") |>
   ↪  nrow())) |>
  ungroup() |>
  arrange(desc(Prop.missings)) |>
  filter(Prop.missings > .15) |> pull(Variable) #|> paste(collapse = '", "')




list(
  CAWI_only = setdiff(manymissings_CAWI, manymissings_PAPI),
  PAPI_only = setdiff(manymissings_PAPI, manymissings_CAWI)
)
```

```
$CAWI_only
 [1] "hpname10" "hpname09" "hpname08" "hpname07" "hpname06" "hpname05"
 [7] "hpname04" "hpname03" "hpname02" "hpname01"

$PAPI_only
[1] "pnoinc"
```

```
# variables excluded from the item nonresponse counts:

exclusion_items <- unique(c(manymissings_CAWI, manymissings_PAPI, problem.vars)) #95 items

mis_items_w1 <-
  mis_df_w1 |>
  select(

    -c(lfd, Interviewmode, #qmethod,
       contains("interview"),
       contains("Nonresp"),

       all_of(exclusion_items))) |>
  colnames()
```

```
mis_df_w1 <-
  mis_df_w1 |>
  rowwise() |>
  mutate(
  Nonresponses = sum(across(all_of(mis_items_w1)) %in% c(-1, -9), na.rm = TRUE),
  N_allNonrespOpportunities_woBO = sum(across(all_of(mis_items_w1), ~!. %in% c(-2, -10)), na.rm
    ↪  = TRUE),
 # BO (-10) counted as "not shown", i.e., not a missed opportunity"
  NonrespRatio = Nonresponses/N_allNonrespOpportunities_woBO) |>
  ungroup()

mis_df_w1 |> pull(N_allNonrespOpportunities_woBO) |> summary()
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    2.0   130.0   133.0   131.8   136.0   143.0
```

```
#    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
#     2.0   130.0   133.0   131.8   136.0   143.0


w1p <-
   w1p |>
   left_join(mis_df_w1 |>
               select(lfd, NonrespRatio), by = "lfd")
```

```
mis_df_w1 |> filter(Interviewmode == "PAPI") |>
  summarise_all(~sum(.  %in% c(-1, -9))) |>
  pivot_longer(
    everything(), names_to = "Variable", values_to = "Numb.missings") |>
  rowwise() |>
  mutate(Prop.missings = Numb.missings/(mis_df_w1 |> filter(Interviewmode == "PAPI") |>
    ↪  nrow())) |>
  ungroup() |>
  arrange(desc(Prop.missings)) |>
  filter(Prop.missings > .15)
```

```
# A tibble: 74 x 3
   Variable        Numb.missings Prop.missings
   <chr>                   <int>         <dbl>
 1 hpname01_check1          8003         0.758
 2 q53_01_check2            8003         0.758
 3 hpall01                  8003         0.758
 4 q53_01_check3            8003         0.758
 5 q53_01_check4            8003         0.758
 6 q53_01_check5            8003         0.758
 7 hpname02_check1          8003         0.758
 8 q53_02_check2            8003         0.758
 9 hpall02                  8003         0.758
10 q53_02_check3            8003         0.758
# i 64 more rows
```

3.9 w1 Extreme/Middle

- 8 items with 5 level likert scales
- 8 items with 11 level likert scales

For comparison of extreme vs. middle response distributions by interviewmode see Figure 8

```r
extrm5_items <-
  w1p |> select(

    # 36 shared with `Nondiff`
    pcohes01:pcohes12,
    pfamexp01:pfamexp06,
    pworkexp01:pworkexp06,
    pneighexp01:pneighexp06,
    punivtrad01:punivtrad06,
    ppopul01:ppopul09,


    # 16
    pcov01:pcov04, #5
    pdemocfit, #5
    pclimsor, #5
    pclimcon,  #5
    preldepriv #5

  ) |> colnames() #n = 53

extrm11_items <-
  w1p |> select(

    # 28 shared with `Nondiff`
    pinsttrust01:pinsttrust11,
    pgrpsymp01:pgrpsymp17,

    plifeuse, #11
    plifesat, #11
    ppolatt,  #11
    pcovmeas, #11
    pcovlim,   #11
    pgentrust, #11

    pdemocsat, #11

    pladder01 #11

  ) |> colnames() #n = 36


extrm7_items <-
  w1p |> select(
    # 17 shared with `Nondiff`
    pgrpknow01:pgrpknow17
  ) |> colnames() #n = 17


t1 <- Sys.time()


w1p <-
  w1p |>
  rowwise() |>
  mutate(
```

```
    # all extreme values on 5-likert
    E5 = sum(across(all_of(extrm5_items)) %in% c(1, 5), na.rm = TRUE),
    # all middle values on 5-likert
    M5 = sum(across(all_of(extrm5_items)) == 3, na.rm = TRUE),

    # all extreme and middle values on 5-likert
    EM.S5 = sum(E5, M5, na.rm = TRUE),


     # all extreme values on 7-likert
    E7 = sum(across(all_of(extrm7_items)) %in% c(1, 6), na.rm = TRUE),
    # all middle values on 7-likert
    M7 = sum(across(all_of(extrm7_items)) %in% c(7), na.rm = TRUE), # 7: "Don't know"

    # all extreme and middle values on 7-likert
    EM.S7 = sum(E7, M7, na.rm = TRUE),


    # all extreme values on 11-likert
    E11 = sum(across(all_of(extrm11_items)) %in% c(0, 10), na.rm = TRUE),
    # all middle values on 11-likert
    M11 = sum(across(all_of(extrm11_items)) %in% c(5), na.rm = TRUE),

    # all extreme and middle values on 11-likert
    EM.S11 = sum(E11, M11, na.rm = TRUE),



    N5_all = sum(across(all_of(extrm5_items)) > 0, na.rm = TRUE),
    N7_all = sum(across(all_of(extrm7_items)) > 0, na.rm = TRUE),
    N11_all = sum(across(all_of(extrm11_items)) >= 0, na.rm = TRUE),



    # all extreme values
    E.S5S7S11 = sum(E5, E7, E11, na.rm = TRUE),
    # all middle values
    M.S5S7S11 = sum(M5, M7, M11, na.rm = TRUE),

    # all items responded
    N_allExMidOpportunities = sum(N5_all, N7_all, N11_all),



    Ratio_Extr = ifelse(N_allExMidOpportunities == 0, NA, E.S5S7S11/N_allExMidOpportunities),
    Ratio_Mid = ifelse(N_allExMidOpportunities == 0, NA, M.S5S7S11/N_allExMidOpportunities),

    RatioExMid = sum(E.S5S7S11, M.S5S7S11, na.rm = TRUE)/N_allExMidOpportunities

  ) |> ungroup()

print(paste("Total duration (Extreme/Middle, mins):", as.numeric(Sys.time()-t1, "mins")))
```

```
[1] "Total duration (Extreme/Middle, mins): 21.1635898510615"
```

```r
cor.test(w1p$Ratio_Mid, w1p$Ratio_Extr) #-0.3700904, t = -51.83, df = 16927, p-value < 2.2e-16
```

```
    Pearson's product-moment correlation

data:  w1p$Ratio_Mid and w1p$Ratio_Extr
t = -51.83, df = 16927, p-value < 2.2e-16
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.3830190 -0.3570168
sample estimates:
       cor
-0.3700904
```

```r
#
```

```r
  w1p |> select(lfd, Interviewmode, Ratio_Extr, Ratio_Mid, RatioExMid) |>
  pivot_longer(cols = 3:5) |>

  ggplot(aes(Interviewmode, value, color = Interviewmode, group = Interviewmode)) +

  geom_violin(scale = "count", linewidth = 0.8, alpha = .9) +

  geom_boxplot(width = 0.15, color = "black", aes(fill = Interviewmode)) +

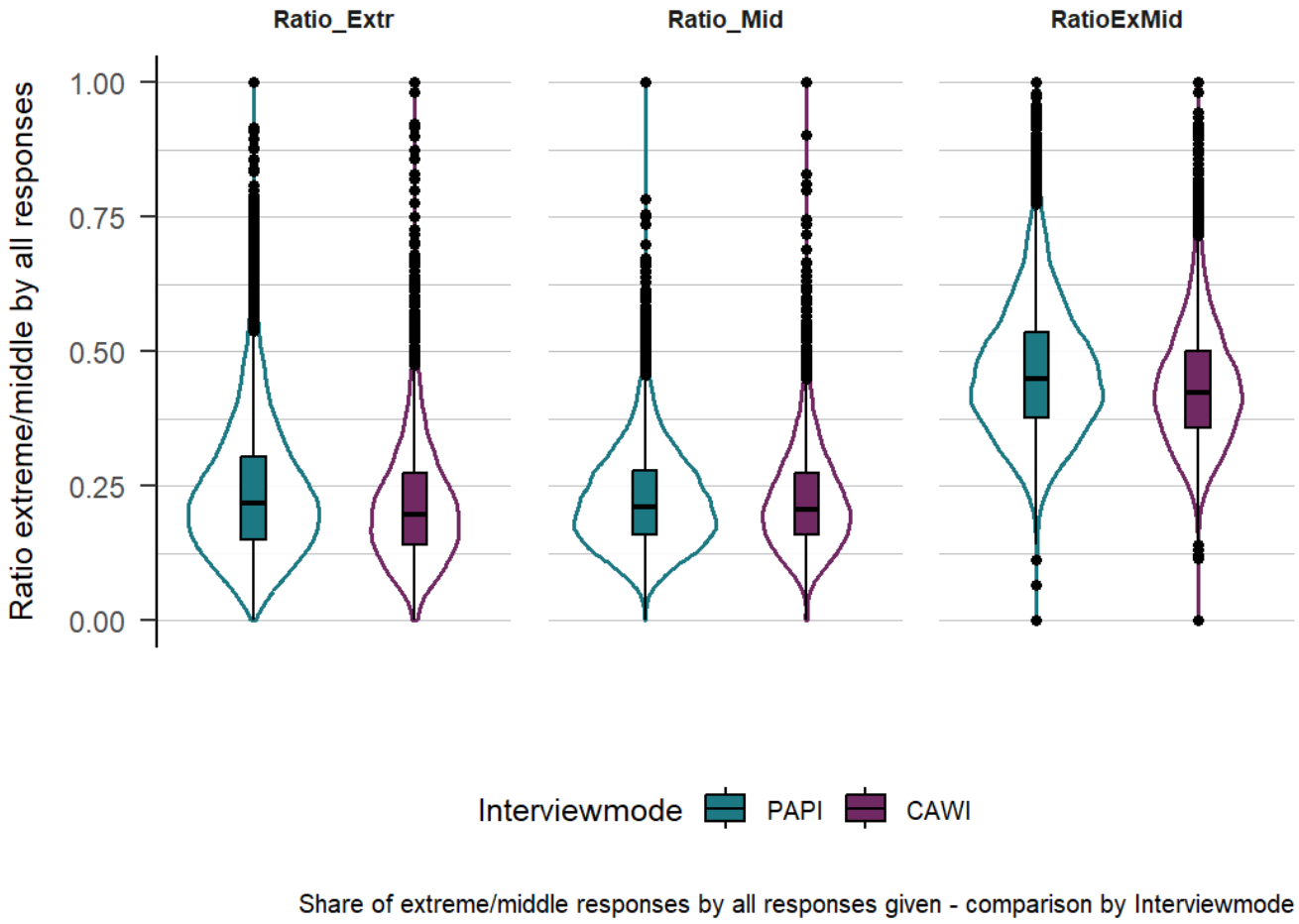
  scale_fill_manual(values = mode_palette) +
  scale_color_manual(values = mode_palette) +
   labs(caption = "Share of extreme/middle responses by all responses given - comparison by
    ↪  Interviewmode",
        x = "",
        y = "Ratio extreme/middle by all responses",
        fill = "Interviewmode") +
  facet_wrap(~ factor(name, c("Ratio_Extr", "Ratio_Mid", "RatioExMid")), ncol = 5) +

   guides(color = "none") +


  papaja::theme_apa() +

  theme(axis.text.x = element_blank(), #element_text(size = 9, angle = 15),
        axis.ticks.x = element_blank(),
        axis.line.x = element_blank(),
        legend.position="bottom",
        strip.text.x = element_text(size = 9, face = "bold"),
        panel.grid.major.y = element_line(color = "grey", linewidth = 0.01),
        panel.grid.minor.y = element_line(color = "grey", linewidth = 0.01))
```

Figure 8: Share of all extreme/middle responses by all responses given - comparison by Interviewmode



- Participants in the category "Above-average" had answered on average 64.5% of the considered questions using the extreme or middle category (SD = 0.08).

3.10 Categorize `Extr` & `Mid`

```
thresholds_ExM_sensan <-
  w1p |>

  group_by(Interviewmode) |>
  summarise(
    thresh_Extr_Ratio = mean(Ratio_Extr, na.rm = TRUE) + sd(Ratio_Extr, na.rm = TRUE),
    thresh_Mid_Ratio = mean(Ratio_Mid, na.rm = TRUE) + sd(Ratio_Mid, na.rm = TRUE),

    percentile_Extr = mean(Ratio_Extr <= thresh_Extr_Ratio, na.rm = TRUE) * 100,
    percentile_Mid = mean(Ratio_Mid <= thresh_Mid_Ratio, na.rm = TRUE) * 100,

    prop_above_Extr = mean(Ratio_Extr > thresh_Extr_Ratio, na.rm = TRUE) * 100,
    prop_above_Mid = mean(Ratio_Mid > thresh_Mid_Ratio, na.rm = TRUE) * 100
  )
```

```r
thresholds_ExM_Sample <-
  w1p |>
  filter(Sample == "Sample") |>
  group_by(Interviewmode) |>
  summarise(
    thresh_Extr_Ratio = mean(Ratio_Extr, na.rm = TRUE) + sd(Ratio_Extr, na.rm = TRUE),
    thresh_Mid_Ratio = mean(Ratio_Mid, na.rm = TRUE) + sd(Ratio_Mid, na.rm = TRUE),

    percentile_Extr = mean(Ratio_Extr <= thresh_Extr_Ratio, na.rm = TRUE) * 100,
    percentile_Mid = mean(Ratio_Mid <= thresh_Mid_Ratio, na.rm = TRUE) * 100,

    prop_above_Extr = mean(Ratio_Extr > thresh_Extr_Ratio, na.rm = TRUE) * 100,
    prop_above_Mid = mean(Ratio_Mid > thresh_Mid_Ratio, na.rm = TRUE) * 100
  )



w1p <-
  w1p |>
  mutate(Extr_sensan =
           case_when(
             is.na(Ratio_Extr) ~ "Not applicable",
             Interviewmode == "PAPI" & Ratio_Extr > thresholds_ExM_sensan |>
 ↪  filter(Interviewmode == "PAPI") |> pull(thresh_Extr_Ratio) ~ "High",
             Interviewmode == "CAWI" & Ratio_Extr > thresholds_ExM_sensan |>
 ↪  filter(Interviewmode == "CAWI") |> pull(thresh_Extr_Ratio) ~ "High",
             TRUE ~ "Not high"),
         Extr_sensan = factor(Extr_sensan, levels = c("Not high", "High", "Not applicable")),
         Mid_sensan =
           case_when(
             is.na(Ratio_Mid) ~ "Not applicable",
             Interviewmode == "PAPI" & Ratio_Mid > thresholds_ExM_sensan |>
 ↪  filter(Interviewmode == "PAPI") |> pull(thresh_Mid_Ratio) ~ "High",
             Interviewmode == "CAWI" & Ratio_Mid > thresholds_ExM_sensan |>
 ↪  filter(Interviewmode == "CAWI") |> pull(thresh_Mid_Ratio) ~ "High",
             TRUE ~ "Not high"),

         Mid_sensan = factor(Mid_sensan, levels = c("Not high", "High", "Not applicable")))


w1p <-
  w1p |>
  mutate(Extr_Sample =
           case_when(
             Sample == "sensan" ~NA,
             is.na(Ratio_Extr) ~ "Not applicable",
             Interviewmode == "PAPI" & Ratio_Extr > thresholds_ExM_Sample |>
 ↪  filter(Interviewmode == "PAPI") |> pull(thresh_Extr_Ratio) ~ "High",
             Interviewmode == "CAWI" & Ratio_Extr > thresholds_ExM_Sample |>
 ↪  filter(Interviewmode == "CAWI") |> pull(thresh_Extr_Ratio) ~ "High",
             TRUE ~ "Not high"),
         Extr_Sample = factor(Extr_Sample, levels = c("Not high", "High", "Not applicable")),

         Mid_Sample =
           case_when(
             Sample == "sensan" ~NA,
```

```
            is.na(Ratio_Mid) ~ "Not applicable",
            Interviewmode == "PAPI" & Ratio_Mid > thresholds_ExM_Sample |>
 ↪  filter(Interviewmode == "PAPI") |> pull(thresh_Mid_Ratio) ~ "High",
            Interviewmode == "CAWI" & Ratio_Mid > thresholds_ExM_Sample |>
 ↪  filter(Interviewmode == "CAWI") |> pull(thresh_Mid_Ratio) ~ "High",
            TRUE ~ "Not high"),
        Mid_Sample = factor(Mid_Sample, levels = c("Not high", "High", "Not applicable"))) |>
  select(lfd, Extr_Sample, Mid_Sample) |>
  right_join(w1p, by = "lfd")



w1p |> filter(!(is.na(Extr_Sample) & is.na(Mid_Sample))) |> count(Extr_Sample, Mid_Sample)
```

```
# A tibble: 4 x 3
  Extr_Sample Mid_Sample     n
  <fct>       <fct>      <int>
1 Not high    Not high    6758
2 Not high    High        1327
3 High        Not high    1302
4 High        High          60
```

3.11 w1 Nondifferentiation

3.11.1 A) Sensitivity Analysis only

```
w1 <- w1p

pcohes.items <- w1 |> select(pcohes01:pcohes12) |> colnames()
pinsttrust.items <- w1 |> select(pinsttrust01:pinsttrust11) |> colnames()
pfamexp.items <- w1 |> select(pfamexp01:pfamexp06) |> colnames()
pworkexp.items <- w1 |> select(pworkexp01:pworkexp06) |> colnames()    # Category "Unemployed"
pneighexp.items <- w1 |> select(pneighexp01:pneighexp06) |> colnames()
punivtrad.items <- w1 |> select(punivtrad01:punivtrad06) |> colnames()
pgrpsymp.items <- w1 |> select(pgrpsymp01:pgrpsymp17) |> colnames()
pgrpknow.items <- w1 |> select(pgrpknow01:pgrpknow17) |> colnames()
pfrspcons.items <- w1 |> select(pfrspcons01:pfrspcons07) |> colnames()
ppopul.items <- w1 |> select(ppopul01:ppopul09) |> colnames()

items_list <- list(pcohes.items, pinsttrust.items,
                   pfamexp.items, pworkexp.items,
                   pneighexp.items, punivtrad.items, pgrpsymp.items,
                   pgrpknow.items, pfrspcons.items, ppopul.items)

tND <- Sys.time()

# extract all items from items_list
all_items <- unlist(items_list)

w1_papi <-
  w1 |>
  filter(Interviewmode == "PAPI") |>
  select(lfd, all_of(all_items)) |>  # select only the item columns
  meanRootPairs(items_list) |>
  ungroup()
```

```
Duration for pcohes : 0.5161638 minutes
Duration for pinsttrust : 0.5349367 minutes
Duration for pfamexp : 0.4046811 minutes
Duration for pworkexp : 0.3527302 minutes
Duration for pneighexp : 0.3757414 minutes
Duration for punivtrad : 0.3808442 minutes
Duration for pgrpsymp : 0.6549037 minutes
Duration for pgrpknow : 0.7342638 minutes
Duration for pfrspcons : 0.4540633 minutes
Duration for ppopul : 0.4785579 minutes
All done!
Total duration for all item packs: 4.887139 minutes.
```

```
w1_cawi  <-
  w1 |>
  filter(Interviewmode == "CAWI") |>
  select(lfd, all_of(all_items)) |>
   meanRootPairs(items_list) |> ungroup()
```

```
Duration for pcohes : 0.235919 minutes
Duration for pinsttrust : 0.3114202 minutes
Duration for pfamexp : 0.2519551 minutes
Duration for pworkexp : 0.2225699 minutes
Duration for pneighexp : 0.2371921 minutes
Duration for punivtrad : 0.3016769 minutes
Duration for pgrpsymp : 0.4679533 minutes
Duration for pgrpknow : 0.456792 minutes
Duration for pfrspcons : 0.2797093 minutes
Duration for ppopul : 0.3205139 minutes
All done!
Total duration for all item packs: 3.086086 minutes.
```

```
# merge PAPI and CAWI
w1_meanroot <-
  bind_rows(w1_papi, w1_cawi)

# join onto the original w1 dataset
w1 <- w1 |>
  left_join(w1_meanroot |>
              select(lfd, starts_with("temp_index"), starts_with("temp_max"),
                     starts_with("MeanRootPairs_")), by = "lfd")


#10.75397 minutes
```

```
p_load(patchwork)


# prepare the data for the WK variables (long format)
wk_data <- w1 |>
  select(lfd, Interviewmode, MeanRootPairs_pinsttrust, MeanRootPairs_pfrspcons,
         MeanRootPairs_ppopul, MeanRootPairs_pgrpsymp) |>
  pivot_longer(c(MeanRootPairs_pinsttrust:MeanRootPairs_pgrpsymp),
               names_to = "variable",
               values_to = "mrp_value") |>
  mutate(variable = case_when(
    variable == "MeanRootPairs_pinsttrust" ~ "pinsttrust",
```

```
    variable == "MeanRootPairs_pfrspcons" ~ "pfrspcons",
    variable == "MeanRootPairs_ppopul" ~ "ppopul",
    variable == "MeanRootPairs_pgrpsymp" ~ "pgrpsymp",
    TRUE ~ variable
  ))

# density plot with a different color per variable
a <- ggplot(wk_data, aes(x = mrp_value, color = variable)) +
  geom_density(alpha = 0.6) +  # density instead of points (better at N = 10k)
  scale_color_manual(values = c("pinsttrust" = "#E31A1C",
                                "pfrspcons" = "#1F78B4",
                                "ppopul" = "#33A02C",
                                "pgrpsymp" = "#FF7F00")) +
  facet_wrap(~Interviewmode) +
  labs(title = "MRP Distribution: WEAK Straightlining Batteries (sensan)",
       x = "Mean Root Pairs",
       y = "Density",
       color = "Battery") +
  theme_minimal() +
  theme(legend.position = "bottom")



# ST data, long format
st_data <- w1 |>
  select(lfd, Interviewmode, MeanRootPairs_pcohes, MeanRootPairs_pfamexp,
         MeanRootPairs_pworkexp, MeanRootPairs_pneighexp,
         MeanRootPairs_punivtrad, MeanRootPairs_pgrpknow) |>
  pivot_longer(c(MeanRootPairs_pcohes:MeanRootPairs_pgrpknow), names_to = "variable", values_to
   ↪  = "mrp_value") |>
  mutate(variable = case_when(
    variable == "MeanRootPairs_pcohes" ~ "pcohes",
    variable == "MeanRootPairs_pfamexp" ~ "pfamexp",
    variable == "MeanRootPairs_pworkexp" ~ "pworkexp",
    variable == "MeanRootPairs_pneighexp" ~ "pneighexp",
    variable == "MeanRootPairs_punivtrad" ~ "punivtrad",
    variable == "MeanRootPairs_pgrpknow" ~ "pgrpknow"
  ))

#  density plot for ST
b <- ggplot(st_data, aes(x = mrp_value, color = variable)) +
  geom_density(alpha = 0.6) +
  scale_color_manual(values = c("pcohes" = "#E31A1C",
                                "pfamexp" = "#1F78B4",
                                "pworkexp" = "#33A02C",
                                "pneighexp" = "#FF7F00",
                                "punivtrad" = "#6A3D9A",
                                "pgrpknow" = "#FDBF6F")) +
  facet_wrap(~Interviewmode) +
  labs(title = "MRP Distribution: STRONG Straightlining Batteries (sensan)",
       x = "Mean Root Pairs",
       color = "Battery") +
  theme_minimal() +
  theme(legend.position = "bottom")
```

In Wave 1, anchor persons and household members were asked to respond to 52 questions. The question for

panelconsent, the household panel (only for anker persons) and the questions for the device used followed (CAWI only). From question number 34, all questions were demography. The only exception made question number 47 (psunvote).

We calculated the level on nondifferentiations for a variety of questions that were presented in the form of 10 item grids from the whole survey namely questions 1, 14, 23, 24, 25, 26, 27, 28, 30 and 31. The scales include a total of 97 items.

```
chart.Correlation(w1 |> select(contains("MeanRootPairs")), histogram = TRUE)
```

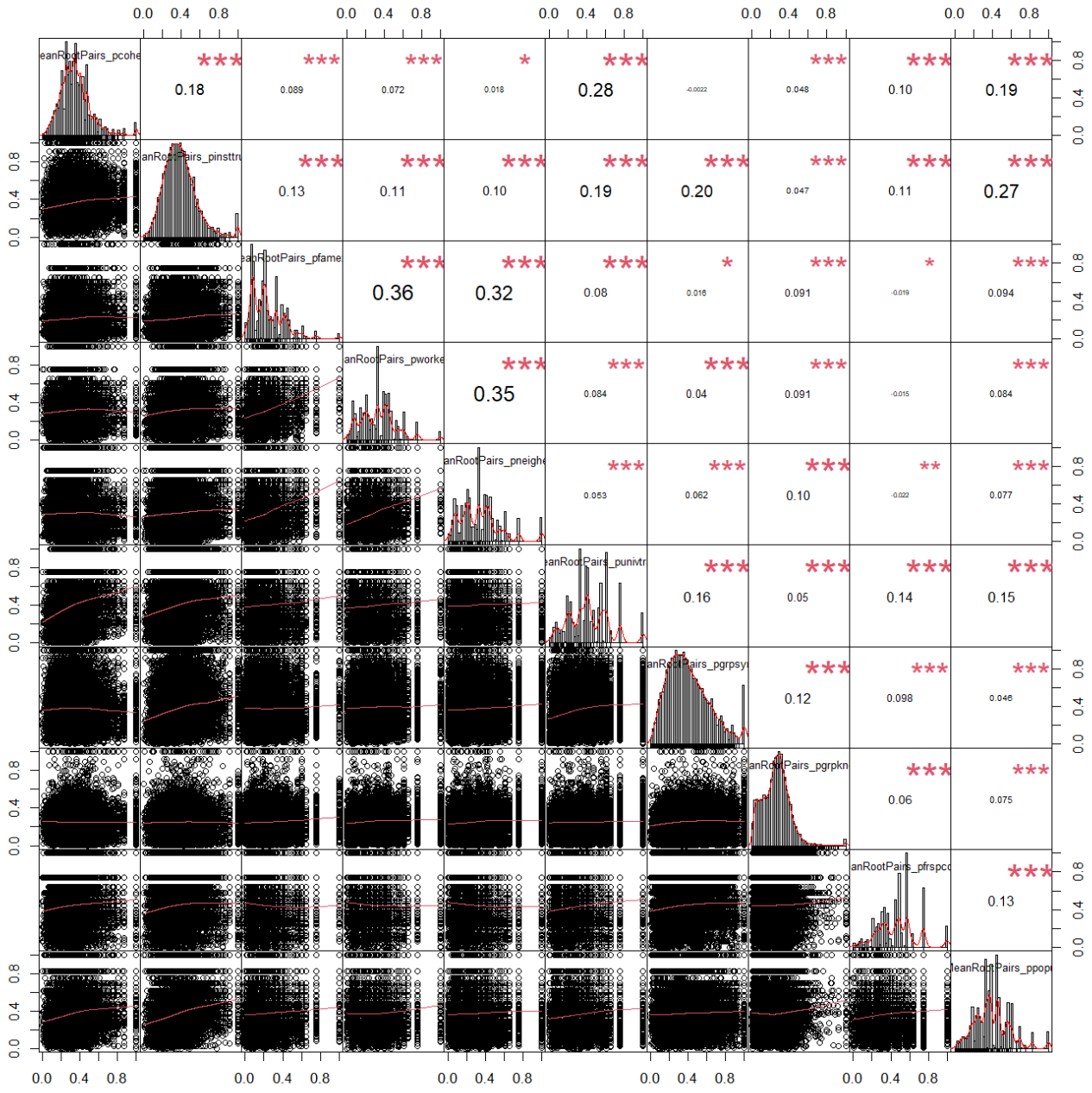


Figure 9: Correlations of Nondifferentiation Items

3.11.2 w1 Nd SEM using CT-C(M − 1) Model

- Ideally, the latent scores would be used. However, they can only be estimated for records with complete MeanRootPairs values, which would entail a very high amount of missingness.
- We therefore use the manifest scores, aggregated as the mean across all MeanRootPairs values (=> almost no missingness), and then check the agreement between the two measures.
- The latent scores correlate with the aggregated manifest scores at .86*** (WK) and .82*** (ST), respectively

```
p_load("lavaan", "tidySEM", "semPlot")


# correlated traits: WK (weak) and ST (strong)
# correlated methods-1: Reference-scale: 11-likert

w1LatND_mod <- "

#NondiffWeak
WK =~
MeanRootPairs_pinsttrust + MeanRootPairs_pfrspcons + MeanRootPairs_ppopul +
 ↪  MeanRootPairs_pgrpsymp

#NonidffStrong
ST =~
MeanRootPairs_pcohes + MeanRootPairs_pfamexp + MeanRootPairs_pworkexp + MeanRootPairs_pneighexp
 ↪  + MeanRootPairs_punivtrad + MeanRootPairs_pgrpknow

#grids with scales with less-than-11-point-likert scales
u11_sensan =~
MeanRootPairs_pcohes + MeanRootPairs_pfamexp + MeanRootPairs_pworkexp + MeanRootPairs_pneighexp
 ↪  + MeanRootPairs_punivtrad + MeanRootPairs_ppopul + MeanRootPairs_pfrspcons +
 ↪  MeanRootPairs_pgrpknow

WK ~~ 0 * u11_sensan
ST ~~ 0 * u11_sensan

"

# simple model without groups
w1LatND_fit_sensan <- sem(w1LatND_mod, w1, meanstructure = TRUE)
summary(w1LatND_fit_sensan, fit = TRUE)
```

```
lavaan 0.6-21 ended normally after 215 iterations

  Estimator                                         ML
  Optimization method                           NLMINB
  Number of model parameters                        39

                                                  Used       Total
  Number of observations                         10148       17023

Model Test User Model:

  Test statistic                               489.380
  Degrees of freedom                               26
  P-value (Chi-square)                          0.000

Model Test Baseline Model:
```

```
  Test statistic                                7558.058
  Degrees of freedom                                 45
  P-value                                         0.000

User Model versus Baseline Model:

  Comparative Fit Index (CFI)                     0.938
  Tucker-Lewis Index (TLI)                        0.893

Loglikelihood and Information Criteria:

  Loglikelihood user model (H0)               32892.354
  Loglikelihood unrestricted model (H1)       33137.044

  Akaike (AIC)                               -65706.708
  Bayesian (BIC)                             -65424.932
  Sample-size adjusted Bayesian (SABIC)      -65548.868

Root Mean Square Error of Approximation:

  RMSEA                                           0.042
  90 Percent confidence interval - lower          0.039
  90 Percent confidence interval - upper          0.045
  P-value H_0: RMSEA <= 0.050                     1.000
  P-value H_0: RMSEA >= 0.080                     0.000

Standardized Root Mean Square Residual:

  SRMR                                            0.024

Parameter Estimates:

  Standard errors                             Standard
  Information                                 Expected
  Information saturated (h1) model          Structured

Latent Variables:
                   Estimate  Std.Err  z-value  P(>|z|)
  WK =~
    MnRtPrs_pnsttr    1.000
    MnRtPrs_pfrspc    0.586    0.033   17.972    0.000
    MeanRtPrs_pppl    0.898    0.037   24.553    0.000
    MnRtPrs_pgrpsy    0.633    0.035   18.272    0.000
  ST =~
    MeanRtPrs_pchs    1.000
    MenRtPrs_pfmxp    0.658    0.048   13.579    0.000
    MnRtPrs_pwrkxp    0.746    0.059   12.682    0.000
    MnRtPrs_pnghxp    0.647    0.061   10.624    0.000
    MnRtPrs_pnvtrd    1.532    0.071   21.625    0.000
    MnRtPrs_pgrpkn    0.319    0.027   11.753    0.000
  u11_sensan =~
    MeanRtPrs_pchs    1.000
    MenRtPrs_pfmxp   -4.142    0.717   -5.780    0.000
    MnRtPrs_pwrkxp   -5.531    0.945   -5.856    0.000
    MnRtPrs_pnghxp   -5.538    0.938   -5.906    0.000
```

```
    MnRtPrs_pnvtrd    1.582    0.231    6.850    0.000
    MeanRtPrs_pppl    0.077    0.137    0.566    0.572
    MnRtPrs_pfrspc    1.194    0.210    5.677    0.000
    MnRtPrs_pgrpkn   -0.627    0.161   -3.906    0.000

Covariances:
                   Estimate  Std.Err  z-value  P(>|z|)
  WK ~~
    u11_sensan        0.000
  ST ~~
    u11_sensan        0.000
  WK ~~
    ST                0.005    0.000   21.655    0.000

Intercepts:
                   Estimate  Std.Err  z-value  P(>|z|)
   .MnRtPrs_pnsttr    0.380    0.002  233.908    0.000
   .MnRtPrs_pfrspc    0.458    0.002  227.962    0.000
   .MeanRtPrs_pppl    0.393    0.002  216.304    0.000
   .MnRtPrs_pgrpsy    0.414    0.002  191.372    0.000
   .MeanRtPrs_pchs    0.338    0.002  223.262    0.000
   .MenRtPrs_pfmxp    0.231    0.002  141.590    0.000
   .MnRtPrs_pwrkxp    0.332    0.002  181.662    0.000
   .MnRtPrs_pnghxp    0.346    0.002  161.635    0.000
   .MnRtPrs_pnvtrd    0.409    0.002  194.334    0.000
   .MnRtPrs_pgrpkn    0.270    0.001  198.751    0.000

Variances:
                   Estimate  Std.Err  z-value  P(>|z|)
   .MnRtPrs_pnsttr    0.018    0.000   40.939    0.000
   .MnRtPrs_pfrspc    0.037    0.001   65.118    0.000
   .MeanRtPrs_pppl    0.026    0.000   54.357    0.000
   .MnRtPrs_pgrpsy    0.044    0.001   66.539    0.000
   .MeanRtPrs_pchs    0.017    0.000   46.035    0.000
   .MenRtPrs_pfmxp    0.018    0.000   48.905    0.000
   .MnRtPrs_pwrkxp    0.020    0.001   36.752    0.000
   .MnRtPrs_pnghxp    0.033    0.001   50.594    0.000
   .MnRtPrs_pnvtrd    0.031    0.001   38.414    0.000
   .MnRtPrs_pgrpkn    0.018    0.000   69.607    0.000
    WK                0.009    0.000   19.913    0.000
    ST                0.005    0.000   16.183    0.000
    u11_sensan        0.000    0.000    3.193    0.001
```

```r
fscores <- lavPredict(w1LatND_fit_sensan)  # matrix, not a list
idx     <- lavInspect(w1LatND_fit_sensan, "case.idx")  # single vector

# merge the scores back in, following the lavaan documentation for single-group models:
for (fs in colnames(fscores)) {
  w1[idx, fs] <- fscores[, fs]
}



# rename to wave suffix (latent scores WK/ST are not carried on)
w1 <-
  w1 |>
  rename_with(
```

```
    ~ str_c(., "_sensan"),
    starts_with("MeanRootPairs_"),
    c(WK, ST)
  ) |> select(-starts_with("temp_"), -u11_sensan)

w1 |> names()
```

```
  [1] "lfd"                       "Extr_Sample"
  [3] "Mid_Sample"                "EquivIncome_Sample"
  [5] "Age_cat_Sample"            "pid"
  [7] "hid"                       "gebjahr.weights"
  [9] "hh_lfd"                    "ZG"
 [11] "interviewsprache"          "interviewdatum"
 [13] "interviewdauer"            "qmethod"
 [15] "intvollst"                 "pcohes01"
 [17] "pcohes02"                  "pcohes03"
 [19] "pcohes04"                  "pcohes05"
 [21] "pcohes06"                  "pcohes07"
 [23] "pcohes08"                  "pcohes09"
 [25] "pcohes10"                  "pcohes11"
 [27] "pcohes12"                  "plifeuse"
 [29] "plifesat"                  "ppolatt"
 [31] "pgender"                   "pbirthmnth"
 [33] "pbirthyear_w1"             "pfamstat"
 [35] "pcohab"                    "pcohabgebyear"
 [37] "pcohabgender"              "pcov01"
 [39] "pcov02"                    "pcov03"
 [41] "pcov04"                    "pcovmeas"
 [43] "pcovlim"                   "psuppo01"
 [45] "psuppo02"                  "psuppo03"
 [47] "psuppo04"                  "pgentrust"
 [49] "pinsttrust01"              "pinsttrust02"
 [51] "pinsttrust03"              "pinsttrust04"
 [53] "pinsttrust05"              "pinsttrust06"
 [55] "pinsttrust07"              "pinsttrust08"
 [57] "pinsttrust09"              "pinsttrust10"
 [59] "pinsttrust11"              "porigin"
 [61] "pothcount"                 "pgermstay"
 [63] "pnation01"                 "pnation02"
 [65] "pnatoth"                   "pmigback01"
 [67] "pmigback02"                "preligion"
 [69] "pdemocfit"                 "pdemocsat"
 [71] "pclimsor"                  "pclimcon"
 [73] "pfamexp01"                 "pfamexp02"
 [75] "pfamexp03"                 "pfamexp04"
 [77] "pfamexp05"                 "pfamexp06"
 [79] "pworkexp01"                "pworkexp02"
 [81] "pworkexp03"                "pworkexp04"
 [83] "pworkexp05"                "pworkexp06"
 [85] "pneighexp01"               "pneighexp02"
 [87] "pneighexp03"               "pneighexp04"
 [89] "pneighexp05"               "pneighexp06"
 [91] "punivtrad01"               "punivtrad02"
 [93] "punivtrad03"               "punivtrad04"
 [95] "punivtrad05"               "punivtrad06"
 [97] "pgrpsymp01"                "pgrpsymp02"
```

```
 [99] "pgrpsymp03"                "pgrpsymp04"
[101] "pgrpsymp05"                "pgrpsymp06"
[103] "pgrpsymp07"                "pgrpsymp08"
[105] "pgrpsymp09"                "pgrpsymp10"
[107] "pgrpsymp11"                "pgrpsymp12"
[109] "pgrpsymp13"                "pgrpsymp14"
[111] "pgrpsymp15"                "pgrpsymp16"
[113] "pgrpsymp17"                "pgrpknow01"
[115] "pgrpknow02"                "pgrpknow03"
[117] "pgrpknow04"                "pgrpknow05"
[119] "pgrpknow06"                "pgrpknow07"
[121] "pgrpknow08"                "pgrpknow09"
[123] "pgrpknow10"                "pgrpknow11"
[125] "pgrpknow12"                "pgrpknow13"
[127] "pgrpknow14"                "pgrpknow15"
[129] "pgrpknow16"                "pgrpknow17"
[131] "pfreespeech"               "pfrspcons01"
[133] "pfrspcons02"               "pfrspcons03"
[135] "pfrspcons04"               "pfrspcons05"
[137] "pfrspcons06"               "pfrspcons07"
[139] "ppopul01"                  "ppopul02"
[141] "ppopul03"                  "ppopul04"
[143] "ppopul05"                  "ppopul06"
[145] "ppopul07"                  "ppopul08"
[147] "ppopul09"                  "pladder01"
[149] "preldepriv"                "pedu01"
[151] "pedu02"                    "pempstat1"
[153] "pgainemp"                  "pempcond01"
[155] "pempcond02"                "pempcond03"
[157] "pcontract"                 "poccupation"
[159] "poccupation_isco08"        "poccstat"
[161] "pselfemp"                  "pcivil"
[163] "pempl"                     "pworker"
[165] "plabincbrut"               "plabincnet"
[167] "pnoinc"                    "q46_check1"
[169] "q46_check2"                "q46_check5"
[171] "psundvote"                 "hhhtyp"
[173] "hhhsize"                   "q49_check1"
[175] "hhhage01"                  "hhhage02"
[177] "hhhage03"                  "q50_check1"
[179] "q50_check2"                "hhhincnet"
[181] "q51_check1"                "q51_check2"
[183] "hhhbasup"                  "hpname01"
[185] "hpname01_check1"           "hpsex01"
[187] "hprelat01"                 "hpbmnth01"
[189] "hpbyear01"                 "q53_01_check2"
[191] "hpall01"                   "q53_01_check3"
[193] "q53_01_check4"             "q53_01_check5"
[195] "hpname02"                  "hpname02_check1"
[197] "hpsex02"                   "hprelat02"
[199] "hpbmnth02"                 "hpbyear02"
[201] "q53_02_check2"             "hpall02"
[203] "q53_02_check3"             "q53_02_check4"
[205] "q53_02_check5"             "hpname03"
[207] "hpsex03"                   "hprelat03"
[209] "hpbmnth03"                 "hpbyear03"
```

```
[211] "q53_03_check2"                "hpall03"
[213] "q53_03_check3"                "q53_03_check4"
[215] "q53_03_check5"                "hpname04"
[217] "hpsex04"                      "hprelat04"
[219] "hpbmnth04"                    "hpbyear04"
[221] "q53_04_check2"                "hpall04"
[223] "q53_04_check3"                "q53_04_check4"
[225] "q53_04_check5"                "hpname05"
[227] "hpsex05"                      "hprelat05"
[229] "hpbmnth05"                    "hpbyear05"
[231] "hpall05"                      "q53_05_check5"
[233] "hpname06"                     "hpsex06"
[235] "hprelat06"                    "hpbmnth06"
[237] "hpbyear06"                    "hpall06"
[239] "q53_06_check5"                "hpname07"
[241] "hpsex07"                      "hprelat07"
[243] "hpbmnth07"                    "hpbyear07"
[245] "hpall07"                      "q53_07_check5"
[247] "hpname08"                     "hpsex08"
[249] "hprelat08"                    "hpbmnth08"
[251] "hpbyear08"                    "hpall08"
[253] "q53_08_check3"                "q53_08_check5"
[255] "hpname09"                     "hpsex09"
[257] "hprelat09"                    "hpbmnth09"
[259] "hpbyear09"                    "hpall09"
[261] "q53_09_check5"                "hpname10"
[263] "hpsex10"                      "hprelat10"
[265] "hpbmnth10"                    "hpbyear10"
[267] "pdevice01"                    "pdevice02"
[269] "pdevice03"                    "pdevice04"
[271] "pdevice05"                    "pothdevice"
[273] "ppanelcons"                   "q46_check3"
[275] "q46_check4"                   "pbirthyear_w2"
[277] "intvollst_w2"                 "plifesat01_w2"
[279] "pbirthyear_w3"                "intvollst_w3"
[281] "psundvoteOthWhich"            "peduOthWhich"
[283] "psundvoteOthWhich.text"       "peduOthWhich.text"
[285] "PartWaves"                    "byear"
[287] "bmnth"                        "myob"
[289] "ymd_interview"                "Age"
[291] "AgeConsistency"               "Sample"
[293] "Age_cat_sensan"               "BreakOff"
[295] "cid"                          "syear"
[297] "wave"                         "pgnation_scp"
[299] "pgfamstd_scp"                 "pglabgro"
[301] "pgi1labgro"                   "pgi2labgro"
[303] "pgi3labgro"                   "pgi4labgro"
[305] "pgi5labgro"                   "pgi6labgro"
[307] "pgi7labgro"                   "pgi8labgro"
[309] "pgi9labgro"                   "pgi10labgro"
[311] "pgi11labgro"                  "pgi12labgro"
[313] "pgi13labgro"                  "pgi14labgro"
[315] "pgi15labgro"                  "pgimpgro"
[317] "pglabnet"                     "pgi1labnet"
[319] "pgi2labnet"                   "pgi3labnet"
[321] "pgi4labnet"                   "pgi5labnet"
```

```
[323] "pgi6labnet"                      "pgi7labnet"
[325] "pgi8labnet"                      "pgi9labnet"
[327] "pgi10labnet"                     "pgi11labnet"
[329] "pgi12labnet"                     "pgi13labnet"
[331] "pgi14labnet"                     "pgi15labnet"
[333] "pgimpnet"                        "pgstib_scp"
[335] "pgstib1_scp"                     "pgemplst_scp"
[337] "pglfs_scp"                       "pglfs1_scp"
[339] "pgisco08_scp"                    "pgmisco08_scp"
[341] "pgvisco08_scp"                   "pgisco88_scp"
[343] "pgmisco88_scp"                   "pgvisco88_scp"
[345] "pgisei08_scp"                    "pgmisei08_scp"
[347] "pgvisei08_scp"                   "pgisei88_scp"
[349] "pgmisei88_scp"                   "pgvisei88_scp"
[351] "pgmps08_scp"                     "pgmmps08_scp"
[353] "pgvmps08_scp"                    "pgsiops08_scp"
[355] "pgmsiops08_scp"                  "pgvsiops08_scp"
[357] "pgsiops88_scp"                   "pgmsiops88_scp"
[359] "pgvsiops88_scp"                  "pgegp88_scp"
[361] "pgegp88_1_scp"                   "pgmegp88_scp"
[363] "pgvegp88_scp"                    "pgiscolj08_scp"
[365] "pgiscolj88_scp"                  "pgbetr_scp"
[367] "pgisced11_scp"                   "pgisced97_scp"
[369] "pgcasmin_scp"                    "pgbilzeit_scp"
[371] "pgmonth"                         "pgpiyear"
[373] "pgmode"                          "pfoccupation_isco08"
[375] "pmoccupation_isco08"             "Education3"
[377] "Education_i"                     "Education_method"
[379] "Education"                       "Gender"
[381] "Employment"                      "ChildHH"
[383] "AllWavesComplete"                "Mig"
[385] "hhsize"                          "u14"
[387] "ue14"                            "equivSize"
[389] "EquivIncome_num"                 "EquivIncome_sensan"
[391] "ProportionSpeeding_sensan"       "ProportionSpeeding_Sample"
[393] "Speeding_sensan"                 "Speeding_Sample"
[395] "Interviewmode"                   "TargetGroup"
[397] "freetextmis"                     "N_allFreeTitems"
[399] "FrTMisRatio"                     "FrTMis_sensan"
[401] "FrTMis_Sample"                   "NonrespRatio"
[403] "E5"                              "M5"
[405] "EM.S5"                           "E7"
[407] "M7"                              "EM.S7"
[409] "E11"                             "M11"
[411] "EM.S11"                          "N5_all"
[413] "N7_all"                          "N11_all"
[415] "E.S5S7S11"                       "M.S5S7S11"
[417] "N_allExMidOpportunities"         "Ratio_Extr"
[419] "Ratio_Mid"                       "RatioExMid"
[421] "Extr_sensan"                     "Mid_sensan"
[423] "MeanRootPairs_pcohes_sensan"     "MeanRootPairs_pinsttrust_sensan"
[425] "MeanRootPairs_pfamexp_sensan"    "MeanRootPairs_pworkexp_sensan"
[427] "MeanRootPairs_pneighexp_sensan"  "MeanRootPairs_punivtrad_sensan"
[429] "MeanRootPairs_pgrpsymp_sensan"   "MeanRootPairs_pgrpknow_sensan"
[431] "MeanRootPairs_pfrspcons_sensan"  "MeanRootPairs_ppopul_sensan"
[433] "WK"                              "ST"
```

```r
w1s <- w1 |> filter(Sample == "Sample") |> select(c("lfd", "Extr_Sample":"Mid_sensan"))

pcohes.items <- w1s|> select(pcohes01:pcohes12) |> colnames()
pinsttrust.items <- w1s|> select(pinsttrust01:pinsttrust11) |> colnames()
pfamexp.items <- w1s|> select(pfamexp01:pfamexp06) |> colnames()
pworkexp.items <- w1s|> select(pworkexp01:pworkexp06) |> colnames()    # Category "Unemployed"
pneighexp.items <- w1s|> select(pneighexp01:pneighexp06) |> colnames()
punivtrad.items <- w1s|> select(punivtrad01:punivtrad06) |> colnames()
pgrpsymp.items <- w1s|> select(pgrpsymp01:pgrpsymp17) |> colnames()
pgrpknow.items <- w1s|> select(pgrpknow01:pgrpknow17) |> colnames()
pfrspcons.items <- w1s|> select(pfrspcons01:pfrspcons07) |> colnames()
ppopul.items <- w1s|> select(ppopul01:ppopul09) |> colnames()

items_list <- list(pcohes.items, pinsttrust.items,
                   pfamexp.items, pworkexp.items,
                   pneighexp.items, punivtrad.items, pgrpsymp.items,
                   pgrpknow.items, pfrspcons.items, ppopul.items)

tND <- Sys.time()

# extract all items from items_list
all_items <- unlist(items_list)

w1s_papi <-
  w1s|>
  filter(Interviewmode == "PAPI") |>
  select(lfd, all_of(all_items)) |>  # select only the item columns
  meanRootPairs(items_list) |>
  ungroup()
```

```
Duration for pcohes : 0.2580687 minutes
Duration for pinsttrust : 0.2943765 minutes
Duration for pfamexp : 0.2342652 minutes
Duration for pworkexp : 0.223498 minutes
Duration for pneighexp : 0.2429087 minutes
Duration for punivtrad : 0.2619737 minutes
Duration for pgrpsymp : 0.4009406 minutes
Duration for pgrpknow : 0.4259431 minutes
Duration for pfrspcons : 0.2677344 minutes
Duration for ppopul : 0.2569893 minutes
All done!
Total duration for all item packs: 2.866911 minutes.
```

```r
w1s_cawi  <-
  w1s|>
  filter(Interviewmode == "CAWI") |>
  select(lfd, all_of(all_items)) |>
   meanRootPairs(items_list) |> ungroup()
```

```
Duration for pcohes : 0.1851843 minutes
Duration for pinsttrust : 0.2381775 minutes
Duration for pfamexp : 0.165717 minutes
Duration for pworkexp : 0.181799 minutes
Duration for pneighexp : 0.1568253 minutes
Duration for punivtrad : 0.1450705 minutes
Duration for pgrpsymp : 0.3268179 minutes
```

```
Duration for pgrpknow : 0.3219236 minutes
Duration for pfrspcons : 0.1849515 minutes
Duration for ppopul : 0.2217563 minutes
All done!
Total duration for all item packs: 2.128393 minutes.
```

```r
# merge PAPI and CAWI
w1s_meanroot <- bind_rows(w1s_papi, w1s_cawi) |> select(lfd, starts_with("temp_"),
 ↪  starts_with("MeanRoot"))

# join onto the original w1s dataset
w1s<- w1s|>
  left_join(w1s_meanroot, by = "lfd")
```

```r
# prepare the data for the WK variables (long format)
wk_datas <- w1s |>
  select(lfd, Interviewmode, MeanRootPairs_pinsttrust, MeanRootPairs_pfrspcons,
         MeanRootPairs_ppopul, MeanRootPairs_pgrpsymp) |>
  pivot_longer(c(MeanRootPairs_pinsttrust:MeanRootPairs_pgrpsymp),
               names_to = "variable",
               values_to = "mrp_value") |>
  mutate(variable = case_when(
    variable == "MeanRootPairs_pinsttrust" ~ "pinsttrust",
    variable == "MeanRootPairs_pfrspcons" ~ "pfrspcons",
    variable == "MeanRootPairs_ppopul" ~ "ppopul",
    variable == "MeanRootPairs_pgrpsymp" ~ "pgrpsymp",
    TRUE ~ variable
  ))

# density plot with a different color per variable
a <- ggplot(wk_datas, aes(x = mrp_value, color = variable)) +
  geom_density(alpha = 0.6) +  # density instead of points (better at N = 10k)
  scale_color_manual(values = c("pinsttrust" = "#E31A1C",
                                "pfrspcons" = "#1F78B4",
                                "ppopul" = "#33A02C",
                                "pgrpsymp" = "#FF7F00")) +
  facet_wrap(~Interviewmode) +
  labs(title = "MRP Distribution: WEAK Straightlining Batteries (Analysis Sample)",
       x = "Mean Root Pairs",
       y = "Density",
       color = "Battery") +
  theme_minimal() +
  theme(legend.position = "bottom")



# ST data, long format
st_datas <- w1s |>
  select(lfd, Interviewmode, MeanRootPairs_pcohes, MeanRootPairs_pfamexp,
         MeanRootPairs_pworkexp, MeanRootPairs_pneighexp,
         MeanRootPairs_punivtrad, MeanRootPairs_pgrpknow) |>
  pivot_longer(c(MeanRootPairs_pcohes:MeanRootPairs_pgrpknow), names_to = "variable", values_to
   ↪  = "mrp_value") |>
  mutate(variable = case_when(
    variable == "MeanRootPairs_pcohes" ~ "pcohes",
    variable == "MeanRootPairs_pfamexp" ~ "pfamexp",
    variable == "MeanRootPairs_pworkexp" ~ "pworkexp",
```

```
    variable == "MeanRootPairs_pneighexp" ~ "pneighexp",
    variable == "MeanRootPairs_punivtrad" ~ "punivtrad",
    variable == "MeanRootPairs_pgrpknow" ~ "pgrpknow"
  ))

#  density plot for ST
b <- ggplot(st_datas, aes(x = mrp_value, color = variable)) +
  geom_density(alpha = 0.6) +
  scale_color_manual(values = c("pcohes" = "#E31A1C",
                                "pfamexp" = "#1F78B4",
                                "pworkexp" = "#33A02C",
                                "pneighexp" = "#FF7F00",
                                "punivtrad" = "#6A3D9A",
                                "pgrpknow" = "#FDBF6F")) +
  facet_wrap(~Interviewmode) +
  labs(title = "MRP Distribution: STRONG Straightlining Batteries (Analysis Sample)",
       x = "Mean Root Pairs",
       color = "Battery") +
  theme_minimal() +
  theme(legend.position = "bottom")
```

```
chart.Correlation(w1s |> select(contains("MeanRootPairs")), histogram = TRUE)
```

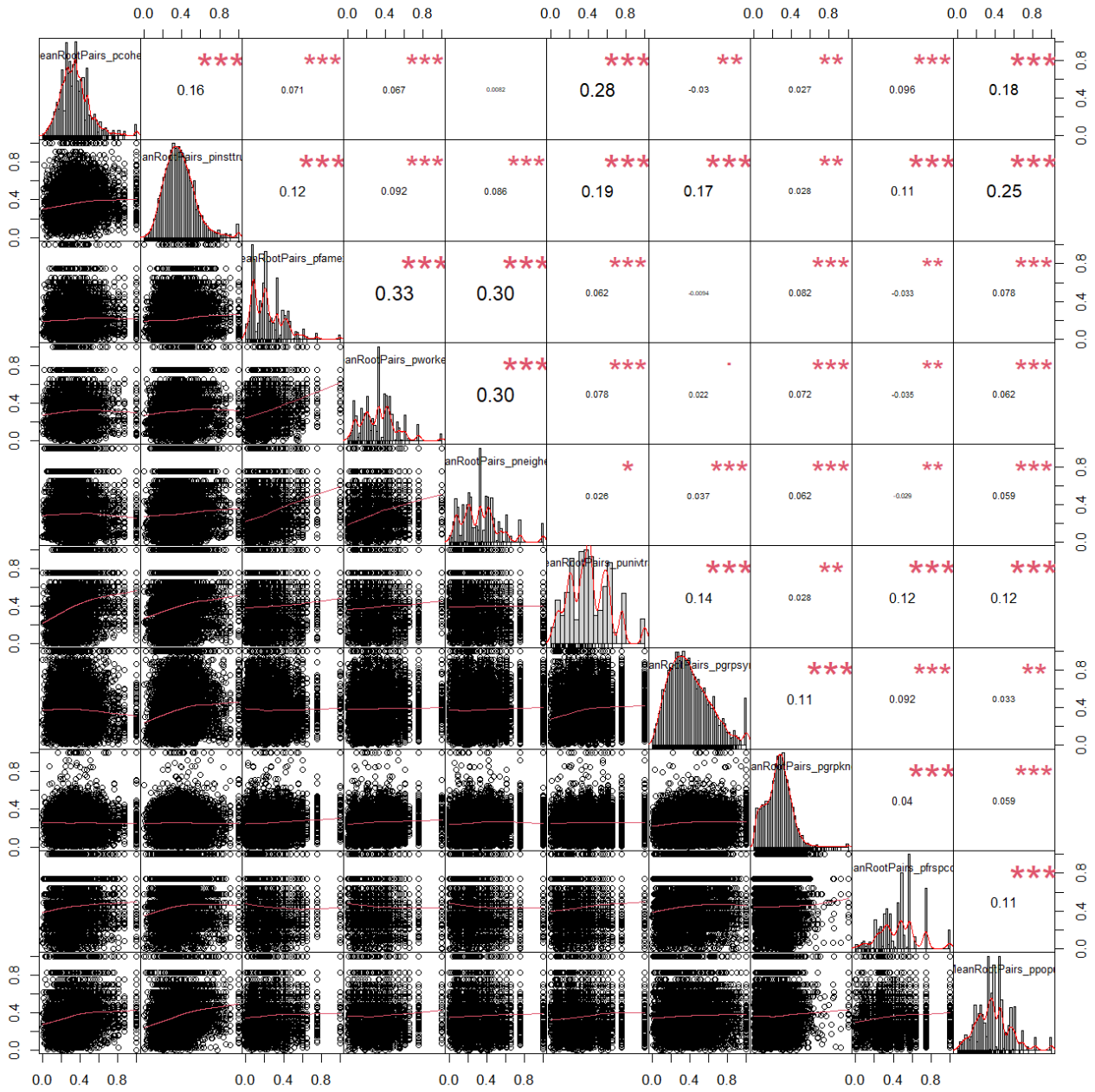


Figure 10: Correlations of Nondifferentiation Items (Sample)

3.11.3 w1 Nd SEM using CT-C(M − 1) Model (analysis sample)

```
# correlated traits: WK (weak) and ST (strong)
# correlated methods-1: Reference-scale: 11-likert

w1sLatND_mod <- "

#NondiffWeak
WK =~
MeanRootPairs_pinsttrust + MeanRootPairs_pfrspcons + MeanRootPairs_ppopul +
 ↪  MeanRootPairs_pgrpsymp
```

```
#NonidffStrong
ST =~
MeanRootPairs_pcohes + MeanRootPairs_pfamexp + MeanRootPairs_pworkexp + MeanRootPairs_pneighexp
 ↪  + MeanRootPairs_punivtrad + MeanRootPairs_pgrpknow

#grids with scales with less-than-11-point-likert scales
u11 =~
MeanRootPairs_pcohes + MeanRootPairs_pfamexp + MeanRootPairs_pworkexp + MeanRootPairs_pneighexp
 ↪  + MeanRootPairs_punivtrad + MeanRootPairs_ppopul + MeanRootPairs_pfrspcons +
 ↪  MeanRootPairs_pgrpknow

WK ~~ 0 * u11
ST ~~ 0 * u11

"

# simple model without groups
w1sLatND_fit <- sem(w1sLatND_mod, w1s, meanstructure = TRUE)
summary(w1sLatND_fit, fit = TRUE)
```

```
lavaan 0.6-21 ended normally after 213 iterations

  Estimator                                         ML
  Optimization method                           NLMINB
  Number of model parameters                        39

                                                  Used       Total
  Number of observations                          6003        9447

Model Test User Model:

  Test statistic                               309.018
  Degrees of freedom                                26
  P-value (Chi-square)                           0.000

Model Test Baseline Model:

  Test statistic                              3724.317
  Degrees of freedom                                45
  P-value                                        0.000

User Model versus Baseline Model:

  Comparative Fit Index (CFI)                    0.923
  Tucker-Lewis Index (TLI)                       0.867

Loglikelihood and Information Criteria:

  Loglikelihood user model (H0)              21170.842
  Loglikelihood unrestricted model (H1)      21325.351

  Akaike (AIC)                              -42263.685
  Bayesian (BIC)                            -42002.384
  Sample-size adjusted Bayesian (SABIC)     -42126.315
```

```
Root Mean Square Error of Approximation:

  RMSEA                                          0.043
  90 Percent confidence interval - lower         0.038
  90 Percent confidence interval - upper         0.047
  P-value H_0: RMSEA <= 0.050                    0.998
  P-value H_0: RMSEA >= 0.080                    0.000

Standardized Root Mean Square Residual:

  SRMR                                           0.025

Parameter Estimates:

  Standard errors                             Standard
  Information                                 Expected
  Information saturated (h1) model          Structured

Latent Variables:
                   Estimate  Std.Err  z-value  P(>|z|)
  WK =~
    MnRtPrs_pnsttr    1.000
    MnRtPrs_pfrspc    0.550    0.046   12.024    0.000
    MeanRtPrs_pppl    0.835    0.052   16.184    0.000
    MnRtPrs_pgrpsy    0.532    0.047   11.328    0.000
  ST =~
    MeanRtPrs_pchs    1.000
    MenRtPrs_pfmxp    0.582    0.068    8.594    0.000
    MnRtPrs_pwrkxp    0.664    0.079    8.411    0.000
    MnRtPrs_pnghxp    0.501    0.081    6.147    0.000
    MnRtPrs_pnvtrd    1.564    0.105   14.875    0.000
    MnRtPrs_pgrpkn    0.237    0.034    6.962    0.000
  u11 =~
    MeanRtPrs_pchs    1.000
    MenRtPrs_pfmxp   -4.735    1.238   -3.825    0.000
    MnRtPrs_pwrkxp   -5.729    1.492   -3.841    0.000
    MnRtPrs_pnghxp   -5.930    1.518   -3.907    0.000
    MnRtPrs_pnvtrd    1.732    0.360    4.806    0.000
    MeanRtPrs_pppl    0.046    0.200    0.230    0.818
    MnRtPrs_pfrspc    1.526    0.384    3.978    0.000
    MnRtPrs_pgrpkn   -0.553    0.219   -2.527    0.011

Covariances:
                   Estimate  Std.Err  z-value  P(>|z|)
  WK ~~
    u11               0.000
  ST ~~
    u11               0.000
  WK ~~
    ST                0.004    0.000   15.243    0.000

Intercepts:
                   Estimate  Std.Err  z-value  P(>|z|)
   .MnRtPrs_pnsttr    0.372    0.002  186.992    0.000
   .MnRtPrs_pfrspc    0.457    0.003  177.291    0.000
   .MeanRtPrs_pppl    0.379    0.002  165.774    0.000
```

```
   .MnRtPrs_pgrpsy    0.405    0.003  147.012    0.000
   .MeanRtPrs_pchs    0.334    0.002  176.842    0.000
   .MenRtPrs_pfmxp    0.223    0.002  108.249    0.000
   .MnRtPrs_pwrkxp    0.327    0.002  141.702    0.000
   .MnRtPrs_pnghxp    0.338    0.003  126.843    0.000
   .MnRtPrs_pnvtrd    0.402    0.003  149.044    0.000
   .MnRtPrs_pgrpkn    0.269    0.002  162.592    0.000

Variances:
                   Estimate  Std.Err  z-value  P(>|z|)
   .MnRtPrs_pnsttr    0.015    0.001   26.903    0.000
   .MnRtPrs_pfrspc    0.037    0.001   50.164    0.000
   .MeanRtPrs_pppl    0.026    0.001   42.482    0.000
   .MnRtPrs_pgrpsy    0.043    0.001   52.083    0.000
   .MeanRtPrs_pchs    0.016    0.000   33.260    0.000
   .MenRtPrs_pfmxp    0.018    0.001   35.148    0.000
   .MnRtPrs_pwrkxp    0.021    0.001   30.968    0.000
   .MnRtPrs_pnghxp    0.032    0.001   39.016    0.000
   .MnRtPrs_pnvtrd    0.030    0.001   27.603    0.000
   .MnRtPrs_pgrpkn    0.016    0.000   53.911    0.000
    WK                0.008    0.001   13.869    0.000
    ST                0.005    0.000   11.630    0.000
    u11               0.000    0.000    2.080    0.037
```

```
fscores <- lavPredict(w1sLatND_fit)  # matrix, not a list
idx     <- lavInspect(w1sLatND_fit, "case.idx")  # single vector

# merge the scores back in, following the lavaan documentation for single-group models:
for (fs in colnames(fscores)) {
  w1s[idx, fs] <- fscores[, fs]
}

#
#
```

```
# model used as the basis for the coefficient table
SEM_w1 <- table_results(w1sLatND_fit)|>
    mutate(label = label |>
                str_replace("MeanRootPairs_", "Mrp.") |>
                str_replace("=~", ".BY.") |>
                str_replace("~~", ".WITH.") |>
                str_replace("~1", "") |>
                str_replace("^\\.", "Means.")) |>
  filter(!str_starts(label, "Variances"),
         !str_starts(label, "Means")) |>
  select(label, est_sig, se, confint)


SEM_w1|>
  kable(
    format = "latex",
    booktabs = TRUE,
    col.names = c("Label",
                  "Est.", "SE", "95\\% CI"),
    escape = FALSE
  )
```

Label	Est.	SE	95% CI
WK.BY.Mrp.pinsttrust	1.00	0.00	[1.00, 1.00]
WK.BY.Mrp.pfrspcons	0.55***	0.05	[0.46, 0.64]
WK.BY.Mrp.ppopul	0.84***	0.05	[0.73, 0.94]
WK.BY.Mrp.pgrpsymp	0.53***	0.05	[0.44, 0.62]
ST.BY.Mrp.pcohes	1.00	0.00	[1.00, 1.00]
ST.BY.Mrp.pfamexp	0.58***	0.07	[0.45, 0.71]
ST.BY.Mrp.pworkexp	0.66***	0.08	[0.51, 0.82]
ST.BY.Mrp.pneighexp	0.50***	0.08	[0.34, 0.66]
ST.BY.Mrp.punivtrad	1.56***	0.11	[1.36, 1.77]
ST.BY.Mrp.pgrpknow	0.24***	0.03	[0.17, 0.30]
u11.BY.Mrp.pcohes	1.00	0.00	[1.00, 1.00]
u11.BY.Mrp.pfamexp	-4.73***	1.24	[-7.16, -2.31]
u11.BY.Mrp.pworkexp	-5.73***	1.49	[-8.65, -2.81]
u11.BY.Mrp.pneighexp	-5.93***	1.52	[-8.90, -2.96]
u11.BY.Mrp.punivtrad	1.73***	0.36	[1.03, 2.44]
u11.BY.Mrp.ppopul	0.05	0.20	[-0.35, 0.44]
u11.BY.Mrp.pfrspcons	1.53***	0.38	[0.77, 2.28]
u11.BY.Mrp.pgrpknow	-0.55*	0.22	[-0.98, -0.12]
WK.WITH.u11	0.00	0.00	[0.00, 0.00]
ST.WITH.u11	0.00	0.00	[0.00, 0.00]
WK.WITH.ST	0.00***	0.00	[0.00, 0.00]

```
w1s |>
select(lfd, Interviewmode,

       "Q1: pcohes\n12 items, 5-likert, multi" = MeanRootPairs_pcohes,
       "Q14: pinsttrust\n11 items, 11-likert, uni" = MeanRootPairs_pinsttrust,
       "Q23: pfamexp\n6 items, 5-likert, multi" = MeanRootPairs_pfamexp,
       "Q24: pworkexp\n6 items, 5-likert, multi" = MeanRootPairs_pworkexp,
       "Q25: pneighexp\n6 items, 5-likert, multi" = MeanRootPairs_pneighexp,
       "Q26: punivtrad\n6 items, 5-likert, multi" = MeanRootPairs_punivtrad,
       "Q27: pgrpsymp\n17 items, 11-likert, uni" = MeanRootPairs_pgrpsymp,
       "Q28: pgrpknow\n17 items, 7-likert, multi" = MeanRootPairs_pgrpknow,
       "Q30: pfrspcons\n7 items, 4-likert, uni" = MeanRootPairs_pfrspcons,
       "Q31: ppopul\n9 items, 5-likert, uni" = MeanRootPairs_ppopul

       ) |>

# long format
pivot_longer(cols = 3:12) |>



ggplot(aes(Interviewmode, value, color = Interviewmode, group = Interviewmode)) +

geom_violin(scale = "count", linewidth = 0.8) +

geom_boxplot(width = 0.15, color = "black", aes(fill = Interviewmode)) +

scale_fill_manual(values = mode_palette) +
scale_color_manual(values = mode_palette) +
 labs(caption = "Wave 1: Nondifferentiation (Mean Root of Pairs) by Interviewmode (Analysis
  ↪  Sample)",
```

```
      x = "",
      y = "Nondifferentiation (mean root of pairs)",
      fill = "Interviewmode") +


facet_wrap(~ factor(name, c(

  "Q1: pcohes\n12 items, 5-likert, multi",
  "Q14: pinsttrust\n11 items, 11-likert, uni",
  "Q23: pfamexp\n6 items, 5-likert, multi",
  "Q24: pworkexp\n6 items, 5-likert, multi",
  "Q25: pneighexp\n6 items, 5-likert, multi",
  "Q26: punivtrad\n6 items, 5-likert, multi",
  "Q27: pgrpsymp\n17 items, 11-likert, uni",
  "Q28: pgrpknow\n17 items, 7-likert, multi",
  "Q30: pfrspcons\n7 items, 4-likert, uni",
  "Q31: ppopul\n9 items, 5-likert, uni")), ncol = 5) +

 guides(color = "none") +


papaja::theme_apa() +

theme(axis.text.x = element_blank(),
      axis.text.y = element_text(size = 10),
      axis.ticks.x = element_blank(),
      axis.title.y = element_text(size = 14),
      axis.line.x = element_blank(),
      legend.position="bottom",
      strip.text.x = element_text(size = 12, face = "bold"),
      panel.grid.major.y = element_line(color = "#404040", linewidth = 0.02),
      panel.grid.minor.y = element_line(color = "#404040", linewidth = 0.02),
      plot.caption = element_text(size = 14))
```

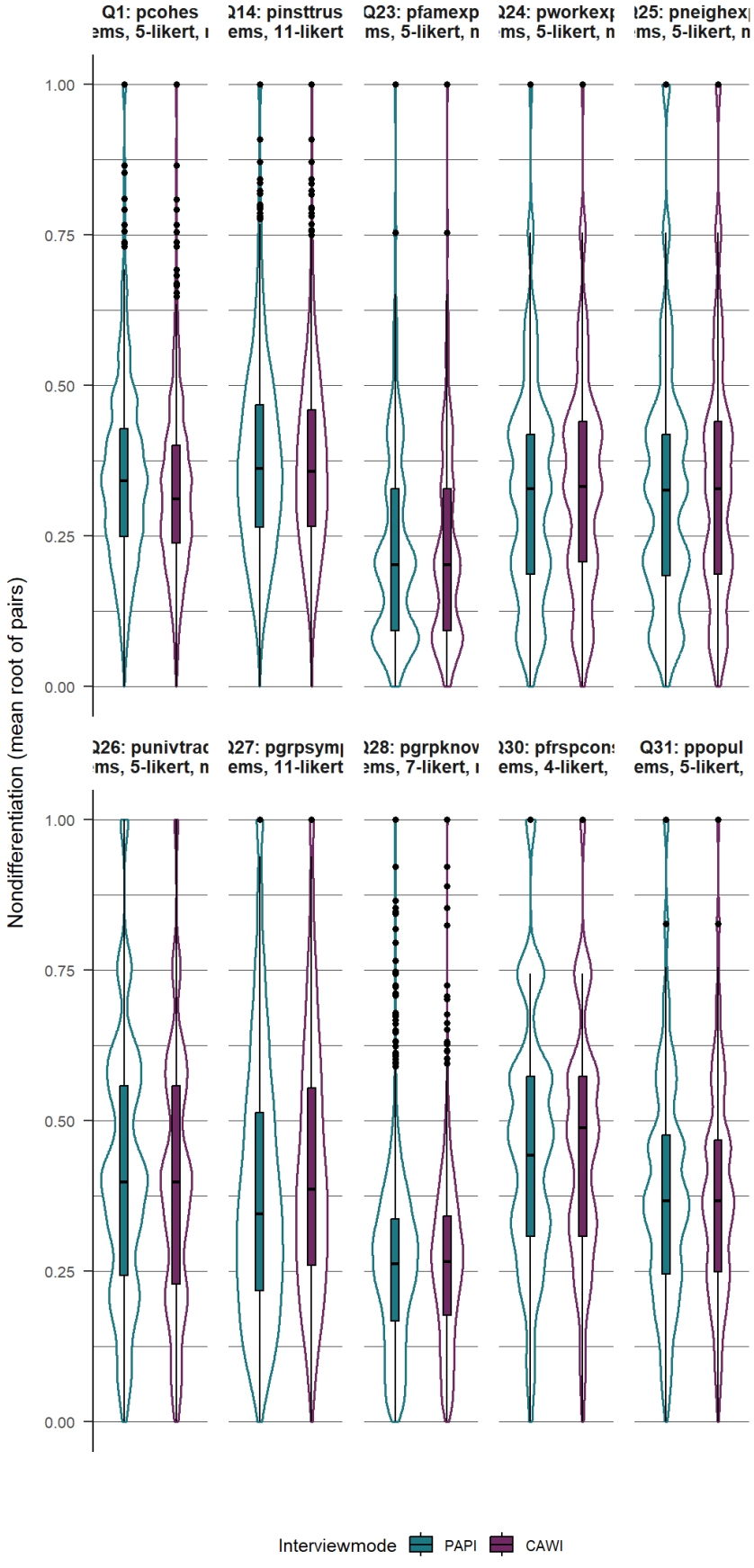


Figure 11: Wave 1 Nondifferentiation by Interviewmode.

```r
w1s |>
select(lfd,
       "Q1: pcohes\n12 items, 5-likert, multi" = MeanRootPairs_pcohes,
       "Q14: pinsttrust\n11 items, 11-likert, uni" = MeanRootPairs_pinsttrust,
       "Q23: pfamexp\n6 items, 5-likert, multi" = MeanRootPairs_pfamexp,
       "Q24: pworkexp\n6 items, 5-likert, multi" = MeanRootPairs_pworkexp,
       "Q25: pneighexp\n6 items, 5-likert, multi" = MeanRootPairs_pneighexp,
       "Q26: punivtrad\n6 items, 5-likert, multi" = MeanRootPairs_punivtrad,
       "Q27: pgrpsymp\n17 items, 11-likert, uni" = MeanRootPairs_pgrpsymp,
       "Q28: pgrpknow\n17 items, 7-likert, multi" = MeanRootPairs_pgrpknow,
       "Q30: pfrspcons\n7 items, 4-likert, uni" = MeanRootPairs_pfrspcons,
       "Q31: ppopul\n9 items, 5-likert, uni" = MeanRootPairs_ppopul

       ) |>

# long format
pivot_longer(cols = 2:11) |> #slice(0) |> glimpse()



ggplot(aes(name, value)) +

geom_violin(scale = "count", linewidth = 0.8) +

geom_boxplot(width = 0.15, color = "black") +


 labs(caption = "Wave 1: Nondifferentiation (Mean Root of Pairs)",
      x = "",
      y = "Mean root of pairs") +

papaja::theme_apa() +

theme(axis.text.x = element_text(angle = 90),
      axis.text.y = element_text(size = 10),
      #axis.ticks.x = element_blank(),
      axis.title.y = element_text(size = 14),
      axis.line.x = element_blank(),
      panel.grid.major.y = element_line(color = "#404040", linewidth = 0.02),
      panel.grid.minor.y = element_line(color = "#404040", linewidth = 0.02),
      plot.caption = element_text(size = 14))
```

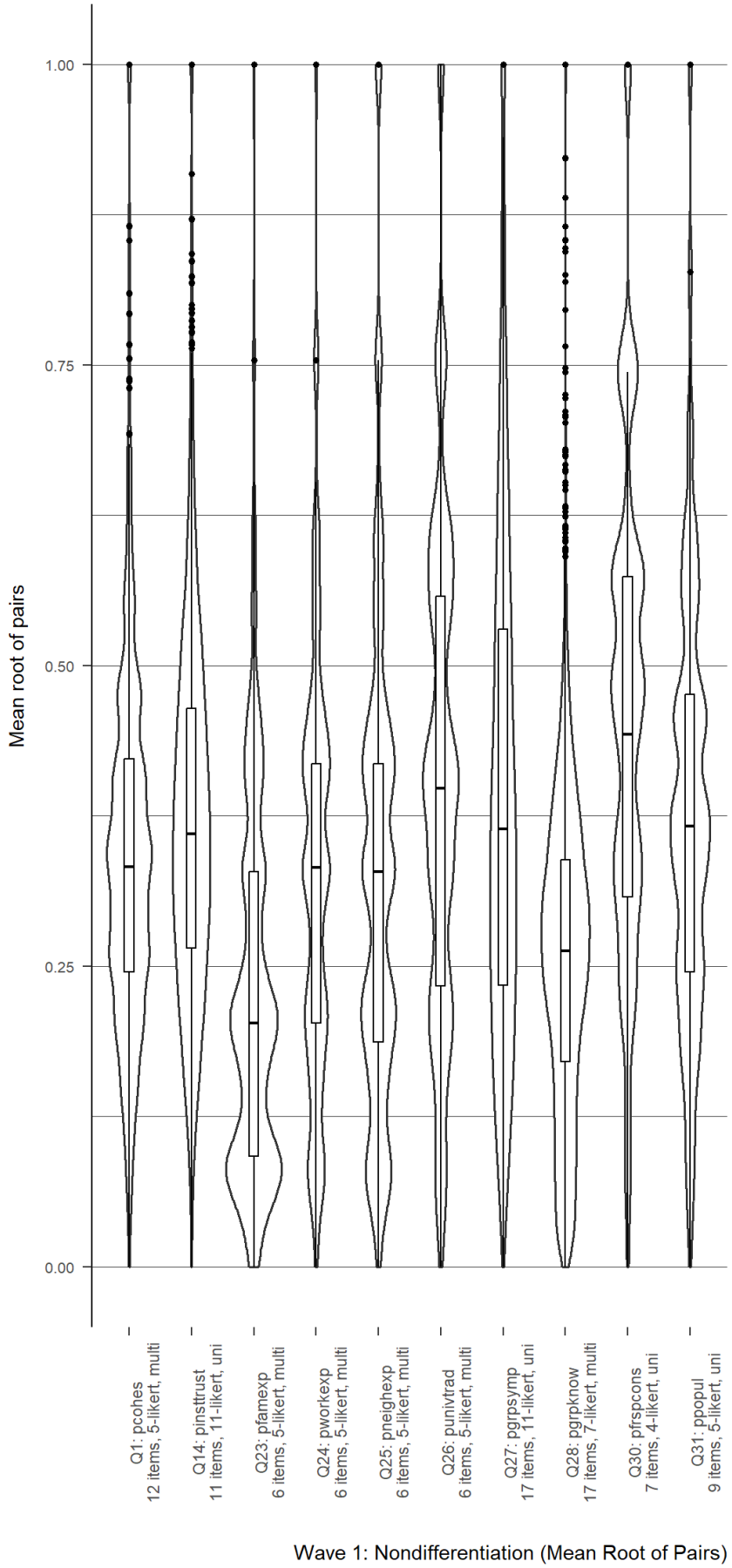


Figure 12: Wave 1 Nondifferentiation (Analysis Sample Only)

```r
w1s <-
  w1s |>
  rename_with(
    ~ str_c(., "_Sample"),
      starts_with("MeanRootPairs_") |
      c(WK, ST)

  ) |> select(-
     starts_with("temp_"))

w1s |> names()
```

```
  [1] "lfd"                   "Extr_Sample"
  [3] "Mid_Sample"            "EquivIncome_Sample"
  [5] "Age_cat_Sample"        "pid"
  [7] "hid"                   "gebjahr.weights"
  [9] "hh_lfd"                "ZG"
 [11] "interviewsprache"      "interviewdatum"
 [13] "interviewdauer"        "qmethod"
 [15] "intvollst"             "pcohes01"
 [17] "pcohes02"              "pcohes03"
 [19] "pcohes04"              "pcohes05"
 [21] "pcohes06"              "pcohes07"
 [23] "pcohes08"              "pcohes09"
 [25] "pcohes10"              "pcohes11"
 [27] "pcohes12"              "plifeuse"
 [29] "plifesat"              "ppolatt"
 [31] "pgender"               "pbirthmnth"
 [33] "pbirthyear_w1"         "pfamstat"
 [35] "pcohab"                "pcohabgebyear"
 [37] "pcohabgender"          "pcov01"
 [39] "pcov02"                "pcov03"
 [41] "pcov04"                "pcovmeas"
 [43] "pcovlim"               "psuppo01"
 [45] "psuppo02"              "psuppo03"
 [47] "psuppo04"              "pgentrust"
 [49] "pinsttrust01"          "pinsttrust02"
 [51] "pinsttrust03"          "pinsttrust04"
 [53] "pinsttrust05"          "pinsttrust06"
 [55] "pinsttrust07"          "pinsttrust08"
 [57] "pinsttrust09"          "pinsttrust10"
 [59] "pinsttrust11"          "porigin"
 [61] "pothcount"             "pgermstay"
 [63] "pnation01"             "pnation02"
 [65] "pnatoth"               "pmigback01"
 [67] "pmigback02"            "preligion"
 [69] "pdemocfit"             "pdemocsat"
 [71] "pclimsor"              "pclimcon"
 [73] "pfamexp01"             "pfamexp02"
 [75] "pfamexp03"             "pfamexp04"
 [77] "pfamexp05"             "pfamexp06"
 [79] "pworkexp01"            "pworkexp02"
 [81] "pworkexp03"            "pworkexp04"
 [83] "pworkexp05"            "pworkexp06"
 [85] "pneighexp01"           "pneighexp02"
 [87] "pneighexp03"           "pneighexp04"
```

```
 [89] "pneighexp05"                "pneighexp06"
 [91] "punivtrad01"                "punivtrad02"
 [93] "punivtrad03"                "punivtrad04"
 [95] "punivtrad05"                "punivtrad06"
 [97] "pgrpsymp01"                 "pgrpsymp02"
 [99] "pgrpsymp03"                 "pgrpsymp04"
[101] "pgrpsymp05"                 "pgrpsymp06"
[103] "pgrpsymp07"                 "pgrpsymp08"
[105] "pgrpsymp09"                 "pgrpsymp10"
[107] "pgrpsymp11"                 "pgrpsymp12"
[109] "pgrpsymp13"                 "pgrpsymp14"
[111] "pgrpsymp15"                 "pgrpsymp16"
[113] "pgrpsymp17"                 "pgrpknow01"
[115] "pgrpknow02"                 "pgrpknow03"
[117] "pgrpknow04"                 "pgrpknow05"
[119] "pgrpknow06"                 "pgrpknow07"
[121] "pgrpknow08"                 "pgrpknow09"
[123] "pgrpknow10"                 "pgrpknow11"
[125] "pgrpknow12"                 "pgrpknow13"
[127] "pgrpknow14"                 "pgrpknow15"
[129] "pgrpknow16"                 "pgrpknow17"
[131] "pfreespeech"                "pfrspcons01"
[133] "pfrspcons02"                "pfrspcons03"
[135] "pfrspcons04"                "pfrspcons05"
[137] "pfrspcons06"                "pfrspcons07"
[139] "ppopul01"                   "ppopul02"
[141] "ppopul03"                   "ppopul04"
[143] "ppopul05"                   "ppopul06"
[145] "ppopul07"                   "ppopul08"
[147] "ppopul09"                   "pladder01"
[149] "preldepriv"                 "pedu01"
[151] "pedu02"                     "pempstat1"
[153] "pgainemp"                   "pempcond01"
[155] "pempcond02"                 "pempcond03"
[157] "pcontract"                  "poccupation"
[159] "poccupation_isco08"         "poccstat"
[161] "pselfemp"                   "pcivil"
[163] "pempl"                      "pworker"
[165] "plabincbrut"                "plabincnet"
[167] "pnoinc"                     "q46_check1"
[169] "q46_check2"                 "q46_check5"
[171] "psundvote"                  "hhhtyp"
[173] "hhhsize"                    "q49_check1"
[175] "hhhage01"                   "hhhage02"
[177] "hhhage03"                   "q50_check1"
[179] "q50_check2"                 "hhhincnet"
[181] "q51_check1"                 "q51_check2"
[183] "hhhbasup"                   "hpname01"
[185] "hpname01_check1"            "hpsex01"
[187] "hprelat01"                  "hpbmnth01"
[189] "hpbyear01"                  "q53_01_check2"
[191] "hpall01"                    "q53_01_check3"
[193] "q53_01_check4"              "q53_01_check5"
[195] "hpname02"                   "hpname02_check1"
[197] "hpsex02"                    "hprelat02"
[199] "hpbmnth02"                  "hpbyear02"
```

```
[201] "q53_02_check2"                "hpall02"
[203] "q53_02_check3"                "q53_02_check4"
[205] "q53_02_check5"                "hpname03"
[207] "hpsex03"                      "hprelat03"
[209] "hpbmnth03"                    "hpbyear03"
[211] "q53_03_check2"                "hpall03"
[213] "q53_03_check3"                "q53_03_check4"
[215] "q53_03_check5"                "hpname04"
[217] "hpsex04"                      "hprelat04"
[219] "hpbmnth04"                    "hpbyear04"
[221] "q53_04_check2"                "hpall04"
[223] "q53_04_check3"                "q53_04_check4"
[225] "q53_04_check5"                "hpname05"
[227] "hpsex05"                      "hprelat05"
[229] "hpbmnth05"                    "hpbyear05"
[231] "hpall05"                      "q53_05_check5"
[233] "hpname06"                     "hpsex06"
[235] "hprelat06"                    "hpbmnth06"
[237] "hpbyear06"                    "hpall06"
[239] "q53_06_check5"                "hpname07"
[241] "hpsex07"                      "hprelat07"
[243] "hpbmnth07"                    "hpbyear07"
[245] "hpall07"                      "q53_07_check5"
[247] "hpname08"                     "hpsex08"
[249] "hprelat08"                    "hpbmnth08"
[251] "hpbyear08"                    "hpall08"
[253] "q53_08_check3"                "q53_08_check5"
[255] "hpname09"                     "hpsex09"
[257] "hprelat09"                    "hpbmnth09"
[259] "hpbyear09"                    "hpall09"
[261] "q53_09_check5"                "hpname10"
[263] "hpsex10"                      "hprelat10"
[265] "hpbmnth10"                    "hpbyear10"
[267] "pdevice01"                    "pdevice02"
[269] "pdevice03"                    "pdevice04"
[271] "pdevice05"                    "pothdevice"
[273] "ppanelcons"                   "q46_check3"
[275] "q46_check4"                   "pbirthyear_w2"
[277] "intvollst_w2"                 "plifesat01_w2"
[279] "pbirthyear_w3"                "intvollst_w3"
[281] "psundvoteOthWhich"            "peduOthWhich"
[283] "psundvoteOthWhich.text"       "peduOthWhich.text"
[285] "PartWaves"                    "byear"
[287] "bmnth"                        "myob"
[289] "ymd_interview"                "Age"
[291] "AgeConsistency"               "Sample"
[293] "Age_cat_sensan"               "BreakOff"
[295] "cid"                          "syear"
[297] "wave"                         "pgnation_scp"
[299] "pgfamstd_scp"                 "pglabgro"
[301] "pgi1labgro"                   "pgi2labgro"
[303] "pgi3labgro"                   "pgi4labgro"
[305] "pgi5labgro"                   "pgi6labgro"
[307] "pgi7labgro"                   "pgi8labgro"
[309] "pgi9labgro"                   "pgi10labgro"
[311] "pgi11labgro"                  "pgi12labgro"
```

```
[313] "pgi13labgro"                    "pgi14labgro"
[315] "pgi15labgro"                    "pgimpgro"
[317] "pglabnet"                       "pgi1labnet"
[319] "pgi2labnet"                     "pgi3labnet"
[321] "pgi4labnet"                     "pgi5labnet"
[323] "pgi6labnet"                     "pgi7labnet"
[325] "pgi8labnet"                     "pgi9labnet"
[327] "pgi10labnet"                    "pgi11labnet"
[329] "pgi12labnet"                    "pgi13labnet"
[331] "pgi14labnet"                    "pgi15labnet"
[333] "pgimpnet"                       "pgstib_scp"
[335] "pgstib1_scp"                    "pgemplst_scp"
[337] "pglfs_scp"                      "pglfs1_scp"
[339] "pgisco08_scp"                   "pgmisco08_scp"
[341] "pgvisco08_scp"                  "pgisco88_scp"
[343] "pgmisco88_scp"                  "pgvisco88_scp"
[345] "pgisei08_scp"                   "pgmisei08_scp"
[347] "pgvisei08_scp"                  "pgisei88_scp"
[349] "pgmisei88_scp"                  "pgvisei88_scp"
[351] "pgmps08_scp"                    "pgmmps08_scp"
[353] "pgvmps08_scp"                   "pgsiops08_scp"
[355] "pgmsiops08_scp"                 "pgvsiops08_scp"
[357] "pgsiops88_scp"                  "pgmsiops88_scp"
[359] "pgvsiops88_scp"                 "pgegp88_scp"
[361] "pgegp88_1_scp"                  "pgmegp88_scp"
[363] "pgvegp88_scp"                   "pgiscolj08_scp"
[365] "pgiscolj88_scp"                 "pgbetr_scp"
[367] "pgisced11_scp"                  "pgisced97_scp"
[369] "pgcasmin_scp"                   "pgbilzeit_scp"
[371] "pgmonth"                        "pgpiyear"
[373] "pgmode"                         "pfoccupation_isco08"
[375] "pmoccupation_isco08"            "Education3"
[377] "Education_i"                    "Education_method"
[379] "Education"                      "Gender"
[381] "Employment"                     "ChildHH"
[383] "AllWavesComplete"               "Mig"
[385] "hhsize"                         "u14"
[387] "ue14"                           "equivSize"
[389] "EquivIncome_num"                "EquivIncome_sensan"
[391] "ProportionSpeeding_sensan"      "ProportionSpeeding_Sample"
[393] "Speeding_sensan"                "Speeding_Sample"
[395] "Interviewmode"                  "TargetGroup"
[397] "freetextmis"                    "N_allFreeTitems"
[399] "FrTMisRatio"                    "FrTMis_sensan"
[401] "FrTMis_Sample"                  "NonrespRatio"
[403] "E5"                             "M5"
[405] "EM.S5"                          "E7"
[407] "M7"                             "EM.S7"
[409] "E11"                            "M11"
[411] "EM.S11"                         "N5_all"
[413] "N7_all"                         "N11_all"
[415] "E.S5S7S11"                      "M.S5S7S11"
[417] "N_allExMidOpportunities"        "Ratio_Extr"
[419] "Ratio_Mid"                      "RatioExMid"
[421] "Extr_sensan"                    "Mid_sensan"
[423] "MeanRootPairs_pcohes_Sample"    "MeanRootPairs_pinsttrust_Sample"
```

```
[425] "MeanRootPairs_pfamexp_Sample"    "MeanRootPairs_pworkexp_Sample"
[427] "MeanRootPairs_pneighexp_Sample"  "MeanRootPairs_punivtrad_Sample"
[429] "MeanRootPairs_pgrpsymp_Sample"   "MeanRootPairs_pgrpknow_Sample"
[431] "MeanRootPairs_pfrspcons_Sample"  "MeanRootPairs_ppopul_Sample"
[433] "WK_Sample"                       "ST_Sample"
[435] "u11"
```

```
w1 <-
  w1 |>
  left_join(w1s |> select(lfd,
                          starts_with("MeanRootPairs"),
                          starts_with("WK_"),
                          starts_with("ST_")
                          ), by = "lfd")
```

4 Wave 2

```
w2p_bo |> nrow()
```

```
[1] 9169
```

```
w2_lfds <-  #8900
  w1 |> filter(Sample == "Sample") |>
  inner_join(w2p_bo, by = "lfd") |>
  select(lfd)

w2 <-
  w2_lfds |>
  inner_join(w2p_bo, by = "lfd")
w2 |> nrow()
```

```
[1] 8900
```

5 w2 Build `BreakOff`

```
w2 <-
  w2 |>
  mutate(BreakOff = case_when(
    intvollst %in% c(2,3) ~ "Break-off",
    qmethod == 3 ~ "PAPI",              #PAPI
    (qmethod == 4 & intvollst == 1) ~ "No Break-off"
  ),
    BreakOff = factor(BreakOff, levels = c("No Break-off", "Break-off", "PAPI")))

frq(w2$BreakOff)
```

```
x <categorical>
# total N=8900 valid N=8900 mean=2.15 sd=0.98

Value        |    N | Raw % | Valid % | Cum. %
---------------------------------------------
No Break-off | 3730 | 41.91 |   41.91 |  41.91
Break-off    |   65 |  0.73 |    0.73 |  42.64
PAPI         | 5105 | 57.36 |   57.36 | 100.00
<NA>         |    0 |  0.00 |    <NA> |   <NA>
```

```
# 65 cases
```

6 w2 Build Speeding

See fig-w1p2Speedingcategorization

```
w2 |>
  filter(qmethod == 4) |>
  select(lfd, BreakOff, interviewdauer) |>
  #left_join(w2_dur, by = "lfd") |>
  left_join(w1 |> select(lfd, Age_cat = Age_cat_Sample), by = "lfd") |>

  #select(-plifesat03_dur, -plifesat04_dur, -plifesat10_dur, -contains("prelfund")) |>


  mutate(


    # not include into calculation of quantile borders: N = 54 Break-off cases
    # not include into calculation of quantile borders: N = 54 Break-off cases
  interviewdauer = case_when(BreakOff == "Break-off" ~ NA,
                             TRUE ~ interviewdauer)) |>

  group_by(Age_cat) |>


  summarise(
    `1` = quantile(interviewdauer, probs = 0.01, na.rm = TRUE),
    `3` = quantile(interviewdauer, probs = 0.03, na.rm = TRUE),
    `5` = quantile(interviewdauer, probs = 0.05, na.rm = TRUE),
    `10` = quantile(interviewdauer, probs = 0.1, na.rm = TRUE),
    `15` = quantile(interviewdauer, probs = 0.15, na.rm = TRUE),
    `20` = quantile(interviewdauer, probs = 0.2, na.rm = TRUE),
    `25` = quantile(interviewdauer, probs = 0.25, na.rm = TRUE),
    `33` = quantile(interviewdauer, probs = 0.33, na.rm = TRUE),
    `50` = quantile(interviewdauer, probs = 0.5, na.rm = TRUE),
    `66` = quantile(interviewdauer, probs = 0.66, na.rm = TRUE),
    `70` = quantile(interviewdauer, probs = 0.7, na.rm = TRUE),
    `75` = quantile(interviewdauer, probs = 0.75, na.rm = TRUE),
    `80` = quantile(interviewdauer, probs = 0.80, na.rm = TRUE),
    `85` = quantile(interviewdauer, probs = 0.85, na.rm = TRUE),
    `95` = quantile(interviewdauer, probs = 0.95, na.rm = TRUE)
  ) |>
  gather(key = "Quantile", value = "Value", -Age_cat) |>
  mutate(Quantile = as.numeric(Quantile)) |>
  ggplot(aes(y = Quantile,
             x = Value, color = Age_cat)) +
  geom_line() +
  scale_y_continuous(breaks = seq(0,100, by = 5)) +
  labs(x = "Interview duration", y = "Quantile",
       title = "w2: Quantiles of interview duration per age group") +
  papaja::theme_apa()
```

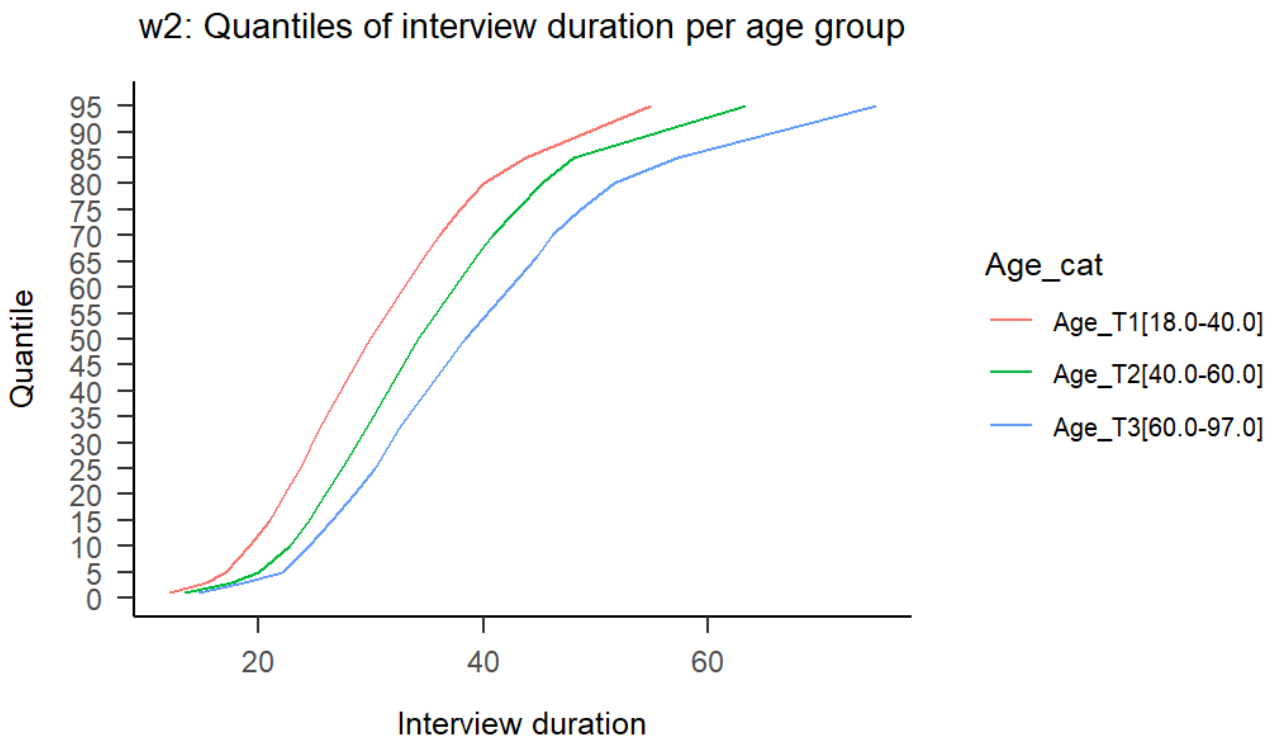


Figure 13: Speeding at w2

```
#|label: w2-build-speeding-variable
#|eval: true


w2_dur <-
  w2_dur |>
  left_join(w1 |> select(lfd, Age_cat = Age_cat_Sample)) |>
    #inner_join(w2_lfds, by = "lfd") |>
  # Step 1: set 0 to NA in all item columns
  mutate(across(ends_with("_dur"),
           ~ if_else(.x == 0, NA_real_, .x))
  )  |>

    # -110 cases not in the analysis sample due to age inconsistency (100) or w1-only
     ↪  participation (10)
    filter(!is.na(Age_cat)) |>


  group_by(Age_cat) |>
  # Step 2: compute medians (0/NA excluded)
  mutate(
    across(ends_with("_dur"),
           list(md = ~median(.x, na.rm = TRUE)),
           .names = "{.col}_md")
  ) |>

  # Step 3: speeding indicators (_s) with group-specific medians
  mutate(
    across(
```

```
      .cols = ends_with("_dur"),
      .fns = ~ case_when(
        is.na(.x) ~ NA_real_,
        .x < 0.4 * get(paste0(cur_column(), "_md")) ~ 1,
        TRUE ~ 0
      ),
      .names = "{.col}_s"
    )
  ) |>
  ungroup() |>

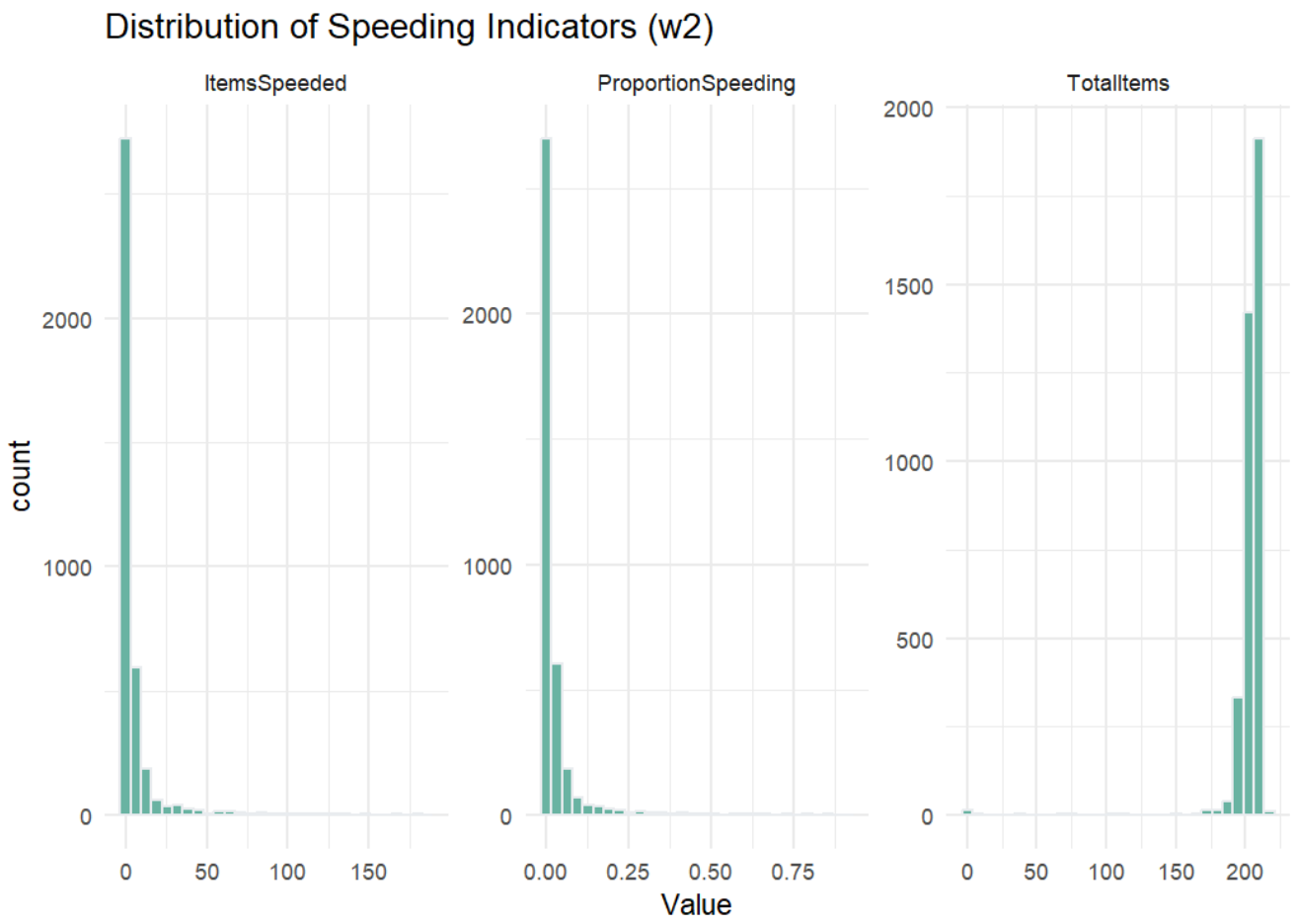
  # Step 4: sum scores
  rowwise() |>
  mutate(

    # number of items whose duration is < 0.4 * the item's median duration across all
     ↪ respondents in the same age group (can be 0 if the respondent did not speed on any
     ↪ item)
    ItemsSpeeded = sum(c_across(ends_with("_s")), na.rm = TRUE),
    TotalItems = sum(!is.na(c_across(ends_with("_s")))),
    ProportionSpeeding = ItemsSpeeded / TotalItems)|>
  ungroup()



w2_dur |>
  select(ItemsSpeeded, TotalItems, ProportionSpeeding) |>
  pivot_longer(everything(), names_to = "Variable", values_to = "Value") |>
  ggplot(aes(x = Value)) +
      geom_histogram(fill="#69b3a2", color="#e9ecef") +

 # geom_density(fill = "steelblue", alpha = 0.5) +
  facet_wrap(~Variable, scales = "free") +
  theme_minimal() +
  labs(title = "Distribution of Speeding Indicators (w2)")
```




```
# Complete sample (sensitivity analysis only)
w2_dur |> filter(ProportionSpeeding>0.000) |> nrow()
```

```
[1] 2730
```

```
w2_dur |> filter(ProportionSpeeding >0 & ProportionSpeeding<0.014286) |> pull(ItemsSpeeded) |>
 ↪  summary()  #1-3 Items - marginal
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  1.000   1.000   1.000   1.457   2.000   3.000
```

```
# upper median range:
w2_dur |> filter(ProportionSpeeding>=0.014286) |> pull(ItemsSpeeded) |> summary()  #1-186 Items
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   1.00    4.00    6.00   16.33   14.00  186.00
```

```
#### TOP 10 MOST SPEEDED ITEMS

 # filter the _s columns
 s_cols <- w2_dur |>
     select(ends_with("_s")) |>
     names()

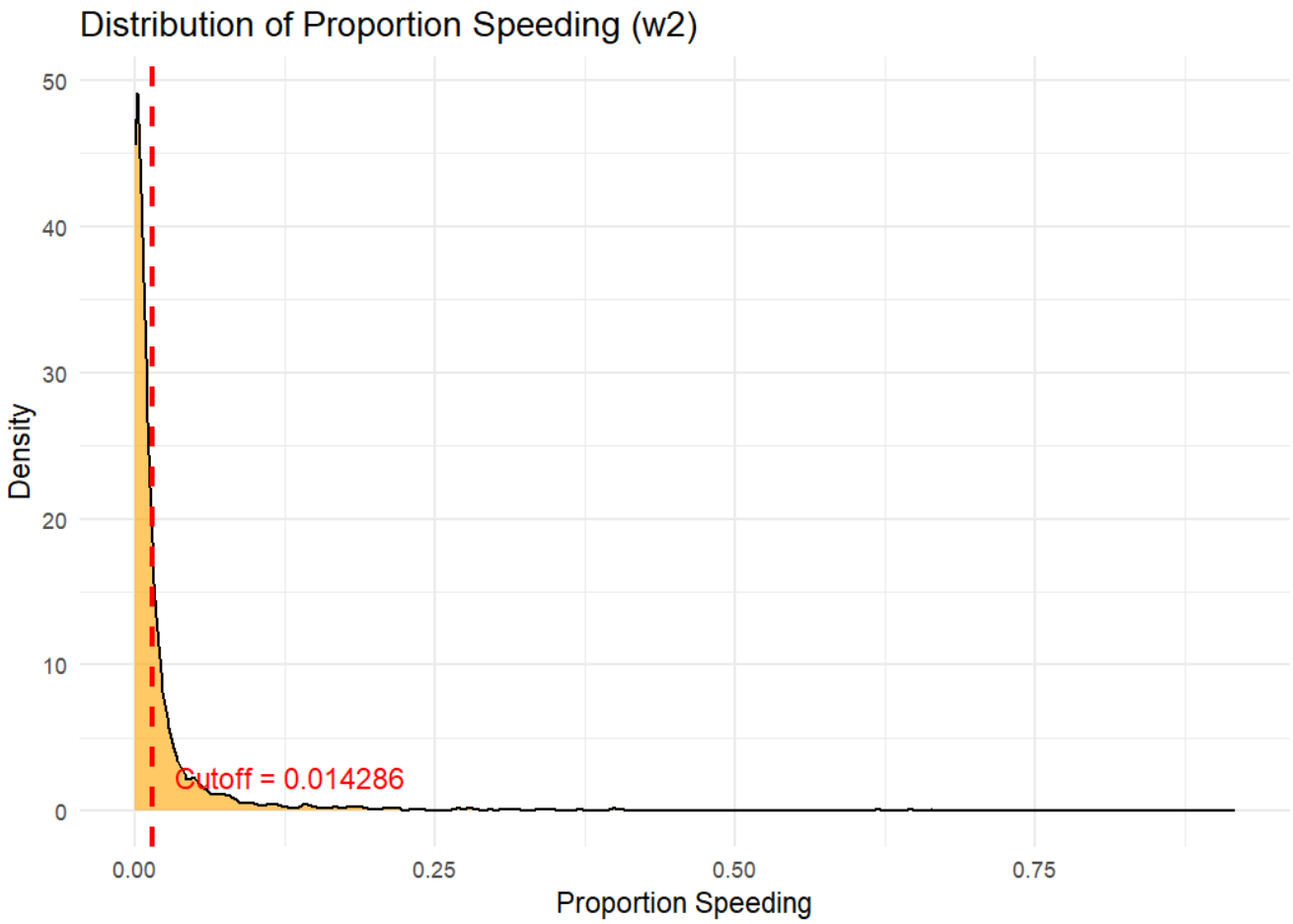
 # compute column sums (number of 1s, ignoring NAs)
 w2_dur |>
     summarise(across(all_of(s_cols), ~ sum(.x == 1, na.rm = TRUE))) |>
```

```
    pivot_longer(everything(), names_to = "variable", values_to = "sum_1er") |>
    arrange(desc(sum_1er)) |>
  head(10)
```

```
# A tibble: 10 x 2
   variable            sum_1er
   <chr>                 <int>
 1 pdiscrimoth_dur_s       460
 2 pmotherocc01_dur_s      395
 3 plabinc_dur_s           385
 4 pmotherocc03_dur_s      341
 5 ppartytend_dur_s        315
 6 pdiscrim12_dur_s        277
 7 pfollow02_dur_s         270
 8 pinform02_dur_s         253
 9 pcovidcosts01_dur_s     244
10 plifesat02_dur_s        225
```

```
# variable  sum_1er
# pdiscrimoth_dur_s 460
# pmotherocc01_dur_s    395
# plabinc_dur_s 385
# pmotherocc03_dur_s    341
# ppartytend_dur_s  315
# pdiscrim12_dur_s  277
# pfollow02_dur_s   270
# pinform02_dur_s   253
# pcovidcosts01_dur_s   244
# plifesat02_dur_s  225
```

```
w2_dur |>
  #filter(Sample == "Sample") |>
  # with cutoff line
ggplot(aes(x = ProportionSpeeding)) +
  geom_density(alpha = 0.6, fill = "orange") +
  geom_vline(xintercept = 0.014286, linetype = "dashed", color = "red", linewidth = 1) +
  annotate("text", x = 0.014286, y = 0, label = "Cutoff = 0.014286",
           hjust = -0.1, vjust = -1, color = "red") +
  labs(title = "Distribution of Proportion Speeding (w2)",
       x = "Proportion Speeding",
       y = "Density") +
  theme_minimal()
```




```
# consistent threshold: mean + 1 SD per group

thresholds_speeding_w2 <-
   w2_dur |>
   group_by(Age_cat) |>
   summarise(
     thresh_Speed = mean(ProportionSpeeding, na.rm = TRUE) + sd(ProportionSpeeding, na.rm =
      ↪  TRUE),
     percentile_sp = mean(ProportionSpeeding <= thresh_Speed, na.rm = TRUE) * 100,
     prop_above_sp = mean(ProportionSpeeding > thresh_Speed, na.rm = TRUE) * 100) |> ungroup()

w2_dur <-
  w2_dur |>
  mutate(
    Speeding = case_when(
      is.na(ProportionSpeeding) ~NA,
      Age_cat == "Age_T1[18.0-40.0]" & ProportionSpeeding > thresholds_speeding_w2 |>
 ↪  filter(Age_cat == "Age_T1[18.0-40.0]") |> pull(thresh_Speed) ~ "Speeding",
      Age_cat == "Age_T2[40.0-60.0]" & ProportionSpeeding > thresholds_speeding_w2 |>
 ↪  filter(Age_cat == "Age_T2[40.0-60.0]") |> pull(thresh_Speed) ~ "Speeding",
      Age_cat == "Age_T3[60.0-97.0]" & ProportionSpeeding > thresholds_speeding_w2 |>
 ↪  filter(Age_cat == "Age_T3[60.0-97.0]") |> pull(thresh_Speed) ~ "Speeding",
      TRUE ~"No speeding"
    ),
    Speeding = factor(Speeding, levels = c("No speeding", "Speeding"))
  )
```

```r
# join to w2
w2 <-
  w2 |>
  left_join(
    w2_dur |> select(lfd, ProportionSpeeding, Speeding), by = "lfd"
  )
```

6.1 w2 Build `Interviewmode`

```r
w2 <-
  w2 |>
   mutate(
    Interviewmode = case_when(
      qmethod == 3 ~ "PAPI",
      qmethod == 4 ~ "CAWI"
    ),
    Interviewmode = factor(Interviewmode, levels = c("PAPI", "CAWI")))
```

6.2 w2 Build `FreeTextMis`

- adjust raw free text variables

```r
w2 <-

  w2up |>

  mutate(
    pothpartend2 = case_when(

      # only applies to those who answered "Other"
      ppartytend != 8 ~ -2,

      # responded by naming a political party
      ppartytend == 8 &
        # longest: "Partei Mensch, Umwelt, Tierschutz"
        (nchar(pothpartend) >= 3 & nchar(pothpartend) <= 35 &

        !(pothpartend %in% c("...", "....", "xxx", "XXX", "abc", "ABC", "K.a.", "k.A.")) &
        !grepl("-1", pothpartend) &
        !grepl("-9", pothpartend) &
        !grepl("KEIN", pothpartend) &
        !grepl("Kein", pothpartend) &
        !grepl("Frage", pothpartend) &
        !grepl("nicht", pothpartend) &
        !grepl("NICHT", pothpartend) &
        !grepl("UMGÜLTIG", pothpartend) &
        !grepl("GEHEIM", pothpartend)) ~ 1,
```

```
    ppartytend == 8 & !(

      (nchar(pothpartend) >= 3 & nchar(pothpartend) <= 35) &

      !(pothpartend %in% c("...", "....", "xxx", "XXX", "abc", "ABC", "K.a.", "k.A.")) &
      !grepl("-1", pothpartend) &
      !grepl("-9", pothpartend) &
      !grepl("KEIN", pothpartend) &
      !grepl("Kein", pothpartend) &
      !grepl("Frage", pothpartend) &
      !grepl("nicht", pothpartend) &
      !grepl("NICHT", pothpartend) &
      !grepl("UMGÜLTIG", pothpartend) &
      !grepl("GEHEIM", pothpartend)) ~ -1),




  pdiscrimoth2 = case_when(


    # did not select "other discrim experience"
    !pdiscrim12 %in% c(2:5) ~ -2,


    # selected "other discrim experience"
    pdiscrim12 %in% c(2:5) &

      (nchar(pdiscrimoth) >= 3  &

      (!(pdiscrimoth %in% c("...", "....", "xxx", "XXX", "abc", "ABC", "K.a.", "k.A.")) &
      !grepl("-1", pdiscrimoth) &
      !grepl("-9", pdiscrimoth) &
      !grepl("KEIN", pdiscrimoth) &
      !grepl("Kein", pdiscrimoth) &
      !grepl("Frage", pdiscrimoth) &
      !grepl("nicht", pdiscrimoth) &
      !grepl("NICHT", pdiscrimoth) &
      !grepl("UMGÜLTIG", pdiscrimoth) &
      !grepl("GEHEIM", pdiscrimoth)))  ~ 1,


    pdiscrim12 %in% c(2:5) &

      !(nchar(pdiscrimoth) >= 3  &

      (!(pdiscrimoth %in% c("...", "....", "xxx", "XXX", "abc", "ABC", "K.a.", "k.A.")) &
      !grepl("-1", pdiscrimoth) &
      !grepl("-9", pdiscrimoth) &
      !grepl("KEIN", pdiscrimoth) &
      !grepl("Kein", pdiscrimoth) &
      !grepl("Frage", pdiscrimoth) &
      !grepl("nicht", pdiscrimoth) &
      !grepl("NICHT", pdiscrimoth) &
```

```
        !grepl("UMGÜLTIG", pdiscrimoth) &
        !grepl("GEHEIM", pdiscrimoth)))  ~ -1)) |>

  select(lfd, pothpartend2, pdiscrimoth2) |>

  right_join(w2, by = "lfd")
```

```
w2 <-
  w2 |> mutate(

    # apply standard missing codes to pfoccupation_isco08
    pfoccupation_isco08 = case_when(pfoccupation_isco08 == 0            ~ -2,
                                    pfoccupation_isco08 %in% c(8, 9)    ~ -1,
                                    TRUE                               ~ pfoccupation_isco08),

    pmoccupation_isco08 = case_when(pmoccupation_isco08 == 0            ~ -2,
                                    pmoccupation_isco08 %in% c(8, 9)    ~ -1,
                                    TRUE                               ~ pmoccupation_isco08))


FreeTextItems_w2 <-

  w2 |> select(
    pfatherbirth,
    pfoccupation_isco08,
    pmotherbirth,
    pmoccupation_isco08,
    plabincbrut, plabincnet,
    potherinc, phiremonth,
    phireyear, pworkhrscontr,
    pworkhrsact, pbirthyear,
    pothpartend2, pdiscrimoth2, pdisabdegree) |> colnames()


w2 <-

  w2 |>

  rowwise() |>
  mutate(

    freetextmissings = sum(across(all_of(FreeTextItems_w2)) %in% c(-1, -9), na.rm = TRUE),

    N_allFreeTitems = sum(across(all_of(FreeTextItems_w2), ~!. %in% c(-2, -10)), na.rm = TRUE),

    FrTMisRatio = freetextmissings/N_allFreeTitems) |>

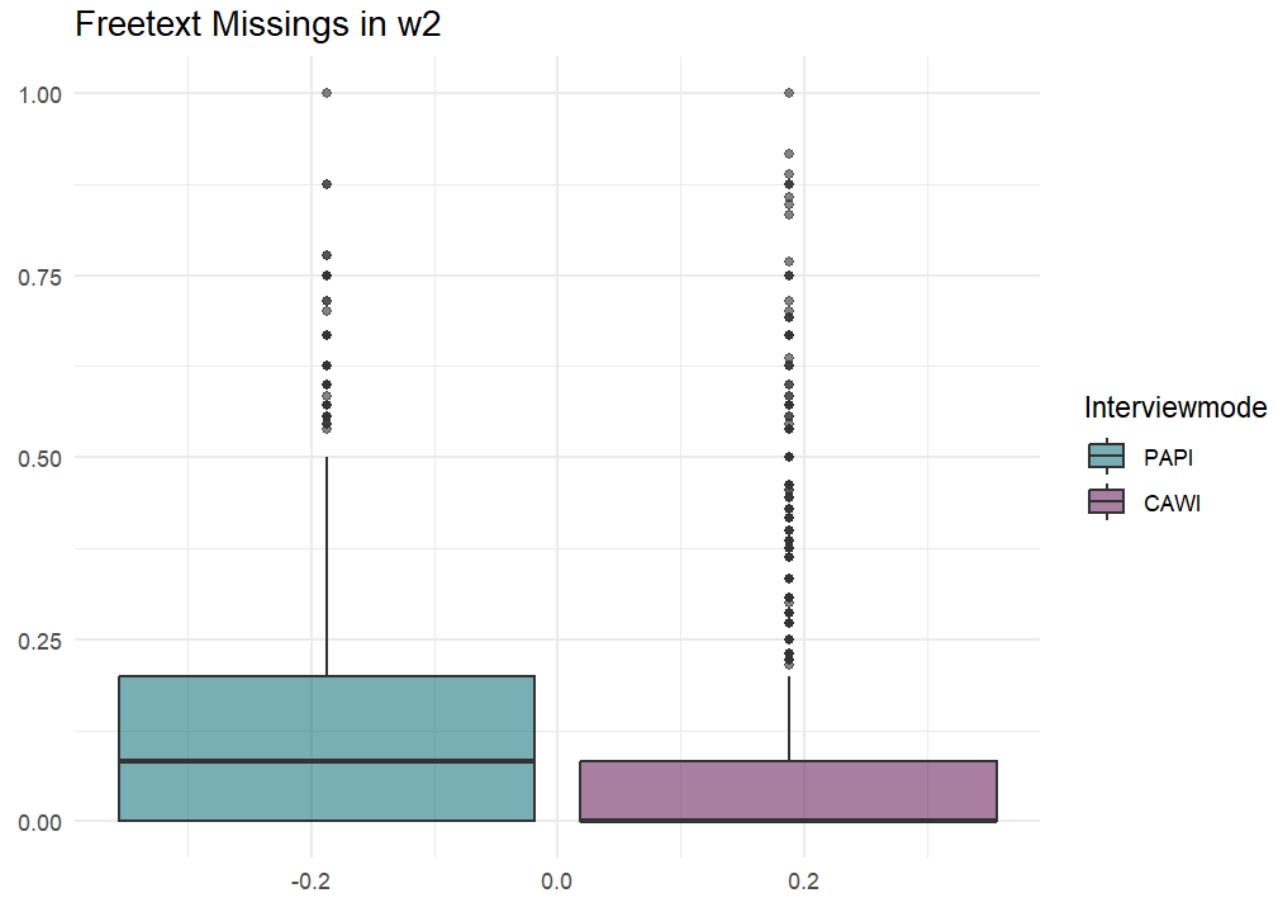
  ungroup()

#   mutate(
#   FreeTextMissings = case_when(FrTMisRatio > 0 ~ "Free text missings",
#                   FrTMisRatio == 0 ~ "No free text missings",
#
#                   freetextmissings == 0 & N_allFreeTitems == 0 ~ "No free text missings"),
#   FreeTextMissings = factor(FreeTextMissings, levels = c("No free text missings", "Free text
 ↪  missings")))
```

```
#
# w2 |> frq(FreeTextMissings)

w2 |>

ggplot(aes(y = FrTMisRatio, fill = Interviewmode)) +
  geom_boxplot(alpha = 0.6) +
 # geom_vline(xintercept = 0.014, linetype = "dashed", color = "red", linewidth = 1) +
 # annotate("text", x = 0.014, y = 0, label = "Cutoff = 0.014",
  #         hjust = -0.1, vjust = -1, color = "red") +
  scale_fill_manual(values = mode_palette) +
  labs(title = "Freetext Missings in w2",
       x = "",
       y = "") +
  theme_minimal()
```




```
# consistent threshold: mean + 1 SD per group

thresholds_FrTMis_w2 <-
   w2 |>
   group_by(Interviewmode) |>
   summarise(
     thresh_FrTMis = mean(FrTMisRatio, na.rm = TRUE) + sd(FrTMisRatio, na.rm = TRUE),
     percentile_sp = mean(FrTMisRatio <= thresh_FrTMis, na.rm = TRUE) * 100,
     prop_above_sp = mean(FrTMisRatio > thresh_FrTMis, na.rm = TRUE) * 100) |> ungroup()
```

```
w2 <-
  w2 |>
  mutate(FrTMis =
           case_when(
             is.na(FrTMisRatio) ~ "Not applicable",
             Interviewmode == "PAPI" & FrTMisRatio > thresholds_FrTMis_w2 |>
 ↪  filter(Interviewmode == "PAPI") |> pull(thresh_FrTMis) ~ "High",
             Interviewmode == "CAWI" & FrTMisRatio > thresholds_FrTMis_w2 |>
 ↪  filter(Interviewmode == "CAWI") |> pull(thresh_FrTMis) ~ "High",
             TRUE ~ "Not high"),
         FrTMis = factor(FrTMis, levels = c("Not high", "High", "Not applicable")))

w2 |> group_by(Interviewmode) |> frq(FrTMis)
```

```
FrTMis <categorical>
# grouped by: PAPI
# total N=5105 valid N=5105 mean=1.16 sd=0.37

Value          |    N | Raw % | Valid % | Cum. %
-----------------------------------------------
Not high       | 4263 | 83.51 |   83.51 |  83.51
High           |  842 | 16.49 |   16.49 | 100.00
Not applicable |    0 |  0.00 |    0.00 | 100.00
<NA>           |    0 |  0.00 |    <NA> |   <NA>

FrTMis <categorical>
# grouped by: CAWI
# total N=3795 valid N=3795 mean=1.11 sd=0.33

Value          |    N | Raw % | Valid % | Cum. %
-----------------------------------------------
Not high       | 3381 | 89.09 |   89.09 |  89.09
High           |  394 | 10.38 |   10.38 |  99.47
Not applicable |   20 |  0.53 |    0.53 | 100.00
<NA>           |    0 |  0.00 |    <NA> |   <NA>
```

```
w2 |>
     count(Interviewmode, FrTMis, name="n") |>
     group_by(Interviewmode) |>
     mutate(share = n/sum(n)) |>
     ggplot(aes(Interviewmode, share, fill=FrTMis)) +
  labs(caption = "Free Text Missings by and Mode (15 items over w2)",
       x = "",
     y = "Proportion") +
     geom_col(position="stack")  +
     papaja::theme_apa()
```

```
w2 |>
  group_by(Interviewmode) |>
   count(FrTMisRatio, name="n") |>
```

```
    mutate(share = n/sum(n)) |>
    ggplot(aes(Interviewmode, share, fill=FrTMisRatio)) +
    geom_col(position="stack")   +
labs(caption = "Free Text Missings Ratio by and Mode (15 items over w2)",
       x = "",
     y = "Proportion") +
 papaja::theme_apa()  +
 theme(#axis.text.x = element_text(angle = 30, vjust = 0.5),
       panel.grid.major.y = element_line(color = "grey", linewidth = 0.1))
```

Figure 14: Free text missings by Interviewmode

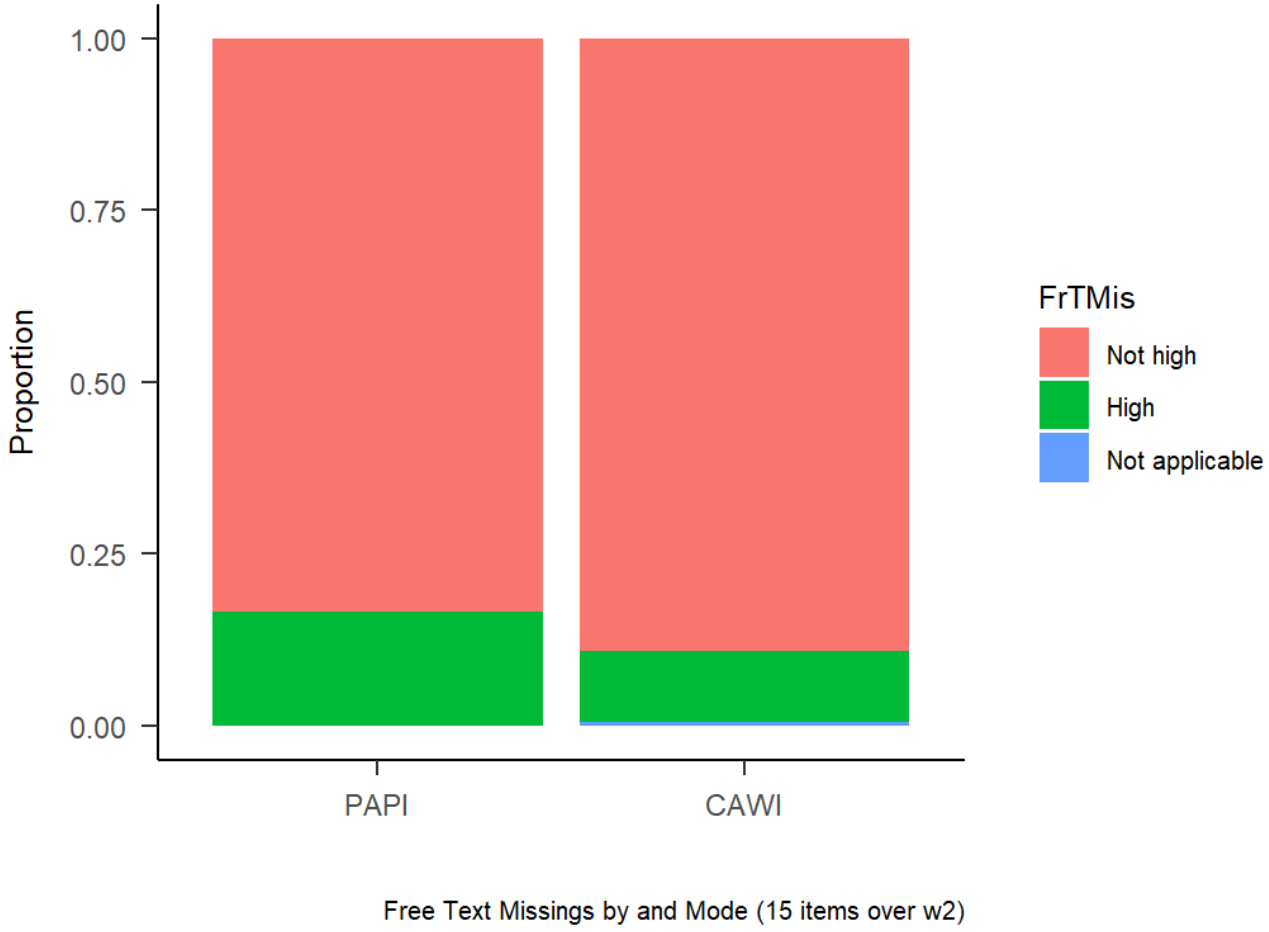
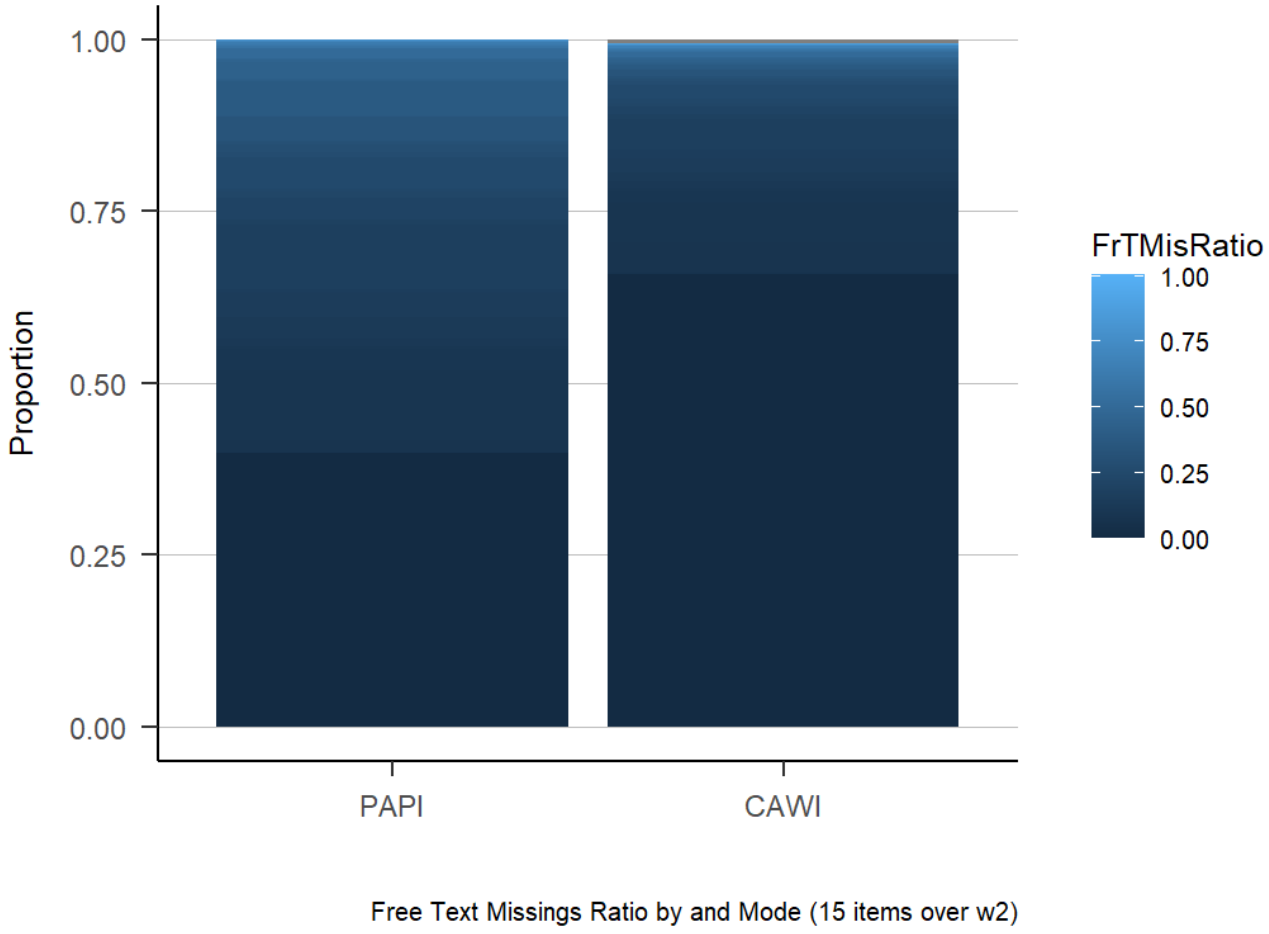


Figure 15: Free text missings by Interviewmode



7 w2 ItemNonresponse

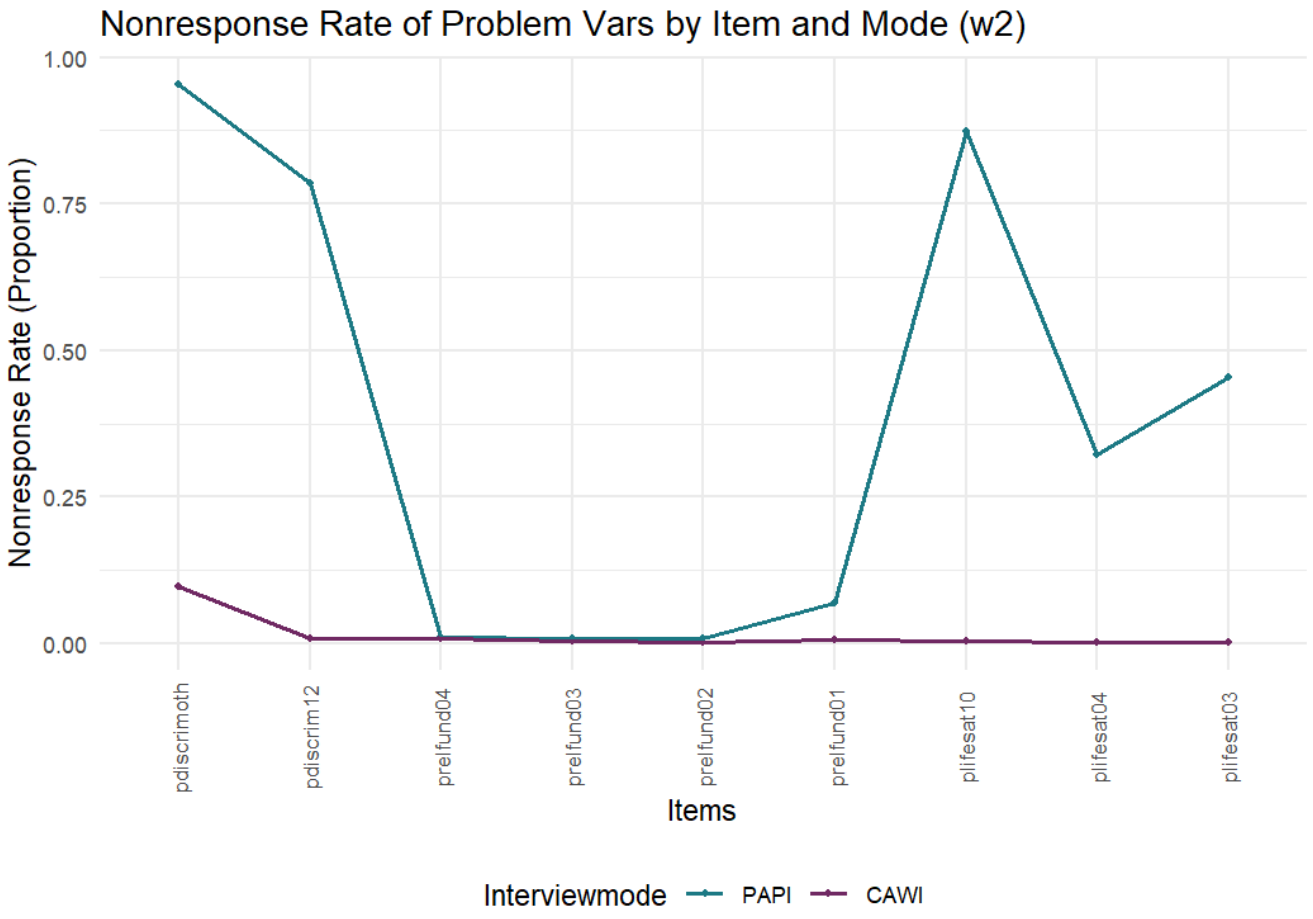
7.0.1 Problem Vars only

```
problem.vars <- c("plifesat03", "plifesat04", "plifesat10",
                  "prelfund01", "prelfund02", "prelfund03", "prelfund04",
                  "pdiscrim12", "pdiscrimoth", "ppertrust04", "plifesat12")



w2 |>
  select(lfd, Interviewmode,
         all_of(problem.vars)) |> #names()
 # select(all_of(mis_items_w2), Interviewmode) |>
  group_by(Interviewmode) |>
  summarise(across(plifesat03:pdiscrimoth,
                   ~ sum(.x %in% c(-1, -9), na.rm = TRUE) / n(),
                   .names = "PropMiss_{.col}")) |>
  pivot_longer(-Interviewmode, names_to = c("dummy", "Variable"),
               names_pattern = "(PropMiss_)(.*)", values_to = "Prop") |>
  select(-dummy) |>
  mutate(Variable = factor(Variable, levels = rev(unique(Variable)))) |>
```

```
ggplot(aes(x = Variable, y = Prop, colour = Interviewmode, group = Interviewmode)) +
  geom_point(size = 1) +
  geom_line(linewidth = 0.8) +
  scale_colour_manual(values = mode_palette) +
  labs(
    x = "Items",
    y = "Nonresponse Rate (Proportion)",
    colour = "Interviewmode",
    title = "Nonresponse Rate of Problem Vars by Item and Mode (w2)"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(size = 8, angle = 90, vjust = 0.5),
    legend.position = "bottom"
  )
```




```
w2 |>
    select(lfd, Interviewmode,
           all_of(problem.vars)) |>
  group_by(Interviewmode) |>
  summarise(across(plifesat03:pdiscrimoth,
                   ~ sum(.x %in% c(-1, -9), na.rm = TRUE) / n(),
                   .names = "PropMiss_{.col}"))
```

```
# A tibble: 2 x 10
  Interviewmode PropMiss_plifesat03 PropMiss_plifesat04 PropMiss_plifesat10
```

```
  <fct>                       <dbl>               <dbl>               <dbl>
1 PAPI                      0.455               0.322               0.874
2 CAWI                      0.00264             0.00105             0.00422
# i 6 more variables: PropMiss_prelfund01 <dbl>, PropMiss_prelfund02 <dbl>,
#   PropMiss_prelfund03 <dbl>, PropMiss_prelfund04 <dbl>,
#   PropMiss_pdiscrim12 <dbl>, PropMiss_pdiscrimoth <dbl>
```

```r
# Absolute counts + N per group
counts <- w2 |>
  select(lfd, Interviewmode, all_of(problem.vars)) |>
  group_by(Interviewmode) |>
  summarise(
    n = n(),
    across(all_of(problem.vars),
           ~ sum(.x %in% c(-1, -9), na.rm = TRUE),
           .names = "miss_{.col}")
  )

# Chi-square test per item
results <- map_dfr(problem.vars, function(var) {
  col <- paste0("miss_", var)

  miss  <- counts[[col]]        # c(n_miss_PAPI, n_miss_CAWI)
  n     <- counts$n             # c(n_PAPI, n_CAWI)
  notmiss <- n - miss

  mat <- matrix(c(miss, notmiss), nrow = 2,
                dimnames = list(c("PAPI", "CAWI"), c("Miss", "NotMiss")))

  # Fisher when expected cell counts < 5, otherwise chi-squared
  test <- if (any(chisq.test(mat)$expected < 5)) {
    fisher.test(mat)
  } else {
    chisq.test(mat, correct = FALSE)
  }

  tibble(
    Item       = var,
    p_value    = test$p.value,
    method     = test$method,
    prop_PAPI  = miss[1] / n[1],
    prop_CAWI  = miss[2] / n[2]
  )
}) |>
  mutate(
    p_adj      = p.adjust(p_value, method = "BH"),  # Benjamini-Hochberg
    significant = p_adj < 0.05
  ) |>
  arrange(p_adj)

print(results)
```

```
# A tibble: 11 x 7
   Item          p_value method        prop_PAPI prop_CAWI     p_adj significant
   <chr>           <dbl> <chr>             <dbl>     <dbl>     <dbl> <lgl>
 1 plifesat03  0         Pearson's Ch~   0.455    0.00264  0         TRUE
 2 plifesat04  0         Pearson's Ch~   0.322    0.00105  0         TRUE
```

```
 3 plifesat10  0         Pearson's Ch~   0.874    0.00422  0         TRUE
 4 pdiscrim12  0         Pearson's Ch~   0.784    0.00870  0         TRUE
 5 pdiscrimoth 0         Pearson's Ch~   0.953    0.0962   0         TRUE
 6 ppertrust04 1.47e-273 Pearson's Ch~   0.279    0.000791 2.70e-273 TRUE
 7 plifesat12  2.72e-173 Pearson's Ch~   0.210    0.0111   4.28e-173 TRUE
 8 prelfund01  1.13e- 46 Pearson's Ch~   0.0699   0.00738  1.56e- 46 TRUE
 9 prelfund02  1.98e-  5 Pearson's Ch~   0.00881  0.00184  2.42e-  5 TRUE
10 prelfund03  1.81e-  3 Pearson's Ch~   0.00921  0.00369  1.99e-  3 TRUE
11 prelfund04  8.68e-  2 Pearson's Ch~   0.0116   0.00791  8.68e-  2 FALSE
```

```
# all significant except "prelfund04"


# Satisfaction: childcare provision (plifesat10); trust: coworkers (ppertrust04); satisfaction:
 ↪  school or vocational education (plifesat12); satisfaction: household tasks (plifesat04)
problem.vars <- c("plifesat03", "plifesat04", "plifesat10",
                  "prelfund01", "prelfund02", "prelfund03", #"prelfund04",
                  "pdiscrim12", "pdiscrimoth", "ppertrust04", "plifesat12")
```

7.0.2 All other items

```
w2 |>
  select(lfd, Interviewmode,
         plifesat01:hsavprecaut,
       # exclude the wave-2 free-text items (measured separately as FreeTextMissings)

       -(any_of(FreeTextItems_w2))) |>
  group_by(Interviewmode) |>
  summarise(across(plifesat01:hsavprecaut,
                   ~ sum(.x %in% c(-1, -9), na.rm = TRUE) / n(),
                   .names = "PropMiss_{.col}")) |>
  pivot_longer(-Interviewmode, names_to = c("dummy", "Variable"),
               names_pattern = "(PropMiss_)(.*)", values_to = "Prop") |>
  select(-dummy) |>
  mutate(Variable = factor(Variable, levels = rev(unique(Variable)))) |>

ggplot(aes(x = Variable, y = Prop, colour = Interviewmode, group = Interviewmode)) +
  geom_point(size = 1) +
  geom_line(linewidth = 0.8) +
  scale_colour_manual(values = mode_palette) +
  labs(
    x = "Items",
    y = "Nonresponse Rate (Proportion)",
    colour = "Interviewmode",
    title = "Nonresponse Rate by Item and Mode (w2)"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(size = 3, angle = 90, vjust = 0.5),
    legend.position = "bottom"
  )
```

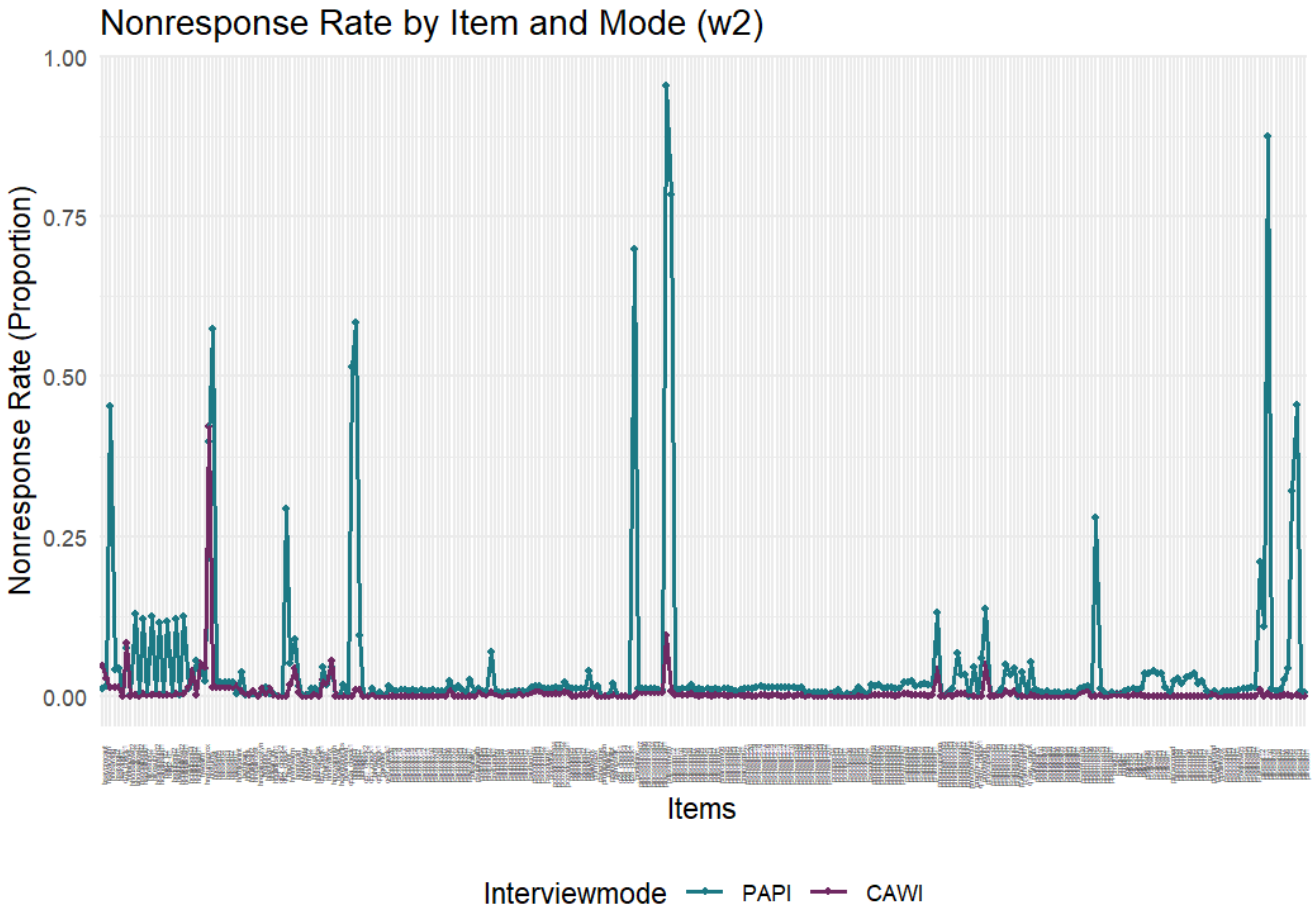


Figure 16: Nonresponse Items w2

```
mis_df_w2 <-
  w2 |>
  select(lfd, Interviewmode,
         plifesat01:hsavprecaut,
         # exclude the wave-2 free-text items (measured separately as FreeTextMissings)

         -(any_of(FreeTextItems_w2)),
         all_of(problem.vars))




# w2 items with high nonresponse rates

manymissings_CAWI <- mis_df_w2 |> filter(Interviewmode == "CAWI") |>
  summarise_all(~sum(.  %in% c(-1, -9))) |>
  pivot_longer(
    everything(), names_to = "Variable", values_to = "Numb.missings") |>
  rowwise() |>
  mutate(Prop.missings = Numb.missings/(mis_df_w2 |> filter(Interviewmode == "CAWI") |>
   ↪  nrow())) |>
  ungroup() |>
  arrange(desc(Prop.missings)) |>
```

```
  filter(Prop.missings > .15) |> pull(Variable) #|> paste(collapse = '", "')

manymissings_PAPI <- mis_df_w2 |> filter(Interviewmode == "PAPI") |>
  summarise_all(~sum(.  %in% c(-1, -9))) |>
  pivot_longer(
    everything(), names_to = "Variable", values_to = "Numb.missings") |>
  rowwise() |>
  mutate(Prop.missings = Numb.missings/(mis_df_w2 |> filter(Interviewmode == "PAPI") |>
   ↪  nrow())) |>
  ungroup() |>
  arrange(desc(Prop.missings)) |>
  filter(Prop.missings > .15) |> pull(Variable) #|> paste(collapse = '", "')


list(
  CAWI_only = setdiff(manymissings_CAWI, manymissings_PAPI),
  PAPI_only = setdiff(manymissings_PAPI, manymissings_CAWI)
)
```

```
$CAWI_only
character(0)

$PAPI_only
 [1] "pdiscrimoth" "plifesat10"  "pdiscrim12"  "pnolabinc"   "hhhage02"
 [6] "hnoasset"    "hhhage03"    "plifesat03"  "hnosaving"   "plifesat04"
[11] "hnorent"     "ppertrust04" "plifesat12"
```

```
exclusion_items <- unique(c(manymissings_CAWI, manymissings_PAPI, problem.vars)) #17 items
 ↪  excluded

mis_items_w2 <-
  mis_df_w2 |>
  select(

    -c(lfd, Interviewmode, #qmethod,
       contains("interview"),
       contains("Nonresp"),

       all_of(exclusion_items))) |>
  colnames()

tNR1 <- Sys.time()



mis_df_w2 <-
  mis_df_w2 |>
  rowwise() |>
  mutate(
  Nonresponses = sum(across(all_of(mis_items_w2)) %in% c(-1, -9), na.rm = TRUE),
  N_allNonrespOpportunities_woBO = sum(across(all_of(mis_items_w2), ~!. %in% c(-2, -10)), na.rm
   ↪  = TRUE),
 # BO (-10) counted as "not shown", i.e., not a missed opportunity"
  NonrespRatio = Nonresponses/N_allNonrespOpportunities_woBO) |>
  ungroup()
```

```r
paste("Total duration (NonrespRatio):", time_length(interval(tNR1, Sys.time()), "minute"),
 ↪  "minutes")
```

```
[1] "Total duration (NonrespRatio): 3.24554696877797 minutes"
```

```r
w2 <-
  mis_df_w2 |> select(lfd, NonrespRatio) |>
  right_join(w2, by = "lfd")
```

8 w2 Extreme/Middle

```r
# 129 items
extrm5_items <-
  w2 |> select(

    # shared with `Nondiff`
    plifesat01:plifesat12, #-3 = 9 items
    pinform01:pinform09,
    pedugoal01:pedugoal11,
    pfathbeh01:pfathbeh08,
    pmothbeh01:pmothbeh08,
    psocdes01:psocdes06,
    pnarcism01:pnarcism06,
    pdiscrim01:pdiscrim12,
    pgender01:pgender05,
    pauthor01:pauthor09,
    pactivity01:pactivity20,



    # exclusively
    pinfsocmed,
    pfeelnow01:pfeelnow05,
    psoccoh01:psoccoh04,
    pconpec01:pconpec04,
    pcovidcosts01:pcovidcosts04,
    psocdom01:psocdom04,
    psubjhealth

  ) |> colnames()

# 21
extrm6_items <-
  w2 |> select(
    pschwartz01:pschwartz21
  ) |> colnames()

# 17
extrm11_items <-
  w2 |> select(

    # shared with `Nondiff`
    pglob01:pglob05,
    pmig01:pmig03,
```

```
    ppertrust01:ppertrust08,

    # exclusively
    pcovidtax

  ) |> colnames()


t1 <- Sys.time()


w2 <-
  w2 |>
  rowwise() |>
  mutate(

    # all extreme values on 5-likert
    E5 = sum(across(all_of(extrm5_items)) %in% c(1, 5), na.rm = TRUE),
    # all middle values on 5-likert
    M5 = sum(across(all_of(extrm5_items)) == 3, na.rm = TRUE),

    # all extreme and middle values on 5-likert
    EM.S5 = sum(E5, M5, na.rm = TRUE),


     # all extreme values on 6-likert
    E6 = sum(across(all_of(extrm6_items)) %in% c(1, 6), na.rm = TRUE),
    # no middle category

    # all extreme and middle values on 6-likert: see E6



    # all extreme values on 11-likert
    E11 = sum(across(all_of(extrm11_items)) %in% c(0, 10), na.rm = TRUE),
    # all middle values on 11-likert
    M11 = sum(across(all_of(extrm11_items)) %in% c(5), na.rm = TRUE),

    # all extreme and middle values on 11-likert
    EM.S11 = sum(E11, M11, na.rm = TRUE),



    N5_all = sum(across(all_of(extrm5_items)) > 0, na.rm = TRUE),
    N6_all = sum(across(all_of(extrm6_items)) > 0, na.rm = TRUE),
    N11_all = sum(across(all_of(extrm11_items)) >= 0, na.rm = TRUE),



    # all extreme values
    E.S5S6S11 = sum(E5, E6, E11, na.rm = TRUE),
    # all middle values (there are no M6)
    M.S5S11 = sum(M5, M11, na.rm = TRUE),

    # all items responded
    N_allExMidOpportunities = sum(N5_all, N6_all, N11_all),
```

```
    # proportion of extreme and middle values on 5-likert by all responses
    Ratio_S5 = ifelse(N5_all == 0, NA, EM.S5/N5_all),
    # proportion of extreme and middle values on 6-likert by all responses
    Ratio_S6 = ifelse(N6_all == 0, NA, E6/N6_all),
    # proportion of extreme and middle values on 11-likert by all responses
    Ratio_S11 = ifelse(N11_all == 0, NA, EM.S11/N11_all),


    Ratio_Extr = ifelse(N_allExMidOpportunities == 0, NA, E.S5S6S11/N_allExMidOpportunities),
    Ratio_Mid = ifelse(N_allExMidOpportunities == 0, NA, M.S5S11/N_allExMidOpportunities),

    RatioExMid = sum(E.S5S6S11, M.S5S11, na.rm = TRUE)/N_allExMidOpportunities

  ) |> ungroup()

print(paste("Total duration (Extreme/Middle, mins):", as.numeric(Sys.time()-t1, "mins")))
```

```
[1] "Total duration (Extreme/Middle, mins): 26.5243004480998"
```

```
cor.test(w2$Ratio_Mid, w2$Ratio_Extr)
```

```

    Pearson's product-moment correlation

data:  w2$Ratio_Mid and w2$Ratio_Extr
t = -56.627, df = 8888, p-value < 2.2e-16
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.5300172 -0.4994609
sample estimates:
       cor
-0.5149026
```

8.1 w2 Categorize Extr & Mid

```
thresholds_ExM <-
  w2 |>

  group_by(Interviewmode) |>
  summarise(
    thresh_Extr_Ratio = mean(Ratio_Extr, na.rm = TRUE) + sd(Ratio_Extr, na.rm = TRUE),
    thresh_Mid_Ratio = mean(Ratio_Mid, na.rm = TRUE) + sd(Ratio_Mid, na.rm = TRUE),

    percentile_Extr = mean(Ratio_Extr <= thresh_Extr_Ratio, na.rm = TRUE) * 100,
    percentile_Mid = mean(Ratio_Mid <= thresh_Mid_Ratio, na.rm = TRUE) * 100,

    prop_above_Extr = mean(Ratio_Extr > thresh_Extr_Ratio, na.rm = TRUE) * 100,
    prop_above_Mid = mean(Ratio_Mid > thresh_Mid_Ratio, na.rm = TRUE) * 100
  )

w2 <-
  w2 |>
  mutate(Extr =
           case_when(
             is.na(Ratio_Extr) ~ "Not applicable",
```

```
            Interviewmode == "PAPI" & Ratio_Extr > thresholds_ExM |> filter(Interviewmode ==
↪  "PAPI") |> pull(thresh_Extr_Ratio) ~ "High",
            Interviewmode == "CAWI" & Ratio_Extr > thresholds_ExM |> filter(Interviewmode ==
↪  "CAWI") |> pull(thresh_Extr_Ratio) ~ "High",
            TRUE ~ "Not high"),
        Extr = factor(Extr, levels = c("Not high", "High", "Not applicable")),
        Mid =
          case_when(
            is.na(Ratio_Mid) ~ "Not applicable",
            Interviewmode == "PAPI" & Ratio_Mid > thresholds_ExM |> filter(Interviewmode ==
↪  "PAPI") |> pull(thresh_Mid_Ratio) ~ "High",
            Interviewmode == "CAWI" & Ratio_Mid > thresholds_ExM |> filter(Interviewmode ==
↪  "CAWI") |> pull(thresh_Mid_Ratio) ~ "High",
            TRUE ~ "Not high"),

        Mid = factor(Mid, levels = c("Not high", "High", "Not applicable")))


w2 |> count(Interviewmode, Extr); w2 |> count(Interviewmode, Mid)
```

```
# A tibble: 5 x 3
  Interviewmode Extr              n
  <fct>         <fct>         <int>
1 PAPI          Not high       4319
2 PAPI          High            786
3 CAWI          Not high       3207
4 CAWI          High            578
5 CAWI          Not applicable   10
```

```
# A tibble: 5 x 3
  Interviewmode Mid               n
  <fct>         <fct>         <int>
1 PAPI          Not high       4394
2 PAPI          High            711
3 CAWI          Not high       3282
4 CAWI          High            503
5 CAWI          Not applicable   10
```

9 w2 Nondifferentiation

See Figure 17

```
plifesat.items <- w2 |> select(plifesat01:plifesat12) |> colnames()
pinform.items <- w2 |> select(pinform01:pinform09) |> colnames()
pglob.items <- w2 |> select(pglob01:pglob05) |> colnames()
ppertrust.items <- w2 |> select(ppertrust01:ppertrust08) |> colnames()
pedugoal.items <- w2 |> select(pedugoal01:pedugoal11) |> colnames()
pfathbeh.items <- w2 |> select(pfathbeh01:pfathbeh08) |> colnames()
pmothbeh.items <- w2 |> select(pmothbeh01:pmothbeh08) |> colnames()
psocdes.items <- w2 |> select(psocdes01:psocdes06) |> colnames()
pschwartz.items <- w2 |> select(pschwartz01:pschwartz21) |> colnames()
pnarcism.items <- w2 |> select(pnarcism01:pnarcism06) |> colnames()
pdiscrim.items <- w2 |> select(pdiscrim01:pdiscrim12) |> colnames()
pgender.items <- w2 |> select(pgender01:pgender05) |> colnames()
pauthor.items <- w2 |> select(pauthor01:pauthor09) |> colnames()
pactivity.items <- w2 |> select(pactivity01:pactivity20) |> colnames()
```

```
items_list_w2 <- list(plifesat.items, pinform.items, pglob.items,
                  ppertrust.items, pedugoal.items, pfathbeh.items,
                  pmothbeh.items, psocdes.items, pschwartz.items,
                  pnarcism.items, pdiscrim.items, pgender.items,
                  pauthor.items, pactivity.items)


tND <- Sys.time()



# extract all items from items_list
all_items_w2 <- unlist(items_list_w2)

w2_papi <-
  w2 |>
  filter(Interviewmode == "PAPI") |>
  select(lfd, all_of(all_items_w2)) |>  # select only the item columns
  meanRootPairs(items_list_w2) |>
  ungroup()
```

```
Duration for plifesat : 0.3974216 minutes
Duration for pinform : 0.4343862 minutes
Duration for pglob : 0.3973524 minutes
Duration for ppertrust : 0.4151595 minutes
Duration for pedugoal : 0.4587621 minutes
Duration for pfathbeh : 0.3813006 minutes
Duration for pmothbeh : 0.3414672 minutes
Duration for psocdes : 0.3741338 minutes
Duration for pschwartz : 0.7417203 minutes
Duration for pnarcism : 0.413005 minutes
Duration for pdiscrim : 0.4859522 minutes
Duration for pgender : 0.3528834 minutes
Duration for pauthor : 0.3821406 minutes
Duration for pactivity : 0.7264682 minutes
All done!
Total duration for all item packs: 6.302579 minutes.
```

```
w2_cawi  <-
  w2 |>
  filter(Interviewmode == "CAWI") |>
  select(lfd, all_of(all_items_w2)) |>
   meanRootPairs(items_list_w2) |> ungroup()
```

```
Duration for plifesat : 0.3177907 minutes
Duration for pinform : 0.2741584 minutes
Duration for pglob : 0.2493619 minutes
Duration for ppertrust : 0.3019107 minutes
Duration for pedugoal : 0.3414717 minutes
Duration for pfathbeh : 0.2956513 minutes
Duration for pmothbeh : 0.3365396 minutes
Duration for psocdes : 0.2924233 minutes
Duration for pschwartz : 0.5998028 minutes
Duration for pnarcism : 0.2694065 minutes
```

```
Duration for pdiscrim : 0.3882664 minutes
Duration for pgender : 0.3214659 minutes
Duration for pauthor : 0.3322784 minutes
Duration for pactivity : 0.5315312 minutes
All done!
Total duration for all item packs: 4.85245 minutes.
```

```r
# merge PAPI and CAWI
w2_meanroot <-
  bind_rows(w2_papi, w2_cawi)

# join onto the original w2 dataset
w2 <- w2 |>
  left_join(w2_meanroot |>
              select(lfd, starts_with("temp_index"), starts_with("temp_max"),
                ↪  starts_with("MeanRootPairs_")), by = "lfd")



Sys.time() - tND
```

```
Time difference of 11.15671 mins
```

- 14 item packages

```r
chart.Correlation(w2 |> select(contains("MeanRootPairs")), histogram = TRUE)
```

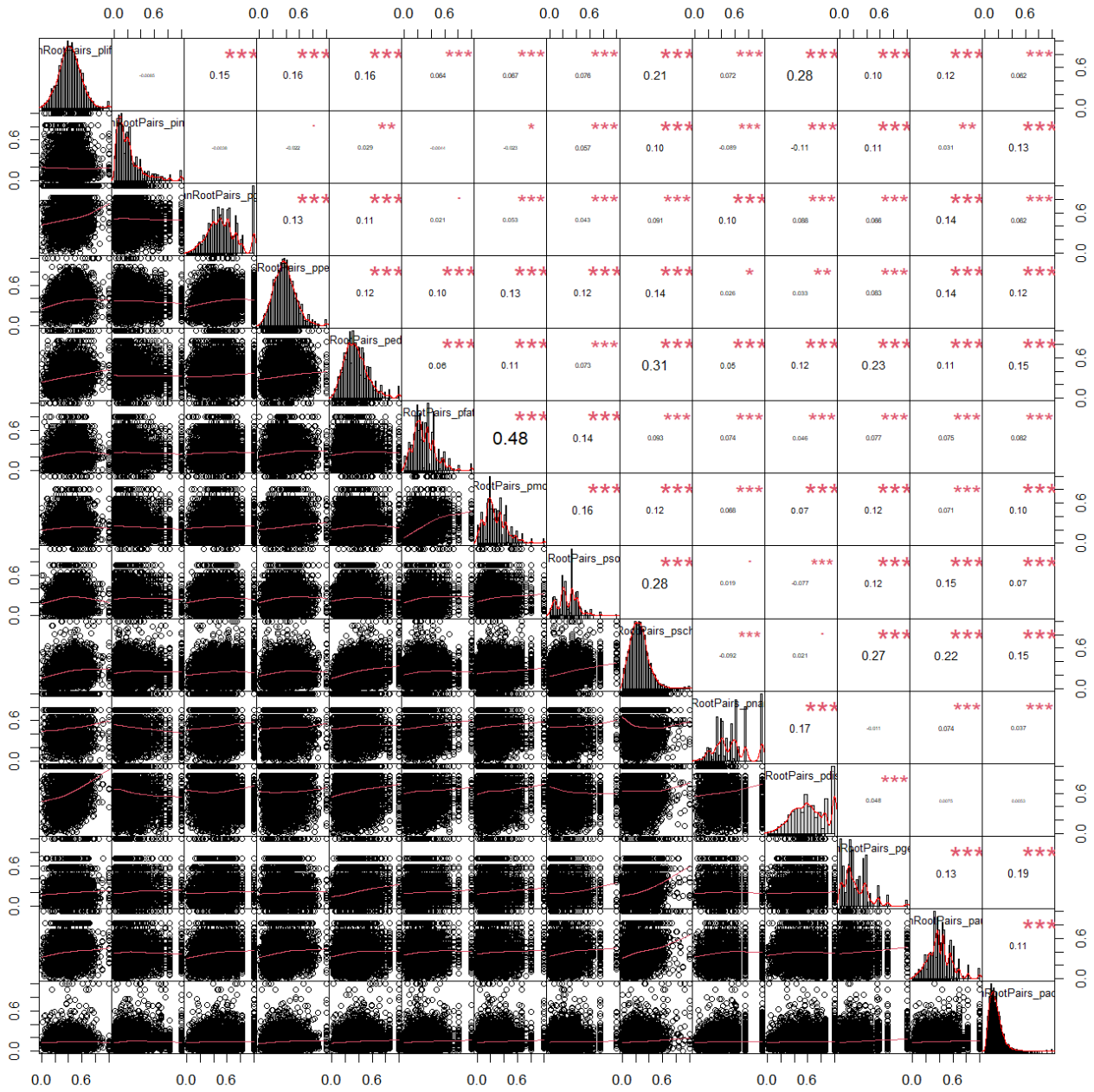


Figure 17: Correlations of Nondifferentiation Items

9.0.1 w2 Nd SEM using CT-C(M-1) Model

```
# correlated traits: WK (weak), ST (strong), Biographical (BIO)
# correlated methods-1: Reference-Scale: 11-likert
w2LatND_mod <- "

WK =~
MeanRootPairs_plifesat +  MeanRootPairs_pglob  + MeanRootPairs_pedugoal +
 ↪  MeanRootPairs_pgender + MeanRootPairs_pauthor + MeanRootPairs_pactivity +
 ↪  MeanRootPairs_pnarcism + MeanRootPairs_pdiscrim + MeanRootPairs_pinform
```

```
ST =~
 MeanRootPairs_psocdes + MeanRootPairs_pschwartz + MeanRootPairs_ppertrust


BIO =~
MeanRootPairs_pfathbeh + MeanRootPairs_pmothbeh

#grids with scales with less-than-11-point-likert scales
u11 =~
MeanRootPairs_pinform + MeanRootPairs_pedugoal + MeanRootPairs_pfathbeh +
 ↪  MeanRootPairs_pmothbeh + MeanRootPairs_psocdes + MeanRootPairs_pschwartz +
 ↪  MeanRootPairs_pnarcism + MeanRootPairs_pdiscrim + MeanRootPairs_pgender +
 ↪  MeanRootPairs_pauthor + MeanRootPairs_pactivity

BIO ~~ 0 * u11
WK ~~ 0 * u11
ST ~~ 0 * u11

"


# simple model without groups
w2LatND_fit <- sem(w2LatND_mod, w2, meanstructure = TRUE)
summary(w2LatND_fit, fit = TRUE)
```

```
lavaan 0.6-21 ended normally after 206 iterations

  Estimator                                         ML
  Optimization method                           NLMINB
  Number of model parameters                        56

                                                  Used       Total
  Number of observations                          3452        8900

Model Test User Model:

  Test statistic                               447.264
  Degrees of freedom                               63
  P-value (Chi-square)                          0.000

Model Test Baseline Model:

  Test statistic                              4168.362
  Degrees of freedom                               91
  P-value                                       0.000

User Model versus Baseline Model:

  Comparative Fit Index (CFI)                    0.906
  Tucker-Lewis Index (TLI)                       0.864

Loglikelihood and Information Criteria:

  Loglikelihood user model (H0)             19110.763
  Loglikelihood unrestricted model (H1)     19334.396
```

```
  Akaike (AIC)                                -38109.527
  Bayesian (BIC)                              -37765.311
  Sample-size adjusted Bayesian (SABIC)       -37943.250

Root Mean Square Error of Approximation:

  RMSEA                                           0.042
  90 Percent confidence interval - lower          0.038
  90 Percent confidence interval - upper          0.046
  P-value H_0: RMSEA <= 0.050                     1.000
  P-value H_0: RMSEA >= 0.080                     0.000

Standardized Root Mean Square Residual:

  SRMR                                            0.030

Parameter Estimates:

  Standard errors                             Standard
  Information                                 Expected
  Information saturated (h1) model          Structured

Latent Variables:
                   Estimate  Std.Err  z-value  P(>|z|)
  WK =~
    MenRtPrs_plfst    1.000
    MeanRtPrs_pglb    0.809    0.072   11.162    0.000
    MeanRtPrs_pdgl    0.850    0.071   11.958    0.000
    MenRtPrs_pgndr    0.635    0.072    8.868    0.000
    MeanRtPrs_pthr    0.716    0.065   11.086    0.000
    MnRtPrs_pctvty    0.307    0.036    8.631    0.000
    MnRtPrs_pnrcsm    0.792    0.083    9.531    0.000
    MnRtPrs_pdscrm    1.282    0.104   12.317    0.000
    MenRtPrs_pnfrm    0.002    0.061    0.033    0.974
  ST =~
    MenRtPrs_pscds    1.000
    MnRtPrs_pschwr    1.226    0.107   11.478    0.000
    MnRtPrs_pprtrs    1.177    0.125    9.407    0.000
  BIO =~
    MnRtPrs_pfthbh    1.000
    MnRtPrs_pmthbh    1.161    0.116   10.005    0.000
  u11 =~
    MenRtPrs_pnfrm    1.000
    MeanRtPrs_pdgl    0.952    0.107    8.883    0.000
    MnRtPrs_pfthbh   -0.027    0.073   -0.377    0.706
    MnRtPrs_pmthbh    0.222    0.076    2.907    0.004
    MenRtPrs_pscds    0.656    0.086    7.656    0.000
    MnRtPrs_pschwr    1.132    0.109   10.341    0.000
    MnRtPrs_pnrcsm   -0.712    0.122   -5.827    0.000
    MnRtPrs_pdscrm   -1.036    0.154   -6.707    0.000
    MenRtPrs_pgndr    1.386    0.136   10.160    0.000
    MeanRtPrs_pthr    0.465    0.085    5.481    0.000
    MnRtPrs_pctvty    0.481    0.057    8.445    0.000

Covariances:
```

```
                   Estimate  Std.Err  z-value  P(>|z|)
  BIO ~~
    u11               0.000
  WK ~~
    u11               0.000
  ST ~~
    u11               0.000
  WK ~~
    ST                0.003    0.000   10.205    0.000
    BIO               0.002    0.000    7.674    0.000
  ST ~~
    BIO               0.002    0.000    7.548    0.000

Intercepts:
                   Estimate  Std.Err  z-value  P(>|z|)
   .MenRtPrs_plfst    0.423    0.003  159.767    0.000
   .MeanRtPrs_pglb    0.525    0.004  146.963    0.000
   .MeanRtPrs_pdgl    0.341    0.003  117.320    0.000
   .MenRtPrs_pgndr    0.226    0.003   73.728    0.000
   .MeanRtPrs_pthr    0.413    0.003  139.368    0.000
   .MnRtPrs_pctvty    0.155    0.002   92.615    0.000
   .MnRtPrs_pnrcsm    0.510    0.004  125.022    0.000
   .MnRtPrs_pdscrm    0.614    0.004  146.498    0.000
   .MenRtPrs_pnfrm    0.222    0.003   69.982    0.000
   .MenRtPrs_pscds    0.281    0.003  106.996    0.000
   .MnRtPrs_pschwr    0.267    0.002  118.750    0.000
   .MnRtPrs_pprtrs    0.371    0.003  142.315    0.000
   .MnRtPrs_pfthbh    0.283    0.003  102.377    0.000
   .MnRtPrs_pmthbh    0.259    0.003   92.393    0.000

Variances:
                   Estimate  Std.Err  z-value  P(>|z|)
   .MenRtPrs_plfst    0.018    0.001   30.616    0.000
   .MeanRtPrs_pglb    0.040    0.001   38.526    0.000
   .MeanRtPrs_pdgl    0.022    0.001   34.317    0.000
   .MenRtPrs_pgndr    0.024    0.001   31.510    0.000
   .MeanRtPrs_pthr    0.027    0.001   38.347    0.000
   .MnRtPrs_pctvty    0.008    0.000   37.882    0.000
   .MnRtPrs_pnrcsm    0.052    0.001   35.878    0.000
   .MnRtPrs_pdscrm    0.048    0.002   26.548    0.000
   .MenRtPrs_pnfrm    0.032    0.001   36.843    0.000
   .MenRtPrs_pscds    0.020    0.001   37.153    0.000
   .MnRtPrs_pschwr    0.010    0.000   22.533    0.000
   .MnRtPrs_pprtrs    0.020    0.001   34.080    0.000
   .MnRtPrs_pfthbh    0.016    0.001   13.612    0.000
   .MnRtPrs_pmthbh    0.012    0.001    8.527    0.000
    WK                0.006    0.001   10.442    0.000
    ST                0.002    0.000    6.251    0.000
    BIO               0.011    0.001    9.097    0.000
    u11               0.003    0.001    6.276    0.000
```

```r
fscores <- lavPredict(w2LatND_fit)  # matrix, not a list
idx     <- lavInspect(w2LatND_fit, "case.idx")  # single vector

# merge the scores back in, following the lavaan documentation for single-group models:
for (fs in colnames(fscores)) {
  w2[idx, fs] <- fscores[, fs]
```

```
}
```

```
# model used as the basis for the coefficient table
SEM_w2 <- table_results(w2LatND_fit)|>
    mutate(label = label |>
                str_replace("MeanRootPairs_", "Mrp.") |>
                str_replace("=~", ".BY.") |>
                str_replace("~~", ".WITH.") |>
                str_replace("~1", "") |>
                str_replace("^\\.", "Means.")) |>

  filter(!str_starts(label, "Variances"),
         !str_starts(label, "Means")) |>
  select(label, est_sig, se, confint)


SEM_w2|>
  kable(
    format = "latex",
    booktabs = TRUE,
    digits = 3,
    col.names = c("Label",
                  "Est.", "SE", "95\\% CI"),
    escape = FALSE
  )
```

Label	Est.	SE	95% CI
WK.BY.Mrp.plifesat	1.00	0.00	[1.00, 1.00]
WK.BY.Mrp.pglob	0.81***	0.07	[0.67, 0.95]
WK.BY.Mrp.pedugoal	0.85***	0.07	[0.71, 0.99]
WK.BY.Mrp.pgender	0.63***	0.07	[0.49, 0.78]
WK.BY.Mrp.pauthor	0.72***	0.06	[0.59, 0.84]
WK.BY.Mrp.pactivity	0.31***	0.04	[0.24, 0.38]
WK.BY.Mrp.pnarcism	0.79***	0.08	[0.63, 0.95]
WK.BY.Mrp.pdiscrim	1.28***	0.10	[1.08, 1.49]
WK.BY.Mrp.pinform	0.00	0.06	[-0.12, 0.12]
ST.BY.Mrp.psocdes	1.00	0.00	[1.00, 1.00]
ST.BY.Mrp.pschwartz	1.23***	0.11	[1.02, 1.44]
ST.BY.Mrp.ppertrust	1.18***	0.13	[0.93, 1.42]
BIO.BY.Mrp.pfathbeh	1.00	0.00	[1.00, 1.00]
BIO.BY.Mrp.pmothbeh	1.16***	0.12	[0.93, 1.39]
u11.BY.Mrp.pinform	1.00	0.00	[1.00, 1.00]
u11.BY.Mrp.pedugoal	0.95***	0.11	[0.74, 1.16]
u11.BY.Mrp.pfathbeh	-0.03	0.07	[-0.17, 0.12]
u11.BY.Mrp.pmothbeh	0.22**	0.08	[0.07, 0.37]
u11.BY.Mrp.psocdes	0.66***	0.09	[0.49, 0.82]
u11.BY.Mrp.pschwartz	1.13***	0.11	[0.92, 1.35]
u11.BY.Mrp.pnarcism	-0.71***	0.12	[-0.95, -0.47]
u11.BY.Mrp.pdiscrim	-1.04***	0.15	[-1.34, -0.73]
u11.BY.Mrp.pgender	1.39***	0.14	[1.12, 1.65]
u11.BY.Mrp.pauthor	0.47***	0.08	[0.30, 0.63]
u11.BY.Mrp.pactivity	0.48***	0.06	[0.37, 0.59]
BIO.WITH.u11	0.00	0.00	[0.00, 0.00]
WK.WITH.u11	0.00	0.00	[0.00, 0.00]
ST.WITH.u11	0.00	0.00	[0.00, 0.00]
WK.WITH.ST	0.00***	0.00	[0.00, 0.00]
WK.WITH.BIO	0.00***	0.00	[0.00, 0.00]
ST.WITH.BIO	0.00***	0.00	[0.00, 0.00]

10 Wave 3

```
w3p_full |> nrow()
```

```
[1] 8369
```

```
w3_lfds <-  #8127
  w1 |> filter(Sample == "Sample") |>
  inner_join(w3p_full, by = "pid") |>
  pull(pid)

w3 <-
  w3p_full |>
  filter(pid %in% w3_lfds)
w3 |> nrow()
```

```
[1] 8127
```

10.1 w3 Build Breakoff

```
w3 <-
  w3 |>
  mutate(BreakOff = case_when(
    intvollst %in% c(2,3) ~ "Break-off",
    qmethod == 3 ~ "PAPI",              #PAPI
    (qmethod == 4 & intvollst == 1) ~ "No Break-off"
  ),
    BreakOff = factor(BreakOff, levels = c("No Break-off", "Break-off", "PAPI")))

w3 |> frq(BreakOff)
```

```
BreakOff <categorical>
# total N=8127 valid N=8127 mean=1.65 sd=0.93

Value        |    N | Raw % | Valid % | Cum. %
---------------------------------------------
No Break-off | 5439 | 66.93 |   66.93 |  66.93
Break-off    |  129 |  1.59 |    1.59 |  68.51
PAPI         | 2559 | 31.49 |   31.49 | 100.00
<NA>         |    0 |  0.00 |    <NA> |   <NA>
```

```
# 129 cases
```

11 w3 Speeder

See fig-w3Speedingcategorization

```
w3 |>
  filter(qmethod == 4) |>

  select(pid, BreakOff) |>
  left_join(w3_dur |> select(pid,  interviewdauer), by = "pid") |>
  #left_join(w3_dur, by = "pid") |>
  left_join(w1 |> select(pid, Age_cat = Age_cat_Sample), by = "pid") |>

  #select(-plifesat03_dur, -plifesat04_dur, -plifesat10_dur, -contains("prelfund")) |>


  mutate(


    # not include into calculation of quantile borders: N = 54 Break-off cases
    # not include into calculation of quantile borders: N = 54 Break-off cases
  interviewdauer = case_when(BreakOff == "Break-off" ~ NA,
                             TRUE ~ interviewdauer)) |>

  group_by(Age_cat) |>


  summarise(
    `1` = quantile(interviewdauer, probs = 0.01, na.rm = TRUE),
    `3` = quantile(interviewdauer, probs = 0.03, na.rm = TRUE),
    `5` = quantile(interviewdauer, probs = 0.05, na.rm = TRUE),
    `10` = quantile(interviewdauer, probs = 0.1, na.rm = TRUE),
```

```
    `15` = quantile(interviewdauer, probs = 0.15, na.rm = TRUE),
    `20` = quantile(interviewdauer, probs = 0.2, na.rm = TRUE),
    `25` = quantile(interviewdauer, probs = 0.25, na.rm = TRUE),
    `33` = quantile(interviewdauer, probs = 0.33, na.rm = TRUE),
    `50` = quantile(interviewdauer, probs = 0.5, na.rm = TRUE),
    `66` = quantile(interviewdauer, probs = 0.66, na.rm = TRUE),
    `70` = quantile(interviewdauer, probs = 0.7, na.rm = TRUE),
    `75` = quantile(interviewdauer, probs = 0.75, na.rm = TRUE),
    `80` = quantile(interviewdauer, probs = 0.80, na.rm = TRUE),
    `85` = quantile(interviewdauer, probs = 0.85, na.rm = TRUE),
    `95` = quantile(interviewdauer, probs = 0.95, na.rm = TRUE)
  ) |>
  gather(key = "Quantile", value = "Value", -Age_cat) |>
  mutate(Quantile = as.numeric(Quantile)) |>
  ggplot(aes(y = Quantile,
             x = Value, color = Age_cat)) +
  geom_line() +
  scale_y_continuous(breaks = seq(0,100, by = 5)) +
  labs(x = "Interview duration", y = "Quantile",
       title = "w3: Quantiles of interview duration per age group (w3)") +
  papaja::theme_apa()
```

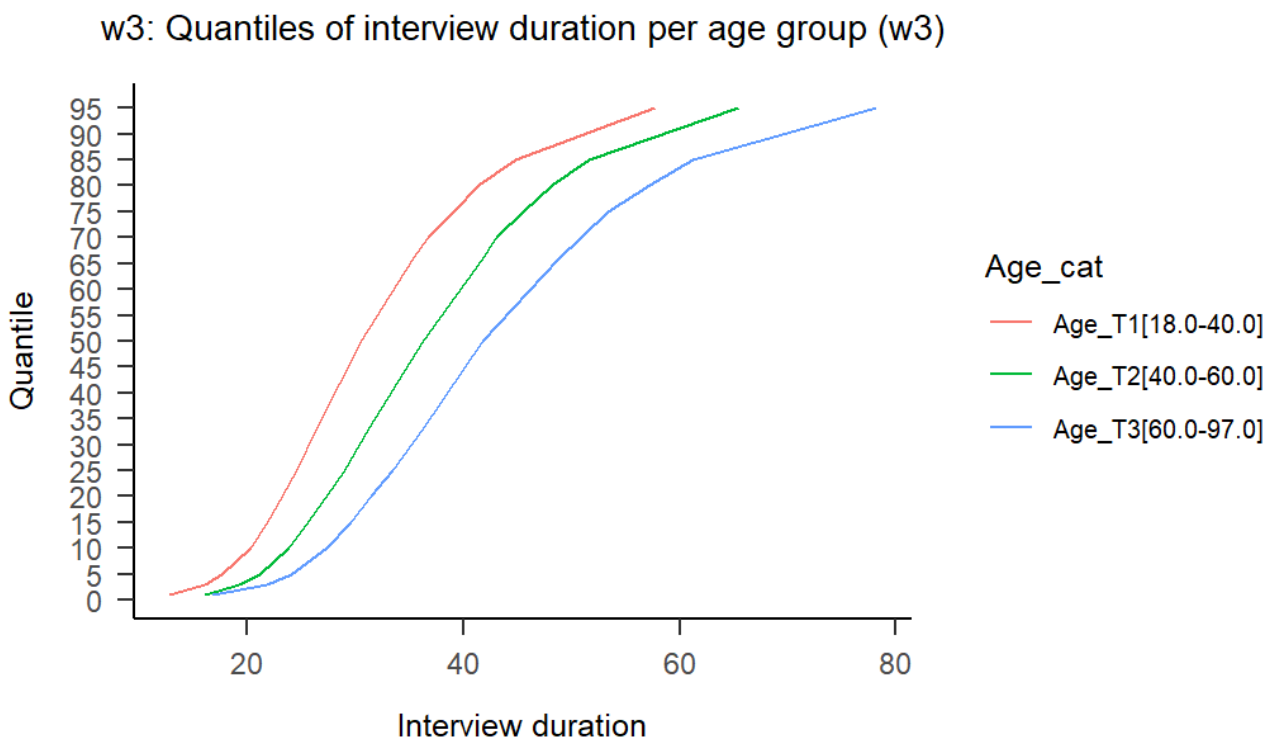


Figure 18: Speeding at w3

```
#|label: w3-build-speeding-variable
#|eval: true


w3_dur <-
  w3_dur |>
```

```
  left_join(w1 |> select(pid, Age_cat = Age_cat_Sample), by = "pid") |>

    #inner_join(w3_lfds, by = "lfd") |>
  # Step 1: set 0 to NA in all item columns
  mutate(across(ends_with("_dur"),
           ~ if_else(.x == 0, NA_real_, .x))
  )  |>

    # -0 cases not in the analysis sample due to age inconsistency (0) or w1/w2-only
     ↪  participation (0)
    filter(!is.na(Age_cat)) |>


  group_by(Age_cat) |>
  # Step 2: compute medians (0/NA excluded)
  mutate(
    across(ends_with("_dur"),
           list(md = ~median(.x, na.rm = TRUE)),
           .names = "{.col}_md")
  ) |>

  # Step 3: speeding indicators (_s) with group-specific medians
  mutate(
    across(
      .cols = ends_with("_dur"),
      .fns = ~ case_when(
        is.na(.x) ~ NA_real_,
        .x < 0.4 * get(paste0(cur_column(), "_md")) ~ 1,
        TRUE ~ 0
      ),
      .names = "{.col}_s"
    )
  ) |>
  ungroup() |>

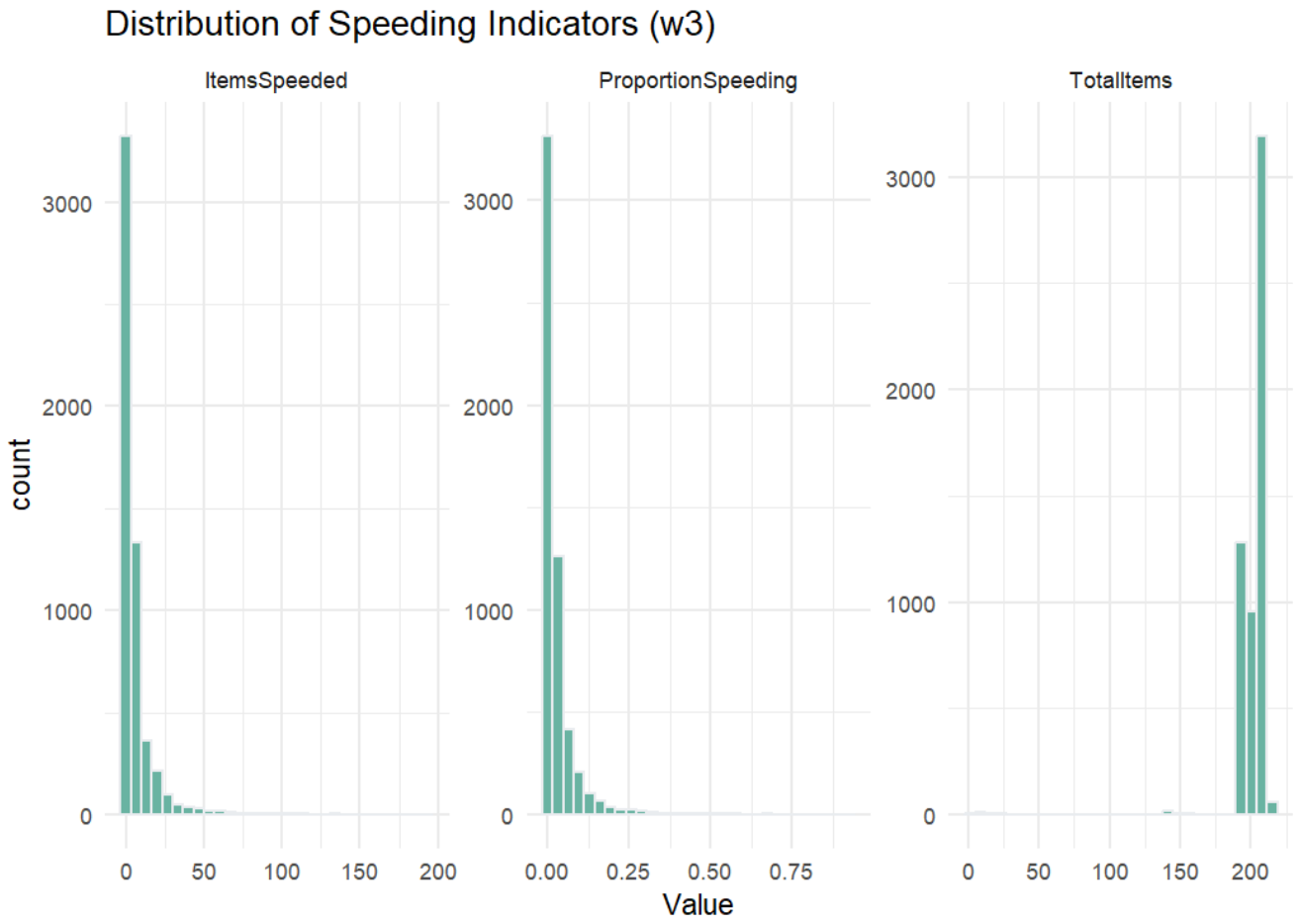
  # Step 4: sum scores
  rowwise() |>
  mutate(

    # number of items whose duration is < 0.4 * the item's median duration across all
     ↪  respondents in the same age group (can be 0 if the respondent did not speed on any
     ↪  item)
    ItemsSpeeded = sum(c_across(ends_with("_s")), na.rm = TRUE),
    TotalItems = sum(!is.na(c_across(ends_with("_s")))),
    ProportionSpeeding = ItemsSpeeded / TotalItems)|>
  ungroup()




w3_dur |>
  select(ItemsSpeeded, TotalItems, ProportionSpeeding) |>
  pivot_longer(everything(), names_to = "Variable", values_to = "Value") |>
  ggplot(aes(x = Value)) +
      geom_histogram(fill="#69b3a2", color="#e9ecef") +
```

```
# geom_density(fill = "steelblue", alpha = 0.5) +
 facet_wrap(~Variable, scales = "free") +
 theme_minimal() +
 labs(title = "Distribution of Speeding Indicators (w3)")
```




```
# Complete sample (sensitivity analysis only)
w3_dur |> filter(ProportionSpeeding>0.000) |> nrow()
```

```
[1] 4308
```

```
w3_dur |> filter(ProportionSpeeding >0 & ProportionSpeeding<0.019324) |> pull(ItemsSpeeded) |>
 ↪  summary()  #1-4 Items - marginal
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  1.000   1.000   2.000   1.913   3.000   4.000
```

```
# upper median range:
w3_dur |> filter(ProportionSpeeding>=0.019324) |> pull(ItemsSpeeded) |> summary()  #1-193 Items
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1.0     6.0     9.0    16.1    17.0   194.0
```

```
#### TOP 10 MOST SPEEDED ITEMS

 # filter the _s columns
 s_cols <- w3_dur |>
```

```
    select(ends_with("_s")) |>
    names()

# compute column sums (number of 1s, ignoring NAs)
w3_dur |>
    summarise(across(all_of(s_cols), ~ sum(.x == 1, na.rm = TRUE))) |>
    pivot_longer(everything(), names_to = "variable", values_to = "sum_1er") |>
    arrange(desc(sum_1er)) |>
  head(10)
```

```
# A tibble: 10 x 2
   variable          sum_1er
   <chr>               <int>
 1 pweathev03_dur_s      652
 2 ptaxparty06_dur_s     617
 3 pedu02_dur_s          613
 4 pweathev02_dur_s      589
 5 ptaxparty02_dur_s     582
 6 pladder01_dur_s       576
 7 ptaxego_dur_s         545
 8 pcoleffic02_dur_s     544
 9 pweathev01_dur_s      538
10 pgendego_dur_s        503
```

```
# pweathev03_dur_s  652
# ptaxparty06_dur_s 617
# pedu02_dur_s  613
# pweathev02_dur_s  589
# ptaxparty02_dur_s 582
# pladder01_dur_s   576
# ptaxego_dur_s 545
# pcoleffic02_dur_s 544
# pweathev01_dur_s  538
# pgendego_dur_s    503
```

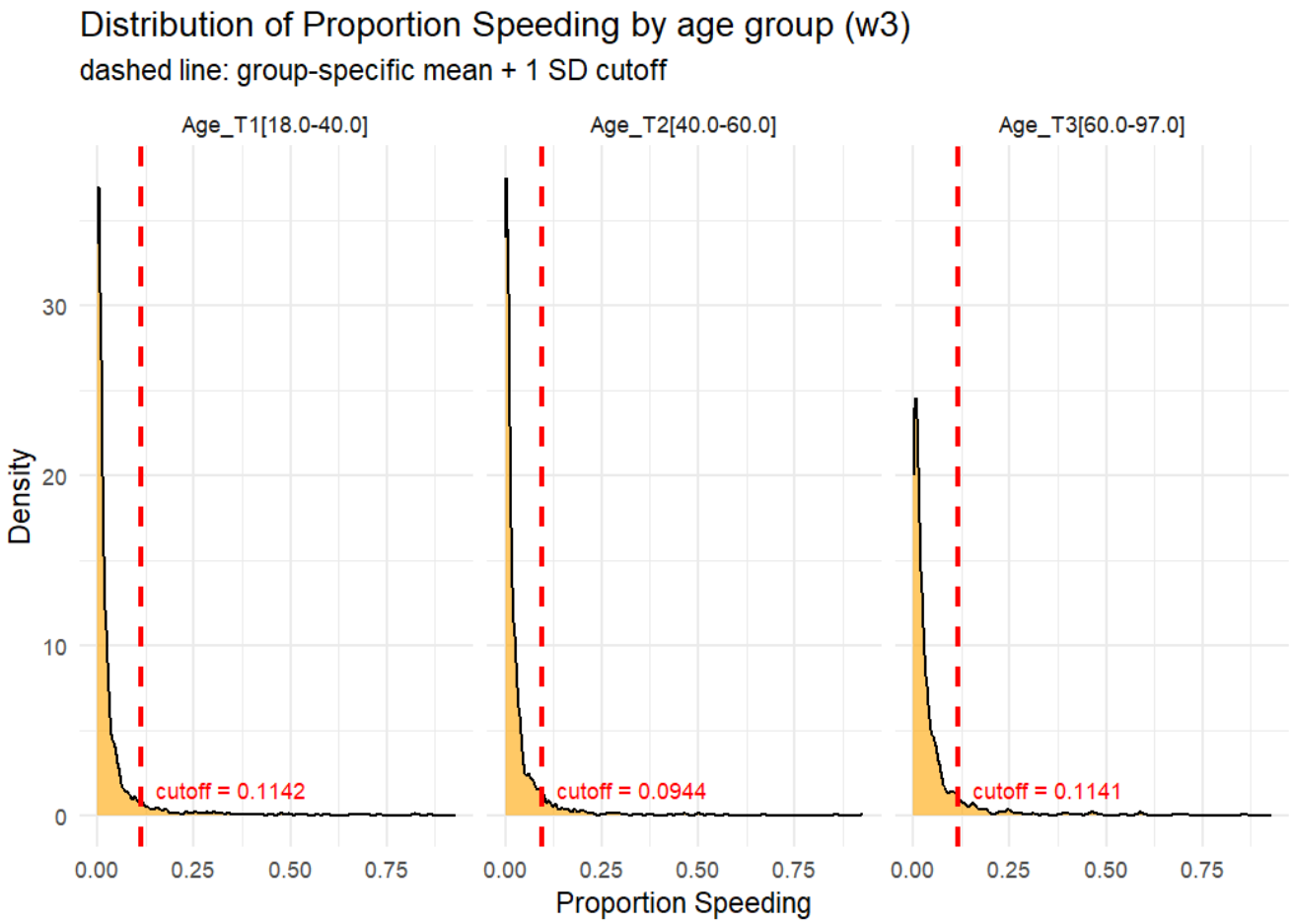
```
# consistent threshold: mean + 1 SD per group

thresholds_speeding_w3 <-
   w3_dur |>
   group_by(Age_cat) |>
   summarise(
     thresh_Speed = mean(ProportionSpeeding, na.rm = TRUE) + sd(ProportionSpeeding, na.rm =
      ↪  TRUE),
     percentile_sp = mean(ProportionSpeeding <= thresh_Speed, na.rm = TRUE) * 100,
     prop_above_sp = mean(ProportionSpeeding > thresh_Speed, na.rm = TRUE) * 100) |> ungroup()

w3_dur <-
  w3_dur |>
  mutate(
    Speeding = case_when(
      is.na(ProportionSpeeding) ~NA,
      Age_cat == "Age_T1[18.0-40.0]" & ProportionSpeeding > thresholds_speeding_w3 |>
 ↪  filter(Age_cat == "Age_T1[18.0-40.0]") |> pull(thresh_Speed) ~ "Speeding",
```

```
      Age_cat == "Age_T2[40.0-60.0]" & ProportionSpeeding > thresholds_speeding_w3 |>
  ↪  filter(Age_cat == "Age_T2[40.0-60.0]") |> pull(thresh_Speed) ~ "Speeding",
      Age_cat == "Age_T3[60.0-97.0]" & ProportionSpeeding > thresholds_speeding_w3 |>
  ↪  filter(Age_cat == "Age_T3[60.0-97.0]") |> pull(thresh_Speed) ~ "Speeding",
      TRUE ~"No speeding"
    ),
    Speeding = factor(Speeding, levels = c("No speeding", "Speeding"))
  )

w3_dur |>
  ggplot(aes(x = ProportionSpeeding)) +
  geom_density(alpha = 0.6, fill = "orange") +
  geom_vline(data = thresholds_speeding_w3,
             aes(xintercept = thresh_Speed),
             linetype = "dashed", color = "red", linewidth = 1) +
  geom_text(data = thresholds_speeding_w3,
            aes(x = thresh_Speed, y = 0,
                label = paste0("cutoff = ", round(thresh_Speed, 4))),
            hjust = -0.1, vjust = -1, color = "red", size = 3) +
  facet_wrap(~ Age_cat) +
  labs(title = "Distribution of Proportion Speeding by age group (w3)",
       subtitle = "dashed line: group-specific mean + 1 SD cutoff",
       x = "Proportion Speeding",
       y = "Density") +
  theme_minimal()
```



```
# join to w3
w3 <-
  w3 |>
  left_join(
    w3_dur |> select(pid, ProportionSpeeding, Speeding), by = "pid"
  )
```

11.1 w3 Interviewmode

```
w3 <-
  w3 |>
 mutate(
    Interviewmode = case_when(
      qmethod == 3 ~ "PAPI",
      qmethod == 4 ~ "CAWI"
    ),
    Interviewmode = factor(Interviewmode, levels = c("PAPI", "CAWI")))
```

11.2 w3 FreeTextMis

```
w3_up <-
  w3_up |>
  mutate(

    peduOth = case_when(
      pedu01.raw != 6  ~ -2,

      pedu01.raw == 6 & (nchar(peduoth01.raw) >= 3 & #nchar(peduoth0.raw1) <= 20 &

        !(peduoth01.raw %in% c("...", "....", "xxx", "XXX", "abc", "ABC", "K.a.", "k.A.")) &
        !grepl("-1", peduoth01.raw) &
        !grepl("-9", peduoth01.raw)) ~ 1,


      pedu01.raw == 6 & !(
        nchar(peduoth01.raw) >= 3 & #nchar(peduoth01.raw) <= 20 &

        !(peduoth01.raw %in% c("...", "....", "xxx", "XXX", "abc", "ABC", "K.a.", "k.A.")) &
        !grepl("-1", peduoth01.raw) &
        !grepl("-9", peduoth01.raw)) ~ -1),


    # apply standard missing codes to poccupation_isco08
    poccupation_isco08_2 = case_when(poccupation_isco08 == 0            ~ -2,
                                     poccupation_isco08 %in% c(8, 9)    ~ -1,
                                     TRUE                               ~ poccupation_isco08),



    plifeventOth = case_when(
      plifevent14.raw != 1  ~ -2,

      plifevent14.raw == 1 & (nchar(plifevoth14.raw) >= 3 & #nchar(plifevoth14.raw) <= 20 &
```

```
    !(plifevoth14.raw %in% c("...", "....", "xxx", "XXX", "abc", "ABC", "K.a.", "k.A.")) &
    !grepl("-1", plifevoth14.raw) &
    !grepl("-9", plifevoth14.raw)) ~ 1,


  plifevent14.raw == 1 & !(
    nchar(plifevoth14.raw) >= 3 & #nchar(plifevoth14.raw) <= 20 &

    !(plifevoth14.raw %in% c("...", "....", "xxx", "XXX", "abc", "ABC", "K.a.", "k.A.")) &
    !grepl("-1", plifevoth14.raw) &
    !grepl("-9", plifevoth14.raw)) ~ -1),




pnetmodwill0th = case_when(
  pnetmodwill02.raw != 5  ~ -2,

  pnetmodwill02.raw == 5 & (nchar(pnetmodwill03) >= 3 & #nchar(pnetmodwill03) <= 20 &

    !(pnetmodwill03 %in% c("...", "....", "xxx", "XXX", "abc", "ABC", "K.a.", "k.A.")) &
    !grepl("-1", pnetmodwill03) &
    !grepl("-9", pnetmodwill03)) ~ 1,


  pnetmodwill02.raw == 5 & !(
    nchar(pnetmodwill03) >= 3 & #nchar(pnetmodwill03) <= 20 &

    !(pnetmodwill03 %in% c("...", "....", "xxx", "XXX", "abc", "ABC", "K.a.", "k.A.")) &
    !grepl("-1", pnetmodwill03) &
    !grepl("-9", pnetmodwill03)) ~ -1),


pdeviceOth = case_when(
  pdevice05.raw != 1 ~ -2,

  pdevice05.raw == 1 & (nchar(pothdevice.raw) >= 3 &

    !(pothdevice.raw %in% c("...", "....", "xxx", "XXX", "abc", "ABC", "K.a.", "k.A.")) &
    !grepl("-1", pothdevice.raw) &
    !grepl("-9", pothdevice.raw)) ~ 1,


  pdevice05.raw == 1 & !(nchar(pothdevice.raw) >= 3 &

    !(pothdevice.raw %in% c("...", "....", "xxx", "XXX", "abc", "ABC", "K.a.", "k.A.")) &
    !grepl("-1", pothdevice.raw) &
    !grepl("-9", pothdevice.raw)) ~ -1),


pmemberOth = case_when(

!(pmember11.raw %in% c(1, 2, 3)) ~ -2,

pmember11.raw %in% c(1, 2, 3) & (nchar(pmemberoth.raw) >= 3 &
```

```
        !(pmemberoth.raw %in% c("...", "....", "xxx", "XXX", "abc", "ABC", "K.a.", "k.A.")) &
        !grepl("-1", pmemberoth.raw) &
        !grepl("-9", pmemberoth.raw)) ~ 1,

    pmember11.raw %in% c(1, 2, 3) & !(nchar(pmemberoth.raw) >= 3 &


        !(pmemberoth.raw %in% c("...", "....", "xxx", "XXX", "abc", "ABC", "K.a.", "k.A.")) &
        !grepl("-1", pmemberoth.raw) &
        !grepl("-9", pmemberoth.raw)) ~ -1


    ))



w3 <-
  w3 |>
  left_join(w3_up |> select(pid,
                            peduOth,
                            poccupation_isco08_2,
                            plifeventOth,
                            pnetmodwillOth,
                            pdeviceOth,
                            pmemberOth), by = "pid")


FreeTextItemsw3 <-

  w3 |> select(
    pmemberOth,
    pbirthmnth, pbirthyear,
    hhhsize,
    hhhage01, hhhage02,
    peduOth,
    plabincbrut, plabincnet, potherinc,
    poccupation_isco08_2,
    phiremonth, phireyear,
    pworkhrscontr, pworkhrsact,
    pnetmodwillOth,
    pdeviceOth,
    plifeventOth


    ) |> colnames()


w3 <-

  w3 |>

  rowwise() |>
  mutate(

    freetextmissings = sum(across(all_of(FreeTextItemsw3)) %in% c(-1, -9), na.rm = TRUE),
```

```
    N_allFreeTitems = sum(across(all_of(FreeTextItemsw3), ~!. %in% c(-2, -10)), na.rm = TRUE),

    FrTMisRatio = freetextmissings/N_allFreeTitems) |>

   ungroup()



thresholds_FrTMis_w3 <-
   w3 |>
   group_by(Interviewmode) |>
   summarise(
     thresh_FrTMis = mean(FrTMisRatio, na.rm = TRUE) + sd(FrTMisRatio, na.rm = TRUE),
     percentile_sp = mean(FrTMisRatio <= thresh_FrTMis, na.rm = TRUE) * 100,
     prop_above_sp = mean(FrTMisRatio > thresh_FrTMis, na.rm = TRUE) * 100) |> ungroup()

thresholds_FrTMis_w3
```

```
# A tibble: 2 x 4
  Interviewmode thresh_FrTMis percentile_sp prop_above_sp
  <fct>                <dbl>         <dbl>         <dbl>
1 PAPI                 0.356          83.7          16.3
2 CAWI                 0.164          84.8          15.2
```

```
w3 <-
  w3 |>
  mutate(FrTMis =
           case_when(
             is.na(FrTMisRatio) ~ "Not applicable",
             Interviewmode == "PAPI" & FrTMisRatio > thresholds_FrTMis_w3 |>
 ↪  filter(Interviewmode == "PAPI") |> pull(thresh_FrTMis) ~ "High",
             Interviewmode == "CAWI" & FrTMisRatio > thresholds_FrTMis_w3 |>
 ↪  filter(Interviewmode == "CAWI") |> pull(thresh_FrTMis) ~ "High",
             TRUE ~ "Not high"),
         FrTMis = factor(FrTMis, levels = c("Not high", "High", "Not applicable")))

w3 |> group_by(Interviewmode) |> frq(FrTMis)
```

```
FrTMis <categorical>
# grouped by: PAPI
# total N=2559 valid N=2559 mean=1.16 sd=0.37

Value          |    N | Raw % | Valid % | Cum. %
-----------------------------------------------
Not high       | 2142 | 83.70 |   83.70 |  83.70
High           |  417 | 16.30 |   16.30 | 100.00
Not applicable |    0 |  0.00 |    0.00 | 100.00
<NA>           |    0 |  0.00 |    <NA> |   <NA>

FrTMis <categorical>
# grouped by: CAWI
# total N=5568 valid N=5568 mean=1.18 sd=0.41

Value          |    N | Raw % | Valid % | Cum. %
-----------------------------------------------
Not high       | 4659 | 83.67 |   83.67 |  83.67
```

```
High           |  835 | 15.00 |   15.00 |  98.67
Not applicable |   74 |  1.33 |    1.33 | 100.00
<NA>           |    0 |  0.00 |    <NA> |   <NA>
```

```
posn_d <- position_dodge(0.4)



  w3 |> select(pid, Interviewmode, FrTMisRatio) |>
  pivot_longer(cols = 3) |>

  ggplot(aes(Interviewmode, value, group = Interviewmode)) +


  geom_jitter(aes(Interviewmode, value, color = Interviewmode, group = Interviewmode),
              stat = "identity",
              position = "jitter",
              alpha = .3) +

  stat_summary(fun = mean,
               geom = "point",
               position = posn_d,
               size = 4,
               color = "black") +

  labs (y = "Ratio Free Text Item Missings",
        color = "Interviewmode", caption = "Free Text Missings by Interviewmode (w3)") +
  scale_color_manual(name = "Interviewmode", values = mode_palette) +


   papaja::theme_apa() +

   theme(axis.text.x = element_blank(), #element_text(size = 9, angle = 15),
        axis.ticks.x = element_blank(),
        axis.line.x = element_blank(),
        legend.position="bottom")



w3 |> group_by(Interviewmode) |>
  summarise(mean.FrTMisRatio = mean(FrTMisRatio, na.rm = TRUE),
            sd.FrTMisRatio = sd(FrTMisRatio, na.rm = TRUE))
```

```
# A tibble: 2 x 3
  Interviewmode mean.FrTMisRatio sd.FrTMisRatio
  <fct>                    <dbl>          <dbl>
1 PAPI                    0.188          0.168
2 CAWI                    0.0510         0.114
```

```
w3 |> group_by(Interviewmode) |>
  summarise(mean.FrTMisRatio = mean(FrTMisRatio, na.rm = TRUE),
            sd.FrTMisRatio = sd(FrTMisRatio, na.rm = TRUE))
```

```
# A tibble: 2 x 3
  Interviewmode mean.FrTMisRatio sd.FrTMisRatio
  <fct>                    <dbl>          <dbl>
1 PAPI                    0.188          0.168
```

```
2 CAWI                    0.0510          0.114
```

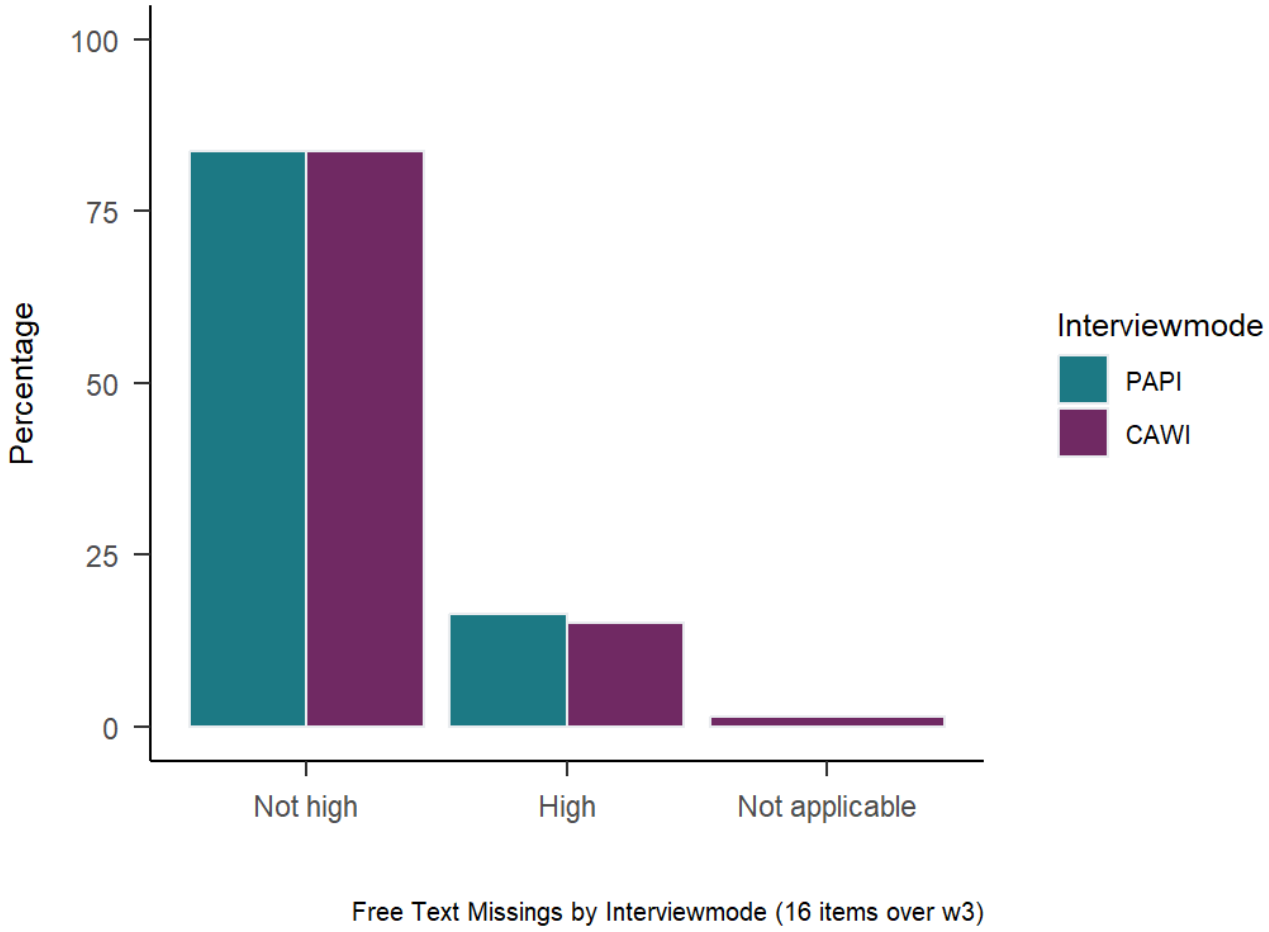
Figure 19: Free text missings by Interviewmode (w3)




```
w3 |>
  group_by(Interviewmode) |>
  count(FrTMis) |>
  mutate(share = (n*100)/sum(n)) |>


  ggplot(aes(x = FrTMis, y = share, fill = Interviewmode)) +
  geom_bar(stat = "identity", position = position_dodge(),color="#e9ecef") +
  labs(caption = "Free Text Missings by Interviewmode (16 items over w3)",
       x = "",
       y = "Percentage",
       fill = "Interviewmode") +
  scale_y_continuous(limits = c(0,100)) +
  scale_fill_manual(values = mode_palette) +
 papaja::theme_apa()
```

Figure 20: Free text missings by Interviewmode (w3)



12 w3 Extreme/Middle

For comparison of extreme vs. middle response distributions by interviewmode see Figure 8

```
#106
extrm5_itemsw3 <-
  w3 |> select(

    pinequal01, pinequal02,
    pcoleffic01, pcoleffic02,
    pselfeffic01, pselfeffic02,
    pclimsor,
    pclimpolcon01:pclimpolcon03,
    pclimcons01:pclimcons03,
    pundesinfra2,
    pislamphob01:pislamphob04,
    pimmsal,
    ptaxsal,
    pclimsal,
    pgendsal,

    # shared with nondiff:
    pbsjo01:pbsjo05,
```

```
    pbsjperc01:pbsjperc05,
    puss01:puss08,
    pconsp01:pconsp05,
    pgendeq01:pgendeq06,
    pregident01:pregident05,
    pinfrasat01:pinfrasat18,
    pclimattit01:pclimattit06,
    pcliminfra01:pcliminfra05,
    ppopul10:ppopul15,
    pantisem01:pantisem06,
    prefmove01:prefmove06,
    psubjhealth,
    pnetmodwill01, pnetmodwill02

  ) |> colnames()

#54
extrm11_itemsw3 <-
  w3 |> select(

    pweathev01:pweathev03,
    pundesinfra1,
    pimmego,
    ptaxego,
    pclimego,
    pgendego,
    plifesat,

    # shared with nondiff:
    pladder01:pladder05,
    pconcern01:pconcern12,
    pimmparty01:pimmparty07,
    ptaxparty01:ptaxparty07,
    pclimparty01:pclimparty07,
    pgendparty01:pgendparty07

  ) |> colnames()




t1 <- Sys.time()


w3 <-
  w3 |>
  rowwise() |>
  mutate(

    # all extreme values on 5-likert
    E5 = sum(across(all_of(extrm5_itemsw3)) %in% c(1, 5), na.rm = TRUE),
    # all middle values on 5-likert
    M5 = sum(across(all_of(extrm5_itemsw3)) == 3, na.rm = TRUE),

    # all extreme and middle values on 5-likert
```

```
    EM.S5 = sum(E5, M5, na.rm = TRUE),



    # all extreme values on 11-likert
    E11 = sum(across(all_of(extrm11_itemsw3)) %in% c(0, 10), na.rm = TRUE),
    # all middle values on 11-likert
    M11 = sum(across(all_of(extrm11_itemsw3)) %in% c(5), na.rm = TRUE),

    # all extreme and middle values on 11-likert
    EM.S11 = sum(E11, M11, na.rm = TRUE),



    N5_all = sum(across(all_of(extrm5_itemsw3)) > 0, na.rm = TRUE),
    N11_all = sum(across(all_of(extrm11_itemsw3)) >= 0, na.rm = TRUE),



    # all extreme values
    E.S5S11 = sum(E5, E11, na.rm = TRUE),
    # all middle values
    M.S5S11 = sum(M5, M11, na.rm = TRUE),

    # all items responded
    N_allExMidOpportunities = sum(N5_all, N11_all),

    # proportion of extreme and middle values on 5-likert by all responses
    Ratio_S5 = ifelse(N5_all == 0, NA, EM.S5/N5_all),

    # proportion of extreme and middle values on 11-likert by all responses
    Ratio_S11 = ifelse(N11_all == 0, NA, EM.S11/N11_all),


    Ratio_Extr = ifelse(N_allExMidOpportunities == 0, NA, E.S5S11/N_allExMidOpportunities),
    Ratio_Mid = ifelse(N_allExMidOpportunities == 0, NA, M.S5S11/N_allExMidOpportunities),

    RatioExMid = sum(E.S5S11, M.S5S11, na.rm = TRUE)/N_allExMidOpportunities

  ) |> ungroup()

print(paste("Total duration (Extreme/Middle, mins):", as.numeric(Sys.time()-t1, "mins")))
```

```
[1] "Total duration (Extreme/Middle, mins): 19.1358497341474"
```

```
cor.test(w3$Ratio_Mid, w3$Ratio_Extr)
```

```
    Pearson's product-moment correlation

data:  w3$Ratio_Mid and w3$Ratio_Extr
t = -51.351, df = 8125, p-value < 2.2e-16
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.5112414 -0.4784087
sample estimates:
```

```
       cor
-0.4950017
```

12.0.1 w3 Categorization ExtrMi

```
thresholds_ExM <-
  w3 |>

  group_by(Interviewmode) |>
  summarise(
    thresh_Extr_Ratio = mean(Ratio_Extr, na.rm = TRUE) + sd(Ratio_Extr, na.rm = TRUE),
    thresh_Mid_Ratio = mean(Ratio_Mid, na.rm = TRUE) + sd(Ratio_Mid, na.rm = TRUE),

    percentile_Extr = mean(Ratio_Extr <= thresh_Extr_Ratio, na.rm = TRUE) * 100,
    percentile_Mid = mean(Ratio_Mid <= thresh_Mid_Ratio, na.rm = TRUE) * 100,

    prop_above_Extr = mean(Ratio_Extr > thresh_Extr_Ratio, na.rm = TRUE) * 100,
    prop_above_Mid = mean(Ratio_Mid > thresh_Mid_Ratio, na.rm = TRUE) * 100
  )

w3 <-
  w3 |>
  mutate(Extr =
           case_when(
             is.na(Ratio_Extr) ~ "Not applicable",
             Interviewmode == "PAPI" & Ratio_Extr > thresholds_ExM |> filter(Interviewmode ==
 ↪  "PAPI") |> pull(thresh_Extr_Ratio) ~ "High",
             Interviewmode == "CAWI" & Ratio_Extr > thresholds_ExM |> filter(Interviewmode ==
 ↪  "CAWI") |> pull(thresh_Extr_Ratio) ~ "High",
             TRUE ~ "Not high"),
         Extr = factor(Extr, levels = c("Not high", "High", "Not applicable")),
         Mid =
           case_when(
             is.na(Ratio_Mid) ~ "Not applicable",
             Interviewmode == "PAPI" & Ratio_Mid > thresholds_ExM |> filter(Interviewmode ==
 ↪  "PAPI") |> pull(thresh_Mid_Ratio) ~ "High",
             Interviewmode == "CAWI" & Ratio_Mid > thresholds_ExM |> filter(Interviewmode ==
 ↪  "CAWI") |> pull(thresh_Mid_Ratio) ~ "High",
             TRUE ~ "Not high"),

         Mid = factor(Mid, levels = c("Not high", "High", "Not applicable")))


w3 |> count(Interviewmode, Extr); w3 |> count(Interviewmode, Mid)
```

```
# A tibble: 4 x 3
  Interviewmode Extr         n
  <fct>         <fct>    <int>
1 PAPI          Not high  2157
2 PAPI          High       402
3 CAWI          Not high  4735
4 CAWI          High       833
```

```
# A tibble: 4 x 3
  Interviewmode Mid          n
  <fct>         <fct>    <int>
1 PAPI          Not high  2177
```

```
2 PAPI          High       382
3 CAWI          Not high  4774
4 CAWI          High       794
```

13 w3 ItemNonresponse

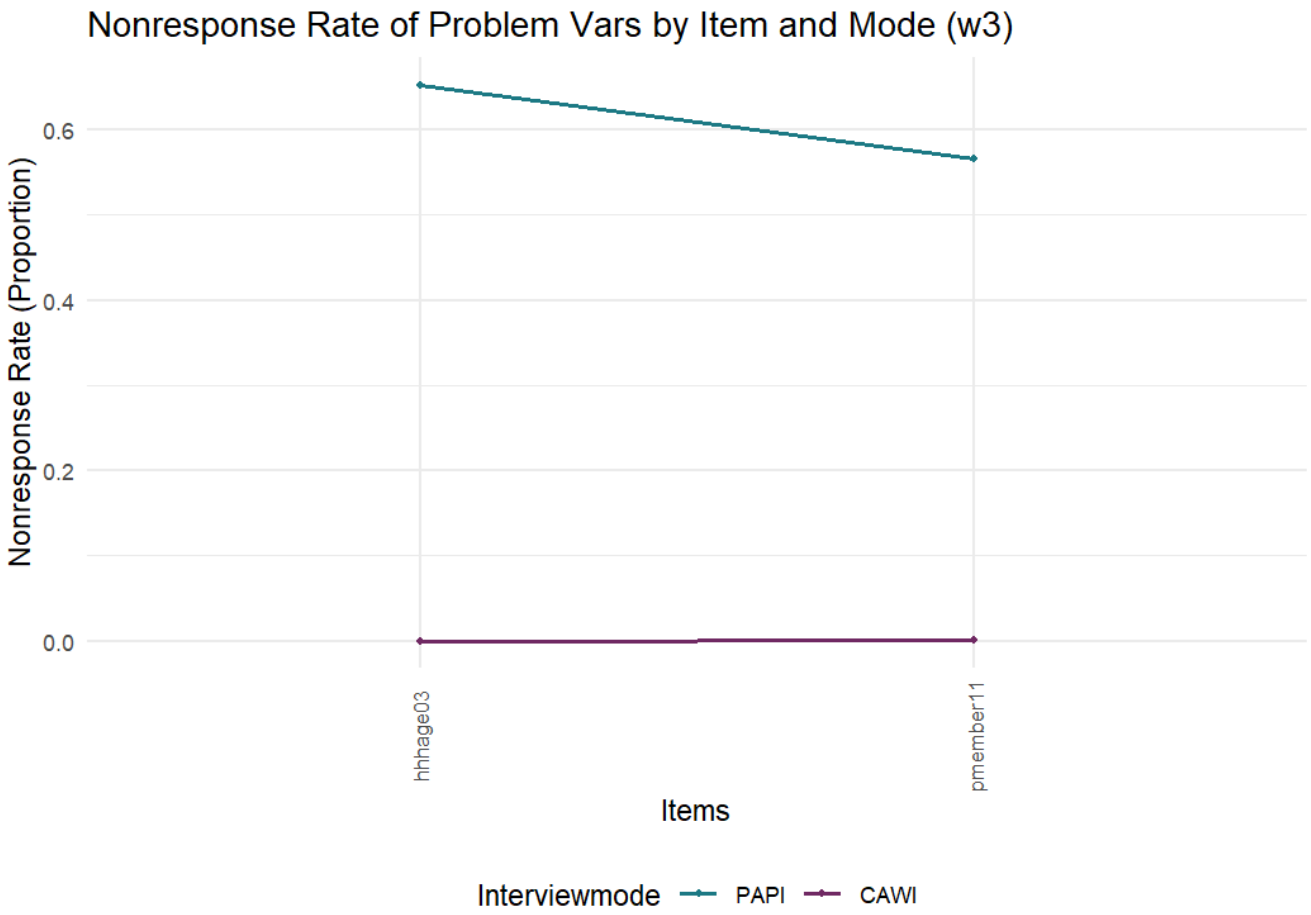
13.0.1 Problem Vars only

```
problem.vars <- c("pmember11", "hhhage03")



w3 |>
  select(pid, Interviewmode,
         all_of(problem.vars)) |> #names()
 # select(all_of(mis_items_w3), Interviewmode) |>
  group_by(Interviewmode) |>
  summarise(across(pmember11:hhhage03,
                   ~ sum(.x %in% c(-1, -9), na.rm = TRUE) / n(),
                   .names = "PropMiss_{.col}")) |>
  pivot_longer(-Interviewmode, names_to = c("dummy", "Variable"),
               names_pattern = "(PropMiss_)(.*)", values_to = "Prop") |>
  select(-dummy) |>
  mutate(Variable = factor(Variable, levels = rev(unique(Variable)))) |>

ggplot(aes(x = Variable, y = Prop, colour = Interviewmode, group = Interviewmode)) +
  geom_point(size = 1) +
  geom_line(linewidth = 0.8) +
  scale_colour_manual(values = mode_palette) +
  labs(
    x = "Items",
    y = "Nonresponse Rate (Proportion)",
    colour = "Interviewmode",
    title = "Nonresponse Rate of Problem Vars by Item and Mode (w3)"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(size = 8, angle = 90, vjust = 0.5),
    legend.position = "bottom"
  )
```




```
w3 |>
    select(pid, Interviewmode,
           all_of(problem.vars)) |>
  group_by(Interviewmode) |>
  summarise(across(pmember11:hhhage03,
                   ~ sum(.x %in% c(-1, -9), na.rm = TRUE) / n(),
                   .names = "PropMiss_{.col}"))
```

```
# A tibble: 2 x 3
  Interviewmode PropMiss_pmember11 PropMiss_hhhage03
  <fct>                      <dbl>             <dbl>
1 PAPI                     0.566               0.652
2 CAWI                     0.000718            0
```

```
# Absolute counts + N per group
counts <- w3 |>
  select(pid, Interviewmode, all_of(problem.vars)) |>
  group_by(Interviewmode) |>
  summarise(
    n = n(),
    across(all_of(problem.vars),
           ~ sum(.x %in% c(-1, -9), na.rm = TRUE),
           .names = "miss_{.col}")
  )

# Chi-square test per item
```

```
results <- map_dfr(problem.vars, function(var) {
  col <- paste0("miss_", var)

  miss  <- counts[[col]]        # c(n_miss_PAPI, n_miss_CAWI)
  n     <- counts$n             # c(n_PAPI, n_CAWI)
  notmiss <- n - miss

  mat <- matrix(c(miss, notmiss), nrow = 2,
                dimnames = list(c("PAPI", "CAWI"), c("Miss", "NotMiss")))

  # Fisher when expected cell counts < 5, otherwise chi-squared
  test <- if (any(chisq.test(mat)$expected < 5)) {
    fisher.test(mat)
  } else {
    chisq.test(mat, correct = FALSE)
  }

  tibble(
    Item       = var,
    p_value    = test$p.value,
    method     = test$method,
    prop_PAPI  = miss[1] / n[1],
    prop_CAWI  = miss[2] / n[2]
  )
}) |>
  mutate(
    p_adj      = p.adjust(p_value, method = "BH"),  # Benjamini-Hochberg
    significant = p_adj < 0.05
  ) |>
  arrange(p_adj)

print(results)
```

```
# A tibble: 2 x 7
  Item      p_value method                 prop_PAPI prop_CAWI p_adj significant
  <chr>       <dbl> <chr>                      <dbl>     <dbl> <dbl> <lgl>
1 pmember11       0 Pearson's Chi-squared~     0.566  0.000718     0 TRUE
2 hhhage03        0 Pearson's Chi-squared~     0.652  0            0 TRUE
```

```
# all significant => problem_vars remain unchanged
```

13.0.2 All other items

```
w3 |>
  select(pid, Interviewmode,
         pbsjo03:pdevice05,
       -(any_of(FreeTextItemsw3))) |>
  group_by(Interviewmode) |>
  summarise(across(pbsjo03:pdevice05,
                   ~ sum(.x %in% c(-1, -9), na.rm = TRUE) / n(),
                   .names = "PropMiss_{.col}")) |>
  pivot_longer(-Interviewmode, names_to = c("dummy", "Variable"),
               names_pattern = "(PropMiss_)(.*)", values_to = "Prop") |>
  select(-dummy) |>
  mutate(Variable = factor(Variable, levels = rev(unique(Variable)))) |>
```

```
ggplot(aes(x = Variable, y = Prop, colour = Interviewmode, group = Interviewmode)) +
  geom_point(size = 1) +
  geom_line(linewidth = 0.8) +
  scale_colour_manual(values = mode_palette) +
  labs(
    x = "Items",
    y = "Nonresponse Rate (Proportion)",
    colour = "Interviewmode",
    title = "Nonresponse Rate by Item and Mode (w3)"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(size = 3, angle = 90, vjust = 0.5),
    legend.position = "bottom"
  )
```

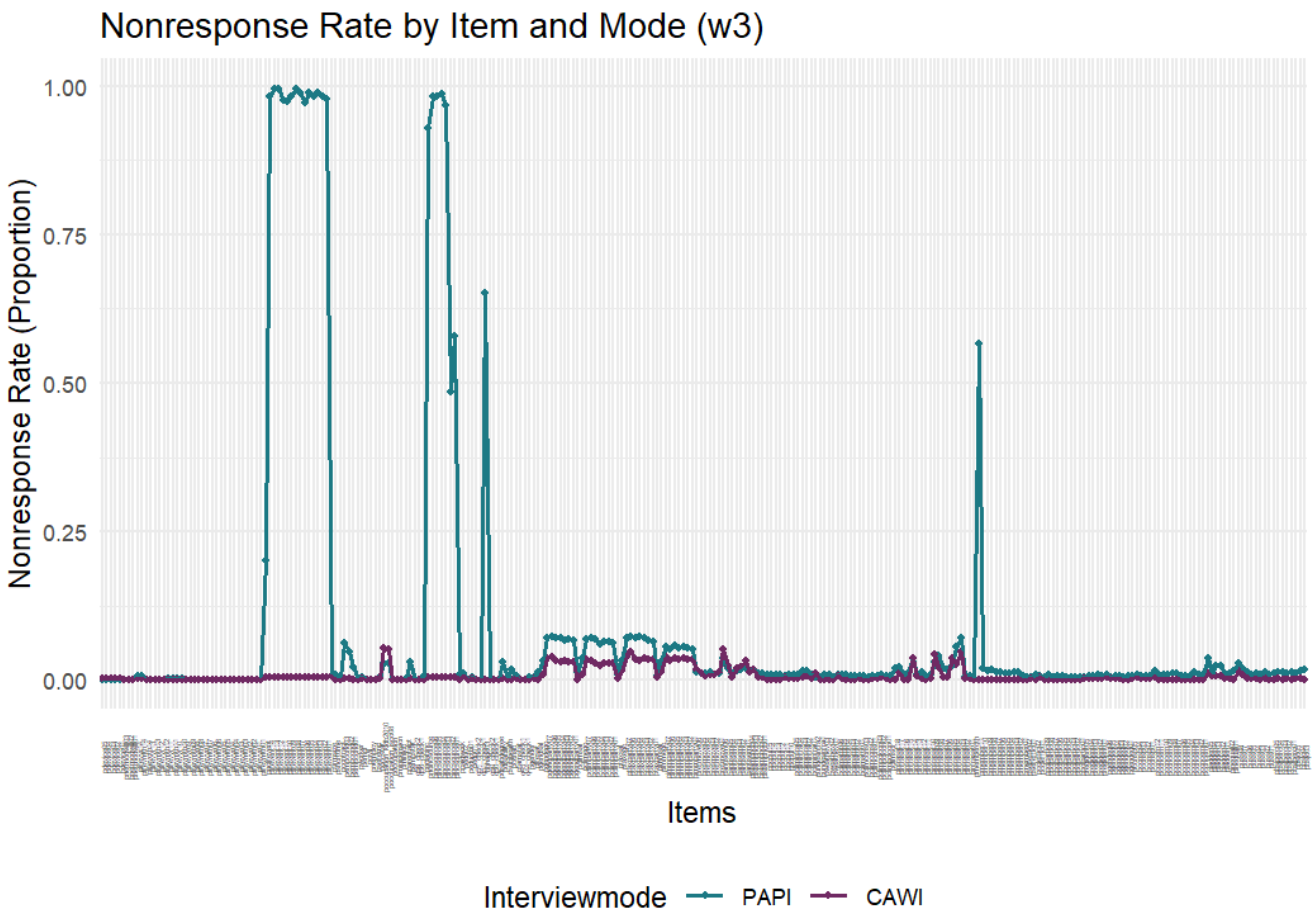


Figure 21: Nonresponse Items w3

```
mis_df_w3 <-
  w3 |>

  # if a number of persons <14yrs in household is stated, there'd be no reason to check the box
  mutate(
```

```
  hhhage03_mod = as.double(hhhage03),

   hhhage03_mod = case_when(


     #if number of children > 14 > 0, no need to check hhhage03
     hhhage02 > 0 ~ 0, #& hhhage03_mod < 0 ~ 0,
   #  hhhage02 == -2 ~ -2,

     TRUE ~ hhhage03_mod),



      pnoincsource = case_when(
    # if any of the pincsource items are selected, there'd be no reason to check the box
  (pincsource01 == 1 | pincsource02 == 1 | pincsource03 == 1 | pincsource04 == 1 |
↪  pincsource05 == 1 | pincsource06 == 1) & Interviewmode == "PAPI" ~ 0,
  TRUE ~ pnoincsource
))  |>

 # for all "pincsource"-items:
mutate(across(starts_with("pincsource"),

              ~ case_when(

                # if all items are missing and pnoincsource is also missing, treat it as
                 ↪  missing
                all(c_across(pincsource01:pincsource06) %in% c(-1, -9) & pnoincsource %in%
                 ↪  c(-1, -9)) ~ -1,
                # otherwise, leave everything as is
                TRUE ~ .
              )
             )) |>

mutate(
       # for the plifevent01 item in PAPI conditions
       pincsource01 = case_when(

         # if this item is labelled as missing, but other plifevent items are well filled in,
          ↪  let's treat it a "no" instead of a missing
         pincsource01 %in% c(-1, -9) &
           (pincsource02 == 1 | pincsource03 == 1 | pincsource04 == 1 | pincsource05 == 1 |
↪  pincsource06 == 1) & Interviewmode == "PAPI" ~ 0,
       TRUE ~ pincsource01),

       pincsource02 = case_when(

         pincsource02 %in% c(-1, -9) &
           (pincsource01 == 1 | pincsource03 == 1 | pincsource04 == 1 | pincsource05 == 1 |
↪  pincsource06 == 1) & Interviewmode == "PAPI" ~ 0,
       TRUE ~ pincsource02),

       pincsource03 = case_when(

         pincsource03 %in% c(-1, -9) &
           (pincsource02 == 1 | pincsource01 == 1 | pincsource04 == 1 | pincsource05 == 1 |
↪  pincsource06 == 1) & Interviewmode == "PAPI" ~ 0,
```

```
        TRUE ~ pincsource03),


        pincsource04 = case_when(

          pincsource04 %in% c(-1, -9) &
            (pincsource02 == 1 | pincsource03 == 1 | pincsource01 == 1 | pincsource05 == 1 |
↪  pincsource06 == 1) & Interviewmode == "PAPI" ~ 0,
        TRUE ~ pincsource04),


        pincsource05 = case_when(

          pincsource05 %in% c(-1, -9) &
            (pincsource02 == 1 | pincsource03 == 1 | pincsource04 == 1 | pincsource01 == 1 |
↪  pincsource06 == 1) & Interviewmode == "PAPI" ~ 0,
        TRUE ~ pincsource05),


        pincsource06 = case_when(

          pincsource06 %in% c(-1, -9) &
            (pincsource02 == 1 | pincsource03 == 1 | pincsource04 == 1 | pincsource05 == 1 |
↪  pincsource01 == 1) & Interviewmode == "PAPI" ~ 0,
        TRUE ~ pincsource06)) |>

 mutate(
   across(all_of(w3 |> select(plifevent01:plifevent14) |> colnames()),

          # if pnolifevent was selected,
          ~ ifelse(pnolifevent == 1, 0, .)),



    # for the plifevent01 item in PAPI conditions
        plifevent01 = case_when(

          # if this item is labelled as missing, but other plifevent items are well filled in,
           ↪  let's treat it a "no" instead of a missing
          plifevent01 %in% c(-1, -9) &
            (plifevent02 == 1 | plifevent03 == 1 | plifevent04 == 1 | plifevent05 == 1 |
               plifevent06 == 1 | plifevent07 == 1 | plifevent08 == 1 | plifevent09 == 1 |
↪  plifevent10 == 1 |
               plifevent11 == 1 | plifevent12 == 1 | plifevent13 == 1 | plifevent14 == 1 ) &
↪  Interviewmode == "PAPI" ~ 0,
        TRUE ~ plifevent01),


        plifevent02 = case_when(

          # if this item is labelled as missing, but other plifevent items are well filled in,
           ↪  let's treat it a "no" instead of a missing
          plifevent02 %in% c(-1, -9) &
            (plifevent01 == 1  | plifevent03 == 1 | plifevent04 == 1 | plifevent05 == 1 |
               plifevent06 == 1 | plifevent07 == 1 | plifevent08 == 1 | plifevent09 == 1 |
↪  plifevent10 == 1 |
```

```
              plifevent11 == 1 | plifevent12 == 1 | plifevent13 == 1 | plifevent14 == 1 ) &
↪  Interviewmode == "PAPI" ~ 0,
      TRUE ~ plifevent02),



      plifevent03 = case_when(

        # if this item is labelled as missing, but other plifevent items are well filled in,
         ↪  let's treat it a "no" instead of a missing
        plifevent03 %in% c(-1, -9) &
          (plifevent01 == 1 | plifevent02 == 1  | plifevent04 == 1 | plifevent05 == 1 |
             plifevent06 == 1 | plifevent07 == 1 | plifevent08 == 1 | plifevent09 == 1 |
↪  plifevent10 == 1 |
             plifevent11 == 1 | plifevent12 == 1 | plifevent13 == 1 | plifevent14 == 1 ) &
↪  Interviewmode == "PAPI" ~ 0,
      TRUE ~ plifevent03),

   plifevent04 = case_when(

        # if this item is labelled as missing, but other plifevent items are well filled in,
         ↪  let's treat it a "no" instead of a missing
        plifevent04 %in% c(-1, -9) &
          (plifevent01 == 1 | plifevent02 == 1 | plifevent03 == 1 |  plifevent05 == 1 |
             plifevent06 == 1 | plifevent07 == 1 | plifevent08 == 1 | plifevent09 == 1 |
↪  plifevent10 == 1 |
             plifevent11 == 1 | plifevent12 == 1 | plifevent13 == 1 | plifevent14 == 1 ) &
↪  Interviewmode == "PAPI" ~ 0,
      TRUE ~ plifevent04),

   plifevent05 = case_when(

        # if this item is labelled as missing, but other plifevent items are well filled in,
         ↪  let's treat it a "no" instead of a missing
        plifevent05 %in% c(-1, -9) &
          (plifevent01 == 1 | plifevent02 == 1 | plifevent03 == 1 | plifevent04 == 1 |
             plifevent06 == 1 | plifevent07 == 1 | plifevent08 == 1 | plifevent09 == 1 |
↪  plifevent10 == 1 |
             plifevent11 == 1 | plifevent12 == 1 | plifevent13 == 1 | plifevent14 == 1 ) &
↪  Interviewmode == "PAPI" ~ 0,
      TRUE ~ plifevent05),

   plifevent06 = case_when(

        # if this item is labelled as missing, but other plifevent items are well filled in,
         ↪  let's treat it a "no" instead of a missing
        plifevent06 %in% c(-1, -9) &
          (plifevent01 == 1 | plifevent02 == 1 | plifevent03 == 1 | plifevent04 == 1 |
↪  plifevent05 == 1 |
             plifevent07 == 1 | plifevent08 == 1 | plifevent09 == 1 | plifevent10 == 1 |
             plifevent11 == 1 | plifevent12 == 1 | plifevent13 == 1 | plifevent14 == 1 ) &
↪  Interviewmode == "PAPI" ~ 0,
      TRUE ~ plifevent06),

   plifevent07 = case_when(
```

```
        # if this item is labelled as missing, but other plifevent items are well filled in,
         ↪ let's treat it a "no" instead of a missing
        plifevent07 %in% c(-1, -9) &
          (plifevent01 == 1 | plifevent02 == 1 | plifevent03 == 1 | plifevent04 == 1 |
↪ plifevent05 == 1 |
             plifevent06 == 1 |  plifevent08 == 1 | plifevent09 == 1 | plifevent10 == 1 |
             plifevent11 == 1 | plifevent12 == 1 | plifevent13 == 1 | plifevent14 == 1 ) &
↪ Interviewmode == "PAPI" ~ 0,
      TRUE ~ plifevent07),

  plifevent08 = case_when(

        # if this item is labelled as missing, but other plifevent items are well filled in,
         ↪ let's treat it a "no" instead of a missing
        plifevent08 %in% c(-1, -9) &
          (plifevent01 == 1 | plifevent02 == 1 | plifevent03 == 1 | plifevent04 == 1 |
↪ plifevent05 == 1 |
             plifevent06 == 1 | plifevent07 == 1 |plifevent09 == 1 | plifevent10 == 1 |
             plifevent11 == 1 | plifevent12 == 1 | plifevent13 == 1 | plifevent14 == 1 ) &
↪ Interviewmode == "PAPI" ~ 0,
      TRUE ~ plifevent08),

  plifevent09 = case_when(

        # if this item is labelled as missing, but other plifevent items are well filled in,
         ↪ let's treat it a "no" instead of a missing
        plifevent09 %in% c(-1, -9) &
          (plifevent01 == 1 | plifevent02 == 1 | plifevent03 == 1 | plifevent04 == 1 |
↪ plifevent05 == 1 |
             plifevent06 == 1 | plifevent07 == 1 | plifevent08 == 1 |  plifevent10 == 1 |
             plifevent11 == 1 | plifevent12 == 1 | plifevent13 == 1 | plifevent14 == 1 ) &
↪ Interviewmode == "PAPI" ~ 0,
      TRUE ~ plifevent09),

  plifevent10 = case_when(

        # if this item is labelled as missing, but other plifevent items are well filled in,
         ↪ let's treat it a "no" instead of a missing
        plifevent10 %in% c(-1, -9) &
          (plifevent01 == 1 | plifevent02 == 1 | plifevent03 == 1 | plifevent04 == 1 |
↪ plifevent05 == 1 |
             plifevent06 == 1 | plifevent07 == 1 | plifevent08 == 1 | plifevent09 == 1 |
             plifevent11 == 1 | plifevent12 == 1 | plifevent13 == 1 | plifevent14 == 1 ) &
↪ Interviewmode == "PAPI" ~ 0,
      TRUE ~ plifevent10),

  plifevent11 = case_when(

        # if this item is labelled as missing, but other plifevent items are well filled in,
         ↪ let's treat it a "no" instead of a missing
        plifevent11 %in% c(-1, -9) &
          (plifevent01 == 1 | plifevent02 == 1 | plifevent03 == 1 | plifevent04 == 1 |
↪ plifevent05 == 1 |
             plifevent06 == 1 | plifevent07 == 1 | plifevent08 == 1 | plifevent09 == 1 |
↪ plifevent10 == 1 |
             plifevent12 == 1 | plifevent13 == 1 | plifevent14 == 1 ) & Interviewmode ==
↪ "PAPI" ~ 0,
```

```
        TRUE ~ plifevent11),

   plifevent12 = case_when(

          # if this item is labelled as missing, but other plifevent items are well filled in,
           ↪  let's treat it a "no" instead of a missing
          plifevent12 %in% c(-1, -9) &
            (plifevent01 == 1 | plifevent02 == 1 | plifevent03 == 1 | plifevent04 == 1 |
↪  plifevent05 == 1 |
               plifevent06 == 1 | plifevent07 == 1 | plifevent08 == 1 | plifevent09 == 1 |
↪  plifevent10 == 1 |
               plifevent11 == 1 | plifevent13 == 1 | plifevent14 == 1 ) & Interviewmode ==
↪  "PAPI" ~ 0,
        TRUE ~ plifevent12),

   plifevent13 = case_when(

          # if this item is labelled as missing, but other plifevent items are well filled in,
           ↪  let's treat it a "no" instead of a missing
          plifevent13 %in% c(-1, -9) &
            (plifevent01 == 1 | plifevent02 == 1 | plifevent03 == 1 | plifevent04 == 1 |
↪  plifevent05 == 1 |
               plifevent06 == 1 | plifevent07 == 1 | plifevent08 == 1 | plifevent09 == 1 |
↪  plifevent10 == 1 |
               plifevent11 == 1 | plifevent12 == 1 | plifevent14 == 1 ) & Interviewmode ==
↪  "PAPI" ~ 0,
        TRUE ~ plifevent13),

   plifevent14 = case_when(

          # if this item is labelled as missing, but other plifevent items are well filled in,
           ↪  let's treat it a "no" instead of a missing
          plifevent14 %in% c(-1, -9) &
            (plifevent01 == 1 | plifevent02 == 1 | plifevent03 == 1 | plifevent04 == 1 |
↪  plifevent05 == 1 |
               plifevent06 == 1 | plifevent07 == 1 | plifevent08 == 1 | plifevent09 == 1 |
↪  plifevent10 == 1 |
               plifevent11 == 1 | plifevent12 == 1 | plifevent13 == 1) & Interviewmode ==
↪  "PAPI" ~ 0,
        TRUE ~ plifevent14),

 pnolabinc_mod = as.double(pnolabinc),
   pnolabinc_mod =
     case_when(
       (plabincbrut > 0 | plabincnet > 0) ~ 0,

       plabincbrut  %in% c(-1, -9) &
         plabincnet %in%  c(-1, -9)  &
         !pnolabinc_mod %in% c(0, 1) ~ -1,


       TRUE ~ pnolabinc_mod
     )) |>


   select(c(pid, Interviewmode, pbsjo01:pnetmodwill02,
```

```
            pnolabinc_mod, hhhage03_mod),
          -hhhage03, -pnolabinc,
            -any_of(FreeTextItemsw3),

          # these items have extremely high missing rates, and it cannot be see what share of
           ↪  this is false coding and what share is nonresponse => items are excluded
          -pmember11, -hhhage03_mod)



w3 |> filter(pmember11 != -10) |>  group_by(Interviewmode) |> frq(pmember11)
```

```
Mitglied: Sonstiges (pmember11) <numeric>
# grouped by: PAPI
# total N=2559 valid N=2559 mean=1.02 sd=2.37

Value |                                                      Label |    N | Raw %
------------------------------------------------------------------------------
  -10 |                                              [-10] Abbruch |    0 |  0.00
   -9 |           [-9] Moechte ich nicht beantworten [nur CAWI] |    0 |  0.00
   -8 | [-8] Frage in diesem Jahr nicht Teil des Frageprogramms |    0 |  0.00
   -7 | [-7] Nur in weniger eingeschraenkter Edition verfuegbar |    0 |  0.00
   -6 |   [-6] Fragebogenversion mit geaenderter Filterfuehrung |    0 |  0.00
   -5 |               [-5] In Fragebogenversion nicht enthalten |    0 |  0.00
   -4 |                       [-4] Unzulaessige Mehrfachantwort |    0 |  0.00
   -3 |                                    [-3] Nicht valide |    0 |  0.00
   -2 |                                 [-2] Trifft nicht zu |    0 |  0.00
   -1 |                                   [-1] Keine Angabe | 1448 | 56.58
    1 |                     [1] Aktives Mitglied mit Ehrenamt |   76 |  2.97
    2 |                   [2] Aktives Mitglied ohne Ehrenamt |   63 |  2.46
    3 |                               [3] Inaktives Mitglied |   38 |  1.48
    4 |                                    [4] Kein Mitglied |  934 | 36.50
 <NA> |                                                 <NA> |    0 |  0.00

Value | Valid % | Cum. %
-----------------------
  -10 |    0.00 |   0.00
   -9 |    0.00 |   0.00
   -8 |    0.00 |   0.00
   -7 |    0.00 |   0.00
   -6 |    0.00 |   0.00
   -5 |    0.00 |   0.00
   -4 |    0.00 |   0.00
   -3 |    0.00 |   0.00
   -2 |    0.00 |   0.00
   -1 |   56.58 |  56.58
    1 |    2.97 |  59.55
    2 |    2.46 |  62.02
    3 |    1.48 |  63.50
    4 |   36.50 | 100.00
 <NA> |    <NA> |   <NA>

Mitglied: Sonstiges (pmember11) <numeric>
# grouped by: CAWI
# total N=5533 valid N=5533 mean=3.78 sd=0.71
```

Value	Label	N	Raw %
-10	[-10] Abbruch	0	0.00
-9	[-9] Moechte ich nicht beantworten [nur CAWI]	2	0.04
-8	[-8] Frage in diesem Jahr nicht Teil des Frageprogramms	0	0.00
-7	[-7] Nur in weniger eingeschraenkter Edition verfuegbar	0	0.00
-6	[-6] Fragebogenversion mit geaenderter Filterfuehrung	0	0.00
-5	[-5] In Fragebogenversion nicht enthalten	0	0.00
-4	[-4] Unzulaessige Mehrfachantwort	0	0.00
-3	[-3] Nicht valide	0	0.00
-2	[-2] Trifft nicht zu	0	0.00
-1	[-1] Keine Angabe	2	0.04
1	[1] Aktives Mitglied mit Ehrenamt	178	3.22
2	[2] Aktives Mitglied ohne Ehrenamt	217	3.92
3	[3] Inaktives Mitglied	213	3.85
4	[4] Kein Mitglied	4921	88.94
<NA>	<NA>	0	0.00

Value	Valid %	Cum. %
-10	0.00	0.00
-9	0.04	0.04
-8	0.00	0.04
-7	0.00	0.04
-6	0.00	0.04
-5	0.00	0.04
-4	0.00	0.04
-3	0.00	0.04
-2	0.00	0.04
-1	0.04	0.07
1	3.22	3.29
2	3.92	7.21
3	3.85	11.06
4	88.94	100.00
<NA>	<NA>	<NA>

```
w3 |> filter(hhhage03 != -10) |>  group_by(Interviewmode) |> frq(hhhage03)
```

```
Keine HH-Mitglieder<14 J (hhhage03) <numeric>
# grouped by: PAPI
# total N=2559 valid N=2559 mean=-0.31 sd=0.96
```

Value	Label	N	Raw %
-10	[-10] Abbruch	0	0.00
-9	[-9] Moechte ich nicht beantworten [nur CAWI]	0	0.00
-8	[-8] Frage in diesem Jahr nicht Teil des Frageprogramms	0	0.00
-7	[-7] Nur in weniger eingeschraenkter Edition verfuegbar	0	0.00
-6	[-6] Fragebogenversion mit geaenderter Filterfuehrung	0	0.00
-5	[-5] In Fragebogenversion nicht enthalten	0	0.00
-4	[-4] Unzulaessige Mehrfachantwort	0	0.00
-3	[-3] Nicht valide	3	0.12
-2	[-2] Trifft nicht zu	0	0.00
-1	[-1] Keine Angabe	1668	65.18
0	[0] Nein	0	0.00
1	[1] Ja	888	34.70
<NA>	<NA>	0	0.00

```
Value | Valid % | Cum. %
-----------------------
  -10 |    0.00 |   0.00
   -9 |    0.00 |   0.00
   -8 |    0.00 |   0.00
   -7 |    0.00 |   0.00
   -6 |    0.00 |   0.00
   -5 |    0.00 |   0.00
   -4 |    0.00 |   0.00
   -3 |    0.12 |   0.12
   -2 |    0.00 |   0.12
   -1 |   65.18 |  65.30
    0 |    0.00 |  65.30
    1 |   34.70 | 100.00
 <NA> |    <NA> |   <NA>
```

```
Keine HH-Mitglieder<14 J (hhhage03) <numeric>
# grouped by: CAWI
# total N=5489 valid N=5489 mean=-0.13 sd=1.45
```

```
Value |                                                                        Label |    N | Raw %
-------------------------------------------------------------------------------------------------
  -10 |                                                                [-10] Abbruch |    0 |  0.00
   -9 |                           [-9] Moechte ich nicht beantworten [nur CAWI] |    0 |  0.00
   -8 | [-8] Frage in diesem Jahr nicht Teil des Frageprogramms |    0 |  0.00
   -7 | [-7] Nur in weniger eingeschraenkter Edition verfuegbar |    0 |  0.00
   -6 |   [-6] Fragebogenversion mit geaenderter Filterfuehrung |    0 |  0.00
   -5 |                [-5] In Fragebogenversion nicht enthalten |    0 |  0.00
   -4 |                        [-4] Unzulaessige Mehrfachantwort |    0 |  0.00
   -3 |                                             [-3] Nicht valide |    0 |  0.00
   -2 |                                          [-2] Trifft nicht zu | 2063 | 37.58
   -1 |                                             [-1] Keine Angabe |    0 |  0.00
    0 |                                                      [0] Nein |    0 |  0.00
    1 |                                                        [1] Ja | 3426 | 62.42
 <NA> |                                                          <NA> |    0 |  0.00
```

```
Value | Valid % | Cum. %
-----------------------
  -10 |    0.00 |   0.00
   -9 |    0.00 |   0.00
   -8 |    0.00 |   0.00
   -7 |    0.00 |   0.00
   -6 |    0.00 |   0.00
   -5 |    0.00 |   0.00
   -4 |    0.00 |   0.00
   -3 |    0.00 |   0.00
   -2 |   37.58 |  37.58
   -1 |    0.00 |  37.58
    0 |    0.00 |  37.58
    1 |   62.42 | 100.00
 <NA> |    <NA> |   <NA>
```

```r
# item columns 3:261
mis_items_w3 <-
  mis_df_w3 |>
  select(
```

```
    c(3:261)) |> colnames()

# number of respondents per mode (denominators for the missing-rate check below)
n_cawi <- mis_df_w3 |> filter(Interviewmode == "CAWI") |> nrow()
n_papi <- mis_df_w3 |> filter(Interviewmode == "PAPI") |> nrow()

# missing counts per item and mode (denominators: mode-specific N)
manymissings_CAWI <- mis_df_w3 |> filter(Interviewmode == "CAWI") |>
  summarise_all(~sum(.  %in% c(-1, -9))) |>
  pivot_longer(
    everything(), names_to = "Variable", values_to = "Numb.missings") |>
  rowwise() |>
  mutate(Prop.missings = Numb.missings/n_cawi) |>
  ungroup() |>
  arrange(desc(Prop.missings))
manymissings_CAWI
```

```
# A tibble: 270 x 3
   Variable             Numb.missings Prop.missings
   <chr>                        <int>         <dbl>
 1 pnolabinc_mod                  331        0.0594
 2 poccupation_kldb2010           296        0.0532
 3 pantisem06                     290        0.0521
 4 poccupation_isco08             282        0.0506
 5 pinfrasat03                    268        0.0481
 6 ptaxparty06                    262        0.0471
 7 pinfrasat09                    245        0.0440
 8 pimmparty07                    216        0.0388
 9 ptaxparty07                    215        0.0386
10 pgendparty06                   210        0.0377
# i 260 more rows
```

```
manymissings_PAPI <- mis_df_w3 |> filter(Interviewmode == "PAPI") |>
  summarise_all(~sum(.  %in% c(-1, -9))) |>
  pivot_longer(
    everything(), names_to = "Variable", values_to = "Numb.missings") |>
  rowwise() |>
  mutate(Prop.missings = Numb.missings/n_papi) |>
  ungroup() |>
  arrange(desc(Prop.missings))
manymissings_PAPI
```

```
# A tibble: 270 x 3
   Variable     Numb.missings Prop.missings
   <chr>                <int>         <dbl>
 1 pnolifevent            515         0.201
 2 ptaxparty06            189         0.0739
 3 pgendparty06           186         0.0727
 4 ptaxparty04            185         0.0723
 5 ptaxparty07            183         0.0715
 6 pgendparty07           183         0.0715
 7 pincsource01           183         0.0715
 8 pincsource02           183         0.0715
 9 pincsource03           183         0.0715
10 pincsource04           183         0.0715
# i 260 more rows
```

```
mis_df_w3 <-
  mis_df_w3 |>
  rowwise() |>
  mutate(
  Nonresponses = sum(across(all_of(mis_items_w3)) %in% c(-1, -9), na.rm = TRUE),
  N_allNonrespOpportunities = sum(across(all_of(mis_items_w3), ~!. %in% c(-2,  -10)), na.rm =
   ↪  TRUE),

  NonrespRatio = Nonresponses/N_allNonrespOpportunities) |>
  ungroup()
```

```
w3 <-
  mis_df_w3 |> select(pid, NonrespRatio) |>
  right_join(w3, by = "pid")
```

```
quantile(w3$NonrespRatio, prob = c(0.33, 0.66))
```

```
        33%         66%
0.000000000 0.004504505
```

14 w3 Nondifferentiation

See Figure 22

```
pbsjo.items <- w3 |> select(pbsjo01:pbsjo05) |> colnames()
pbsjperc.items <- w3 |> select(pbsjperc01:pbsjperc05) |> colnames()
puss.items <- w3 |> select(puss01:puss08) |> colnames()
pladder.items <- w3 |> select(pladder01:pladder05) |> colnames()
pconcern.items <- w3 |> select(pconcern01:pconcern12) |> colnames()
pconsp.items <- w3 |> select(pconsp01:pconsp05) |> colnames()
pgendeq.items <- w3 |> select(pgendeq01:pgendeq06) |> colnames()
pregident.items <- w3 |> select(pregident01:pregident05) |> colnames()
pinfrasat.items <- w3 |> select(pinfrasat01:pinfrasat18) |> colnames()
pclimattit.items <- w3 |> select(pclimattit01:pclimattit06) |> colnames()
pcliminfra.items <- w3 |> select(pcliminfra01:pcliminfra05) |> colnames()
ppopul.items <- w3 |> select(ppopul10:ppopul15) |> colnames()
pantisem.items <- w3 |> select(pantisem01:pantisem06) |> colnames()
prefmove.items <- w3 |> select(prefmove01:prefmove06) |> colnames()
pimmparty.items <- w3 |> select(pimmparty01:pimmparty07) |> colnames()
ptaxparty.items <- w3 |> select(ptaxparty01:ptaxparty07) |> colnames()
pclimparty.items <- w3 |> select(pclimparty01:pclimparty07) |> colnames()
pgendparty.items <- w3 |> select(pgendparty01:pgendparty07) |> colnames()
```

```
items_list_w3 <- list(pbsjo.items, pbsjperc.items, puss.items,
                      pladder.items, pconcern.items, pconsp.items,
                      pgendeq.items, pregident.items, pinfrasat.items,
                      pclimattit.items, pcliminfra.items, ppopul.items,
```

```
                  pantisem.items, prefmove.items, pimmparty.items,
                  ptaxparty.items, pclimparty.items, pgendparty.items)



tND <- Sys.time()



# extract all items from items_list
all_items_w3 <- unlist(items_list_w3)

w3_papi <-
  w3 |>
  filter(Interviewmode == "PAPI") |>
  select(pid, all_of(all_items_w3)) |>  # select only the item columns
  meanRootPairs(items_list_w3) |>
  ungroup()
```

```
Duration for pbsjo : 0.08217342 minutes
Duration for pbsjperc : 0.08914947 minutes
Duration for puss : 0.09969347 minutes
Duration for pladder : 0.08714117 minutes
Duration for pconcern : 0.1282858 minutes
Duration for pconsp : 0.0882391 minutes
Duration for pgendeq : 0.0872236 minutes
Duration for pregident : 0.08750345 minutes
Duration for pinfrasat : 0.1378351 minutes
Duration for pclimattit : 0.0887127 minutes
Duration for pcliminfra : 0.09051327 minutes
Duration for ppopul : 0.09131827 minutes
Duration for pantisem : 0.09170783 minutes
Duration for prefmove : 0.09559954 minutes
Duration for pimmparty : 0.09450278 minutes
Duration for ptaxparty : 0.09660045 minutes
Duration for pclimparty : 0.09540532 minutes
Duration for pgendparty : 0.09745805 minutes
All done!
Total duration for all item packs: 1.729483 minutes.
```

```
w3_cawi  <-
  w3 |>
  filter(Interviewmode == "CAWI") |>
  select(pid, all_of(all_items_w3)) |>
   meanRootPairs(items_list_w3) |> ungroup()
```

```
Duration for pbsjo : 0.1706389 minutes
Duration for pbsjperc : 0.1819197 minutes
Duration for puss : 0.2070617 minutes
Duration for pladder : 0.184294 minutes
Duration for pconcern : 0.2409568 minutes
Duration for pconsp : 0.1905254 minutes
Duration for pgendeq : 0.2018189 minutes
Duration for pregident : 0.1891375 minutes
Duration for pinfrasat : 0.3074862 minutes
Duration for pclimattit : 0.2015421 minutes
```

```
Duration for pcliminfra : 0.1914052 minutes
Duration for ppopul : 0.2034849 minutes
Duration for pantisem : 0.2094935 minutes
Duration for prefmove : 0.2083178 minutes
Duration for pimmparty : 0.2174188 minutes
Duration for ptaxparty : 0.2130556 minutes
Duration for pclimparty : 0.2063323 minutes
Duration for pgendparty : 0.2066614 minutes
All done!
Total duration for all item packs: 3.7318 minutes.
```

```r
# merge PAPI and CAWI
w3_meanroot <-
  bind_rows(w3_papi, w3_cawi)

# join onto the original w3 dataset
w3 <- w3 |>
  left_join(w3_meanroot |>
              select(pid, starts_with("temp_index"), starts_with("temp_max"),
                ↪  starts_with("MeanRootPairs_")), by = "pid")


#10.75397 minutes

Sys.time() - tND
```

```
Time difference of 5.461875 mins
```

- 18 item packages

```r
chart.Correlation(w3 |> select(contains("MeanRootPairs")), histogram = TRUE)
```

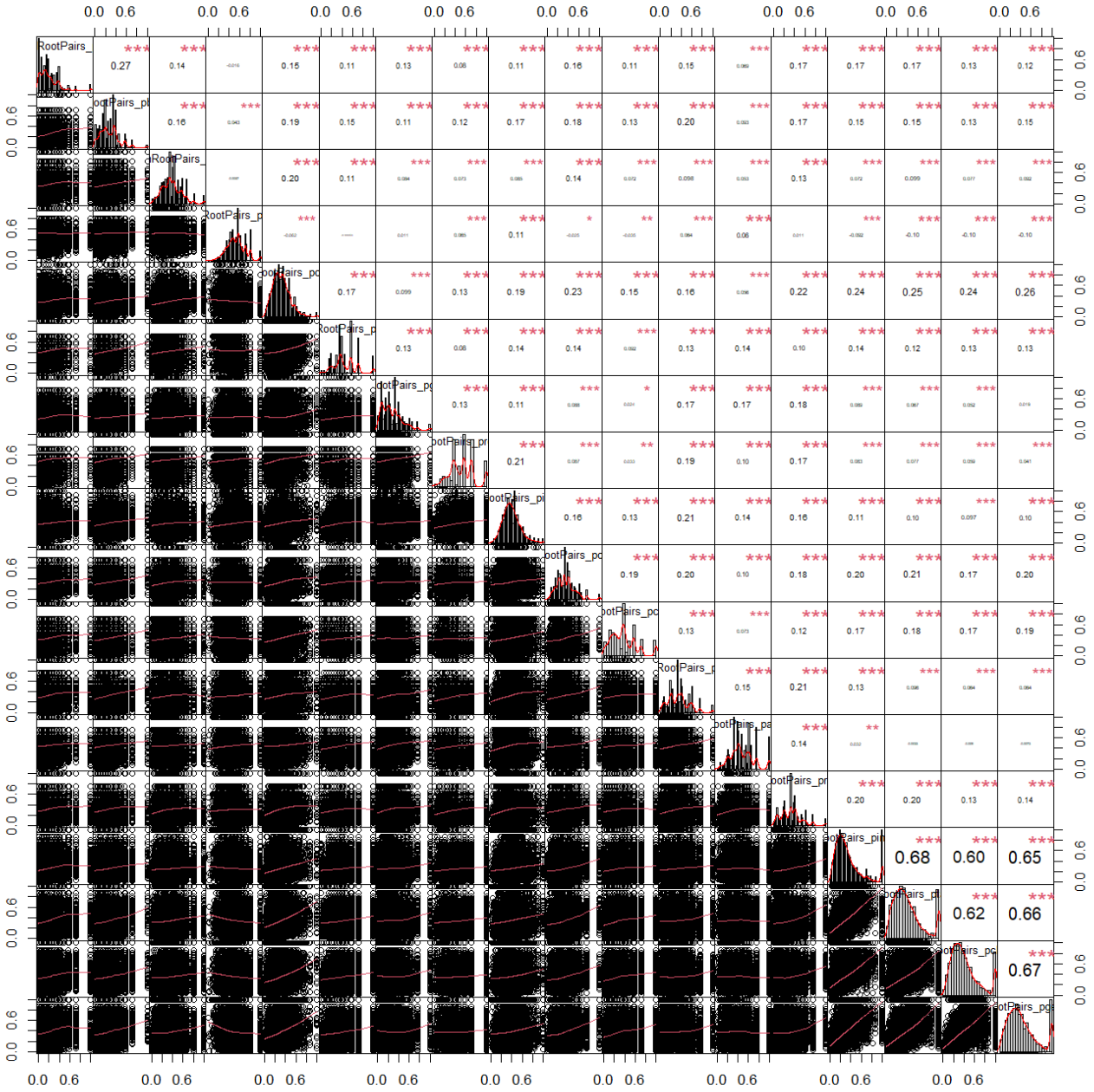


Figure 22: Correlations of Nondifferentiation Items

14.0.1 w3 Nd SEM using CT-C(M-1) Model

```
# correlated traits: WK (weak), ST (strong), Selfexperienced (SE)
# correlated methods-1: Reference-Scale: 11-likert
w3LatND_mod <- "


WK =~
MeanRootPairs_puss + MeanRootPairs_pconcern + MeanRootPairs_pconsp + MeanRootPairs_pgendeq +
 ↪  MeanRootPairs_pregident + MeanRootPairs_pinfrasat + MeanRootPairs_pcliminfra +
 ↪  MeanRootPairs_ppopul
```

```
ST =~
 MeanRootPairs_pbsjo + MeanRootPairs_pbsjperc + MeanRootPairs_pladder +
 ↪  MeanRootPairs_pclimattit + MeanRootPairs_pantisem + MeanRootPairs_prefmove +
 ↪  MeanRootPairs_pimmparty + MeanRootPairs_ptaxparty + MeanRootPairs_pclimparty +
 ↪  MeanRootPairs_pgendparty


# nondiff for scales with fewer than 11 Likert points
u11 =~
MeanRootPairs_pbsjo + MeanRootPairs_pbsjperc + MeanRootPairs_puss + MeanRootPairs_pconsp +
 ↪  MeanRootPairs_pgendeq + MeanRootPairs_pregident + MeanRootPairs_pinfrasat +
 ↪  MeanRootPairs_pclimattit + MeanRootPairs_pcliminfra + MeanRootPairs_ppopul +
 ↪  MeanRootPairs_pantisem + MeanRootPairs_prefmove




WK ~~ 0 * u11
ST ~~ 0 * u11"

# simple model without groups
w3LatND_fit <- sem(w3LatND_mod, w3, meanstructure = TRUE)
summary(w3LatND_fit, fit = TRUE)
```

```
lavaan 0.6-21 ended normally after 231 iterations

  Estimator                                         ML
  Optimization method                           NLMINB
  Number of model parameters                        67

                                                  Used       Total
  Number of observations                          6307        8127

Model Test User Model:

  Test statistic                              1498.995
  Degrees of freedom                               122
  P-value (Chi-square)                           0.000

Model Test Baseline Model:

  Test statistic                             20981.364
  Degrees of freedom                               153
  P-value                                        0.000

User Model versus Baseline Model:

  Comparative Fit Index (CFI)                    0.934
  Tucker-Lewis Index (TLI)                       0.917

Loglikelihood and Information Criteria:

  Loglikelihood user model (H0)              28731.917
  Loglikelihood unrestricted model (H1)      29481.415
```

```
  Akaike (AIC)                                -57329.835
  Bayesian (BIC)                              -56877.624
  Sample-size adjusted Bayesian (SABIC)       -57090.532

Root Mean Square Error of Approximation:

  RMSEA                                           0.042
  90 Percent confidence interval - lower          0.040
  90 Percent confidence interval - upper          0.044
  P-value H_0: RMSEA <= 0.050                     1.000
  P-value H_0: RMSEA >= 0.080                     0.000

Standardized Root Mean Square Residual:

  SRMR                                            0.035

Parameter Estimates:

  Standard errors                             Standard
  Information                                 Expected
  Information saturated (h1) model          Structured

Latent Variables:
                   Estimate  Std.Err  z-value  P(>|z|)
  WK =~
    MeanRotPrs_pss    1.000
    MnRtPrs_pcncrn    2.238    0.208   10.769    0.000
    MenRtPrs_pcnsp    1.375    0.148    9.323    0.000
    MenRtPrs_pgndq    0.476    0.092    5.180    0.000
    MnRtPrs_prgdnt    0.745    0.119    6.244    0.000
    MnRtPrs_pnfrst    0.910    0.102    8.957    0.000
    MnRtPrs_pclmnf    1.850    0.187    9.878    0.000
    MeanRtPrs_pppl    0.850    0.109    7.792    0.000
  ST =~
    MeanRtPrs_pbsj    1.000
    MnRtPrs_pbsjpr    0.861    0.078   11.058    0.000
    MenRtPrs_plddr   -0.554    0.072   -7.732    0.000
    MnRtPrs_pclmtt    1.243    0.098   12.642    0.000
    MenRtPrs_pntsm   -0.005    0.080   -0.068    0.946
    MenRtPrs_prfmv    1.164    0.098   11.824    0.000
    MnRtPrs_pmmprt    5.160    0.354   14.596    0.000
    MnRtPrs_ptxprt    5.974    0.409   14.623    0.000
    MnRtPrs_pclmpr    5.317    0.365   14.566    0.000
    MnRtPrs_pgndpr    6.093    0.417   14.618    0.000
  u11 =~
    MeanRtPrs_pbsj    1.000
    MnRtPrs_pbsjpr    1.061    0.061   17.393    0.000
    MeanRotPrs_pss    0.588    0.052   11.330    0.000
    MenRtPrs_pcnsp    0.752    0.062   12.229    0.000
    MenRtPrs_pgndq    0.939    0.061   15.311    0.000
    MnRtPrs_prgdnt    1.050    0.073   14.457    0.000
    MnRtPrs_pnfrst    0.849    0.054   15.849    0.000
    MnRtPrs_pclmtt    0.858    0.055   15.638    0.000
    MnRtPrs_pclmnf    0.564    0.063    8.890    0.000
    MeanRtPrs_pppl    1.367    0.077   17.853    0.000
```

```
    MenRtPrs_pntsm    1.040    0.069   15.021    0.000
    MenRtPrs_prfmv    1.212    0.070   17.216    0.000

Covariances:
                   Estimate  Std.Err  z-value  P(>|z|)
  WK ~~
    u11               0.000
  ST ~~
    u11               0.000
  WK ~~
    ST                0.001    0.000    8.491    0.000

Intercepts:
                   Estimate  Std.Err  z-value  P(>|z|)
   .MeanRotPrs_pss    0.385    0.002  163.551    0.000
   .MnRtPrs_pcncrn    0.316    0.002  155.229    0.000
   .MenRtPrs_pcnsp    0.450    0.003  164.770    0.000
   .MenRtPrs_pgndq    0.264    0.002  110.787    0.000
   .MnRtPrs_prgdnt    0.525    0.003  178.481    0.000
   .MnRtPrs_pnfrst    0.392    0.002  191.595    0.000
   .MnRtPrs_pclmnf    0.377    0.003  123.221    0.000
   .MeanRtPrs_pppl    0.342    0.003  135.845    0.000
   .MeanRtPrs_pbsj    0.202    0.002   85.092    0.000
   .MnRtPrs_pbsjpr    0.266    0.002  124.422    0.000
   .MenRtPrs_plddr    0.527    0.002  249.764    0.000
   .MnRtPrs_pclmtt    0.345    0.002  157.407    0.000
   .MenRtPrs_pntsm    0.473    0.003  172.428    0.000
   .MenRtPrs_prfmv    0.332    0.003  132.114    0.000
   .MnRtPrs_pmmprt    0.321    0.003  108.285    0.000
   .MnRtPrs_ptxprt    0.397    0.003  119.889    0.000
   .MnRtPrs_pclmpr    0.404    0.003  127.687    0.000
   .MnRtPrs_pgndpr    0.421    0.003  123.582    0.000

Variances:
                   Estimate  Std.Err  z-value  P(>|z|)
   .MeanRotPrs_pss    0.032    0.001   53.311    0.000
   .MnRtPrs_pcncrn    0.019    0.001   33.819    0.000
   .MenRtPrs_pcnsp    0.042    0.001   52.259    0.000
   .MenRtPrs_pgndq    0.031    0.001   51.715    0.000
   .MnRtPrs_prgdnt    0.049    0.001   52.518    0.000
   .MnRtPrs_pnfrst    0.022    0.000   50.198    0.000
   .MnRtPrs_pclmnf    0.053    0.001   51.695    0.000
   .MeanRtPrs_pppl    0.031    0.001   46.382    0.000
   .MeanRtPrs_pbsj    0.030    0.001   50.572    0.000
   .MnRtPrs_pbsjpr    0.023    0.000   47.911    0.000
   .MenRtPrs_plddr    0.028    0.000   56.050    0.000
   .MnRtPrs_pclmtt    0.025    0.000   51.127    0.000
   .MenRtPrs_pntsm    0.043    0.001   51.841    0.000
   .MenRtPrs_prfmv    0.031    0.001   48.379    0.000
   .MnRtPrs_pmmprt    0.020    0.000   42.941    0.000
   .MnRtPrs_ptxprt    0.022    0.001   39.811    0.000
   .MnRtPrs_pclmpr    0.026    0.001   45.161    0.000
   .MnRtPrs_pgndpr    0.024    0.001   40.477    0.000
    WK                0.001    0.000    5.916    0.000
    ST                0.001    0.000    7.323    0.000
    u11               0.004    0.000   11.267    0.000
```

```
fscores <- lavPredict(w3LatND_fit)  # matrix, not a list
idx     <- lavInspect(w3LatND_fit, "case.idx")  # single vector

# merge the scores back in, following the lavaan documentation for single-group models:
for (fs in colnames(fscores)) {
  w3[idx, fs] <- fscores[, fs]
}
```

```
# model used as the basis for the coefficient table
SEM_w3 <- table_results(w3LatND_fit)|>
    mutate(label = label |>
                str_replace("MeanRootPairs_", "Mrp.") |>
                str_replace("=~", ".BY.") |>
                str_replace("~~", ".WITH.") |>
                str_replace("~1", "") |>
                str_replace("^\\.", "Means.")) |>
  filter(!str_starts(label, "Variances"),
         !str_starts(label, "Means")) |>
  select(label, est_sig, se, confint)


SEM_w3|>
  kable(
    format = "latex",
    booktabs = TRUE,
    col.names = c("Label",
                  "Est.", "SE", "95\\% CI"),
    escape = FALSE
  )
```

Label	Est.	SE	95% CI
WK.BY.Mrp.puss	1.00	0.00	[1.00, 1.00]
WK.BY.Mrp.pconcern	2.24***	0.21	[1.83, 2.65]
WK.BY.Mrp.pconsp	1.38***	0.15	[1.09, 1.66]
WK.BY.Mrp.pgendeq	0.48***	0.09	[0.30, 0.66]
WK.BY.Mrp.pregident	0.74***	0.12	[0.51, 0.98]
WK.BY.Mrp.pinfrasat	0.91***	0.10	[0.71, 1.11]
WK.BY.Mrp.pcliminfra	1.85***	0.19	[1.48, 2.22]
WK.BY.Mrp.ppopul	0.85***	0.11	[0.64, 1.06]
ST.BY.Mrp.pbsjo	1.00	0.00	[1.00, 1.00]
ST.BY.Mrp.pbsjperc	0.86***	0.08	[0.71, 1.01]
ST.BY.Mrp.pladder	-0.55***	0.07	[-0.69, -0.41]
ST.BY.Mrp.pclimattit	1.24***	0.10	[1.05, 1.44]
ST.BY.Mrp.pantisem	-0.01	0.08	[-0.16, 0.15]
ST.BY.Mrp.prefmove	1.16***	0.10	[0.97, 1.36]
ST.BY.Mrp.pimmparty	5.16***	0.35	[4.47, 5.85]
ST.BY.Mrp.ptaxparty	5.97***	0.41	[5.17, 6.77]
ST.BY.Mrp.pclimparty	5.32***	0.36	[4.60, 6.03]
ST.BY.Mrp.pgendparty	6.09***	0.42	[5.28, 6.91]
u11.BY.Mrp.pbsjo	1.00	0.00	[1.00, 1.00]
u11.BY.Mrp.pbsjperc	1.06***	0.06	[0.94, 1.18]
u11.BY.Mrp.puss	0.59***	0.05	[0.49, 0.69]
u11.BY.Mrp.pconsp	0.75***	0.06	[0.63, 0.87]
u11.BY.Mrp.pgendeq	0.94***	0.06	[0.82, 1.06]
u11.BY.Mrp.pregident	1.05***	0.07	[0.91, 1.19]
u11.BY.Mrp.pinfrasat	0.85***	0.05	[0.74, 0.95]
u11.BY.Mrp.pclimattit	0.86***	0.05	[0.75, 0.97]
u11.BY.Mrp.pcliminfra	0.56***	0.06	[0.44, 0.69]
u11.BY.Mrp.ppopul	1.37***	0.08	[1.22, 1.52]
u11.BY.Mrp.pantisem	1.04***	0.07	[0.90, 1.18]
u11.BY.Mrp.prefmove	1.21***	0.07	[1.07, 1.35]
WK.WITH.u11	0.00	0.00	[0.00, 0.00]
ST.WITH.u11	0.00	0.00	[0.00, 0.00]
WK.WITH.ST	0.00***	0.00	[0.00, 0.00]

```
MRP_w3 <-
 w3 %>%
 select(pid,
         "Q1: pbsjo\n5 items, 5-likert, multi" = MeanRootPairs_pbsjo,
        "Q2: pbsjperc\n5 items, 5-likert, multi" = MeanRootPairs_pbsjperc,
        "Q3: puss\n08 items, 5-likert, uni" = MeanRootPairs_puss,
        "Q5: pladder\n5 items, 11-likert, multi" = MeanRootPairs_pladder,
        "Q10: pconcern\n12 items, 11-likert, uni" = MeanRootPairs_pconcern,
        "Q11: pconsp\n5 items, 5-likert, uni" = MeanRootPairs_pconsp,
        "Q12: pgendeq\n6 items, 5-likert, uni" = MeanRootPairs_pgendeq,
        "Q13: pregident\n5 items, 5-likert, uni" = MeanRootPairs_pregident,
        "Q18: pinfrasat\n18 items, 5-likert, uni" = MeanRootPairs_pinfrasat,
        "Q22: pclimattit\n6 items, 5-likert, multi" = MeanRootPairs_pclimattit,
        "Q26: pcliminfra\n5 items, 5-likert, uni" = MeanRootPairs_pcliminfra,
        "Q27: ppopul\n5 items, 5-likert, uni" = MeanRootPairs_ppopul,
        "Q29: pantisem\n6 items, 5-likert, multi" = MeanRootPairs_pantisem,
        "Q30: prefmove\n6 items, 5-likert, multi" = MeanRootPairs_prefmove,
        "Q31: pimmparty\n7 items, 11-likert, multi" = MeanRootPairs_pimmparty,
        "Q34: ptaxparty\n7 items, 11-likert, multi" = MeanRootPairs_ptaxparty,
```

```
        "Q37: pclimparty\n7 items, 11-likert, multi" = MeanRootPairs_pclimparty,
        "Q40: pgendparty\n7 items, 11-likert, multi" = MeanRootPairs_pgendparty) %>%

  # long format
  pivot_longer(cols = 2:19) %>%


  ggplot(aes(name, value)) +

  geom_violin(scale = "count", linewidth = 0.8) +

  geom_boxplot(width = 0.15, color = "black") +


   labs(caption = "Nondifferentiation (Mean Root of Pairs) in Wave 3",
        x = "",
        y = "Mean root of pairs") +

  papaja::theme_apa() +

  theme(axis.text.x = element_text(angle = 90),
        axis.text.y = element_text(size = 10),
        #axis.ticks.x = element_blank(),
        axis.title.y = element_text(size = 14),
        axis.line.x = element_blank(),
        panel.grid.major.y = element_line(color = "#404040", linewidth = 0.02),
        panel.grid.minor.y = element_line(color = "#404040", linewidth = 0.02),
        plot.caption = element_text(size = 14))



MRP_w3
```

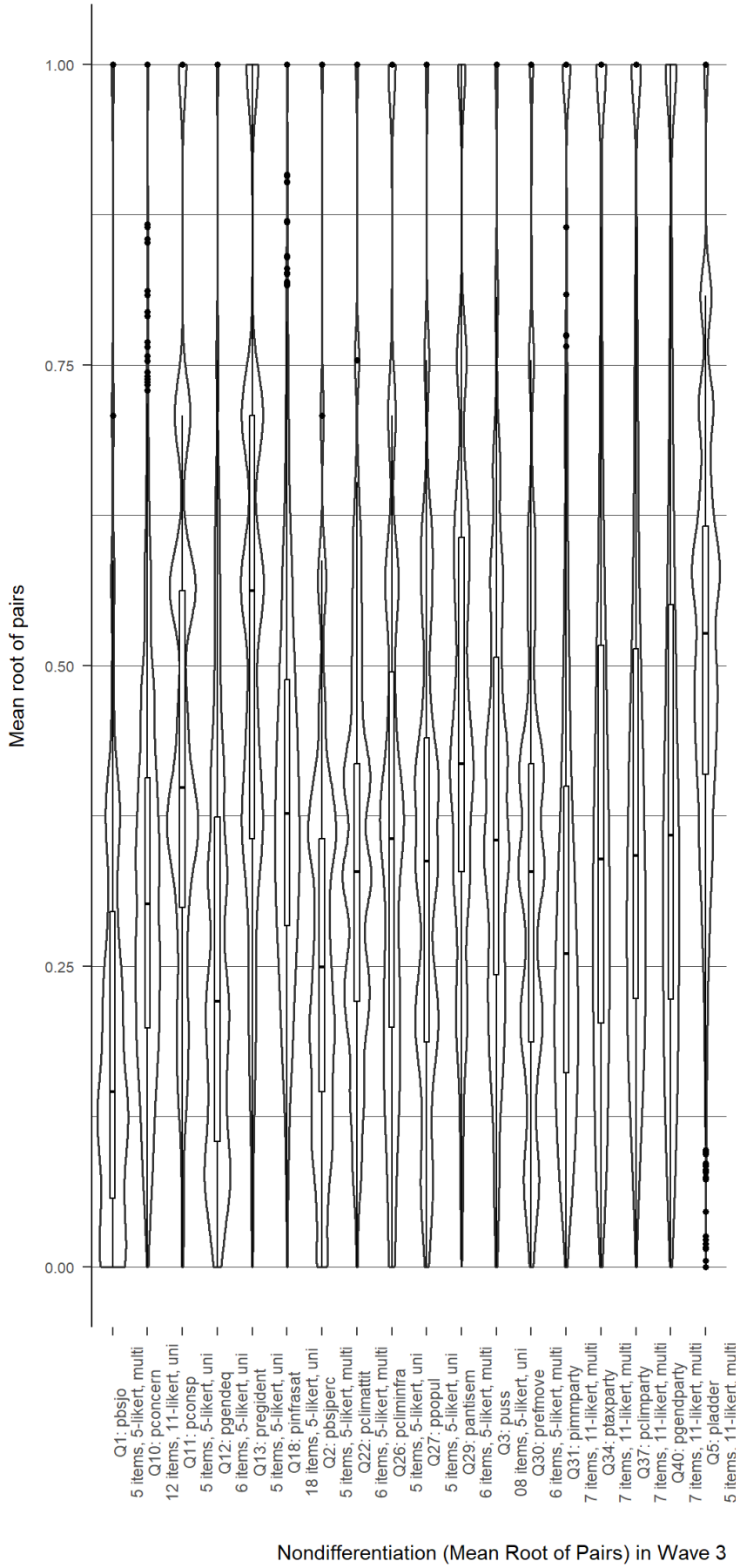


Figure 23: Wave 3 Nondifferentiation

15 Join w1, w2, w3

```
w1t_sensan <-
  w1 |>
  select(lfd, pid, hid, Education, Employment, Gender, ChildHH, Mig, Interviewmode, BreakOff,
   ↪  ppanelcons, NonrespRatio_sensan = NonrespRatio, TargetGroup,
         # "&" intersects the two selections: keep sensan columns that
         # match any of the listed patterns
         contains("sensan") &c(
         contains("Age_cat"), contains("Extr_"), contains("Mid_"), contains("EquivIncome_S"),
          ↪  contains("Speeding_"), contains("Interviewmode"), contains("TargetGr"),
          ↪  contains("FrTMis_"), contains("MeanRootPairs"))
  ) |>
  rename_with(.fn = function(.x){paste0(.x,"_w1")},
              .cols = -c(lfd, pid, hid)) #|>

w1t_Sample <-
  w1 |>
  filter(Sample == "Sample") |>
  select(pid,
         contains("Sample") &c(
         contains("Age_cat"), contains("Extr_"), contains("Mid_"), contains("EquivIncome_S"),
          ↪  contains("Speeding_"), contains("Interviewmode"), contains("FrTMis_"),
          ↪  contains("MeanRootPairs")),
#         starts_with("WK_"), starts_with("ST_")),
         NonrespRatio_Sample = NonrespRatio
  )|>
  rename_with(.fn = function(.x){paste0(.x,"_w1")},
              .cols = -c(pid)) |>
  mutate(Sample = "Sample")




w2t <-
  w2 |>

  select(lfd,  Interviewmode, BreakOff, Extr, Mid, Speeding, FrTMis, NonrespRatio,
   ↪  contains("MeanRootPairs")) |>
        #WK, ST, ,
   mutate(Participation = "Participated") |>
   rename_with(.fn = function(.x){paste0(.x,"_w2")},
              .cols = -c(lfd))

w3t <-
  w3 |>
  select(pid,  Interviewmode, BreakOff, Extr, Mid, Speeding, FrTMis, NonrespRatio,
   ↪  contains("MeanRootPairs")) |>    #WK, ST, ,
    mutate(Participation = "Participated") |>
   rename_with(.fn = function(.x){paste0(.x,"_w3")},
              .cols = -c(pid))

waves <-
  w1t_sensan |>
  left_join(w1t_Sample, by = "pid") |>
```

```
  left_join(w2t, by = "lfd") |>
  left_join(w3t, by = "pid") |>

  mutate(
    Participation_w2 = case_when(

      is.na(Participation_w2) ~ "Not participated",
      TRUE ~ Participation_w2),
    Participation_w2 = factor(Participation_w2, levels = c("Participated", "Not
     ↪  participated")),


    Participation_w3 = case_when(
      is.na(Participation_w3) ~ "Not participated",
      TRUE ~ Participation_w3),
    Participation_w3 = factor(Participation_w3, levels = c("Participated", "Not participated"))
  )

waves |> names()
```

```
 [1] "lfd"                                "pid"
 [3] "hid"                                "Education_w1"
 [5] "Employment_w1"                      "Gender_w1"
 [7] "ChildHH_w1"                         "Mig_w1"
 [9] "Interviewmode_w1"                   "BreakOff_w1"
[11] "ppanelcons_w1"                      "NonrespRatio_sensan_w1"
[13] "TargetGroup_w1"                     "Age_cat_sensan_w1"
[15] "EquivIncome_sensan_w1"              "ProportionSpeeding_sensan_w1"
[17] "Speeding_sensan_w1"                 "FrTMis_sensan_w1"
[19] "Extr_sensan_w1"                     "Mid_sensan_w1"
[21] "MeanRootPairs_pcohes_sensan_w1"     "MeanRootPairs_pinsttrust_sensan_w1"
[23] "MeanRootPairs_pfamexp_sensan_w1"    "MeanRootPairs_pworkexp_sensan_w1"
[25] "MeanRootPairs_pneighexp_sensan_w1"  "MeanRootPairs_punivtrad_sensan_w1"
[27] "MeanRootPairs_pgrpsymp_sensan_w1"   "MeanRootPairs_pgrpknow_sensan_w1"
[29] "MeanRootPairs_pfrspcons_sensan_w1"  "MeanRootPairs_ppopul_sensan_w1"
[31] "Extr_Sample_w1"                     "Mid_Sample_w1"
[33] "EquivIncome_Sample_w1"              "Age_cat_Sample_w1"
[35] "ProportionSpeeding_Sample_w1"       "Speeding_Sample_w1"
[37] "FrTMis_Sample_w1"                   "MeanRootPairs_pcohes_Sample_w1"
[39] "MeanRootPairs_pinsttrust_Sample_w1" "MeanRootPairs_pfamexp_Sample_w1"
[41] "MeanRootPairs_pworkexp_Sample_w1"   "MeanRootPairs_pneighexp_Sample_w1"
[43] "MeanRootPairs_punivtrad_Sample_w1"  "MeanRootPairs_pgrpsymp_Sample_w1"
[45] "MeanRootPairs_pgrpknow_Sample_w1"   "MeanRootPairs_pfrspcons_Sample_w1"
[47] "MeanRootPairs_ppopul_Sample_w1"     "NonrespRatio_Sample_w1"
[49] "Sample"                             "Interviewmode_w2"
[51] "BreakOff_w2"                        "Extr_w2"
[53] "Mid_w2"                             "Speeding_w2"
[55] "FrTMis_w2"                          "NonrespRatio_w2"
[57] "MeanRootPairs_plifesat_w2"          "MeanRootPairs_pinform_w2"
[59] "MeanRootPairs_pglob_w2"             "MeanRootPairs_ppertrust_w2"
[61] "MeanRootPairs_pedugoal_w2"          "MeanRootPairs_pfathbeh_w2"
[63] "MeanRootPairs_pmothbeh_w2"          "MeanRootPairs_psocdes_w2"
[65] "MeanRootPairs_pschwartz_w2"         "MeanRootPairs_pnarcism_w2"
[67] "MeanRootPairs_pdiscrim_w2"          "MeanRootPairs_pgender_w2"
[69] "MeanRootPairs_pauthor_w2"           "MeanRootPairs_pactivity_w2"
[71] "Participation_w2"                   "Interviewmode_w3"
```

```
[73] "BreakOff_w3"                    "Extr_w3"
[75] "Mid_w3"                         "Speeding_w3"
[77] "FrTMis_w3"                      "NonrespRatio_w3"
[79] "MeanRootPairs_pbsjo_w3"         "MeanRootPairs_pbsjperc_w3"
[81] "MeanRootPairs_puss_w3"          "MeanRootPairs_pladder_w3"
[83] "MeanRootPairs_pconcern_w3"      "MeanRootPairs_pconsp_w3"
[85] "MeanRootPairs_pgendeq_w3"       "MeanRootPairs_pregident_w3"
[87] "MeanRootPairs_pinfrasat_w3"     "MeanRootPairs_pclimattit_w3"
[89] "MeanRootPairs_pcliminfra_w3"    "MeanRootPairs_ppopul_w3"
[91] "MeanRootPairs_pantisem_w3"      "MeanRootPairs_prefmove_w3"
[93] "MeanRootPairs_pimmparty_w3"     "MeanRootPairs_ptaxparty_w3"
[95] "MeanRootPairs_pclimparty_w3"    "MeanRootPairs_pgendparty_w3"
[97] "Participation_w3"
```

16 Categorize Nondiff

```
nondiff_ratio <- function(data, item_cols, mode_var,
                          subgroup_var = NULL, subgroup_val = NULL) {

  ref_data <- if (!is.null(subgroup_var)) {
    data |> filter(.data[[subgroup_var]] == subgroup_val)
  } else data

  p66_by_mode <- ref_data |>
    group_by(.data[[mode_var]]) |>
    summarise(across(all_of(item_cols),
                     ~ quantile(.x, 0.66, na.rm = TRUE),
                     .names = "p66_{.col}"),
              .groups = "drop")

  p66_cols <- paste0("p66_", item_cols)

  # Explicit filter before pipeline - avoids '.' placeholder issue
  working_data <- if (!is.null(subgroup_var)) {
    data |> filter(.data[[subgroup_var]] == subgroup_val)
  } else data

  working_data |>
    left_join(p66_by_mode, by = mode_var) |>
    rowwise() |>
    mutate(
      n_high  = sum(c_across(all_of(item_cols)) > c_across(all_of(p66_cols)), na.rm = TRUE),
      n_valid = sum(!is.na(c_across(all_of(item_cols)))),
      ratio   = case_when(          # <- these three lines replace if_else(...)
            n_valid == 0 ~ NA_real_,
            n_high  == 0 ~ 0,
            TRUE         ~ n_high / n_valid
        )
    ) |>
    ungroup() |>
    mutate(nd_cat = factor(case_when(
      is.na(ratio)  ~ NA_character_,  # <- also catch NA here
      ratio == 0    ~ "Never",
      ratio <  0.5  ~ "Sometimes (0-50%)",
      ratio >= 0.5  ~ "Often (>=50%)"
```

```
    ), levels = c("Never", "Sometimes (0-50%)", "Often (>=50%)"))) |>
    select(pid, nd_cat, ratio)
}
# resolve the item vectors to character in advance
nd_w1Str_Sample_raw  <- waves |> select(contains("_Sample") & contains("_w1") &
  (contains("pcohes") | contains("exp") | contains("univtrad") | contains("pgrpknow"))) |>
 ↪  names()

nd_w1Wk_Sample_raw   <- waves |> select(contains("_Sample") & contains("_w1") &
  (contains("insttrust") | contains("pgrpsymp") | contains("pfrspcons") | contains("ppopul")))
 ↪  |> names()

nd_w1Str_sensan_raw <- waves |> select(contains("_sensan") & contains("_w1") &
  (contains("pcohes") | contains("exp") | contains("univtrad") | contains("pgrpknow"))) |>
 ↪  names()

nd_w1Wk_sensan_raw  <- waves |> select(contains("_sensan") & contains("_w1") &
  (contains("insttrust") | contains("pgrpsymp") | contains("pfrspcons") | contains("ppopul")))
 ↪  |> names()

nd_w2Wk_raw <- waves |> select(contains("MeanRoot") & (
  contains("plifesat") | contains("pinform") | contains("pglob") | contains("pedugoal") |
   ↪  contains("pnarcism") | contains("pdiscrim") | contains("pgender") | contains("pauthor") |
   ↪  contains("pactivity"))) |> names()

nd_w2St_raw <- waves |> select(contains("MeanRoot") & (contains("ppertrust") |
 ↪  contains("pfathbeh") | contains("pmothbeh") | contains("psocdes") | contains("pschwartz")))
 ↪  |> names()

nd_w3Wk_raw <- waves |> select(contains("MeanRoot") & (contains("puss_") | contains("pconcern")
 ↪  | contains("pconsp") | contains("pgendeq") | contains("pregident") | contains("pinfrasat")
 ↪  | contains("pcliminfra") | contains("ppopul"))) |> names()

nd_w3St_raw <- waves |> select(contains("MeanRoot") & (contains("pbsjo") | contains("pbsjperc")
 ↪  | contains("pladder") | contains("pclimattit") | contains("pantisem") |
 ↪  contains("prefmove") | contains("pimmparty") | contains("ptaxparty") |
 ↪  contains("pclimparty") | contains("pgendparty")))  |> names()


# function calls with character vectors
waves <- waves |>
  left_join(
    nondiff_ratio(
      waves |> filter(!is.na(Interviewmode_w1)),
      nd_w1Str_Sample_raw, "Interviewmode_w1") |>
      rename(NondiffSt66_Sample_w1 = nd_cat, Ratio_NDStr_w1_Sample = ratio),
    by = "pid") |>
  left_join(
    nondiff_ratio(
      waves |> filter(!is.na(Interviewmode_w1)),
      nd_w1Wk_Sample_raw, "Interviewmode_w1") |>
      rename(NondiffWk66_Sample_w1 = nd_cat, Ratio_NDWk_w1_Sample = ratio),
    by = "pid") |>
  # sensan and the remaining waves as before - no filtering needed
  left_join(
    nondiff_ratio(
```

```
      waves |> filter(!is.na(Interviewmode_w1)),
      nd_w1Str_sensan_raw, "Interviewmode_w1") |>
      rename(NondiffSt66_sensan_w1 = nd_cat, Ratio_NDStr_w1_sensan = ratio),
    by = "pid") |>
  left_join(
    nondiff_ratio(
      waves |> filter(!is.na(Interviewmode_w1)),
      nd_w1Wk_sensan_raw, "Interviewmode_w1") |>
      rename(NondiffWk66_sensan_w1 = nd_cat, Ratio_NDWk_w1_sensan = ratio),
    by = "pid") |>
  left_join(
    nondiff_ratio(waves |> filter(Participation_w2 == "Participated"),
                  nd_w2St_raw, "Interviewmode_w2") |>
      rename(NondiffSt66_w2 = nd_cat, Ratio_NDStr_w2 = ratio),
    by = "pid") |>
  left_join(
    nondiff_ratio(waves |> filter(Participation_w2 == "Participated"),
                  nd_w2Wk_raw, "Interviewmode_w2") |>
      rename(NondiffWk66_w2 = nd_cat, Ratio_NDWk_w2 = ratio),
    by = "pid") |>
  left_join(
    nondiff_ratio(waves |> filter(Participation_w3 == "Participated"),
                  nd_w3St_raw, "Interviewmode_w3") |>
      rename(NondiffSt66_w3 = nd_cat, Ratio_NDStr_w3 = ratio),
    by = "pid") |>
  left_join(
    nondiff_ratio(waves |> filter(Participation_w3 == "Participated"),
                  nd_w3Wk_raw, "Interviewmode_w3") |>
      rename(NondiffWk66_w3 = nd_cat, Ratio_NDWk_w3 = ratio),
    by = "pid")


waves <-
  waves |>
  # Recode non-participants to "Not participated"
  mutate(
    NondiffSt66_w2 = factor(case_when(
      Participation_w2 == "Not participated" ~ "Not participated",
      TRUE ~ as.character(NondiffSt66_w2)), levels = c("Never","Sometimes (0-50%)","Often
      ↪  (>=50%)","Not participated")),
    NondiffWk66_w2 = factor(case_when(
      Participation_w2 == "Not participated" ~ "Not participated",
      TRUE ~ as.character(NondiffWk66_w2)), levels = c("Never","Sometimes (0-50%)","Often
      ↪  (>=50%)","Not participated")),
    NondiffSt66_w3 = factor(case_when(
      Participation_w3 == "Not participated" ~ "Not participated",
      TRUE ~ as.character(NondiffSt66_w3)), levels = c("Never","Sometimes (0-50%)","Often
      ↪  (>=50%)","Not participated")),
    NondiffWk66_w3 = factor(case_when(
      Participation_w3 == "Not participated" ~ "Not participated",
      TRUE ~ as.character(NondiffWk66_w3)), levels = c("Never","Sometimes (0-50%)","Often
      ↪  (>=50%)","Not participated"))
  )


waves |> count(Interviewmode_w2,NondiffSt66_w2)
```

```
# A tibble: 9 x 3
  Interviewmode_w2 NondiffSt66_w2        n
  <fct>            <fct>             <int>
1 PAPI             Never              1229
2 PAPI             Sometimes (0-50%)  2308
3 PAPI             Often (>=50%)      1560
4 PAPI             <NA>                  8
5 CAWI             Never               884
6 CAWI             Sometimes (0-50%)  1898
7 CAWI             Often (>=50%)       990
8 CAWI             <NA>                 23
9 <NA>             Not participated   8123
```

```
waves |> filter(Sample == "Sample") |>filter(is.na(NondiffSt66_Sample_w1)) |>
  select(!contains("w2")& !contains("w3")&!contains("sensan")) |> #select(-c(pid, hid, lfd)) |>
    ↪
  ai()
```

```
Rows: 6
Columns: 32
$ Education_w1                      <fct> Non-tertiary (ISCED 0-4), Tertiary ~
$ Employment_w1                     <fct> "Retirement", "Retirement", "Not cu~
$ Gender_w1                         <fct> Female, Female, Female, Female, Fem~
$ ChildHH_w1                        <fct> No child in hh, No child in hh, mis~
$ Mig_w1                            <fct> No, No, Yes, No, No, No
$ Interviewmode_w1                  <fct> PAPI, PAPI, PAPI, PAPI, PAPI, PAPI
$ BreakOff_w1                       <fct> PAPI, PAPI, PAPI, PAPI, PAPI, PAPI
$ ppanelcons_w1                     <dbl+lbl> 1, 1, 1, 1, 1, 1
$ TargetGroup_w1                    <fct> Anchor, Anchor, Anchor, Anchor, Anc~
$ Extr_Sample_w1                    <fct> Not high, High, Not high, High, Not~
$ Mid_Sample_w1                     <fct> Not high, Not high, High, Not high,~
$ EquivIncome_Sample_w1             <fct> EquivIncome_T1min, EquivIncome_~
$ Age_cat_Sample_w1                 <chr> "Age_T3[60.0-97.0]", "Age_T3[60.0-9~
$ ProportionSpeeding_Sample_w1      <dbl> NA, NA, NA, NA, NA, NA
$ Speeding_Sample_w1                <fct> NA, NA, NA, NA, NA, NA
$ FrTMis_Sample_w1                  <fct> Not high, Not high, High, Not high,~
$ MeanRootPairs_pcohes_Sample_w1    <dbl> NA, NA, NA, NA, NA, NA
$ MeanRootPairs_pinsttrust_Sample_w1 <dbl> NA, NA, 1, NA, NA, NA
$ MeanRootPairs_pfamexp_Sample_w1   <dbl> NA, NA, NA, NA, NA, NA
$ MeanRootPairs_pworkexp_Sample_w1  <dbl> NA, NA, NA, NA, NA, NA
$ MeanRootPairs_pneighexp_Sample_w1 <dbl> NA, NA, NA, NA, NA, NA
$ MeanRootPairs_punivtrad_Sample_w1 <dbl> NA, NA, NA, NA, NA, NA
$ MeanRootPairs_pgrpsymp_Sample_w1  <dbl> NA, NA, NA, NA, NA, NA
$ MeanRootPairs_pgrpknow_Sample_w1  <dbl> NA, NA, NA, NA, NA, NA
$ MeanRootPairs_pfrspcons_Sample_w1 <dbl> NA, NA, NA, NA, NA, NA
$ MeanRootPairs_ppopul_Sample_w1    <dbl> NA, 0.3309709, NA, NA, NA, NA
$ NonrespRatio_Sample_w1            <dbl> 0.6769231, 0.4384615, 0.5384615, 0.~
$ Sample                            <chr> "Sample", "Sample", "Sample", "Samp~
$ NondiffSt66_Sample_w1             <fct> NA, NA, NA, NA, NA, NA
$ Ratio_NDStr_w1_Sample             <dbl> NA, NA, NA, NA, NA, NA
$ NondiffWk66_Sample_w1             <fct> NA, Never, Often (>=50%), NA, NA, NA
$ Ratio_NDWk_w1_Sample              <dbl> NA, 0, 1, NA, NA, NA
```

17 INR66

```
# w1 sensan = all cases, unfiltered by sample
p66_INR_w1_sensan <- waves |>
  group_by(Interviewmode_w1) |>
  summarise(p66_sensan = quantile(NonrespRatio_sensan_w1, 0.66, na.rm = TRUE), .groups =
   ↪  "drop")

# w1 Sample = Sample subgroup only
p66_INR_w1_Sample <- waves |>
  group_by(Interviewmode_w1) |>
  summarise(p66_Sample = quantile(NonrespRatio_Sample_w1, 0.66, na.rm = TRUE), .groups =
   ↪  "drop")

# w2 Sample
p66_INR_w2 <- waves |>
  filter(Participation_w2 == "Participated") |>
  group_by(Interviewmode_w2) |>
  summarise(p66_w2 = quantile(NonrespRatio_w2, 0.66, na.rm = TRUE), .groups = "drop")

# w3 Sample
p66_INR_w3 <- waves |>
  filter(Participation_w3 == "Participated") |>
  group_by(Interviewmode_w3) |>
  summarise(p66_w3 = quantile(NonrespRatio_w3, 0.66, na.rm = TRUE), .groups = "drop")

print(p66_INR_w1_sensan)
```

```
# A tibble: 2 x 2
  Interviewmode_w1 p66_sensan
  <fct>                 <dbl>
1 PAPI                 0.0150
2 CAWI                 0
```

```
print(p66_INR_w1_Sample)
```

```
# A tibble: 2 x 2
  Interviewmode_w1 p66_Sample
  <fct>                  <dbl>
1 PAPI                 0.00775
2 CAWI                 0
```

```
print(p66_INR_w2)
```

```
# A tibble: 2 x 2
  Interviewmode_w2 p66_w2
  <fct>              <dbl>
1 PAPI             0.0161
2 CAWI             0
```

```
print(p66_INR_w3)
```

```
# A tibble: 2 x 2
  Interviewmode_w3 p66_w3
  <fct>              <dbl>
1 PAPI             0.0129
2 CAWI             0
```

```
waves <-
  waves |>
  left_join(p66_INR_w1_sensan, by = "Interviewmode_w1") |>
  left_join(p66_INR_w1_Sample,   by = "Interviewmode_w1") |>
  left_join(p66_INR_w2, by = "Interviewmode_w2") |>
  left_join(p66_INR_w3, by = "Interviewmode_w3") |>
  mutate(
    INR66_sensan_w1 = factor(case_when(
      is.na(NonrespRatio_sensan_w1)     ~ NA_character_,
      NonrespRatio_sensan_w1 > p66_sensan ~ "High",
      TRUE                          ~ "Not high"),
      levels = c("Not high", "High")),

    INR66_Sample_w1 = factor(case_when(
      is.na(NonrespRatio_Sample_w1) ~ NA_character_, #| Sample_w1 != "Sample" ~ NA_character_,
      NonrespRatio_Sample_w1 > p66_Sample ~ "High",
      TRUE                            ~ "Not high"),
      levels = c("Not high", "High")),

    INR66_w2 = factor(case_when(
      Participation_w2 == "Not participated" ~ "Not participated",
      is.na(NonrespRatio_w2)                 ~ NA_character_,
      NonrespRatio_w2 > p66_w2             ~ "High",
      TRUE                                   ~ "Not high"),
      levels = c("Not high", "High", "Not participated")),

    INR66_w3 = factor(case_when(
      Participation_w3 == "Not participated"   ~ "Not participated",
      is.na(NonrespRatio_w3)                   ~ NA_character_,
      NonrespRatio_w3 > p66_w3                 ~ "High",
      TRUE                                     ~ "Not high"),
      levels = c("Not high", "High", "Not participated"))
  ) |>
  select(-p66_sensan, -p66_Sample, -p66_w2, -p66_w3)

waves |> select(starts_with("INR66")) |> summary()
```

```
 INR66_sensan_w1  INR66_Sample_w1             INR66_w2
 Not high:12776   Not high:7245   Not high        :6048
 High    : 4247   High    :2202   High            :2846
                  NA's    :7576   Not participated:8123
                                  NA's            :   6
            INR66_w3
 Not high        :5854
 High            :2273
 Not participated:8896
```

```
waves |> filter(is.na(NondiffSt66_w2)) |>
   select(!contains("w1") & !contains("w3")) |> ai()
```

```
Rows: 6
Columns: 28
$ Sample                    <chr> "Sample", "Sample", "Sample", "Sample", "Sa~
$ Interviewmode_w2          <fct> CAWI, CAWI, CAWI, PAPI, CAWI, CAWI
$ BreakOff_w2               <fct> Break-off, Break-off, Break-off, PAPI, Brea~
```

```
$ Extr_w2                   <fct> Not high, Not high, Not high, High, Not hig~
$ Mid_w2                    <fct> Not high, Not high, Not high, Not high, Hig~
$ Speeding_w2               <fct> Speeding, No speeding, No speeding, NA, No ~
$ FrTMis_w2                 <fct> Not applicable, Not applicable, Not applica~
$ NonrespRatio_w2           <dbl> 0.00000000, 0.02777778, 0.00000000, 0.42600~
$ MeanRootPairs_plifesat_w2 <dbl> NA, 0.4627821, NA, NA, 1.0000000, NA
$ MeanRootPairs_pinform_w2  <dbl> NA, 0.1105657, NA, NA, NA, NA
$ MeanRootPairs_pglob_w2    <dbl> NA, NA, NA, NA, NA, NA
$ MeanRootPairs_ppertrust_w2 <dbl> NA, NA, NA, NA, NA, NA
$ MeanRootPairs_pedugoal_w2 <dbl> NA, NA, NA, NA, NA, NA
$ MeanRootPairs_pfathbeh_w2 <dbl> NA, NA, NA, NA, NA, NA
$ MeanRootPairs_pmothbeh_w2 <dbl> NA, NA, NA, NA, NA, NA
$ MeanRootPairs_psocdes_w2  <dbl> NA, NA, NA, NA, NA, NA
$ MeanRootPairs_pschwartz_w2 <dbl> NA, NA, NA, NA, NA, NA
$ MeanRootPairs_pnarcism_w2 <dbl> NA, NA, NA, 1, NA, NA
$ MeanRootPairs_pdiscrim_w2 <dbl> NA, NA, NA, 1, NA, NA
$ MeanRootPairs_pgender_w2  <dbl> NA, NA, NA, NA, NA, NA
$ MeanRootPairs_pauthor_w2  <dbl> NA, NA, NA, 1, NA, NA
$ MeanRootPairs_pactivity_w2 <dbl> NA, NA, NA, 0.03717545, NA, NA
$ Participation_w2          <fct> Participated, Participated, Participated, P~
$ NondiffSt66_w2            <fct> NA, NA, NA, NA, NA, NA
$ Ratio_NDStr_w2            <dbl> NA, NA, NA, NA, NA, NA
$ NondiffWk66_w2            <fct> NA, Never, NA, Often (>=50%), Often (>=50%)~
$ Ratio_NDWk_w2             <dbl> NA, 0.00, NA, 0.75, 1.00, NA
$ INR66_w2                  <fct> Not high, High, Not high, High, Not high, NA
```

```
waves |> frq(NondiffSt66_w2)
```

```
NondiffSt66_w2 <categorical>
# total N=17023 valid N=16992 mean=2.98 sd=1.11

Value             |    N | Raw % | Valid % | Cum. %
--------------------------------------------------
Never             | 2113 | 12.41 |   12.44 |  12.44
Sometimes (0-50%) | 4206 | 24.71 |   24.75 |  37.19
Often (>=50%)     | 2550 | 14.98 |   15.01 |  52.20
Not participated  | 8123 | 47.72 |   47.80 | 100.00
<NA>              |   31 |  0.18 |    <NA> |   <NA>
```

18 Summary Table

```
p_load(gtsummary)

#   1. Helper function: tbl_summary by interview mode
make_tbl <- function(data, mode_label) {
  data |>
    filter(Interviewmode == mode_label) |>
    mutate(across(
      any_of(c("Extremes", "Midpoint", "FrTMis", "INR66")),
      ~ as.character(.x) |> na_if("Not applicable") |> factor() |> fct_drop()
    )) |>
    select(TargetGroup, Age_3grp, Gender, Migration, Education, ChildHH, EquivIncome,
           Extremes, Midpoint, Speeding, FrTMis, INR66,
           NondiffStrong, NondiffWeak, BreakOff) |>
    tbl_summary(
      statistic = all_categorical() ~ "{n} ({p}%)",
```

```
      missing   = "no",
      label     = list(
        TargetGroup            ~ "Target Group",
        Age_3grp      ~ "Age group",
        Gender        ~ "Gender",
        Migration     ~ "Migration background",
        Education     ~ "Education",
        ChildHH       ~ "Child in Household",
        EquivIncome   ~ "Equivalized Income",
        Extremes      ~ "Extremes Selection",
        Midpoint      ~ "Midpoint Selection",
        Speeding      ~ "Speeding",
        FrTMis        ~ "Open Question Nonresponse",
        INR66         ~ "Closed Question Nonresponse",
        NondiffStrong ~ "Strong Nondifferentiation",
        NondiffWeak   ~ "Weak Nondifferentiation",
        BreakOff      ~ "Break-Off"
      )
    ) |>
    bold_labels() |>
    italicize_levels()
}


#   2. Helper function: reshape and filter by wave
prep_wave <- function(suffix, wave_label,
                      extr, mid, speed, frtmis, inr, ndst, ndwk,
                      age_cat, equivinc,
                      part_var    = NULL,
                      sample_only = FALSE) {
  d <- waves |>
    select(
      TargetGroup = TargetGroup_w1,
      Education     = Education_w1,
      Gender        = Gender_w1,
      ChildHH       = ChildHH_w1,
      Migration     = Mig_w1,
      Age_cat       = all_of(age_cat),
      EquivIncome   = all_of(equivinc),
      Interviewmode = all_of(paste0("Interviewmode_", suffix)),
      BreakOff      = all_of(paste0("BreakOff_", suffix)),
      Extremes      = all_of(extr),
      Midpoint      = all_of(mid),
      Speeding      = all_of(speed),
      FrTMis        = all_of(frtmis),
      INR66         = all_of(inr),
      NondiffStrong = all_of(ndst),
      NondiffWeak   = all_of(ndwk),
      any_of(c(part_var, "Sample"))
    ) |>
    mutate(
      BreakOff = case_when(
        as.character(BreakOff) == "No Break-off" ~ "No",
        as.character(BreakOff) == "Break-off"    ~ "Yes",
        TRUE ~ NA_character_
      ) |> factor(levels = c("No", "Yes")),
```

```
      INR66 = case_when(
        as.character(INR66) == "Yes"              ~ "High",
        as.character(INR66) == "No"               ~ "Not high",
        as.character(INR66) == "High"             ~ "High",
        as.character(INR66) == "Not high"         ~ "Not high",
        as.character(INR66) == "Not participated" ~ NA_character_,
        TRUE ~ NA_character_
      ) |> factor(levels = c("Not high", "High")),

      NondiffStrong = if_else(
        as.character(NondiffStrong) == "Not participated", NA_character_,
        ↪  as.character(NondiffStrong)
      ) |> factor() |> fct_drop(),

      NondiffWeak = if_else(
        as.character(NondiffWeak) == "Not participated", NA_character_,
        ↪  as.character(NondiffWeak)
      ) |> factor() |> fct_drop()
    )

  if (!is.null(part_var)) d <- d |> filter(.data[[part_var]] == "Participated")
  if (sample_only)        d <- d |> filter(Sample == "Sample")

  d |>
    mutate(
      Age_3grp = case_when(
        str_detect(Age_cat, "T1") ~ "18-40 yrs",
        str_detect(Age_cat, "T2") ~ "41-60 yrs",
        str_detect(Age_cat, "T3") ~ "> 60 yrs"
      ) |> factor(levels = c("18-40 yrs", "41-60 yrs", "> 60 yrs")),
      Interviewmode = factor(Interviewmode, levels = c("PAPI", "CAWI"))
    ) |>
    filter(!is.na(Interviewmode))
}

#   3. Helper function: merge PAPI + CAWI with N in the header
build_wave_tbl <- function(d, wave_label) {
  n_papi <- d |> filter(Interviewmode == "PAPI") |> nrow()
  n_cawi <- d |> filter(Interviewmode == "CAWI") |> nrow()

  tbl_merge(
    tbls        = list(make_tbl(d, "PAPI"), make_tbl(d, "CAWI")),
    tab_spanner = c("**PAPI**", "**CAWI**")   # spanning header = mode
  ) |>
    modify_header(
      stat_0_1 ~ paste0("**PAPI**  \nN = ", format(n_papi, big.mark = ",")),
      stat_0_2 ~ paste0("**CAWI**  \nN = ", format(n_cawi, big.mark = ","))
    )
}

#   4. Prepare data
d_w1_sens <- prep_wave(
  suffix = "w1", wave_label = "Wave 1 (sensan)",
  extr  = "Extr_sensan_w1",        mid    = "Mid_sensan_w1",
  speed = "Speeding_sensan_w1",    frtmis = "FrTMis_sensan_w1",
```

```
  inr   = "INR66_sensan_w1",
  ndst  = "NondiffSt66_sensan_w1", ndwk   = "NondiffWk66_sensan_w1",
  age_cat = "Age_cat_sensan_w1",   equivinc = "EquivIncome_sensan_w1"
)

d_w1_samp <- prep_wave(
  suffix = "w1", wave_label = "Wave 1 (Sample)",
  extr  = "Extr_Sample_w1",        mid    = "Mid_Sample_w1",
  speed = "Speeding_Sample_w1",    frtmis = "FrTMis_Sample_w1",
  inr   = "INR66_Sample_w1",
  ndst  = "NondiffSt66_Sample_w1", ndwk   = "NondiffWk66_Sample_w1",
  age_cat = "Age_cat_Sample_w1",   equivinc = "EquivIncome_Sample_w1",
  sample_only = TRUE
)

d_w12 <- prep_wave(
  suffix = "w2", wave_label = "Wave 2",
  extr  = "Extr_w2",     mid    = "Mid_w2",
  speed = "Speeding_w2", frtmis = "FrTMis_w2",
  inr   = "INR66_w2",
  ndst  = "NondiffSt66_w2", ndwk = "NondiffWk66_w2",
  age_cat  = "Age_cat_sensan_w1", equivinc = "EquivIncome_sensan_w1",
  part_var = "Participation_w2"
)

d_w3 <- prep_wave(
  suffix = "w3", wave_label = "Wave 3",
  extr  = "Extr_w3",     mid    = "Mid_w3",
  speed = "Speeding_w3", frtmis = "FrTMis_w3",
  inr   = "INR66_w3",
  ndst  = "NondiffSt66_w3", ndwk = "NondiffWk66_w3",
  age_cat  = "Age_cat_sensan_w1", equivinc = "EquivIncome_sensan_w1",
  part_var = "Participation_w3"
)

#   5. Build tables
tbl_w1_sens <- build_wave_tbl(d_w1_sens, "Wave 1 (sensan)")
tbl_w1_samp <- build_wave_tbl(d_w1_samp, "Wave 1 (Sample)")
tbl_w12     <- build_wave_tbl(d_w12,     "Wave 2")
tbl_w3      <- build_wave_tbl(d_w3,      "Wave 3")

#   6. Merge all waves side by side
tbl_final <- tbl_merge(
  tbls        = list(tbl_w1_sens, tbl_w1_samp, tbl_w12, tbl_w3),
  tab_spanner = c("**Wave 1 - sensan**", "**Wave 1 - Sample**",
                  "**Wave 2**",            "**Wave 3**")
) |>
 # modify_caption("**Sample Characteristics by Wave and Interview Mode**") |>
  modify_header(label ~ "**Sample Characteristics**")

tbl_final
```

```
p_load(kableExtra, stringr)

latex_raw <- tbl_final |>
  as_kable_extra(
```

Sample Characteristics	Wave 1 – sensan		Wave 1 – Sampl	
	PAPI N = 10,556[1]	**CAWI** N = 6,467[1]	**PAPI** N = 5,670[1]	**CAWI**
Target Group				
Anchor	8,080 (77%)	4,972 (77%)	4,029 (71%)	2,709
HH member	2,476 (23%)	1,495 (23%)	1,641 (29%)	1,068
Age group				
18–40 yrs	2,372 (22%)	3,294 (51%)	1,297 (23%)	1,853
41–60 yrs	3,528 (33%)	2,164 (33%)	1,882 (33%)	1,267
> 60 yrs	4,656 (44%)	1,009 (16%)	2,490 (44%)	658
Gender				
Female	5,789 (55%)	2,724 (42%)	3,091 (55%)	1,594
Not female	4,767 (45%)	3,743 (58%)	2,579 (45%)	2,183
Migration background	1,575 (15%)	1,326 (21%)	780 (14%)	684
Education				
Non-tertiary (ISCED 0-4)	6,436 (61%)	3,177 (49%)	3,321 (59%)	1,827
Tertiary (ISCED 5-8)	3,669 (35%)	3,001 (46%)	2,158 (38%)	1,945
missing	451 (4.3%)	289 (4.5%)	191 (3.4%)	5 (
Child in Household				
No child in hh	8,073 (76%)	4,450 (69%)	4,459 (79%)	2,777
Child in hh	1,539 (15%)	1,590 (25%)	828 (15%)	959
missing	944 (8.9%)	427 (6.6%)	383 (6.8%)	41
Equivalized Income				
EquivIncome_T2medium	2,830 (27%)	1,589 (25%)	1,655 (29%)	1,039
EquivIncome_T1min	2,931 (28%)	1,489 (23%)	1,722 (30%)	972
EquivIncome_T3max	2,386 (23%)	2,033 (31%)	1,399 (25%)	1,294
missing	2,409 (23%)	1,356 (21%)	894 (16%)	472
Extremes Selection				
High	1,582 (15%)	806 (13%)	848 (15%)	514
Not high	8,973 (85%)	5,568 (87%)	4,822 (85%)	3,263
Midpoint Selection				
High	1,583 (15%)	851 (13%)	854 (15%)	533
Not high	8,972 (85%)	5,523 (87%)	4,816 (85%)	3,244
Speeding				
No speeding	0 (NA%)	6,059 (95%)	0 (NA%)	3,586
Speeding	0 (NA%)	341 (5.3%)	0 (NA%)	191
Open Question Nonresponse				
High	1,610 (15%)	865 (14%)	845 (15%)	483
Not high	8,946 (85%)	5,445 (86%)	4,825 (85%)	3,294
Closed Question Nonresponse				
High	3,198 (30%)	1,049 (16%)	1,701 (30%)	501
Not high	7,358 (70%)	5,418 (84%)	3,969 (70%)	3,270
Strong Nondifferentiation				
Never	1,781 (17%)	1,010 (16%)	931 (16%)	550
Often (>=50%)	3,055 (29%)	1,939 (31%)	1,612 (28%)	1,153
Sometimes (0-50%)	5,687 (54%)	3,365 (53%)	3,121 (55%)	2,071
Weak Nondifferentiation				
Never	2,689 (26%)	1,607 (26%)	1,423 (25%)	989
Often (>=50%)	4,147 (39%)	2,577 (41%)	2,210 (39%)	1,526
Sometimes (0-50%)	3,669 (35%)	2,092 (33%)	2,028 (36%)	1,250
Break-Off	0 (NA%)	542 (8.4%)	0 (NA%)	121

```
    format   = "latex",
    booktabs = TRUE,
    escape   = FALSE
  ) |>
  as.character()

latex_clean <- latex_raw |>
  # bold spans → \textbf{}
  str_replace_all(
    '<span style=" font-weight: bold;[^"]*"[^>]*>(.*?)</span>',
    '\\\\textbf{\\1}'
  ) |>
  # italic spans → \textit{}
  str_replace_all(
    '<span style="  font-style: italic;[^"]*"[^>]*>(.*?)</span>',
    '\\\\textit{\\1}'
  )

cat(latex_clean)
```

```
\begin{tabular}{lcccccccc}
\toprule
\multicolumn{1}{c}{ } & \multicolumn{2}{c}{\textbf{Wave 1 - sensan}} & \multicolumn{2}{c}{\textbf{Wave 1 -
Sample}} & \multicolumn{2}{c}{\textbf{Wave 2}} & \multicolumn{2}{c}{\textbf{Wave 3}} \\
\cmidrule(l{3pt}r{3pt}){2-3} \cmidrule(l{3pt}r{3pt}){4-5} \cmidrule(l{3pt}r{3pt}){6-7} \cmidrule(l{3pt}r{3
\textbf{Sample Characteristics} & \makecell[c]{\textbf{PAPI}\ \ \\N = 10,556} & \makecell[c]{\textbf{CAWI}
\midrule
\textbf{Target Group} &  &  &  &  &  &  &  & \\
\hspace{1em}\em{Anchor} & 8,080 (77\%) & 4,972 (77\%) & 4,029 (71\%) & 2,709 (72\%) & 3,626 (71\%) & 2,701
\hspace{1em}\em{HH member} & 2,476 (23\%) & 1,495 (23\%) & 1,641 (29\%) & 1,068 (28\%) & 1,479 (29\%) & 1,
\textbf{Age group} &  &  &  &  &  &  &  & \\
\hspace{1em}\em{18-40 yrs} & 2,372 (22\%) & 3,294 (51\%) & 1,297 (23\%) & 1,852 (49\%) & 1,120 (22\%) & 1,
\addlinespace
\hspace{1em}\em{41-60 yrs} & 3,528 (33\%) & 2,164 (33\%) & 1,882 (33\%) & 1,267 (34\%) & 1,672 (33\%) & 1,
\hspace{1em}\em{> 60 yrs} & 4,656 (44\%) & 1,009 (16\%) & 2,490 (44\%) & 658 (17\%) & 2,313 (45\%) & 618 (
\textbf{Gender} &  &  &  &  &  &  &  & \\
\hspace{1em}\em{Female} & 5,789 (55\%) & 2,724 (42\%) & 3,091 (55\%) & 1,594 (42\%) & 2,776 (54\%) & 1,657
\hspace{1em}\em{Not female} & 4,767 (45\%) & 3,743 (58\%) & 2,579 (45\%) & 2,183 (58\%) & 2,329 (46\%) & 2
\addlinespace
\textbf{Migration background} & 1,575 (15\%) & 1,326 (21\%) & 780 (14\%) & 684 (18\%) & 693 (14\%) & 655 (
\textbf{Education} &  &  &  &  &  &  &  & \\
\hspace{1em}\em{Non-tertiary (ISCED 0-4)} & 6,436 (61\%) & 3,177 (49\%) & 3,321 (59\%) & 1,827 (48\%) & 2,
\hspace{1em}\em{Tertiary (ISCED 5-8)} & 3,669 (35\%) & 3,001 (46\%) & 2,158 (38\%) & 1,945 (51\%) & 1,940
\hspace{1em}\em{missing} & 451 (4.3\%) & 289 (4.5\%) & 191 (3.4\%) & 5 (0.1\%) & 171 (3.3\%) & 17 (0.4\%)
\addlinespace
\textbf{Child in Household} &  &  &  &  &  &  &  & \\
\hspace{1em}\em{No child in hh} & 8,073 (76\%) & 4,450 (69\%) & 4,459 (79\%) & 2,777 (74\%) & 4,076 (80\%)
\hspace{1em}\em{Child in hh} & 1,539 (15\%) & 1,590 (25\%) & 828 (15\%) & 959 (25\%) & 707 (14\%) & 952 (2
\hspace{1em}\em{missing} & 944 (8.9\%) & 427 (6.6\%) & 383 (6.8\%) & 41 (1.1\%) & 322 (6.3\%) & 75 (2.0\%)
\textbf{Equivalized Income} &  &  &  &  &  &  &  & \\
\addlinespace
\hspace{1em}\em{EquivIncome\_T2medium} & 2,830 (27\%) & 1,589 (25\%) & 1,655 (29\%) & 1,039 (28\%) & 1,549
\hspace{1em}\em{EquivIncome\_T1min} & 2,931 (28\%) & 1,489 (23\%) & 1,722 (30\%) & 972 (26\%) & 1,447 (28\
\hspace{1em}\em{EquivIncome\_T3max} & 2,386 (23\%) & 2,033 (31\%) & 1,399 (25\%) & 1,294 (34\%) & 1,329 (2
\hspace{1em}\em{missing} & 2,409 (23\%) & 1,356 (21\%) & 894 (16\%) & 472 (12\%) & 780 (15\%) & 493 (13\%)
```

```
\textbf{Extremes Selection} &  &  &  &  &  &  &  & \\
\addlinespace
\hspace{1em}\em{High} & 1,582 (15\%) & 806 (13\%) & 848 (15\%) & 514 (14\%) & 786 (15\%) & 578 (15\%) & 40
\hspace{1em}\em{Not high} & 8,973 (85\%) & 5,568 (87\%) & 4,822 (85\%) & 3,263 (86\%) & 4,319 (85\%) & 3,2
\textbf{Midpoint Selection} &  &  &  &  &  &  &  & \\
\hspace{1em}\em{High} & 1,583 (15\%) & 851 (13\%) & 854 (15\%) & 533 (14\%) & 711 (14\%) & 503 (13\%) & 38
\hspace{1em}\em{Not high} & 8,972 (85\%) & 5,523 (87\%) & 4,816 (85\%) & 3,244 (86\%) & 4,394 (86\%) & 3,2
\addlinespace
\textbf{Speeding} &  &  &  &  &  &  &  & \\
\hspace{1em}\em{No speeding} & 0 (NA\%) & 6,059 (95\%) & 0 (NA\%) & 3,586 (95\%) & 0 (NA\%) & 3,566 (94\%)
\hspace{1em}\em{Speeding} & 0 (NA\%) & 341 (5.3\%) & 0 (NA\%) & 191 (5.1\%) & 0 (NA\%) & 219 (5.8\%) & 0 (
\textbf{Open Question Nonresponse} &  &  &  &  &  &  &  & \\
\hspace{1em}\em{High} & 1,610 (15\%) & 865 (14\%) & 845 (15\%) & 483 (13\%) & 842 (16\%) & 394 (10\%) & 41
\addlinespace
\hspace{1em}\em{Not high} & 8,946 (85\%) & 5,445 (86\%) & 4,825 (85\%) & 3,294 (87\%) & 4,263 (84\%) & 3,3
\textbf{Closed Question Nonresponse} &  &  &  &  &  &  &  & \\
\hspace{1em}\em{High} & 3,198 (30\%) & 1,049 (16\%) & 1,701 (30\%) & 501 (13\%) & 1,732 (34\%) & 1,114 (29
\hspace{1em}\em{Not high} & 7,358 (70\%) & 5,418 (84\%) & 3,969 (70\%) & 3,276 (87\%) & 3,373 (66\%) & 2,6
\textbf{Strong Nondifferentiation} &  &  &  &  &  &  &  & \\
\addlinespace
\hspace{1em}\em{Never} & 1,781 (17\%) & 1,010 (16\%) & 931 (16\%) & 550 (15\%) & 1,229 (24\%) & 884 (23\%)
\hspace{1em}\em{Often (>=50\%)} & 3,055 (29\%) & 1,939 (31\%) & 1,612 (28\%) & 1,153 (31\%) & 1,560 (31\%)
\hspace{1em}\em{Sometimes (0-50\%)} & 5,687 (54\%) & 3,365 (53\%) & 3,121 (55\%) & 2,072 (55\%) & 2,308 (4
\textbf{Weak Nondifferentiation} &  &  &  &  &  &  &  & \\
\hspace{1em}\em{Never} & 2,689 (26\%) & 1,607 (26\%) & 1,423 (25\%) & 989 (26\%) & 348 (6.8\%) & 228 (6.0\
\addlinespace
\hspace{1em}\em{Often (>=50\%)} & 4,147 (39\%) & 2,577 (41\%) & 2,210 (39\%) & 1,526 (40\%) & 1,166 (23\%)
\hspace{1em}\em{Sometimes (0-50\%)} & 3,669 (35\%) & 2,092 (33\%) & 2,028 (36\%) & 1,256 (33\%) & 3,588 (7
\textbf{Break-Off} & 0 (NA\%) & 542 (8.4\%) & 0 (NA\%) & 121 (3.2\%) & 0 (NA\%) & 65 (1.7\%) & 0 (NA\%) &
\bottomrule
\multicolumn{9}{l}{\rule{0pt}{1em}\textsuperscript{1} n (\%)}\\
\end{tabular}
```

18.0.1 Dummyize Sample

```
waves |> select(-c(contains("MeanRoot"), contains("Ratio"), lfd, pid, hid)) |> ai()
```

```
Rows: 6
Columns: 50
$ Education_w1                 <fct> Non-tertiary (ISCED 0-4), Tertiary (ISCED~
$ Employment_w1                <fct> "Part-time or flexible", "Full-time", "No~
$ Gender_w1                    <fct> Not female, Not female, Female, Not femal~
$ ChildHH_w1                   <fct> No child in hh, Child in hh, No child in ~
$ Mig_w1                       <fct> No, Yes, No, Yes, No, No
$ Interviewmode_w1             <fct> PAPI, CAWI, PAPI, PAPI, CAWI, CAWI
$ BreakOff_w1                  <fct> PAPI, No Break-off, PAPI, PAPI, No Break-~
$ ppanelcons_w1                <dbl+lbl> 1, 1, 1, 1, 1, 1
$ TargetGroup_w1               <fct> Anchor, Anchor, Anchor, Anchor, Anchor, A~
$ Age_cat_sensan_w1            <chr> "Age_T1[18.0-41.0]", "Age_T2[41.0-61.0]",~
$ EquivIncome_sensan_w1        <fct> EquivIncome_T1min, EquivIncome_T1min,~
$ ProportionSpeeding_sensan_w1 <dbl> NA, 0.000000000, NA, NA, 0.000000000, 0.0~
$ Speeding_sensan_w1           <fct> NA, No speeding, NA, NA, No speeding, No ~
$ FrTMis_sensan_w1             <fct> Not high, Not high, Not high, High, Not h~
$ Extr_sensan_w1               <fct> Not high, Not high, Not high, Not high, N~
$ Mid_sensan_w1                <fct> Not high, Not high, Not high, Not high, N~
$ Extr_Sample_w1               <fct> Not high, Not high, Not high, Not high, N~
```

```
$ Mid_Sample_w1                <fct> Not high, Not high, Not high, Not high, N~
$ EquivIncome_Sample_w1        <fct> EquivIncome_T1min, EquivIncome_T1min, Equ~
$ Age_cat_Sample_w1            <chr> "Age_T1[18.0-40.0]", "Age_T2[40.0-60.0]",~
$ ProportionSpeeding_Sample_w1 <dbl> NA, 0.000000000, NA, NA, 0.000000000, 0.0~
$ Speeding_Sample_w1           <fct> NA, No speeding, NA, NA, No speeding, No ~
$ FrTMis_Sample_w1             <fct> Not high, Not high, Not high, High, Not h~
$ Sample                       <chr> "Sample", "Sample", "Sample", "Sample", "~
$ Interviewmode_w2             <fct> PAPI, CAWI, PAPI, PAPI, CAWI, CAWI
$ BreakOff_w2                  <fct> PAPI, No Break-off, PAPI, PAPI, No Break-~
$ Extr_w2                      <fct> Not high, Not high, Not high, Not high, N~
$ Mid_w2                       <fct> Not high, Not high, Not high, High, Not h~
$ Speeding_w2                  <fct> NA, No speeding, NA, NA, No speeding, No ~
$ FrTMis_w2                    <fct> Not high, Not high, Not high, High, High,~
$ Participation_w2             <fct> Participated, Participated, Participated,~
$ Interviewmode_w3             <fct> PAPI, CAWI, CAWI, CAWI, CAWI, CAWI
$ BreakOff_w3                  <fct> PAPI, No Break-off, No Break-off, No Brea~
$ Extr_w3                      <fct> High, Not high, Not high, Not high, Not h~
$ Mid_w3                       <fct> Not high, High, High, Not high, Not high,~
$ Speeding_w3                  <fct> NA, No speeding, No speeding, No speeding~
$ FrTMis_w3                    <fct> Not high, Not high, High, High, Not high,~
$ Participation_w3             <fct> Participated, Participated, Participated,~
$ NondiffSt66_Sample_w1        <fct> Never, Sometimes (0-50%), Sometimes (0-50~
$ NondiffWk66_Sample_w1        <fct> Never, Often (>=50%), Sometimes (0-50%), ~
$ NondiffSt66_sensan_w1        <fct> Never, Sometimes (0-50%), Sometimes (0-50~
$ NondiffWk66_sensan_w1        <fct> Never, Often (>=50%), Sometimes (0-50%), ~
$ NondiffSt66_w2               <fct> Sometimes (0-50%), Sometimes (0-50%), Nev~
$ NondiffWk66_w2               <fct> Sometimes (0-50%), Sometimes (0-50%), Som~
$ NondiffSt66_w3               <fct> Sometimes (0-50%), Often (>=50%), Sometim~
$ NondiffWk66_w3               <fct> Sometimes (0-50%), Often (>=50%), Sometim~
$ INR66_sensan_w1              <fct> Not high, Not high, Not high, Not high, N~
$ INR66_Sample_w1              <fct> Not high, Not high, Not high, Not high, N~
$ INR66_w2                     <fct> Not high, Not high, Not high, High, Not h~
$ INR66_w3                     <fct> Not high, Not high, Not high, High, High,~
```

```r
# LCA input dummies, main sample (saved below as waves_dummies_sample.RData
# for the LCA scripts in the wave folders).
# Coding: 1 = reference/no, 2 = yes (Mplus CATEGORICAL, see README legend)
waves_dummies_Sample <-
  waves |>
  select(-contains("_sensan")) |>
  filter(Sample == "Sample") |>
  mutate(
    # --- Demographics ---
    AGET1   = case_when(Age_cat_Sample_w1 == "Age_T1[18.0-40.0]"  ~ 2, TRUE ~ 1),
    AGET3   = case_when(Age_cat_Sample_w1 == "Age_T3[60.0-97.0]"  ~ 2, TRUE ~ 1),
    MIGY    = case_when(Mig_w1            == "Yes"                ~ 2, TRUE ~ 1),
    GENF    = case_when(Gender_w1         == "Female"             ~ 2, TRUE ~ 1),

    EDTERT  = case_when(Education_w1 == "Tertiary (ISCED 5-8)"    ~ 2, TRUE ~ 1),
    EDNA    = case_when(Education_w1 == "missing"                 ~ 2, TRUE ~ 1),

    EMPP    = case_when(Employment_w1 == "Part-time or flexible"             ~ 2, TRUE ~ 1),
    EMPNO   = case_when(Employment_w1 == "Not currently employed, training"  ~ 2, TRUE ~ 1),
    EMPRET  = case_when(Employment_w1 == "Retirement"                        ~ 2, TRUE ~ 1),
    EMPNA   = case_when(Employment_w1 == "missing"                           ~ 2, TRUE ~ 1),

    KIDY    = case_when(ChildHH_w1 == "Child in hh" ~ 2, TRUE ~ 1),
```

```
  KIDNA   = case_when(ChildHH_w1 == "missing"      ~ 2, TRUE ~ 1),

  INCT1   = case_when(EquivIncome_Sample_w1 == "EquivIncome_T1min" ~ 2, TRUE ~ 1),
  INCT3   = case_when(EquivIncome_Sample_w1 == "EquivIncome_T3max" ~ 2, TRUE ~ 1),
  INCNA   = case_when(EquivIncome_Sample_w1 == "missing"           ~ 2, TRUE ~ 1),

  # --- Target Group ---
  TGANC   = case_when(TargetGroup_w1 == "Anchor" ~ 2, TRUE ~ 1),

  # --- Extreme Responding ---
  EXT1    = case_when(Extr_Sample_w1 == "High" ~ 2, TRUE ~ 1),
  EXT2    = case_when(Extr_w2        == "High" ~ 2, TRUE ~ 1),
  EXT3    = case_when(Extr_w3          == "High" ~ 2, TRUE ~ 1),

  # --- Midpoint Responding ---
  MID1    = case_when(Mid_Sample_w1 == "High" ~ 2, TRUE ~ 1),
  MID2    = case_when(Mid_w2        == "High" ~ 2, TRUE ~ 1),
  MID3    = case_when(Mid_w3          == "High" ~ 2, TRUE ~ 1),

  # --- Break-Off ---
  BO1     = case_when(BreakOff_w1 == "Break-off" ~ 2, TRUE ~ 1),
  BO2     = case_when(BreakOff_w2 == "Break-off" ~ 2, TRUE ~ 1),
  BO3     = case_when(BreakOff_w3   == "Break-off" ~ 2, TRUE ~ 1),

  # --- Speeding ---
  SPEE1   = case_when(Speeding_Sample_w1 == "Speeding" ~ 2, TRUE ~ 1),
  SPEE2   = case_when(Speeding_w2        == "Speeding" ~ 2, TRUE ~ 1),
  SPEE3   = case_when(Speeding_w3          == "Speeding" ~ 2, TRUE ~ 1),

  # --- Free Text Missingness ---
  FTM1    = case_when(FrTMis_Sample_w1 == "High" ~ 2, TRUE ~ 1),
  FTM2    = case_when(FrTMis_w2        == "High" ~ 2, TRUE ~ 1),
  FTM3    = case_when(FrTMis_w3          == "High" ~ 2, TRUE ~ 1),

  # --- Item Nonresponse ---
  INR1    = case_when(INR66_Sample_w1 == "High" ~ 2, TRUE ~ 1),
  INR2    = case_when(INR66_w2        == "High" ~ 2, TRUE ~ 1),
  INR3    = case_when(INR66_w3          == "High" ~ 2, TRUE ~ 1),

  # --- Nondiff Strong Sometimes ---
  NDSS1   = case_when(NondiffSt66_Sample_w1 == "Sometimes (0-50%)" ~ 2, TRUE ~ 1),
  NDSS2   = case_when(NondiffSt66_w2        == "Sometimes (0-50%)" ~ 2, TRUE ~ 1),
  NDSS3   = case_when(NondiffSt66_w3          == "Sometimes (0-50%)" ~ 2, TRUE ~ 1),

  # --- Nondiff Weak Often ---
  NDWO1   = case_when(NondiffWk66_Sample_w1 == "Often (>=50%)" ~ 2, TRUE ~ 1),
  NDWO2   = case_when(NondiffWk66_w2        == "Often (>=50%)" ~ 2, TRUE ~ 1),
  NDWO3   = case_when(NondiffWk66_w3          == "Often (>=50%)" ~ 2, TRUE ~ 1),

  # --- Nondiff Strong Often ---
  NDSO1   = case_when(NondiffSt66_Sample_w1 == "Often (>=50%)" ~ 2, TRUE ~ 1),
  NDSO2   = case_when(NondiffSt66_w2        == "Often (>=50%)" ~ 2, TRUE ~ 1),
  NDSO3   = case_when(NondiffSt66_w3          == "Often (>=50%)" ~ 2, TRUE ~ 1)
)
```

```
waves_dummies_Sample |> select(AGET1:NDSO3) |> ai()
```

```
Rows: 6
Columns: 43
$ AGET1  <dbl> 2, 1, 2, 2, 1, 2
$ AGET3  <dbl> 1, 1, 1, 1, 2, 1
$ MIGY   <dbl> 1, 2, 1, 2, 1, 1
$ GENF   <dbl> 1, 1, 2, 1, 2, 1
$ EDTERT <dbl> 1, 2, 1, 2, 2, 1
$ EDNA   <dbl> 1, 1, 1, 1, 1, 1
$ EMPP   <dbl> 2, 1, 1, 1, 1, 1
$ EMPNO  <dbl> 1, 1, 2, 1, 1, 1
$ EMPRET <dbl> 1, 1, 1, 1, 1, 1
$ EMPNA  <dbl> 1, 1, 1, 1, 1, 1
$ KIDY   <dbl> 1, 2, 1, 2, 1, 2
$ KIDNA  <dbl> 1, 1, 1, 1, 1, 1
$ INCT1  <dbl> 2, 2, 1, 1, 1, 1
$ INCT3  <dbl> 1, 1, 1, 1, 2, 1
$ INCNA  <dbl> 1, 1, 1, 2, 1, 1
$ TGANC  <dbl> 2, 2, 2, 2, 2, 2
$ EXT1   <dbl> 1, 1, 1, 1, 1, 1
$ EXT2   <dbl> 1, 1, 1, 1, 1, 1
$ EXT3   <dbl> 2, 1, 1, 1, 1, 1
$ MID1   <dbl> 1, 1, 1, 1, 1, 1
$ MID2   <dbl> 1, 1, 1, 2, 1, 1
$ MID3   <dbl> 1, 2, 2, 1, 1, 1
$ BO1    <dbl> 1, 1, 1, 1, 1, 1
$ BO2    <dbl> 1, 1, 1, 1, 1, 1
$ BO3    <dbl> 1, 1, 1, 1, 1, 1
$ SPEE1  <dbl> 1, 1, 1, 1, 1, 1
$ SPEE2  <dbl> 1, 1, 1, 1, 1, 1
$ SPEE3  <dbl> 1, 1, 1, 1, 1, 1
$ FTM1   <dbl> 1, 1, 1, 2, 1, 1
$ FTM2   <dbl> 1, 1, 1, 2, 2, 1
$ FTM3   <dbl> 1, 1, 2, 2, 1, 1
$ INR1   <dbl> 1, 1, 1, 1, 1, 1
$ INR2   <dbl> 1, 1, 1, 2, 1, 2
$ INR3   <dbl> 1, 1, 1, 2, 2, 2
$ NDSS1  <dbl> 1, 2, 2, 2, 2, 1
$ NDSS2  <dbl> 2, 2, 1, 1, 1, 1
$ NDSS3  <dbl> 2, 1, 2, 2, 1, 2
$ NDWO1  <dbl> 1, 2, 1, 1, 1, 1
$ NDWO2  <dbl> 1, 1, 1, 2, 2, 1
$ NDWO3  <dbl> 1, 2, 1, 1, 1, 1
$ NDSO1  <dbl> 1, 1, 1, 1, 1, 2
$ NDSO2  <dbl> 1, 1, 1, 2, 2, 1
$ NDSO3  <dbl> 1, 2, 1, 1, 2, 1
```

```
# Check 1: all variables contain only the values 1 and 2
waves_dummies_Sample |>
  select(AGET1:NDSO3) |>
  summarise(across(everything(), ~ paste(sort(unique(.)), collapse = ", "))) |>
  t()
```

```
       [,1]
AGET1  "1, 2"
```

```
AGET3  "1, 2"
MIGY   "1, 2"
GENF   "1, 2"
EDTERT "1, 2"
EDNA   "1, 2"
EMPP   "1, 2"
EMPNO  "1, 2"
EMPRET "1, 2"
EMPNA  "1, 2"
KIDY   "1, 2"
KIDNA  "1, 2"
INCT1  "1, 2"
INCT3  "1, 2"
INCNA  "1, 2"
TGANC  "1, 2"
EXT1   "1, 2"
EXT2   "1, 2"
EXT3   "1, 2"
MID1   "1, 2"
MID2   "1, 2"
MID3   "1, 2"
BO1    "1, 2"
BO2    "1, 2"
BO3    "1, 2"
SPEE1  "1, 2"
SPEE2  "1, 2"
SPEE3  "1, 2"
FTM1   "1, 2"
FTM2   "1, 2"
FTM3   "1, 2"
INR1   "1, 2"
INR2   "1, 2"
INR3   "1, 2"
NDSS1  "1, 2"
NDSS2  "1, 2"
NDSS3  "1, 2"
NDWO1  "1, 2"
NDWO2  "1, 2"
NDWO3  "1, 2"
NDSO1  "1, 2"
NDSO2  "1, 2"
NDSO3  "1, 2"
```

```
# Check 2: cross-tabulation as a plausibility check for individual variables
waves_dummies_Sample |> count(AGET1, Age_cat_Sample_w1)
```

```
# A tibble: 4 x 3
  AGET1 Age_cat_Sample_w1     n
  <dbl> <chr>             <int>
1     1 Age_T2[40.0-60.0]  3149
2     1 Age_T3[60.0-97.0]  3148
3     1 <NA>                  1
4     2 Age_T1[18.0-40.0]  3149
```

```
waves_dummies_Sample |> count(MIGY,  Mig_w1)
```

```
# A tibble: 2 x 3
```

```
   MIGY Mig_w1     n
  <dbl> <fct>  <int>
1     1 No      7983
2     2 Yes     1464
```

```
waves_dummies_Sample |> count(INR1,  INR66_Sample_w1)
```

```
# A tibble: 2 x 3
   INR1 INR66_Sample_w1     n
  <dbl> <fct>           <int>
1     1 Not high         7245
2     2 High             2202
```

```
waves_dummies_Sample |> count(NDSS1, NondiffSt66_Sample_w1)
```

```
# A tibble: 4 x 3
  NDSS1 NondiffSt66_Sample_w1     n
  <dbl> <fct>                  <int>
1     1 Never                  1481
2     1 Often (>=50%)          2765
3     1 <NA>                      8
4     2 Sometimes (0-50%)      5193
```

```
waves_dummies_Sample |> count(NDSO1, NondiffSt66_Sample_w1)
```

```
# A tibble: 4 x 3
  NDSO1 NondiffSt66_Sample_w1     n
  <dbl> <fct>                  <int>
1     1 Never                  1481
2     1 Sometimes (0-50%)      5193
3     1 <NA>                      8
4     2 Often (>=50%)          2765
```

```
waves |> filter(Sample == "Sample") |> count(Interviewmode_w1, NondiffSt66_Sample_w1)
```

```
# A tibble: 8 x 3
  Interviewmode_w1 NondiffSt66_Sample_w1     n
  <fct>            <fct>                  <int>
1 PAPI             Never                    931
2 PAPI             Sometimes (0-50%)       3121
3 PAPI             Often (>=50%)           1612
4 PAPI             <NA>                       6
5 CAWI             Never                    550
6 CAWI             Sometimes (0-50%)       2072
7 CAWI             Often (>=50%)           1153
8 CAWI             <NA>                       2
```

```
waves |> filter(Sample == "Sample") |>filter(is.na(NondiffSt66_Sample_w1)) |>
  select(!contains("w2")& !contains("w3")&!contains("sensan")) |> ai()
```

```
Rows: 6
Columns: 33
$ Education_w1                    <fct> Non-tertiary (ISCED 0-4), Tertiary ~
$ Employment_w1                   <fct> "Retirement", "Retirement", "Not cu~
$ Gender_w1                       <fct> Female, Female, Female, Female, Fem~
$ ChildHH_w1                      <fct> No child in hh, No child in hh, mis~
$ Mig_w1                          <fct> No, No, Yes, No, No, No
$ Interviewmode_w1                <fct> PAPI, PAPI, PAPI, PAPI, PAPI, PAPI
$ BreakOff_w1                     <fct> PAPI, PAPI, PAPI, PAPI, PAPI, PAPI
```

```
$ ppanelcons_w1                      <dbl+lbl> 1, 1, 1, 1, 1, 1
$ TargetGroup_w1                     <fct> Anchor, Anchor, Anchor, Anchor, Anc~
$ Extr_Sample_w1                     <fct> Not high, High, Not high, High, Not~
$ Mid_Sample_w1                      <fct> Not high, Not high, High, Not high,~
$ EquivIncome_Sample_w1              <fct> EquivIncome_T1min, EquivIncome_~
$ Age_cat_Sample_w1                  <chr> "Age_T3[60.0-97.0]", "Age_T3[60.0-9~
$ ProportionSpeeding_Sample_w1       <dbl> NA, NA, NA, NA, NA, NA
$ Speeding_Sample_w1                 <fct> NA, NA, NA, NA, NA, NA
$ FrTMis_Sample_w1                   <fct> Not high, Not high, High, Not high,~
$ MeanRootPairs_pcohes_Sample_w1     <dbl> NA, NA, NA, NA, NA, NA
$ MeanRootPairs_pinsttrust_Sample_w1 <dbl> NA, NA, 1, NA, NA, NA
$ MeanRootPairs_pfamexp_Sample_w1    <dbl> NA, NA, NA, NA, NA, NA
$ MeanRootPairs_pworkexp_Sample_w1   <dbl> NA, NA, NA, NA, NA, NA
$ MeanRootPairs_pneighexp_Sample_w1  <dbl> NA, NA, NA, NA, NA, NA
$ MeanRootPairs_punivtrad_Sample_w1  <dbl> NA, NA, NA, NA, NA, NA
$ MeanRootPairs_pgrpsymp_Sample_w1   <dbl> NA, NA, NA, NA, NA, NA
$ MeanRootPairs_pgrpknow_Sample_w1   <dbl> NA, NA, NA, NA, NA, NA
$ MeanRootPairs_pfrspcons_Sample_w1  <dbl> NA, NA, NA, NA, NA, NA
$ MeanRootPairs_ppopul_Sample_w1     <dbl> NA, 0.3309709, NA, NA, NA, NA
$ NonrespRatio_Sample_w1             <dbl> 0.6769231, 0.4384615, 0.5384615, 0.~
$ Sample                             <chr> "Sample", "Sample", "Sample", "Samp~
$ NondiffSt66_Sample_w1              <fct> NA, NA, NA, NA, NA, NA
$ Ratio_NDStr_w1_Sample              <dbl> NA, NA, NA, NA, NA, NA
$ NondiffWk66_Sample_w1              <fct> NA, Never, Often (>=50%), NA, NA, NA
$ Ratio_NDWk_w1_Sample               <dbl> NA, 0, 1, NA, NA, NA
$ INR66_Sample_w1                    <fct> High, High, High, High, High, High
```

```
# save for the LCA R scripts in the wave folders (they load ../waves_dummies_sample.RData)
save(waves_dummies_Sample, file = file.path(userpath, "waves_dummies_sample.RData"))
```

```
waves_dummies_sensan <-
  waves |>
  select(-contains("_Sample")) |>
  mutate(
    # --- Demographics ---
    AGET1   = case_when(Age_cat_sensan_w1 == "Age_T1[18.0-41.0]"  ~ 2, TRUE ~ 1),
    AGET3   = case_when(Age_cat_sensan_w1 == "Age_T3[61.0-101.0]"  ~ 2, TRUE ~ 1),
    MIGY    = case_when(Mig_w1            == "Yes"                  ~ 2, TRUE ~ 1),
    GENF    = case_when(Gender_w1         == "Female"               ~ 2, TRUE ~ 1),

    EDTERT  = case_when(Education_w1 == "Tertiary (ISCED 5-8)"    ~ 2, TRUE ~ 1),
    EDNA    = case_when(Education_w1 == "missing"                 ~ 2, TRUE ~ 1),

    EMPP    = case_when(Employment_w1 == "Part-time or flexible"             ~ 2, TRUE ~ 1),
    EMPNO   = case_when(Employment_w1 == "Not currently employed, training"  ~ 2, TRUE ~ 1),
    EMPRET  = case_when(Employment_w1 == "Retirement"                        ~ 2, TRUE ~ 1),
    EMPNA   = case_when(Employment_w1 == "missing"                           ~ 2, TRUE ~ 1),

    KIDY    = case_when(ChildHH_w1 == "Child in hh" ~ 2, TRUE ~ 1),
    KIDNA   = case_when(ChildHH_w1 == "missing"      ~ 2, TRUE ~ 1),

    INCT1   = case_when(EquivIncome_sensan_w1 == "EquivIncome_T1min" ~ 2, TRUE ~ 1),
    INCT3   = case_when(EquivIncome_sensan_w1 == "EquivIncome_T3max" ~ 2, TRUE ~ 1),
    INCNA   = case_when(EquivIncome_sensan_w1 == "missing"           ~ 2, TRUE ~ 1),

    # --- Target Group ---
    TGANC   = case_when(TargetGroup_w1 == "Anchor" ~ 2, TRUE ~ 1),
```

```
    # --- Extreme Responding ---
    EXT1    = case_when(Extr_sensan_w1 == "High" ~ 2, TRUE ~ 1),
    EXT2    = case_when(Extr_w2        == "High" ~ 2, TRUE ~ 1),
    EXT3    = case_when(Extr_w3          == "High" ~ 2, TRUE ~ 1),

    # --- Midpoint Responding ---
    MID1    = case_when(Mid_sensan_w1 == "High" ~ 2, TRUE ~ 1),

    # --- Break-Off ---
    BO1     = case_when(BreakOff_w1 == "Break-off" ~ 2, TRUE ~ 1),

    # --- Speeding ---
    SPEE1   = case_when(Speeding_sensan_w1 == "Speeding" ~ 2, TRUE ~ 1),

    # --- Free Text Missingness ---
    FTM1    = case_when(FrTMis_sensan_w1 == "High" ~ 2, TRUE ~ 1),

    # --- Item Nonresponse ---
    INR1    = case_when(INR66_sensan_w1 == "High" ~ 2, TRUE ~ 1),

    # --- Nondiff Strong Sometimes ---
    NDSS1   = case_when(NondiffSt66_sensan_w1 == "Sometimes (0-50%)" ~ 2, TRUE ~ 1),


    # --- Nondiff Weak Often ---
    NDWO1   = case_when(NondiffWk66_sensan_w1 == "Often (>=50%)" ~ 2, TRUE ~ 1),


    # --- Nondiff Strong Often ---
    NDSO1   = case_when(NondiffSt66_sensan_w1 == "Often (>=50%)" ~ 2, TRUE ~ 1),

    # Panelconsent
    PCONS = case_when(ppanelcons_w1 == 1 ~ 2, TRUE ~1)
  )

waves_dummies_sensan |> select(AGET1:PCONS) |> ai()
```

```
Rows: 6
Columns: 28
$ AGET1  <dbl> 2, 1, 2, 2, 1, 2
$ AGET3  <dbl> 1, 1, 1, 1, 1, 1
$ MIGY   <dbl> 1, 2, 1, 2, 1, 1
$ GENF   <dbl> 1, 1, 2, 1, 2, 1
$ EDTERT <dbl> 1, 2, 1, 2, 2, 1
$ EDNA   <dbl> 1, 1, 1, 1, 1, 1
$ EMPP   <dbl> 2, 1, 1, 1, 1, 1
$ EMPNO  <dbl> 1, 1, 2, 1, 1, 1
$ EMPRET <dbl> 1, 1, 1, 1, 1, 1
$ EMPNA  <dbl> 1, 1, 1, 1, 1, 1
$ KIDY   <dbl> 1, 2, 1, 2, 1, 2
$ KIDNA  <dbl> 1, 1, 1, 1, 1, 1
$ INCT1  <dbl> 2, 2, 1, 1, 1, 1
$ INCT3  <dbl> 1, 1, 1, 1, 2, 2
$ INCNA  <dbl> 1, 1, 1, 2, 1, 1
$ TGANC  <dbl> 2, 2, 2, 2, 2, 2
```

```
$ EXT1   <dbl> 1, 1, 1, 1, 1, 1
$ EXT2   <dbl> 1, 1, 1, 1, 1, 1
$ EXT3   <dbl> 2, 1, 1, 1, 1, 1
$ MID1   <dbl> 1, 1, 1, 1, 1, 1
$ BO1    <dbl> 1, 1, 1, 1, 1, 1
$ SPEE1  <dbl> 1, 1, 1, 1, 1, 1
$ FTM1   <dbl> 1, 1, 1, 2, 1, 1
$ INR1   <dbl> 1, 1, 1, 1, 1, 1
$ NDSS1  <dbl> 1, 2, 2, 2, 2, 1
$ NDWO1  <dbl> 1, 2, 1, 1, 1, 1
$ NDSO1  <dbl> 1, 1, 1, 1, 1, 2
$ PCONS  <dbl> 2, 2, 2, 2, 2, 2
```

```
# Check 1: all variables contain only the values 1 and 2
waves_dummies_sensan |>
  select(AGET1:PCONS) |>
  summarise(across(everything(), ~ paste(sort(unique(.)), collapse = ", "))) |>
  t()
```

```
       [,1]
AGET1  "1, 2"
AGET3  "1, 2"
MIGY   "1, 2"
GENF   "1, 2"
EDTERT "1, 2"
EDNA   "1, 2"
EMPP   "1, 2"
EMPNO  "1, 2"
EMPRET "1, 2"
EMPNA  "1, 2"
KIDY   "1, 2"
KIDNA  "1, 2"
INCT1  "1, 2"
INCT3  "1, 2"
INCNA  "1, 2"
TGANC  "1, 2"
EXT1   "1, 2"
EXT2   "1, 2"
EXT3   "1, 2"
MID1   "1, 2"
BO1    "1, 2"
SPEE1  "1, 2"
FTM1   "1, 2"
INR1   "1, 2"
NDSS1  "1, 2"
NDWO1  "1, 2"
NDSO1  "1, 2"
PCONS  "1, 2"
```

```
# Check 2: cross-tabulation as a plausibility check for individual variables
waves_dummies_sensan |> count(AGET1, Age_cat_sensan_w1)
```

```
# A tibble: 3 x 3
  AGET1 Age_cat_sensan_w1      n
  <dbl> <chr>              <int>
1     1 Age_T2[41.0-61.0]   5692
2     1 Age_T3[61.0-101.0]  5665
```

```
3     2 Age_T1[18.0-41.0]   5666
```

```
waves_dummies_sensan |> count(MIGY,  Mig_w1)
```

```
# A tibble: 2 x 3
   MIGY Mig_w1     n
  <dbl> <fct>  <int>
1     1 No     14122
2     2 Yes     2901
```

```
waves_dummies_sensan |> count(INR1,  INR66_sensan_w1)
```

```
# A tibble: 2 x 3
   INR1 INR66_sensan_w1     n
  <dbl> <fct>           <int>
1     1 Not high        12776
2     2 High             4247
```

```
waves_dummies_sensan |> count(NDSS1, NondiffSt66_sensan_w1)
```

```
# A tibble: 4 x 3
  NDSS1 NondiffSt66_sensan_w1     n
  <dbl> <fct>                 <int>
1     1 Never                  2791
2     1 Often (>=50%)          4994
3     1 <NA>                    186
4     2 Sometimes (0-50%)      9052
```

```
waves_dummies_sensan |> count(NDSO1, NondiffSt66_sensan_w1)
```

```
# A tibble: 4 x 3
  NDSO1 NondiffSt66_sensan_w1     n
  <dbl> <fct>                 <int>
1     1 Never                  2791
2     1 Sometimes (0-50%)      9052
3     1 <NA>                    186
4     2 Often (>=50%)          4994
```

```
# save for the sensitivity-analysis LCA R scripts in the wave1_*_sensan folders (they load
 ↪  ../waves_dummies_sensan.RData)
save(waves_dummies_sensan, file = file.path(userpath, "waves_dummies_sensan.RData"))
```

```
waves_dummies_Sample |> ai()
```

```
Rows: 6
Columns: 134
$ Education_w1                  <fct> Non-tertiary (ISCED 0-4), Tertiary ~
$ Employment_w1                 <fct> "Part-time or flexible", "Full-time~
$ Gender_w1                     <fct> Not female, Not female, Female, Not~
$ ChildHH_w1                    <fct> No child in hh, Child in hh, No chi~
$ Mig_w1                        <fct> No, Yes, No, Yes, No, No
$ Interviewmode_w1              <fct> PAPI, CAWI, PAPI, PAPI, CAWI, CAWI
$ BreakOff_w1                   <fct> PAPI, No Break-off, PAPI, PAPI, No ~
$ ppanelcons_w1                 <dbl+lbl> 1, 1, 1, 1, 1, 1
$ TargetGroup_w1                <fct> Anchor, Anchor, Anchor, Anchor, Anc~
$ Extr_Sample_w1                <fct> Not high, Not high, Not high, Not h~
$ Mid_Sample_w1                 <fct> Not high, Not high, Not high, N~
$ EquivIncome_Sample_w1         <fct> EquivIncome_T1min, EquivIncome_T1mi~
$ Age_cat_Sample_w1             <chr> "Age_T1[18.0-40.0]", "Age_T2[40.0-6~
```

```
$ ProportionSpeeding_Sample_w1       <dbl> NA, 0.000000000, NA, NA, 0.00000000~
$ Speeding_Sample_w1                 <fct> NA, No speeding, NA, NA, No speedin~
$ FrTMis_Sample_w1                   <fct> Not high, Not high, Not high, High,~
$ MeanRootPairs_pcohes_Sample_w1     <dbl> 0.18488460, 0.29474137, 0.08723639,~
$ MeanRootPairs_pinsttrust_Sample_w1 <dbl> 0.3528966, 0.2878394, 0.2635003, 0.~
$ MeanRootPairs_pfamexp_Sample_w1    <dbl> 0.20752013, 0.39817177, 0.39817177,~
$ MeanRootPairs_pworkexp_Sample_w1   <dbl> 0.1871581, 0.3370856, 0.2075201, 0.~
$ MeanRootPairs_pneighexp_Sample_w1  <dbl> 0.3981718, 0.1355473, 0.2291263, 0.~
$ MeanRootPairs_punivtrad_Sample_w1  <dbl> 0.00000000, 0.41853383, 0.08884133,~
$ MeanRootPairs_pgrpsymp_Sample_w1   <dbl> 0.3085557, 0.9392092, 0.1335752, 0.~
$ MeanRootPairs_pgrpknow_Sample_w1   <dbl> 0.2591710, 0.1856384, 0.2923571, 0.~
$ MeanRootPairs_pfrspcons_Sample_w1  <dbl> 0.33271084, 0.57380381, 0.05829193,~
$ MeanRootPairs_ppopul_Sample_w1     <dbl> 0.1675752, 0.4457581, 0.4826431, 0.~
$ NonrespRatio_Sample_w1             <dbl> 0.007518797, 0.000000000, 0.0000000~
$ Sample                             <chr> "Sample", "Sample", "Sample", "Samp~
$ Interviewmode_w2                   <fct> PAPI, CAWI, PAPI, PAPI, CAWI, CAWI
$ BreakOff_w2                        <fct> PAPI, No Break-off, PAPI, PAPI, No ~
$ Extr_w2                            <fct> Not high, Not high, Not high, Not h~
$ Mid_w2                             <fct> Not high, Not high, Not high, High,~
$ Speeding_w2                        <fct> NA, No speeding, NA, NA, No speedin~
$ FrTMis_w2                          <fct> Not high, Not high, Not high, High,~
$ NonrespRatio_w2                    <dbl> 0.004424779, 0.000000000, 0.0044052~
$ MeanRootPairs_plifesat_w2          <dbl> 0.4348399, 0.1503301, 0.1715455, NA~
$ MeanRootPairs_pinform_w2           <dbl> 0.1717391, 0.4763081, 0.3667624, 0.~
$ MeanRootPairs_pglob_w2             <dbl> 0.4070843, 0.2989322, 0.3020311, 0.~
$ MeanRootPairs_ppertrust_w2         <dbl> 0.5153833, 0.7283167, 0.1598988, 0.~
$ MeanRootPairs_pedugoal_w2          <dbl> 0.19679743, 0.36456403, 0.06445861,~
$ MeanRootPairs_pfathbeh_w2          <dbl> 0.35025775, 0.23486746, 0.06471109,~
$ MeanRootPairs_pmothbeh_w2          <dbl> 0.238819943, 0.336831991, 0.1909951~
$ MeanRootPairs_psocdes_w2           <dbl> 0.3261989, 0.1355473, 0.2435066, 0.~
$ MeanRootPairs_pschwartz_w2         <dbl> 0.1197530, 0.1624897, 0.1099251, 0.~
$ MeanRootPairs_pnarcism_w2          <dbl> 0.2206921, 0.5527051, 0.2206921, 0.~
$ MeanRootPairs_pdiscrim_w2          <dbl> 0.5735085, 0.8655340, 0.2455051, 0.~
$ MeanRootPairs_pgender_w2           <dbl> 0.05340808, 0.35590624, 0.16361588,~
$ MeanRootPairs_pauthor_w2           <dbl> 0.2836375, 0.3066667, 0.3742546, 0.~
$ MeanRootPairs_pactivity_w2         <dbl> 0.1778409, 0.1303047, 0.1156777, 0.~
$ Participation_w2                   <fct> Participated, Participated, Partici~
$ Interviewmode_w3                   <fct> PAPI, CAWI, CAWI, CAWI, CAWI, CAWI
$ BreakOff_w3                        <fct> PAPI, No Break-off, No Break-off, N~
$ Extr_w3                            <fct> High, Not high, Not high, Not high,~
$ Mid_w3                             <fct> Not high, High, High, Not high, Not~
$ Speeding_w3                        <fct> NA, No speeding, No speeding, No sp~
$ FrTMis_w3                          <fct> Not high, Not high, High, High, Not~
$ NonrespRatio_w3                    <dbl> 0.000000000, 0.000000000, 0.0000000~
$ MeanRootPairs_pbsjo_w3             <dbl> 0.10317649, 0.29885849, 0.09614518,~
$ MeanRootPairs_pbsjperc_w3          <dbl> 0.049768409, 0.398643346, 0.0961451~
$ MeanRootPairs_puss_w3              <dbl> 0.4723811, 0.5715852, 0.8063385, 0.~
$ MeanRootPairs_pladder_w3           <dbl> 0.4624456, 0.2322679, 0.3765782, 0.~
$ MeanRootPairs_pconcern_w3          <dbl> 0.06855286, 0.37033295, 0.22997025,~
$ MeanRootPairs_pconsp_w3            <dbl> 0.3559062, 1.0000000, 0.1993217, 0.~
$ MeanRootPairs_pgendeq_w3           <dbl> 0.65239785, 0.07321697, 0.22912631,~
$ MeanRootPairs_pregident_w3         <dbl> 0.70817282, 0.70817282, 0.56225922,~
$ MeanRootPairs_pinfrasat_w3         <dbl> 0.2854572, 0.6180168, 0.2256619, 0.~
$ MeanRootPairs_pclimattit_w3        <dbl> 0.1451898, 0.1822532, 0.2075201, 0.~
$ MeanRootPairs_pcliminfra_w3        <dbl> 0.188650693, 0.587294039, 0.0070313~
$ MeanRootPairs_ppopul_w3            <dbl> 0.2075201, 0.1871581, 0.5575747, 0.~
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$ MeanRootPairs_pantisem_w3         <dbl> 0.6523979, 0.2075201, 0.4401400, NA~
$ MeanRootPairs_prefmove_w3         <dbl> 0.22069207, 0.29266492, 0.32865134,~
$ MeanRootPairs_pimmparty_w3        <dbl> 0.1406323, 1.0000000, 0.4544524, 0.~
$ MeanRootPairs_ptaxparty_w3        <dbl> 0.2287803, 1.0000000, 0.5983903, 0.~
$ MeanRootPairs_pclimparty_w3       <dbl> 0.2880490, 1.0000000, 0.4703746, 0.~
$ MeanRootPairs_pgendparty_w3       <dbl> 0.08366354, 1.00000000, 0.61506511,~
$ Participation_w3                  <fct> Participated, Participated, Partici~
$ NondiffSt66_Sample_w1             <fct> Never, Sometimes (0-50%), Sometimes~
$ Ratio_NDStr_w1_Sample             <dbl> 0.0000000, 0.1666667, 0.1666667, 0.~
$ NondiffWk66_Sample_w1             <fct> Never, Often (>=50%), Sometimes (0-~
$ Ratio_NDWk_w1_Sample              <dbl> 0.00, 0.50, 0.25, 0.25, 0.25, 0.00
$ NondiffSt66_w2                    <fct> Sometimes (0-50%), Sometimes (0-50%~
$ Ratio_NDStr_w2                    <dbl> 0.4, 0.4, 0.0, 0.8, 0.6, 0.0
$ NondiffWk66_w2                    <fct> Sometimes (0-50%), Sometimes (0-50%~
$ Ratio_NDWk_w2                     <dbl> 0.1111111, 0.3333333, 0.1111111, 0.~
$ NondiffSt66_w3                    <fct> Sometimes (0-50%), Often (>=50%), S~
$ Ratio_NDStr_w3                    <dbl> 0.1000000, 0.6000000, 0.4000000, 0.~
$ NondiffWk66_w3                    <fct> Sometimes (0-50%), Often (>=50%), S~
$ Ratio_NDWk_w3                     <dbl> 0.3, 0.6, 0.4, 0.3, 0.4, 0.1
$ INR66_Sample_w1                   <fct> Not high, Not high, Not high, Not h~
$ INR66_w2                          <fct> Not high, Not high, Not high, High,~
$ INR66_w3                          <fct> Not high, Not high, Not high, High,~
$ AGET1                             <dbl> 2, 1, 2, 2, 1, 2
$ AGET3                             <dbl> 1, 1, 1, 1, 2, 1
$ MIGY                              <dbl> 1, 2, 1, 2, 1, 1
$ GENF                              <dbl> 1, 1, 2, 1, 2, 1
$ EDTERT                            <dbl> 1, 2, 1, 2, 2, 1
$ EDNA                              <dbl> 1, 1, 1, 1, 1, 1
$ EMPP                              <dbl> 2, 1, 1, 1, 1, 1
$ EMPNO                             <dbl> 1, 1, 2, 1, 1, 1
$ EMPRET                            <dbl> 1, 1, 1, 1, 1, 1
$ EMPNA                             <dbl> 1, 1, 1, 1, 1, 1
$ KIDY                              <dbl> 1, 2, 1, 2, 1, 2
$ KIDNA                             <dbl> 1, 1, 1, 1, 1, 1
$ INCT1                             <dbl> 2, 2, 1, 1, 1, 1
$ INCT3                             <dbl> 1, 1, 1, 1, 2, 1
$ INCNA                             <dbl> 1, 1, 1, 2, 1, 1
$ TGANC                             <dbl> 2, 2, 2, 2, 2, 2
$ EXT1                              <dbl> 1, 1, 1, 1, 1, 1
$ EXT2                              <dbl> 1, 1, 1, 1, 1, 1
$ EXT3                              <dbl> 2, 1, 1, 1, 1, 1
$ MID1                              <dbl> 1, 1, 1, 1, 1, 1
$ MID2                              <dbl> 1, 1, 1, 2, 1, 1
$ MID3                              <dbl> 1, 2, 2, 1, 1, 1
$ BO1                               <dbl> 1, 1, 1, 1, 1, 1
$ BO2                               <dbl> 1, 1, 1, 1, 1, 1
$ BO3                               <dbl> 1, 1, 1, 1, 1, 1
$ SPEE1                             <dbl> 1, 1, 1, 1, 1, 1
$ SPEE2                             <dbl> 1, 1, 1, 1, 1, 1
$ SPEE3                             <dbl> 1, 1, 1, 1, 1, 1
$ FTM1                              <dbl> 1, 1, 1, 2, 1, 1
$ FTM2                              <dbl> 1, 1, 1, 2, 2, 1
$ FTM3                              <dbl> 1, 1, 2, 2, 1, 1
$ INR1                              <dbl> 1, 1, 1, 1, 1, 1
$ INR2                              <dbl> 1, 1, 1, 2, 1, 2
$ INR3                              <dbl> 1, 1, 1, 2, 2, 2
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$ NDSS1                         <dbl> 1, 2, 2, 2, 2, 1
$ NDSS2                         <dbl> 2, 2, 1, 1, 1, 1
$ NDSS3                         <dbl> 2, 1, 2, 2, 1, 2
$ NDWO1                         <dbl> 1, 2, 1, 1, 1, 1
$ NDWO2                         <dbl> 1, 1, 1, 2, 2, 1
$ NDWO3                         <dbl> 1, 2, 1, 1, 1, 1
$ NDSO1                         <dbl> 1, 1, 1, 1, 1, 2
$ NDSO2                         <dbl> 1, 1, 1, 2, 2, 1
$ NDSO3                         <dbl> 1, 2, 1, 1, 2, 1
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